Targeted Estimation of Difference-in-Differences with Time-Varying Covariates and Staggered Adoption*

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Abstract

Staggered difference-in-differences designs face a control-group trade-off: using only never-treated units discards information, whereas imposing parallel trends against several untreated cohorts creates over-identifying restrictions and a different efficiency problem. We instead pool all not-yet-treated units into a sliding risk set and condition on the frozen pre-adoption history of each target cohort. The resulting sequential conditional-parallel-trends model accommodates dynamic selection through time-varying covariates and past outcomes while remaining exactly identified: apart from restrictions implied by the support of the outcome, it places no restriction on the observed data. We derive efficient influence functions for group-period and prespecified aggregate effects and propose a cross-fitted doubly robust one-step estimator and a pairwise targeted minimum loss-based estimator (TMLE) that updates each group-period target separately. The Horvitz–Thompson-normalized targeted estimator attains efficiency under a product-rate condition on the outcome-regression and inverse-risk-ratio errors, and a root-n bridge transfers the result to Hájek normalization under additional conditions. In a finite-support design, exact population calculations corroborate double-robust consistency, and the proposed one-step and pairwise targeted estimators show near-nominal Monte Carlo coverage at the largest sample size when at least one nuisance working model is correctly specified.

Keywords: staggered adoption; difference-in-differences; semiparametric efficiency; targeted estimation.

1 Introduction

Difference-in-differences (DiD) is a workhorse quasi-experimental method for evaluating policies with panel data, often in settings where treatment is adopted at different times across cohorts (staggered adoption). Because two-way fixed-effects regressions under staggered adoption can assign negative weights to heterogeneous group-period effects (Goodman-Bacon, 2021; Roth et al., 2023), modern staggered DiD analyses define the causal estimand separately from such model-dependent summaries. In many applications, selection into treatment may also depend on time-invariant characteristics and the pre-adoption evolution of time-varying covariates and past outcomes.

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Write $[T] := \{1, \ldots, T\}$. For $t \in \{0\} \cup [T]$, let D_t indicate treatment, and let $G \in [T] \cup \{\infty\}$ be the first treatment period, with $G = \infty$ for never-treated units. Write $Y_t(g)$ for the potential outcome under first treatment at g and $Y_t(\infty)$ for the never-treated path. Our granular target is the average treatment effect on the treated (ATT) of cohort g at period t,

$$\mathrm{ATT}(g,t) = \mathbb{E}\{Y_t(g) - Y_t(\infty) \mid G = g\}, \qquad t \geq g, \tag{1.1}$$

with prespecified event-study, groupwise, periodwise, and overall summaries formed as linear aggregates of these effects (Callaway and Sant'Anna, 2021; Callaway, 2022; Caetano and Callaway, 2024).

The central design problem is how to use untreated cohorts. Restricting the comparison to $G = \infty$ preserves an exactly identified formulation but discards future adopters. Imposing parallel trends simultaneously against several comparison cohorts creates over-identifying restrictions; efficient estimation then pertains to that more restrictive model and generally combines the associated moments using covariance information (Chen et al., 2025). Directly pooling a fixed terminal risk set uses more controls but suppresses the period-specific risk-set structure.

We reformulate conditional parallel trends in two linked ways. First, for target cohort g we *freeze* the conditioning information at a prespecified pre-adoption history $\mathcal{C}_{g-1}$ that may include time-varying covariates and past outcomes. The same history enters the untreated-increment regression and a sequential treatment-timing hazard, permitting adjustment for dynamic selection without conditioning on post-adoption variables. Second, at each increment k we compare cohort g with the single *sliding risk set* $\{G > k\}$. This replaces multiple cross-cohort restrictions by one history-adjusted sequential restriction and preserves exact identification: apart from restrictions implied by the support of the outcome, the model places no restriction on the observed law (Theorem 2.4). Related work permits time-varying covariates under alternative conditional-parallel-trends formulations (Caetano et al., 2022; Caetano and Callaway, 2024). Parallel-trends restrictions have also been used for longitudinal causal effects outside conventional DiD (Shahn et al., 2022; Renson et al., 2023). We study the staggered-adoption group-period ATT and its prespecified aggregates under the *frozen-history risk-set* model.

The paper makes three principal contributions. First, Section 2 establishes identification (Theorem 2.1) and derives the efficient influence functions of group-period and prespecified aggregate effects (Theorems 2.2 and 2.3). Within the maintained observed-data model, the causal assumptions restrict the observed law only through the support of the outcome, so these are also the efficient influence functions of the causal model whenever the identified conditional untreated means lie in the interior of the convex hull of that support at every relevant post-adoption period (Theorem 2.4). Second, building on doubly robust DiD estimation (Sant'Anna and Zhao, 2020), Section 3 develops a cross-fitted one-step estimator whose remainder depends on the product of the control-regression and fitted inverse-risk-ratio errors (Lemma 3.1); the ratio, rather than its two propensity components, is the primitive propensity-side nuisance. The estimator is doubly robust and attains the efficiency bound when both nuisances are consistently estimated and their errors satisfy the product-rate condition (Theorems 3.1 and 3.2). Third, Section 4 develops a pairwise targeted minimum loss-based estimator (TMLE), which updates each group-period target separately on a prespecified grid and reuses the updated estimates for the prespecified aggregates; under the empirical implementation conditions, this solves the granular and aggregate empirical score equations without additional targeting regressions (Theorem 4.1). Under Horvitz–Thompson (HT) and Hájek normalizations, the targeted estimators are doubly robust and attain the efficiency bound under their respective regularity conditions (Theorem 4.2, Proposition 4.1, and Corollary 4.1), with pointwise inference in Section 4.3. Appendix F develops a cohort-wise counterpart.

Section 2.4 also relates the framework to the cohort-by-cohort model of Chen et al. (2025). For a common target and conditioning history, their cohort-by-cohort restrictions imply our pooled risk-set restriction, so their model is a submodel whose efficiency bound is weakly smaller at common laws where the target is pathwise differentiable in the pooled model (Proposition 2.1). The common-history qualification is essential: our history $\mathcal{C}_{g-1}$ may include lagged outcomes and time-varying covariates, whereas their benchmark conditions on baseline Z. Without restrictions linking the two histories, the model inclusion does not follow and no efficiency-bound ordering is implied.

The remainder of the paper is organized as follows. Section 2 introduces the framework, the identifying assumptions, and the efficient influence functions. Sections 3 and 4 develop the cross-fitted one-step and targeted estimators. Section 5 reports Monte Carlo evidence, including a finite-support design in which the population limit of every procedure is computed exactly, and Section 6 concludes. Appendix A collects the principal main-text notation; Appendices B–E give the proofs for Sections 2–4; Appendix F develops cohort-wise targeting and the two-fluctuation diagnostic, with proofs; and Appendices G–J document the simulation implementation and report the complete results.

2 Framework, Assumptions, and Identification

2.1 Observed Data and Maintained Design

For unit i at time t, let Y_{it} denote the observed outcome, D_{it} treatment status, X_{it} a $p_X \times 1$ vector of time-varying covariates, and Z_i a $p_Z \times 1$ vector of time-invariant covariates. The observations $O_i = (Z_i, X_{i0}, \ldots, X_{iT}, Y_{i0}, \ldots, Y_{iT}, D_{i1}, \ldots, D_{iT})$, $i \in [n]$, are independent and identically distributed. Here $[n] := \{1, \ldots, n\}$. Define $G_i = \min\{t \in [T] : D_{it} = 1\}$ and $G_i = \infty$ if the unit is never treated. For integer periods $a \le b$, write $[a, b] = \{a, a+1, \ldots, b\}$; for truncation bounds, brackets denote real intervals. Let $X_{0:s} = (X_0^\top, \ldots, X_s^\top)^\top$. For a function f of O, write $\|f\|_2 := \{\mathbb{E}_{P_0}[f(O)^2]\}^{1/2}$ for its $L_2(P_0)$ norm.

Assumption B-STA (Staggered Treatment Adoption)**.** $D_0 = 0$. *For any* $s \in [T]$, $D_{s-1} = 1$ *implies* $D_s = 1$.

For $j \in [T] \cup \{\infty\}$, let $Y_t(j)$ and $X_t(j)$ denote the potential outcome and potential covariate vector under first adoption at j, with $j = \infty$ the never-treated path.

Assumption B-CON (Consistency)**.** $Y_t = Y_t(G)$ *and* $X_t = X_t(G)$ *for every* $t \in \{0\} \cup [T]$.

Under Assumption B-STA, the adoption date G defined above determines the whole treatment path through $D_s = \mathbb{I}(G \le s)$, $s \in \{0\} \cup [T]$, so the path and G carry the same information. We therefore work on the reduced observed-data sample space and write

$$O = (Z, X_{0:T}, Y_{0:T}, G).$$

This reduction incorporates staggered adoption into the support of the observed law rather than treating it as a moment restriction. Assumption B-CON equates the observed outcomes and covariates with those under the realized treatment path; it does not require the covariates to be unaffected by treatment. For a generic observed-data law P, write $p_g(P) = \mathbb{P}_P(G = g)$ and

$$\mathcal{G}_+(P) = \{g \in [T] : p_g(P) > 0\}, \qquad \mathcal{I}_+(P) = \{(g, t) : g \in \mathcal{G}_+(P),\ g \le t \le T\}.$$

At the true law P_0, abbreviate these sets as $\mathcal{G}_+$ and $\mathcal{I}_+$. Every target collection $\mathcal{S}$ below is a finite subset of $\mathcal{I}_+$. Population expectations and probabilities of observed-data quantities are written $\mathbb{E}_{P_0}$

and $\mathbb{P}_{P_0}$, or $\mathbb{E}_P$ and $\mathbb{P}_P$ when the law varies; an unsubscripted $\mathbb{E}$ or $\mathbb{P}$ refers to the full-data law, whose observed-data marginal is P_0, and it agrees with $\mathbb{E}_{P_0}$ or $\mathbb{P}_{P_0}$ for observed-data quantities. For a function f of O, $\mathbb{P}_n[f] := n^{-1}\sum_{i=1}^n f(O_i)$ denotes the empirical average.

2.2 History-Adjusted Assumptions and Identification

To identify $\mathrm{ATT}(g,t)$ without over-identifying restrictions across untreated cohorts and without conditioning on post-adoption variables, we pair two design elements at each increment $k \in [g,t]$. First, we condition on a prespecified pre-adoption history $\mathcal{C}_{g-1}$ of target cohort g, which stays fixed as k advances. Second, we compare the mean untreated increment of cohort g at k with that of the sliding risk set $\{G > k\}$ of units still untreated at k. We formalize the history and the almost-sure convention before stating the assumptions.

Conditioning history. For each cohort $g \in \mathcal{G}_+$, the analyst fixes before estimation a cutoff period $a_g \le g-1$ and a known measurable map κ_g, and sets

$$\mathcal{C}_{g-1} := \kappa_g\big(Z, X_{0:a_g}, Y_{0:a_g}\big). \tag{2.1}$$

The cutoff a_g is the last period used, and κ_g selects or summarizes the record available through that period. Both are prespecified and shared by every endpoint $t \ge g$ of cohort g. In the leading case κ_g is the identity, so $\mathcal{C}_{g-1}$ is the ordered tuple $(Z, X_{0:a_g}, Y_{0:a_g})$. Other choices retain selected variables, such as (Z, X_{a_g}, Y_{a_g}), or summaries, such as $\big(Z, (a_g+1)^{-1}\sum_{s=0}^{a_g} X_s, Y_{a_g}\big)$. The subscript $g-1$ indicates that the history precedes adoption; its information ends at a_g. The cutoff should precede the earliest plausible anticipatory or treatment-induced change in the conditioning variables. For example, if covariates may respond at $g-1$ to coming adoption, one may choose $a_g = g-2$ (for $g \ge 2$). An earlier cutoff changes the conditioning history, not the baseline period: outcomes are still differenced from period $g-1$, so the conditional-mean no-anticipation restriction at $s = g-1$ (Assumption 1 below) remains. Overlap must also hold for the chosen history. Write c for a generic value of $\mathcal{C}_{g-1}$.

Freezing the history keeps every conditioning variable pre-adoption for all increments and endpoints: an updated history could include outcomes or covariates affected by cohort g's treatment, potentially removing part of the total effect or inducing selection bias (Rosenbaum, 1984). Pre-adoption timing alone does not make a conditioning variable safe; it can still be a collider or make the parallel-trends restriction implausible. The identifying assumptions must therefore be defended for the chosen history. The assumptions below compare conditional outcome means given the same history across different cohort and risk-set events, so we first specify the relevant almost-sure convention.

Almost-sure statements. For a law P and an event B of positive probability, $\mathcal{L}_P(\mathcal{C}_{g-1} \mid B)$ denotes the conditional history distribution, and a property holds *almost surely on* B if its failure at $\mathcal{C}_{g-1}$ has zero joint probability with B. Conditional means are defined up to null sets of this distribution; under the overlap condition for P (the analogue of Assumption 3 below), the history laws given $G = g$, $G \ge g$, and $G > k$, $k \in [g,t]$, have the same null sets.

Assumptions 1–3 are stated for a single target pair $(g,t) \in \mathcal{I}_+$; for a target collection $\mathcal{S}$, they are imposed for every $(g,t) \in \mathcal{S}$.

Assumption 1 (History-Adjusted Conditional No Anticipation)**.** *For every $j \in \mathcal{G}_+$ with $j \ge g$,*

$$\mathbb{E}[Y_s(j) \mid \mathcal{C}_{g-1}, G = j] = \mathbb{E}[Y_s(\infty) \mid \mathcal{C}_{g-1}, G = j]$$

for each $s \in \{g-1, g, \ldots, \min\{t, j-1\}\}$, almost surely on $\{G = j\}$.

Assumption 1 conditions on the same history $\mathcal{C}_{g-1}$ as the parallel-trends restriction below, whereas conventional conditional no anticipation conditions on baseline covariates. It is implied by pointwise no anticipation, $Y_s(j) = Y_s(\infty)$ almost surely, which Caetano and Callaway (2024) impose with covariate-conditional parallel trends, and it implies unconditional no anticipation by iterated expectations; like conditional parallel trends, it is maintained and not testable. When Y_{g-1} belongs to $\mathcal{C}_{g-1}$, setting $(j, s) = (g, g-1)$ and using consistency gives $\mathbb{E}[Y_{g-1}(\infty) \mid \mathcal{C}_{g-1}, G = g] = Y_{g-1}$ almost surely on $\{G = g\}$, a conditional-mean statement rather than pointwise equality of $Y_{g-1}(g)$ and $Y_{g-1}(\infty)$. Excluding Y_{g-1} from $\mathcal{C}_{g-1}$ coarsens this anchor to $\mathbb{E}[Y_{g-1}(\infty) \mid \mathcal{C}_{g-1}, G = g] = \mathbb{E}[Y_{g-1} \mid \mathcal{C}_{g-1}, G = g]$ but does not remove the restriction at $s = g-1$. Because all conditioning is on the observed history, no no-anticipation restriction on the potential covariate processes $X_t(j)$ is needed.

Assumption 2 (History-Adjusted Conditional Parallel Trends)**.** *For any $k \in [g, t]$,*

$$\mathbb{E}[Y_k(\infty) - Y_{k-1}(\infty) \mid \mathcal{C}_{g-1}, G = g] = \mathbb{E}[Y_k(\infty) - Y_{k-1}(\infty) \mid \mathcal{C}_{g-1}, G > k]$$

almost surely on $\{G = g\}$.

Assumption 2 combines two choices. First, the comparison group is the *sliding risk set* $\{G > k\}$: cohorts leave it as they adopt, and it contains all units still untreated at the end of the adjacent-period increment k. It is thus the *largest* eligible untreated comparison pool for that increment; under Assumption 1, its observed increments identify the conditional untreated mean. This is an eligibility statement, not an efficiency ranking across models with different restrictions; Section 2.4 compares efficiency bounds for a common conditioning history. Second, the target-cohort history $\mathcal{C}_{g-1}$ is frozen across increments and endpoints. For any shared increment k, comparison membership depends on k, not on g or the endpoint t; the history and reweighting depend on g, and the target window depends on (g, t).

Under consistency and conditional no anticipation (Assumptions B-CON and 1), with integrable potential outcomes and the overlap condition stated below (Assumption 3), taking $a_g = g - 1$ with Y_{g-1} measurable with respect to $\mathcal{C}_{g-1}$ makes the first increment restriction ($k = g$) equivalent to

$$\mathbb{E}[Y_g(\infty) \mid \mathcal{C}_{g-1}, G = g] = \mathbb{E}[Y_g(\infty) \mid \mathcal{C}_{g-1}, G > g].$$

This is conditional mean exchangeability of the first post-adoption untreated outcome, rather than conditional independence (cf. Ding and Li, 2019); later increments retain the stated frozen-history restrictions. Including the lag changes the identifying restriction, while the estimand remains the cohort ATT in (1.1), averaged over cohort g's history distribution. The restrictions with and without the lag are generally not nested: conditioning on it can induce regression-to-the-mean bias when adoption depends on a persistent unit effect, whereas omitting it can leave bias under selection on transitory pre-adoption shocks; with both features, neither restriction need hold (Chabé-Ferret, 2015; Daw and Hatfield, 2018).

We also require overlap for the chosen history.

Assumption 3 (History-Adjusted Overlap)**.** *There exists a constant $\varepsilon > 0$ such that*

$$\mathbb{P}(G > k \mid \mathcal{C}_{g-1}, G \geq g) \in (\varepsilon, 1 - \varepsilon) \quad \textit{for all } k \in \{g, g+1, \ldots, t\}, \quad \textit{almost surely on } \{G \geq g\}.$$

Write $q_{g,k}(\mathcal{C}_{g-1}) := \mathbb{P}(G > k \mid \mathcal{C}_{g-1}, G \geq g)$. The lower bound ensures a comparison group of later adopters and never-treated units at every $k \in [g, t]$. The upper bound at $k = g$ gives the target cohort positive conditional mass in the entry risk set, $1 - q_{g,g}(\mathcal{C}_{g-1}) = \mathbb{P}(G = g \mid \mathcal{C}_{g-1}, G \geq g) > \varepsilon$. Since $q_{g,k}$ is nonincreasing in k, this upper bound also holds for $k > g$.

With the target-cohort discrete-time hazard

$$\lambda_{g,s}(\mathcal{C}_{g-1}) := \mathbb{P}(G = s \mid \mathcal{C}_{g-1}, G \geq s), \tag{2.2}$$

the conditional survival factors as $q_{g,k} = \prod_{s=g}^{k}\{1 - \lambda_{g,s}(\mathcal{C}_{g-1})\}$, so Assumption 3 holds if, for some $\varepsilon' > 0$, $\lambda_{g,g} \geq \varepsilon'$ and $\lambda_{g,s} \leq 1-\varepsilon'$ for $s \in [g,t]$ almost surely. No lower bound is required on the future hazards $\lambda_{g,s}$, $s > g$, which may be zero, or on the entry probability $e_g(\mathcal{C}_{g-1}) := \mathbb{P}(G \geq g \mid \mathcal{C}_{g-1})$.

As a domain convention, all conditional objects for cohort g—the hazards, $q_{g,k}$, and the inverse-risk ratio $r_{g,k} = \lambda_{g,g}/q_{g,k}$ of Section 3—are defined almost surely on $\{e_g(\mathcal{C}_{g-1}) > 0\}$, where Assumption 3 is imposed. On the complement, $\{G = g\}$ and $\{G > k\}$ are null, so every score, remainder, and norm introduced below vanishes there; norms of these objects are taken on $\{e_g > 0\}$.

Relation to fixed comparison pools. The never-treated and not-yet-treated estimands of Callaway and Sant'Anna (2021) condition on baseline covariates Z and use the never-treated pool $B_t = \{G = \infty\}$ or the terminal pool $B_t = \{G > t\}$ of units not yet treated at t. Identification of a particular endpoint ATT uses the long-difference restriction

$$\mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) \mid Z, G = g] = \mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) \mid Z, B_t].$$

Imposing this equality increment by increment, with the terminal pool B_t in every increment, is sufficient but stronger. When several endpoints t are targeted, those increment-level restrictions equate the means of different terminal pools for the same increment, whereas the long-difference restrictions alone need not; in our formulation the pool for increment k is $\{G > k\}$ whatever the endpoint.

The maintained not-yet-treated assumption of Callaway and Sant'Anna (2021) (their Assumption 5, with zero anticipation) is stronger than an endpoint long-difference restriction: for each relevant increment k, it compares the target cohort with every pool $\{G > s\}$ satisfying $k \leq s \leq T$ and $s < \overline{g}$, where $\overline{g}$ is the largest value of G, infinite if a never-treated cohort exists. With a common conditioning history, its pool $s = k$ gives Assumption 2; the additional pools impose further restrictions. What distinguishes our formulation is therefore not the use of adjacent increments but their pairing with the contemporaneous risk set $\{G > k\}$ and a frozen target-cohort history.

Let $\Delta Y_k := Y_k - Y_{k-1}$ and, for potential outcomes, $\Delta Y_k(j) := Y_k(j) - Y_{k-1}(j)$ for $j \in [T] \cup \{\infty\}$. The following condition collects the identifying assumptions and the integrability they require.

Condition 1 (Identification domain)**.** *For every target pair $(g,t) \in \mathcal{S}$, $p_g > 0$, Assumptions B-STA, B-CON, and 1–3 hold, and each potential-outcome level appearing in Assumptions 1–2 and in* $\mathrm{ATT}(g,t)$ *is integrable.*

Theorem 2.1 generalizes the two-period DiD representation to a sequential risk-set adjustment based on pre-adoption histories.

Theorem 2.1 (Identification)**.** *Suppose Condition 1 holds for $\mathcal{S} = \{(g,t)\}$. Then* $\mathrm{ATT}(g,t)$ *in (1.1) is nonparametrically identified by*

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \sum_{k=g}^{t} \mathbb{E}\Big[\mathbb{E}\big[\Delta Y_k \mid \mathcal{C}_{g-1}, G > k\big] \mid G = g\Big].$$

We aggregate the group-period effects $\tau_{g,t} := \mathrm{ATT}(g,t)$ with a known weighting function $\Omega_{g,t}(O)$, typically a cohort indicator, and $\omega_{g,t} = \mathbb{E}[\Omega_{g,t}(O)]$. Define the total population weight

mass by $W_{\mathcal{S}} := \sum_{(g,t)\in\mathcal{S}} \omega_{g,t}$. Write $\mathcal{G}_{\mathcal{S}} := \{g : \exists\, t \text{ with } (g,t) \in \mathcal{S}\}$ for the represented cohorts. A *generic aggregated parameter* Θ over an index set $\mathcal{S}$ is

$$\Theta := \frac{1}{W_{\mathcal{S}}} \sum_{(g,t)\in\mathcal{S}} \omega_{g,t}\tau_{g,t}, \tag{2.3}$$

an identified normalized linear combination whenever $W_{\mathcal{S}} \neq 0$, and a weighted average when the $\omega_{g,t}$ are nonnegative.

Leading aggregates. With $\Omega_{g,t}(O) = \mathbb{I}(G = g)$ and $\mathcal{S} = \{(g, g+e) : g \in \mathcal{G}_e\}$, where $\mathcal{G}_e = \{g \in \mathcal{G}_+ : g + e \leq T\} \neq \varnothing$, Θ is the event-study parameter

$$\mathrm{ES}(e) := \sum_{g\in\mathcal{G}_e} \frac{\mathbb{P}(G=g)}{\sum_{g'\in\mathcal{G}_e}\mathbb{P}(G=g')}\,\mathrm{ATT}(g,g+e), \qquad e = 0,1,\ldots,T-1;$$

with $\mathcal{S} = \{(g,s) : g \in \mathcal{G}_+,\ g \leq s\}$ it is the periodwise ATT at period s, and with $\mathcal{S} = \mathcal{I}_+$ the overall ATT, $\sum_{g\in\mathcal{G}_+}\sum_{t=g}^{T}\mathbb{P}(G=g)\,\mathrm{ATT}(g,t)/\sum_{g\in\mathcal{G}_+}\mathbb{P}(G=g)(T-g+1)$. With $\Omega_{g,t}(O) = 1$ and $\mathcal{S} = \{(\tilde{g},t) : \tilde{g} \leq t \leq T\}$, Θ is the groupwise ATT $(T-\tilde{g}+1)^{-1}\sum_{t=\tilde{g}}^{T}\mathrm{ATT}(\tilde{g},t)$ of cohort $\tilde{g} \in \mathcal{G}_+$, and the singleton $\mathcal{S} = \{(g,t)\}$ gives $\mathrm{ATT}(g,t)$ itself.

2.3 Efficient Influence Function

The identification theorem expresses the causal target as a functional of the observed-data law. We now derive its canonical gradient, the efficient influence function (EIF), in the observed-data model $\mathcal{M}_{\mathrm{obs}}(\mathcal{S})$ defined below.

Condition 2 (Moments and nondegeneracy)**.** *(i)* $\max_{k\in[g,t]}\mathbb{E}_{P_0}[(\Delta Y_k)^2] < \infty$ *for a granular target; (ii) for a finite aggregate index set* $\mathcal{S}$*, clause (i) holds for every target window,* $\mathbb{E}_{P_0}[\Omega_{g,t}^2] < \infty$ *for every* $(g,t) \in \mathcal{S}$*, and* $W_{\mathcal{S}} \neq 0$*; and (iii) the corresponding influence-function variance lies in* $(0,\infty)$*.*

Let $\mathcal{M}_{\mathrm{obs}}(\mathcal{S})$ be the collection of laws P on the reduced sample space such that, for every $(g,t) \in \mathcal{S}$: $p_g(P) > 0$; for some $\varepsilon_P > 0$, $q_{g,k}^{P}(\mathcal{C}_{g-1}) \in (\varepsilon_P, 1-\varepsilon_P)$ almost surely on $\{G \geq g\}$ for every $k \in [g,t]$; $\mathbb{E}_P[|Y_{g-1}|\,\mathbb{I}(G \geq g)] < \infty$; and the analogues under P of clauses (i)–(ii) of Condition 2 hold as applicable. The law-specific margin ε_P makes the identified functional of Theorem 2.1 well defined on all of $\mathcal{M}_{\mathrm{obs}}(\mathcal{S})$. The variance requirement in clause (iii) is a condition for studentization, not a restriction defining the model. $\mathcal{M}_{\mathrm{obs}}(\mathcal{S})$ is an observed-data model; Section 2.4 relates it to the causal model.

Throughout, pathwise differentiability, canonical gradients, and efficiency bounds are relative to quadratic-mean differentiable paths whose likelihood ratios are locally bounded above and away from zero. Appendix B.1 gives the precise convention and explains how it controls tail perturbations while retaining the bounded tilts used to obtain a full tangent space.

Lemma 2.1 (Reduced-Support Observed-Data Tangent Space)**.** *At every* $P_0 \in \mathcal{M}_{\mathrm{obs}}(\mathcal{S})$*, the tangent space of* $\mathcal{M}_{\mathrm{obs}}(\mathcal{S})$ *is* $L_2^0(P_0)$*, the full space of mean-zero, square-integrable functions of* O*.*

Lemma 2.1 shows that the observed-data model has a full tangent space despite its support, overlap, and moment requirements. Its proof is in Appendix B.3.

Theorem 2.2 (Canonical Gradient for a Granular ATT)**.** *Suppose Condition 1 holds for* $\mathcal{S} = \{(g,t)\}$ *and Condition 2(i) holds. In* $\mathcal{M}_{\text{obs}}(\{(g,t)\})$*, the canonical gradient of the identified functional* $\text{ATT}(g,t)$ *is*

$$\varphi^{\text{eff}}\{\,\text{ATT}(g,t)\} = \sum_{k=g}^{t}\left(\frac{\mathbb{I}(G=g)}{\mathbb{P}(G=g)} - \frac{\mathbb{I}(G>k)}{\mathbb{P}(G=g)}\frac{\mathbb{P}(G=g \mid \mathcal{C}_{g-1}, G\geq g)}{\mathbb{P}(G>k \mid \mathcal{C}_{g-1}, G\geq g)}\right) \times \Big(\Delta Y_k - \mathbb{E}[\Delta Y_k \mid \mathcal{C}_{g-1}, G>k]\Big) - \frac{\mathbb{I}(G=g)}{\mathbb{P}(G=g)}\,\text{ATT}(g,t).$$

The EIF uses the increments ΔY_k of the sliding risk set $\{G>k\}$, reweighted through the fixed pre-adoption history $\mathcal{C}_{g-1}$. To simplify the notation, let $\phi_{g,k}(O)$ denote the uncentered doubly robust increment signal for cohort g at period k, written with the ratio $\mathbb{P}(G=g \mid \mathcal{C}_{g-1})/\mathbb{P}(G>k \mid \mathcal{C}_{g-1})$, which equals the ratio in Theorem 2.2 on $\{e_g>0\}$ because the entry probability $e_g(\mathcal{C}_{g-1}) = \mathbb{P}(G\geq g \mid \mathcal{C}_{g-1})$ cancels:

$$\phi_{g,k}(O) := \left(\frac{\mathbb{I}(G=g)}{\mathbb{P}(G=g)} - \frac{\mathbb{I}(G>k)}{\mathbb{P}(G=g)}\frac{\mathbb{P}(G=g \mid \mathcal{C}_{g-1})}{\mathbb{P}(G>k \mid \mathcal{C}_{g-1})}\right)\Big(\Delta Y_k - \mathbb{E}[\Delta Y_k \mid \mathcal{C}_{g-1}, G>k]\Big). \tag{2.4}$$

The EIF of Theorem 2.2 can then be written as

$$\psi_{g,t}(O) := \varphi^{\text{eff}}\{\tau_{g,t}\}(O) = \sum_{k=g}^{t}\phi_{g,k}(O) - \frac{\mathbb{I}(G=g)}{\mathbb{P}(G=g)}\tau_{g,t}. \tag{2.5}$$

Theorem 2.3 (Canonical Gradient for Aggregated ATTs)**.** *Suppose Conditions 1 and 2(i)–(ii) hold for the finite index set* $\mathcal{S}$*. In* $\mathcal{M}_{\text{obs}}(\mathcal{S})$*, the canonical gradient of the generic aggregated parameter* Θ *defined in (2.3) is*

$$\varphi^{\text{eff}}\{\Theta\} = \frac{1}{\sum_{\mathcal{S}}\omega_{g,t}}\sum_{(g,t)\in\mathcal{S}}\big[\omega_{g,t}\psi_{g,t}(O) + \tau_{g,t}\big(\Omega_{g,t}(O) - \omega_{g,t}\big)\big] - \frac{\Theta}{\sum_{\mathcal{S}}\omega_{g,t}}\sum_{(g,t)\in\mathcal{S}}\big[\Omega_{g,t}(O) - \omega_{g,t}\big]. \tag{2.6}$$

Substituting (2.5) into (2.6) and using $\sum_{(g,t)\in\mathcal{S}}\omega_{g,t}(\tau_{g,t}-\Theta) = 0$, which follows from the definition of Θ, gives the decomposition

$$\varphi^{\text{eff}}\{\Theta\} = \frac{1}{\sum_{\mathcal{S}}\omega_{g,t}}\sum_{(g,t)\in\mathcal{S}}\omega_{g,t}\left[\sum_{k=g}^{t}\phi_{g,k}(O)\right] \tag{2.7}$$

$$+ \frac{1}{\sum_{\mathcal{S}}\omega_{g,t}}\sum_{(g,t)\in\mathcal{S}}\left[\Omega_{g,t}(O) - \omega_{g,t}\frac{\mathbb{I}(G=g)}{\mathbb{P}(G=g)}\right](\tau_{g,t}-\Theta) \tag{2.8}$$

$$- \frac{1}{\sum_{\mathcal{S}}\omega_{g,t}}\left[\sum_{(g,t)\in\mathcal{S}}\omega_{g,t}\frac{\mathbb{I}(G=g)}{\mathbb{P}(G=g)}\right]\Theta. \tag{2.9}$$

In (2.7)–(2.9), the correction (2.8) has mean zero and vanishes pointwise for the leading fixed- and cohort-share-weighted summaries, and with empirical weights its sample average cancels from the point equation. Variance calculations nevertheless retain the unit-level aggregation-weight score

$(\tau_{g,t}-\Theta)\{\Omega_{g,t}(O)-\omega_{g,t}\}$ from (2.6), evaluated at the estimates of Sections 3.2 and 4.3; it is generally nonzero unit by unit (Appendix B.2).

Write $V_\Theta := \mathbb{E}_{P_0}[\varphi^{\text{eff}}\{\Theta\}(O)^2]$ for the variance of the canonical gradient. By Lemma 2.1, V_Θ is the semiparametric efficiency bound for Θ in $\mathcal{M}_{\text{obs}}(\mathcal{S})$, and Condition 2(iii) requires $V_\Theta \in (0,\infty)$; for a singleton $\mathcal{S} = \{(g,t)\}$ it is the bound for $\tau_{g,t}$.

2.4 Outcome Support and the Causal Model

Theorems 2.2 and 2.3 are stated in the observed-data model $\mathcal{M}_{\text{obs}}(\mathcal{S})$; the causal assumptions restrict the observed law further only through the support of the outcome. Let $\mathcal{Y} \subseteq \mathbb{R}$ be a known nonempty closed set with $Y_s \in \mathcal{Y}$ for every s, and let $\text{conv}(\mathcal{Y})$ be its convex hull. For a cohort g represented in $\mathcal{S}$, let $t_{\mathcal{S}}(g) := \max\{t : (g,t) \in \mathcal{S}\}$. For an observed law P, write $m^P_{g,k,0}(c) := \mathbb{E}_P[\Delta Y_k \mid \mathcal{C}_{g-1} = c, G > k]$ for the control regression under P, so that $m^{P_0}_{g,k,0} = m^0_{g,k,0}$ (Section 3.1), and define the untreated prefix means

$$\mu^P_{g,s}(c) := \mathbb{E}_P[Y_{g-1} \mid \mathcal{C}_{g-1} = c, G = g] + \sum_{k=g}^{s} m^P_{g,k,0}(c), \qquad s \in [g, t_{\mathcal{S}}(g)]. \tag{2.10}$$

If $P \in \mathcal{M}_{\text{obs}}(\mathcal{S})$ is the observed marginal of a full-data law satisfying Assumptions B-CON, 1 and 2 and the integrability clause of Condition 1, the argument of Theorem 2.1 shows that $\mu^P_{g,s}(\mathcal{C}_{g-1})$ is the conditional mean of $Y_s(\infty)$ given $\mathcal{C}_{g-1}$ in cohort g; because the history is shared by every endpoint, these are successive means of one untreated path.

We call P *prefix feasible* if, for every represented g and every $s \in [g, t_{\mathcal{S}}(g)]$,

$$\mu^P_{g,s}(\mathcal{C}_{g-1}) \in \text{conv}(\mathcal{Y}) \qquad \mathcal{L}_P(\mathcal{C}_{g-1} \mid G = g)\text{-almost surely}, \tag{2.11}$$

and *strictly prefix feasible* if, in addition, each $\mu^P_{g,s}(\mathcal{C}_{g-1})$ lies $\mathcal{L}_P(\mathcal{C}_{g-1} \mid G = g)$-almost surely in the interior of $\text{conv}(\mathcal{Y})$.

Let $\mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$ be the set of laws $P \in \mathcal{M}_{\text{obs}}(\mathcal{S})$ with $\mathcal{Y}$-valued outcomes, each the observed-data marginal of a full-data law with $\mathcal{Y}$-valued potential outcomes satisfying Assumptions B-STA, B-CON, 1, and 2 and the integrability clause of Condition 1 for every $(g,t) \in \mathcal{S}$.

Theorem 2.4 (Outcome Support and the Causal Model)**.** *Let $\mathcal{Y} \subseteq \mathbb{R}$ be nonempty and closed, and let $\mathcal{S}$ be finite. (i) A law $P \in \mathcal{M}_{\text{obs}}(\mathcal{S})$ with $\mathcal{Y}$-valued outcomes belongs to $\mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$ if and only if it is prefix feasible. (ii) If $\text{conv}(\mathcal{Y}) = \mathbb{R}$, every such law is prefix feasible. (iii) If $P_0 \in \mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$ is strictly prefix feasible, the tangent space of $\mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$ at P_0 is $L^0_2(P_0)$; under Condition 1 together with Condition 2(i) for a granular target or Condition 2(i)–(ii) for an aggregate, the corresponding function displayed in Theorem 2.2 or Theorem 2.3, respectively, is then the unique canonical gradient in $\mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$.*

The restriction (2.11) depends on the outcome only through $\text{conv}(\mathcal{Y})$. For a finite numerical support, each prefix mean must lie between the smallest and largest support points; for nonnegative counts it must be nonnegative; for outcomes in the unit interval it must lie in that interval; and for $\mathcal{Y} = \mathbb{R}$ or $\mathbb{Z}$ the restriction is vacuous, so that $\mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$ contains every law in $\mathcal{M}_{\text{obs}}(\mathcal{S})$ with $\mathcal{Y}$-valued outcomes. Part (iii) needs no uniform margin: at a strictly prefix-feasible law the tangent space is unchanged, so the observed-data bound V_Θ of Section 2.3 is also the causal-model efficiency bound. At a law that is prefix feasible but not strictly so, Theorem 2.1 still identifies the target, but we make no efficiency claim for the causal model. The theorem concerns support restrictions only; restrictions on the untreated paths themselves, such as absorbing states or prohibited transitions, would require further compatibility conditions. Appendix B.4 gives the proof and the special cases, binary outcomes included.

Common-history nesting and efficiency bounds. Imposing parallel trends against every untreated cohort separately (Chen et al., 2025) creates over-identifying restrictions. With a common target and conditioning history, these restrictions imply our pooled restriction. We make this nesting precise and compare the two efficiency bounds.

Fix a supported group-period target $\tau_{g,t}$ or a prespecified scalar aggregate Θ of finitely many such targets, and a conditioning σ-field $\mathcal{H}_g$ generated by observed variables dated at most $g-1$ for every target cohort g. The almost-sure convention of Section 2.2 also applies to $\mathcal{H}_g$. The pooled model imposes

$$\mathbb{E}\{\Delta Y_k(\infty) \mid \mathcal{H}_g, G = g\} = \mathbb{E}\{\Delta Y_k(\infty) \mid \mathcal{H}_g, G > k\}, \qquad k \in [g, t],$$

almost surely on $\{G = g\}$, whereas the cohort-by-cohort model imposes

$$\mathbb{E}\{\Delta Y_k(\infty) \mid \mathcal{H}_g, G = g\} = \mathbb{E}\{\Delta Y_k(\infty) \mid \mathcal{H}_g, G = j\}$$

for every $k \in [g, t]$ and $j \in \{k+1, \ldots, T, \infty\}$, almost surely on histories where $\mathbb{P}(G = j \mid \mathcal{H}_g, G > k) > 0$.

Let $\mathcal{M}_{\text{pool}}(\mathcal{H})$ and $\mathcal{M}_{\text{cbc}}(\mathcal{H})$ denote the observed-data models induced by the pooled and cohort-by-cohort restrictions, respectively. They use the same histories $\mathcal{H} = (\mathcal{H}_g)_g$, observed-data experiment, outcome support, and history-specific version of Condition 1, differing only in the parallel-trends restriction. The applicable moment and aggregation conditions are also shared, with identical prespecified weight functions $\Omega_{g,t}$ in the two models.

Proposition 2.1 (Common-History Nesting and Efficiency-Bound Order)**.** *Under the shared requirements above,*

$$\mathcal{M}_{\text{cbc}}(\mathcal{H}) \subseteq \mathcal{M}_{\text{pool}}(\mathcal{H}).$$

For $m \in \{\text{pool}, \text{cbc}\}$, let $V_m(\theta; P)$ denote the semiparametric variance bound for scalar target θ in model $\mathcal{M}_m(\mathcal{H})$ at P. At a common law P, each inequality below holds whenever the target in that line is pathwise differentiable in the pooled model:

$$V_{\text{cbc}}(\tau_{g,t}; P) \le V_{\text{pool}}(\tau_{g,t}; P),$$
$$V_{\text{cbc}}(\Theta; P) \le V_{\text{pool}}(\Theta; P).$$

The inequalities compare bounds on attainable asymptotic variance; they do not order the finite-sample variances of arbitrary estimators. The proof in Appendix B.5 first identifies the common target for the chosen history, then averages the component-cohort means and projects the canonical gradient onto the smaller tangent space. The efficiency-bound ordering does not require strict prefix feasibility. When $\mathcal{H}_g = \sigma(\mathcal{C}_{g-1})$ and the law is strictly prefix feasible, Theorem 2.4(iii), under its stated conditions, identifies the pooled canonical gradient with the EIF displayed in Theorem 2.2 or 2.3; at the true law $P = P_0$, the aggregate bound is $V_{\text{pool}}(\Theta; P_0) = V_\Theta$. The common-history premise is essential: the cohort-by-cohort benchmark of Chen et al. (2025) conditions on baseline Z, whereas $\mathcal{C}_{g-1}$ may contain lagged outcomes and time-varying covariates. Conditional-mean equality given one history does not imply it given the other, so without restrictions linking the two the inclusion does not follow, and we make no efficiency-bound comparison.

3 Estimation and Inference via the Efficient Influence Function

Theorem 2.3 gives the canonical gradient for a generic aggregate Θ. We use it to construct a cross-fitted one-step estimator and establish double robustness and efficiency. Theorem 2.4 states when the observed-data efficiency bound is also the causal-model bound.

3.1 Nuisance Parameters

Nuisance functions are indexed by the target cohort g and are functions of its history $\mathcal{C}_{g-1}$; the cohort share $p_g = \mathbb{P}(G = g) > 0$ is a separate marginal component, estimated in Section 3.2. Write $\pi_{g|g-1}(\mathcal{C}_{g-1}) := \mathbb{P}(G = g \mid \mathcal{C}_{g-1})$ for the cohort propensity score and $\pi_{>k|g-1}(\mathcal{C}_{g-1}) := \mathbb{P}(G > k \mid \mathcal{C}_{g-1})$ for the probability of still belonging to the sliding risk set. The EIF depends on them only through the target-cohort inverse-risk ratio, in which the entry probability e_g cancels and which can be written in terms of the hazards (2.2):

$$r_{g,k}(\mathcal{C}_{g-1}) := \frac{\pi_{g|g-1}(\mathcal{C}_{g-1})}{\pi_{>k|g-1}(\mathcal{C}_{g-1})} = \frac{\mathbb{P}(G = g \mid \mathcal{C}_{g-1}, G \geq g)}{\mathbb{P}(G > k \mid \mathcal{C}_{g-1}, G \geq g)} = \frac{\lambda_{g,g}(\mathcal{C}_{g-1})}{\prod_{s=g}^{k}\{1 - \lambda_{g,s}(\mathcal{C}_{g-1})\}}. \tag{3.1}$$

The ratio, rather than its two propensity components, enters the EIF and the product-rate condition. We estimate it through the hazards in (3.1); neither propensity component is estimated separately.

For the outcome regression, the EIF depends only on the untreated trend of the sliding risk set. For the decomposition of the EIF used in Section 4, we also define a composite regression, indexed by the target cohort g, with a treated branch:

$$m_{g,k}(\mathcal{C}_{g-1}, D_k, G) := \begin{cases} m_{g,k,0}(\mathcal{C}_{g-1}) := \mathbb{E}[\Delta Y_k \mid \mathcal{C}_{g-1}, G > k], & \text{if } D_k = 0 \text{ (i.e., } G > k), \\ m_{g,k,G}(\mathcal{C}_{g-1}) := \mathbb{E}[\Delta Y_k \mid \mathcal{C}_{g-1}, G], & \text{if } D_k = 1 \text{ (i.e., } G \leq k), \end{cases}$$

where $D_k := \mathbb{I}(G \leq k)$. The *control outcome regression* $m_{g,k,0}$ is the expected period-k increment of a unit still untreated at k, given its history $\mathcal{C}_{g-1}$; it is the benchmark used to impute counterfactual trends. The treated branch $m_{g,k,G}$ conditions on the realized cohort label, so it captures cohort-specific treated trends; the composite is well defined for never-treated units, and the treated branch drops out of the final expressions.

The one-step estimator uses the nuisance collection $\eta_g = (m_{g,k,0}, r_{g,k} : k \in [g, T])$; a target (g, t) uses only $k \in [g, t]$. We call $[g, t]$ the target window and each period k a slice. The treated branch is used only in the decomposition of Section 4, where factual initialization replaces it by observed increments. A superscript 0 denotes evaluation at the true law P_0, as in $m^0_{g,k,0}$, $r^0_{g,k}$, $\lambda^0_{g,s}$, and η^0_g.

3.2 Cross-Fitting and the One-Step Estimator

We use cross-fitting with a fixed number $L \geq 2$ of folds, stratified by realized treatment cohort (Chernozhukov et al., 2018). The nuisance estimate $\widehat{\eta}_{g,-\ell}$ is fitted on the complement $\mathcal{I}^c_\ell$ of the held-out fold $\mathcal{I}_\ell$ and evaluated only on $\mathcal{I}_\ell$. The split is balanced within every positive-probability cohort. The empirical-process arguments condition on the realized partition and proceed stratum by stratum (Appendix C.1, where Figure C.1 gives the schematic).

We also define a full-sample implementation, denoted $L = 1$: $\mathcal{I}_1 = [n]$ and $\widehat{\eta}_{g,-1}$ is fitted and evaluated on all n observations, not on the empty complement of $\mathcal{I}_1$. Under this convention the formulas below apply unchanged. On a fixed finite observed-data support, Corollary C.1 in Appendix C.1 extends the asymptotic guarantees of Sections 3 and 4 to this implementation, retaining every condition other than sample splitting. Initial control regressions may use parametric models or flexible learners, including Super Learners (van der Laan et al., 2007); their required consistency and rates remain assumptions.

For each target cohort g, we estimate the ratio by a frozen-history hazard cascade. Fix constants $0 < \overline{a} < 1$ and $1 < M < \infty$ before estimation, independently of the sample and of the unknown overlap constant. At period s, fit a binary learner $\widehat{\lambda}_{g,s,-\ell}$ on the risk set $\{G \geq s\}$ of the training sample (the full sample when $L = 1$) using the same history $\mathcal{C}_{g-1}$. Its projection onto the real

interval $[0, 1-\overline{a}]$ is $\widehat{\lambda}_{g,s,-\ell} = \Pi_{[0,1-\overline{a}]}(\widetilde{\lambda}_{g,s,-\ell})$, where $\Pi_{[0,1-\overline{a}]}(x) := \max\{0, \min(1-\overline{a}, x)\}$. Assemble the raw ratio and cap it at M:

$$\widehat{r}^{\,\mathrm{raw}}_{g,k,-\ell} = \frac{\widehat{\lambda}_{g,g,-\ell}}{\prod_{s=g}^{k}(1-\widehat{\lambda}_{g,s,-\ell})}, \qquad \widehat{r}_{g,k,-\ell} = \min\{\widehat{r}^{\,\mathrm{raw}}_{g,k,-\ell},\, M\}, \tag{3.2}$$

all functions of $\mathcal{C}_{g-1}$. The raw ratio is the sequential implementation of (3.1); the capped ratio $\widehat{r}_{g,k,-\ell}$ is the fitted nuisance object.

The cohort share is estimated on the full sample, $\widehat{p}_g = \mathbb{P}_n[\mathbb{I}(G=g)]$, so $\mathbb{P}_n[\mathbb{I}(G=g)/\widehat{p}_g] \equiv 1$ for every fold structure; it needs neither regularization nor cross-fitting, and its sampling contribution is carried by the asymptotic expansion. For unit i in $\mathcal{I}_\ell$, so that $\ell = \ell(i)$, the fitted increment signal evaluates (2.4) at $\widehat{\eta}_{g,-\ell} = (\widehat{m}_{g,k,0,-\ell}, \widehat{r}_{g,k,-\ell} : k \in [g,T])$, fitted under the chosen convention with $\widehat{r}_{g,k,-\ell}$ the capped cascade ratio (3.2):

$$\widehat{\phi}_{g,k}(O_i; \widehat{\eta}_{g,-\ell}) = \frac{1}{\widehat{p}_g}\Big\{\mathbb{I}(G_i = g) - \mathbb{I}(G_i > k)\,\widehat{r}_{g,k,-\ell}(\mathcal{C}_{i,g-1})\Big\} \times \Big\{\Delta Y_{i,k} - \widehat{m}_{g,k,0,-\ell}(\mathcal{C}_{i,g-1})\Big\}, \qquad k \geq g. \tag{3.3}$$

Units with $G_i < g$ contribute zero, because both indicators vanish for them. For a candidate target value τ, the HT empirical unit score is

$$\widehat{\psi}_{g,t}(O_i; \tau) := \sum_{k=g}^{t} \widehat{\phi}_{g,k}(O_i; \widehat{\eta}_{g,-\ell(i)}) - \frac{\mathbb{I}(G_i = g)}{\widehat{p}_g}\tau.$$

Setting its empirical mean to zero and solving for τ gives the one-step estimator

$$\widehat{\tau}^{\,\mathrm{os}}_{g,t} = \frac{1}{n}\sum_{\ell=1}^{L}\sum_{i\in\mathcal{I}_\ell}\left[\sum_{k=g}^{t}\widehat{\phi}_{g,k}(O_i; \widehat{\eta}_{g,-\ell})\right]. \tag{3.4}$$

The centered score used for variance estimation sets $\tau = \widehat{\tau}^{\,\mathrm{os}}_{g,t}$; we suppress this argument in $\widehat{\psi}_{g,t}$.

The estimated weight corrections cancel exactly in the empirical mean (Remark B.2 in Appendix B.2), so the one-step estimator of any aggregate is the weighted average of the granular ones:

$$\widehat{\Theta}_{\mathrm{os}} = \frac{1}{\sum_{\mathcal{S}}\widehat{\omega}_{g,t}}\sum_{(g,t)\in\mathcal{S}}\widehat{\omega}_{g,t}\widehat{\tau}^{\,\mathrm{os}}_{g,t}, \tag{3.5}$$

where $\widehat{\omega}_{g,t} = \mathbb{P}_n[\Omega_{g,t}(O)]$.

Normalization-indexed estimating equations. The estimator above, written $\widehat{\tau}^{\,\mathrm{os,HT}}_{g,t}$ when the normalization matters, uses the canonical Horvitz–Thompson (HT) score and divides both treated and control contributions by $\widehat{p}_g$. To compare the two normalizations, write the fitted nuisances as $(\widehat{m}^{\mathrm{ini}}, \widehat{r}^{\mathrm{ini}})$; the superscript identifies the initial fits, before targeting (Section 4) updates the outcome regression while leaving the fitted ratio unchanged. Hájek normalization replaces only the control divisor by its realized weight mass, $B_{n,g,k} = \mathbb{P}_n[\mathbb{I}(G>k)\widehat{r}^{\mathrm{ini}}_{g,k}(\mathcal{C}_{g-1})]$ (Hájek, 1971). Define

$$\widehat{\tau}_{\mathrm{tr},k} := \frac{\mathbb{P}_n[\mathbb{I}(G=g)\{\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{g-1})\}]}{\widehat{p}_g},$$
$$\widehat{\tau}_{\mathrm{ct},k} := \frac{\mathbb{P}_n[\mathbb{I}(G>k)\widehat{r}^{\mathrm{ini}}_{g,k}(\mathcal{C}_{g-1})\{\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{g-1})\}]}{B_{n,g,k}},$$

$\widehat{\tau}_{\mathrm{tr},(g,t)} := \sum_{k=g}^{t} \widehat{\tau}_{\mathrm{tr},k}$ and $\widehat{\tau}_{\mathrm{ct},(g,t)} := \sum_{k=g}^{t} \widehat{\tau}_{\mathrm{ct},k}$.

The self-normalized empirical unit score replaces each control divisor $\widehat{p}_g$ in $\widehat{\psi}_{g,t}(O_i;\tau)$ by $B_{n,g,k}$, retaining $\widehat{p}_g$ in the treated and centering terms. Setting the empirical mean of this score at $(\widehat{m}^{\mathrm{ini}}, \widehat{r}^{\mathrm{ini}})$ to zero and solving for τ gives the Hájek one-step estimator

$$\widehat{\tau}_{g,t}^{\mathrm{os,Háj,\,ini}} := \widehat{\tau}_{\mathrm{tr},(g,t)} - \widehat{\tau}_{\mathrm{ct},(g,t)}. \tag{3.6}$$

Aggregates use the empirical weights in (3.5); write $\widehat{\Theta}_{\mathrm{os}}^{\mathrm{HT}}$ for (3.5) and $\widehat{\Theta}_{\mathrm{os}}^{\mathrm{Háj}}$ for the same weighted average of the Hájek one-step estimators. Self-normalization changes the empirical estimating equation, not the population EIF. Under the conditions of Corollary 3.2(ii) below, the Hájek and HT one-step estimators are root-n equivalent.

3.3 Theoretical Guarantees: Double Robustness and Efficiency

We state double robustness and efficiency for an aggregate Θ over $\mathcal{S}$; each result applies to $\tau_{g,t}^0$ with the singleton $\mathcal{S} = \{(g,t)\}$, for which the aggregate correction vanishes. Neyman orthogonality and sample splitting control the nuisance-estimation remainder without Donsker-class conditions. The results use $L \geq 2$ unless the fixed-finite-support extension of Corollary C.1 in Appendix C.1 is invoked.

Asymptotic regime. All asymptotic statements in this and the next section hold as $n \to \infty$ with fixed horizon T, target collection $\mathcal{S}$, and fold count L. Every target cohort has $p_g > 0$, the cohort-stratified fold proportions converge to positive constants, and the auxiliary fluctuation and aggregate weight vectors have fixed dimensions.

Growing-T or growing-target asymptotics would require uniform versions of the Lipschitz and Gram-invertibility arguments used below and are not covered by our results.

Three overlap requirements are kept separate: identification overlap (Assumption 3), which bounds the population ratio $r_{g,k}^0 = \lambda_{g,g}^0 / q_{g,k}^0$; estimation stability, the fixed projection and cap in (3.2); and a positive probability limit of each Hájek control normalizer $B_{n,g,k}$. Under the nuisance convergence conditions below, $B_{n,g,k} \xrightarrow{p} \overline{B}_{g,k} := \mathbb{E}_{P_0}[\mathbb{I}(G > k)\overline{r}_{g,k}]$, where $\overline{r}_{g,k}$ is the deterministic limit of the fitted ratio in the identified-support $L_2(P_0)$ norms assumed in Theorem 3.1 below. Ratio consistency makes this limit $p_g > 0$; positivity is an additional requirement only when the ratio is misspecified. The treated divisor satisfies $\widehat{p}_g \xrightarrow{p} p_g > 0$ by the law of large numbers. None bounds the entry probability e_g or the propensity components $\pi_{g|g-1}$ and $\pi_{>k|g-1}$ from below.

Condition 3 (Aggregate target validity)**.** *For the fixed aggregation set $\mathcal{S}$, (i) Condition 1 holds for $\mathcal{S}$; (ii) for every $\varsigma = (g,t) \in \mathcal{S}$ the aggregation weights satisfy $\mathbb{E}[|\Omega_\varsigma(O)|] < \infty$, and the population weight mass is nonzero, $W_{\mathcal{S}} \neq 0$; and (iii) for the asymptotic-normality and variance-consistency results only, the requirements $|W_{\mathcal{S}}| \geq c_W > 0$ and $\max_{\varsigma \in \mathcal{S}} \mathbb{E}[\Omega_\varsigma^2(O)] < \infty$ hold.*

Parts (i)–(ii) of Condition 3 are used for identification and double-robust consistency. Part (iii) supplies second moments for asymptotic normality and variance consistency. At a fixed law, $W_{\mathcal{S}} \neq 0$ already implies $|W_{\mathcal{S}}| \geq c_W > 0$ for some constant c_W; the lower bound does not strengthen nonzero weight mass. For a singleton with $\Omega_{g,t} \equiv 1$, all weight requirements hold with $W_{\mathcal{S}} = 1$.

Condition 4 (Truth containment)**.** *For the fixed constants $(\overline{a}, M)$ of (3.2), the truncation region contains the truth on every slice used by the stated result:*

$$0 \leq \lambda_{g,s}^0 \leq 1 - \overline{a} \quad \text{on } \{G \geq s\}, \qquad 0 \leq r_{g,k}^0 \leq M \quad \text{on } \{G > k\}.$$

The fitted hazards and capped ratio are constructed in Section 3.2. No condition is imposed on where the pre-clipping learners or raw ratio fall. Condition 4 ensures that truncation preserves the limit and rate of a consistent learner; it is not a rule for choosing the thresholds from the unknown overlap constant. Projection and capping leave a truth inside the truncation region fixed and are 1-Lipschitz:

$$\begin{aligned} \left|\widehat{\lambda}_{g,s} - \lambda^0_{g,s}\right| &\le \left|\widetilde{\lambda}_{g,s} - \lambda^0_{g,s}\right| \quad \text{on } \{G \ge s\}, \\ \left|\widehat{r}_{g,k} - r^0_{g,k}\right| &\le \left|\widehat{r}^{\,\mathrm{raw}}_{g,k} - r^0_{g,k}\right| \quad \text{on } \{G > k\}. \end{aligned} \tag{3.7}$$

Every result below rests on the following second-order remainder identity.

Lemma 3.1 (Ratio-first remainder)**.** *Fix $(g,t) \in \mathcal{S}$ and $k \in [g,t]$, and for a candidate control regression m and candidate inverse-risk ratio r write*

$$\phi_{g,k}(O; m, r) := p_g^{-1}\{\mathbb{I}(G = g) - \mathbb{I}(G > k)\, r(\mathcal{C}_{g-1})\}\{\Delta Y_k - m(\mathcal{C}_{g-1})\}$$

for the uncentered increment signal in (2.4) evaluated at (m, r). Let $(m^0_{g,k,0}, r^0_{g,k})$ denote the truth, and let $(\widetilde{m}, \widetilde{r})$ be any pair of measurable functions of $\mathcal{C}_{g-1}$ with $\mathbb{E}_{P_0}[\mathbb{I}(G > k)\,\widetilde{m}^2] < \infty$ and $|\widetilde{r}| \le M$ on the risk-set support. Suppose Condition 1 holds for $\mathcal{S} = \{(g,t)\}$ and Condition 2(i) holds. Then

$$\mathbb{E}_{P_0}\left[\phi_{g,k}(O; \widetilde{m}, \widetilde{r}) - \phi_{g,k}(O; m^0_{g,k,0}, r^0_{g,k})\right] = \frac{1}{p_g}\,\mathbb{E}_{P_0}\Big[\mathbb{I}(G > k)\,\{\widetilde{r} - r^0_{g,k}\}\{\widetilde{m} - m^0_{g,k,0}\}\Big],$$

and hence, by the Cauchy–Schwarz inequality,

$$\left|\mathbb{E}_{P_0}\left[\phi_{g,k}(O; \widetilde{m}, \widetilde{r}) - \phi_{g,k}(O; m^0_{g,k,0}, r^0_{g,k})\right]\right| \le \frac{1}{p_g}\left\|\mathbb{I}(G > k)\{\widetilde{r} - r^0_{g,k}\}\right\|_2 \left\|\mathbb{I}(G > k)\{\widetilde{m} - m^0_{g,k,0}\}\right\|_2.$$

Lemma 3.1 rests on the balancing identity $\mathbb{E}[\mathbb{I}(G > k)\, r^0_{g,k} \mid \mathcal{C}_{g-1}] = \mathbb{E}[\mathbb{I}(G = g) \mid \mathcal{C}_{g-1}]$, which follows from (3.1) (see (C.2) in Appendix C.2), and on conditional centering of the control residual; each double-robustness branch makes one factor of the product remainder vanish.

Nuisance-indexed scores. Write $\eta_{\mathcal{S}} := (\eta_g : g \in \mathcal{G}_{\mathcal{S}})$. In population expressions, $\psi_{g,t}(O; \eta_g)$ denotes (2.5) with only the control regressions and inverse-risk ratios replaced by the candidate collection η_g; the scalar components $(\tau_{g,t}, p_g)$ stay fixed at $(\tau^0_{g,t}, p^0_g)$. Likewise, $\varphi^{\mathrm{eff}}\{\Theta^0; \eta_{\mathcal{S}}\}(O)$ denotes (2.6) with these component scores and with $(\tau_{g,t}, \Theta, p_g, \omega_{g,t}, W_{\mathcal{S}})$ fixed at their true-law values. At η^0, these are the canonical gradients; at other candidates, they are estimating functions. Target values, cohort shares and aggregate weight means are separate from η_g. In empirical plug-in scores, all these scalar components are instead replaced by the corresponding procedure's estimates, with $\widehat{W}_{\mathcal{S}} := \sum_{\mathcal{S}} \widehat{\omega}_{g,t}$, and the initial nuisance functions are fitted off the unit's fold (on all units when $L = 1$). For an estimate $\widehat{\theta}$ of a granular or aggregate target, the notation $\varphi^{\mathrm{eff}}\{\widehat{\theta}; \widehat{\eta}\}$ denotes the associated empirical plug-in score. The self-normalized empirical analogue changes only the control divisors as in Section 3.2.

Theorem 3.1 (Double Robustness)**.** *Suppose Condition 3(i)–(ii) holds and $\max_k \mathbb{E}_{P_0}[(\Delta Y_k)^2] < \infty$. Suppose further that, for every $g \in \mathcal{G}_{\mathcal{S}}$ and $k \in [g, t_{\mathcal{S}}(g)]$, there are deterministic measurable functions $\overline{m}_{g,k,0}$ and $\overline{r}_{g,k}$ of $\mathcal{C}_{g-1}$, with $\mathbb{E}_{P_0}[\mathbb{I}(G > k)\,\overline{m}^2_{g,k,0}] < \infty$ and $0 \le \overline{r}_{g,k} \le M$ on $\{G > k\}$, to which the out-of-fold nuisance estimators converge uniformly over the fixed folds in the identified-support norms:*

$$\max_{1 \le \ell \le L} \|\mathbb{I}(G > k)\{\widehat{m}_{g,k,0,-\ell} - \overline{m}_{g,k,0}\}\|_2 \xrightarrow{p} 0, \qquad \max_{1 \le \ell \le L} \|\mathbb{I}(G > k)\{\widehat{r}_{g,k,-\ell} - \overline{r}_{g,k}\}\|_2 \xrightarrow{p} 0.$$

Collect the limits in $\overline{\eta}_{\mathcal{S}} = (\overline{\eta}_g : g \in \mathcal{G}_{\mathcal{S}})$, $\overline{\eta}_g = (\overline{m}_{g,k,0}, \overline{r}_{g,k} : k \in [g, t_{\mathcal{S}}(g)])$. If either

(a) the control outcome regression models are correctly specified, $\overline{m}_{g,k,0} = m^0_{g,k,0}$ *for all* $(g,t) \in \mathcal{S}$ *and* $k \in [g,t]$*; or*

(b) the inverse-risk ratios are correctly specified, $\overline{r}_{g,k} = r^0_{g,k}$ *for all* $(g,t) \in \mathcal{S}$ *and* $k \in [g,t]$*,*

then $\mathbb{E}_{P_0}\left[\varphi^{\text{eff}}\{\Theta^0; \overline{\eta}_{\mathcal{S}}\}\right] = 0$ *and* $\widehat{\Theta}_{\text{os}} \xrightarrow{p} \Theta^0$*.*

The cap ensures $0 \le \overline{r}_{g,k} \le M$; consistency for the true ratio additionally requires that the cap contain it.

The efficiency result requires the product-rate condition displayed next. An $o_p(n^{-1/4})$ rate for each nuisance is sufficient but not necessary.

Theorem 3.2 (Asymptotic Normality and Observed-Data Efficiency)**.** *Suppose the conditions for Theorem 3.1 hold, together with the applicable clauses of Condition 2 and Condition 3(iii) whenever* $\mathcal{S}$ *is not a singleton. Assume that the limits in Theorem 3.1 are the truth,* $\overline{\eta}_{\mathcal{S}} = \eta^0_{\mathcal{S}}$*, and that the outcome-regression and inverse-risk-ratio errors satisfy the product condition:*

$$\max_{1\le \ell \le L,\ (g,t)\in\mathcal{S},\ k\in[g,t]} \left\|\mathbb{I}(G>k)\{\widehat{m}_{g,k,0,-\ell} - m^0_{g,k,0}\}\right\|_2 \times \left\|\mathbb{I}(G>k)\{\widehat{r}_{g,k,-\ell} - r^0_{g,k}\}\right\|_2 = o_p(n^{-1/2}).$$

Then $\widehat{\Theta}_{\text{os}}$ *is asymptotically linear with the canonical gradient in* $\mathcal{M}_{\text{obs}}(\mathcal{S})$*:*

$$\sqrt{n}(\widehat{\Theta}_{\text{os}} - \Theta^0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi^{\text{eff}}\{\Theta^0; \eta^0_{\mathcal{S}}\}(O_i) + o_p(1) \rightsquigarrow \mathcal{N}(0, V_\Theta),$$

attaining the observed-data efficiency bound $V_\Theta = \mathbb{E}_{P_0}\left[(\varphi^{\text{eff}}\{\Theta^0; \eta^0_{\mathcal{S}}\})^2\right]$ *of Section 2.3.*

By Theorem 2.4, V_Θ is also the causal-model efficiency bound at strictly prefix-feasible laws, including every law with $\text{conv}(\mathcal{Y}) = \mathbb{R}$. For binary outcomes, $\text{conv}\{0,1\} = [0,1]$, so the same conclusion holds when

$$0 < \mu^{P_0}_{g,s}(\mathcal{C}_{g-1}) < 1 \quad \text{almost surely on } \{G = g\}, \qquad s \in [g, t_{\mathcal{S}}(g)],$$

for every represented cohort g; no uniform margin is required. For binary outcomes, only the outcome moment conditions hold automatically; the aggregation-weight moments (for an aggregate), overlap, nuisance consistency and the product-rate condition, and $V_\Theta > 0$ remain requirements of the asymptotic result.

Section 4.3 gives the standard error used for this and the targeted estimators, and the resulting pointwise intervals.

Hazard-cascade specialization. The implementation constructs $\widehat{r}_{g,k}$ through a frozen-history binary hazard cascade for each target cohort, and the following corollary gives primitive conditions in terms of the hazards.

Corollary 3.1 (Guarantees under the sequential hazard cascade)**.** *Suppose the conditions of Theorem 3.1 and Condition 4 hold. Use the pre-clipping hazard learners* $\widetilde{\lambda}_{g,s,-\ell}$ *and fitted hazards* $\widehat{\lambda}_{g,s,-\ell}$ *of Section 3.2. Measure hazard errors in* $\|\mathbb{I}(G \ge s)\{\widetilde{\lambda}_{g,s,-\ell} - \lambda^0_{g,s}\}\|_2$ *and ratio errors in* $\|\mathbb{I}(G>k)\{\widehat{r}_{g,k,-\ell} - r^0_{g,k}\}\|_2$*. All convergence and rate statements are uniform over the fixed folds.*

(i) ***Double robustness.*** $\widehat{\Theta}_{\text{os}}$ *is consistent if* either $\overline{m}_{g,k,0} = m^0_{g,k,0}$ *for every* $(g,t) \in \mathcal{S}$ *and* $k \in [g,t]$*,* or $\|\mathbb{I}(G \ge s)\{\widetilde{\lambda}_{g,s,-\ell} - \lambda^0_{g,s}\}\|_2 \xrightarrow{p} 0$ *for every* $g \in \mathcal{G}_{\mathcal{S}}$ *and every* s *entering a target window for* g*. No rate condition is required.*

(ii) ***Efficiency.*** *Suppose, in addition, that the applicable clauses of Condition 2 and, whenever $\mathcal{S}$ is not a singleton, Condition 3(iii) hold, as in Theorem 3.2, and that the outcome regressions and the pre-clipping hazard learners are consistent: each of the two norms in (3.8) converges to zero in probability for every index over which the maximum is taken. If*

$$\max_{\substack{1\le\ell\le L,\ g\in\mathcal{G}_{\mathcal{S}},\,(g,t)\in\mathcal{S},\\ k,s\in[g,t]}} \big\|\mathbb{I}(G>k)\{\widehat{m}_{g,k,0,-\ell}-m^0_{g,k,0}\}\big\|_2 \times \big\|\mathbb{I}(G\ge s)\{\widetilde{\lambda}_{g,s,-\ell}-\lambda^0_{g,s}\}\big\|_2 = o_p(n^{-1/2}), \tag{3.8}$$

then every condition of Theorem 3.2 holds, including its product-rate condition, and $\widehat{\Theta}_{\mathrm{os}}$ attains the efficiency bound V_Θ of Theorem 3.2.

The outcome-correct branch of part (i) uses neither truth-containment clause. The ratio-correct branch uses both: without them, clipping or capping can move the probability limit. Part (ii) requires consistency of both nuisances as well as the product rate; a vanishing product alone does not imply that each factor vanishes. The projection applies no positive lower clip, so a true future hazard $\lambda^0_{g,s}=0$, $s>g$, is permitted. The proof of Corollary 3.1 in Appendix C.2 gives the cascade-error bound.

Corollary 3.2 (Double robustness and efficiency of the Hájek one-step estimator)**.** *Use the Hájek one-step estimator (3.6) and, for an aggregate, the empirical weights in (3.5).*

(i) ***Consistency.*** *Suppose the conditions of Theorem 3.1 hold. In its outcome-correct branch, also require*

$$\overline{B}_{g,k}=\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}]>0 \qquad \text{for every } (g,t)\in\mathcal{S},\ k\in[g,t].$$

In its ratio-correct branch, this positivity holds automatically, because $\overline{B}_{g,k}=p_g>0$. Then the granular and aggregate Hájek one-step estimators are well defined with probability tending to one and are consistent for their respective targets.

(ii) ***First-order equivalence and efficiency.*** *Suppose the conditions of Theorem 3.2 hold with its displayed product condition replaced by the following pair-level condition. Define*

$$\begin{aligned} d_{m,g,t} &:= \max_{\ell\le L,\,k\in[g,t]} \big\|\mathbb{I}(G>k)\{\widehat{m}_{g,k,0,-\ell}-m^0_{g,k,0}\}\big\|_2,\\ d_{r,g,t} &:= \max_{\ell\le L,\,k\in[g,t]} \big\|\mathbb{I}(G>k)\{\widehat{r}_{g,k,-\ell}-r^0_{g,k}\}\big\|_2, \end{aligned} \tag{3.9}$$

and require

$$\max_{(g,t)\in\mathcal{S}} \sqrt{n}\, d_{m,g,t} d_{r,g,t} = o_p(1). \tag{3.10}$$

Both errors converge to zero by the nuisance-consistency conditions of that theorem. Then, for every $(g,t)\in\mathcal{S}$,

$$\widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t}-\widehat{\tau}^{\mathrm{os,HT}}_{g,t}=o_p(n^{-1/2}),$$

and the corresponding aggregate satisfies

$$\sqrt{n}(\widehat{\Theta}^{\mathrm{Háj}}_{\mathrm{os}}-\Theta^0)=\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varphi^{\mathrm{eff}}\{\Theta^0;\eta^0_{\mathcal{S}}\}(O_i)+o_p(1)\rightsquigarrow\mathcal{N}(0,V_\Theta).$$

The canonical gradient is the full unit-level score in (2.6), including the aggregation-weight term

$$\frac{1}{W_{\mathcal{S}}} \sum_{(g,t)\in\mathcal{S}} (\tau^0_{g,t} - \Theta^0)\{\Omega_{g,t}(O) - \omega^0_{g,t}\}.$$

Thus the estimator attains the observed-data efficiency bound V_Θ, and, at a strictly prefix-feasible law, the causal-model bound of Theorem 2.4(iii).

The conditions of part (ii), including consistency of both nuisances and the pair-level product rate, are sufficient for root-n equivalence of the two one-step normalizations; neither consistency alone guarantees this equivalence. Under the nuisance convergence conditions in part (i),

$$B_{n,g,k} - \widehat{p}_g \xrightarrow{p} \mathbb{E}_{P_0}[\mathbb{I}(G > k)\overline{r}_{g,k}] - p_g,$$

and this limit need not be zero. With only the outcome regression correct, the control residual mean can fluctuate at order $n^{-1/2}$ while the normalizer discrepancy can have a nonzero limit. With only the ratio correct, the normalizer discrepancy can fluctuate at order $n^{-1/2}$ while the control residual mean can have a nonzero limit. Neither consistency alone therefore guarantees an $o_p(n^{-1/2})$ Hájek–HT difference.

4 Pairwise Targeted Estimation

The one-step estimators of Section 3 solve their empirical estimating equations directly. Targeted minimum loss-based estimation (van der Laan and Rose, 2011) instead updates an initial outcome regression through an auxiliary regression. We adapt this construction as a *pairwise targeted minimum loss-based estimator (TMLE) with a linear least-squares fluctuation*; under the stated efficiency conditions in Section 4.2, it has the same efficient first-order limit as the one-step estimator.

Applied work often reports several group-period effects, event-study summaries, and an overall effect. An update tailored to one scalar target need not solve the empirical score equations for the others. Estimating each report separately can therefore require additional targeting regressions, even when the reports are aggregates of the same group-period effects.

We provide a unified TMLE workflow by selecting the reported pairs and aggregates before estimation and fixing a finite supported grid containing every pair they require. For a chosen normalization, the construction fits one scalar fluctuation for each (g, t) in this grid and reuses the resulting pair estimates for all prespecified aggregates. No aggregate-specific targeting regression is required: under its empirical implementation conditions, Theorem 4.1 establishes the pair and aggregate score equations, including the aggregation-weight contribution whose empirical mean cancels (Remark B.2). The updated outcome regressions can differ across pairs; the workflow does not require a single shared updated regression. Inference is pointwise for each reported target, as described in Section 4.3.

Appendix F develops cohort-wise targeting and gives exact representations of the one-step and targeted procedures, including the variants used in Section 5.

4.1 Construction

Separating the treated and control contributions of the EIF in Theorem 2.2 and centering each at its own conditional mean gives

$$\varphi^{\text{eff}}\{\text{ATT}(g,t)\} = \sum_{k=1}^{T} H_{(g,t),k}\big(\mathcal{C}_{g-1}, G\big)\big(\Delta Y_k - m_{g,k}(\mathcal{C}_{g-1}, D_k, G)\big) + \frac{\mathbb{I}(G=g)}{p_g}\left[\sum_{k=g}^{t}\big(m_{g,k,G}(\mathcal{C}_{g-1}) - m_{g,k,0}(\mathcal{C}_{g-1})\big) - \text{ATT}(g,t)\right],$$

with the population clever covariate

$$H_{(g,t),k}\big(\mathcal{C}_{g-1}, G\big) := \frac{\mathbb{I}(g \le k \le t)}{p_g}\left(\mathbb{I}(G=g) - (1-D_k)\frac{\pi_{g|g-1}(\mathcal{C}_{g-1})}{\pi_{>k|g-1}(\mathcal{C}_{g-1})}\right). \tag{4.1}$$

The targeting step updates the composite regression $m_{g,k}(\mathcal{C}_{g-1}, D_k, G)$ until the first sum has empirical mean zero; the estimator is then evaluated from the second term.

Empirical normalization. For a normalization $\nu \in \{\text{HT}, \text{Háj}\}$, the control-arm divisor $B^{\nu}_{n,g,k}$ is $\widehat{p}_g$ under HT normalization and $B_{n,g,k} = \mathbb{P}_n[\mathbb{I}(G>k)\widehat{r}^{\text{ini}}_{g,k}(\mathcal{C}_{g-1})]$ under Hájek normalization, and the fitted clever covariate is

$$\widehat{H}^{\text{ini},\nu}_{(g,t),k}(O) := \mathbb{I}(g \le k \le t)\left\{\frac{\mathbb{I}(G=g)}{\widehat{p}_g} - \frac{\mathbb{I}(G>k)\widehat{r}^{\text{ini}}_{g,k}(\mathcal{C}_{g-1})}{B^{\nu}_{n,g,k}}\right\}. \tag{4.2}$$

The HT choice uses the canonical empirical EIF covariate; the Hájek choice uses the control weight mass defined in Section 3.2. Under ratio consistency $B_{n,g,k} \xrightarrow{p} p_g$. Once the normalization is selected, we suppress ν from the notation.

Factual initialization. The EIF depends on the outcome regression only through the control branch $m^0_{g,k,0}$, so no treated-branch regression is fitted. For a pair with target cohort g, the initial offset of row (i,k) is

$$\widehat{m}^{\text{ini}}_{g,(i,k)} := \begin{cases} \Delta Y_{i,k}, & G_i = g,\ k \ge g, \\ \widehat{m}^{\text{ini}}_{g,k,0}(\mathcal{C}_{i,g-1}), & G_i > k, \end{cases} \tag{4.3}$$

and it is immaterial on the remaining rows, whose clever covariate is zero. For empirical evaluations, $\widehat{m}^{\text{ini}}_{g,k}(O_i)$ denotes the same row offset $\widehat{m}^{\text{ini}}_{g,(i,k)}$, with the out-of-fold control fit for unit i understood. Its observation argument is suppressed inside empirical averages and update expressions. On a treated row the offset is the observed increment of that row: a row-level quantity, not an estimate of the treated branch $m_{g,k,G}$ as a function of the history. The initial treated residuals are therefore zero, and only the untreated risk-set trend $\widehat{m}^{\text{ini}}_{g,k,0}$ is estimated.

Pairwise auxiliary regression. Let $\mathcal{Q} := \mathcal{S} \subseteq \mathcal{I}_+$ be the fixed supported grid of reported pairs selected in advance. Each pair has its own regression block, so omitting unreported pairs leaves every reported update unchanged. Conditions 5–7 of Section 4.2 below apply to this same grid; no fit or normalizer is required for an unreported endpoint. For each $(g,t) \in \mathcal{Q}$, the scalar fluctuation $\epsilon_{g,t}$ is estimated by the auxiliary linear regression without intercept

$$\Delta Y_{i,k} = \widehat{m}^{\text{ini}}_{g,(i,k)} + \widehat{H}^{\text{ini}}_{(g,t),k}(O_i)\,\epsilon_{g,t} + u_{i,k}, \quad i \in [n],\ k \in [g,t], \tag{4.4}$$

whose offset is (4.3) and whose lone regressor is the clever covariate. Only the rows of cohort g and of the risk sets $\{G > k\}$, $k \in [g, t]$, have a nonzero clever covariate, so the scalar normal equation targets (g, t) in isolation. With $\widehat{\epsilon}_{g,t}$ the ordinary least squares (OLS) estimate, the updated value of row (i, k) is

$$\widehat{m}^{*}_{(g,t),(i,k)} := \widehat{m}^{\mathrm{ini}}_{g,(i,k)} + \widehat{H}^{\mathrm{ini}}_{(g,t),k}(O_i)\,\widehat{\epsilon}_{g,t}. \tag{4.5}$$

The targeted estimator. Targeting updates only the outcome regression: $\widehat{r}^{*}_{g,k} := \widehat{r}^{\mathrm{ini}}_{g,k}$, so the operative first-order nuisances remain $(m_{g,k,0}, r_{g,k})$. With the treated-arm weight $\beta^{\mathrm{tr}}_{n,g,k} := 1/\widehat{p}_g$ and the control-arm weight $\beta^{\mathrm{ct}}_{n,g,k}(c) := \widehat{r}^{\mathrm{ini}}_{g,k}(c)/B^{\nu}_{n,g,k}$, write $\widehat{H}^{\mathrm{ini,ct}}_{(g,t),k}(c) := -\mathbb{I}(g \le k \le t)\,\beta^{\mathrm{ct}}_{n,g,k}(c)$ for the control branch of the clever covariate, as a function of the history. The targeted *control* branch

$$\widehat{m}^{*}_{(g,t),k,0}(c) := \widehat{m}^{\mathrm{ini}}_{g,k,0}(c) + \widehat{H}^{\mathrm{ini,ct}}_{(g,t),k}(c)\,\widehat{\epsilon}_{g,t}$$

is a measurable function on the maintained common covariate support, which contains the treated units' histories by overlap (Assumption 3), and it equals the updated value (4.5) on every control row. On a treated row of cohort g with $k \in [g, t]$, (4.5) gives the row-level value $\widehat{m}^{*}_{(g,t),(i,k)} = \Delta Y_{i,k} + \beta^{\mathrm{tr}}_{n,g,k}\,\widehat{\epsilon}_{g,t}$. Evaluating the targeted control branch at the treated units' histories preserves double robustness through the outcome regression and gives the TMLE

$$\widehat{\tau}^{\mathrm{tar}}_{g,t} := \frac{1}{n}\sum_{i=1}^{n} \frac{\mathbb{I}(G_i = g)}{\widehat{p}_g} \sum_{k=g}^{t} \left(\widehat{m}^{*}_{(g,t),(i,k)} - \widehat{m}^{*}_{(g,t),k,0}(\mathcal{C}_{i,g-1})\right) \tag{4.6}$$

and, for an aggregate over $\mathcal{S}$,

$$\widehat{\Theta}_{\mathrm{tar}} := \frac{\sum_{(g,t)\in\mathcal{S}} \widehat{\omega}_{g,t}\,\widehat{\tau}^{\mathrm{tar}}_{g,t}}{\sum_{(g,t)\in\mathcal{S}} \widehat{\omega}_{g,t}}, \qquad \widehat{p}_g = \mathbb{P}_n[\mathbb{I}(G = g)], \quad \widehat{\omega}_{g,t} = \mathbb{P}_n[\Omega_{g,t}(O)].$$

Write $\widehat{\eta}^{*}_{g,t}$ for the targeted nuisances of pair (g, t) and $\widehat{\eta}^{*}_{\mathcal{S}}$ for their collection. Barred objects such as $\overline{\eta}_{\mathcal{S}}$ remain probability limits of the initial estimators, as in Theorem 3.1. For an aggregate, $\widehat{\eta}^{*}_{\mathcal{S}}$ is indexed by pair rather than by cohort: component (g, t) of the aggregate score uses $\widehat{\eta}^{*}_{g,t}$, namely the targeted control regressions $\widehat{m}^{*}_{(g,t),k,0}$ and the fitted ratios, and those regressions can differ across endpoints of the same cohort. The fluctuation coefficients themselves are fitted on the full sample. The cohort-wise update of Appendix F instead uses a common targeted regression for each cohort and slice.

The linear fluctuation need not preserve the outcome range. The TMLE admits a support-feasible substitution (plug-in) interpretation only when the implied updated outcome levels remain in $\mathrm{conv}(\mathcal{Y})$ and the implied untreated prefix means satisfy (2.11); these are properties to check in the fitted design.

Algorithm 1: Pairwise TMLE with a linear least-squares fluctuation

Input: Panel data $\{(Y_{it}, D_{it}, X_{it}, Z_i)\}_{i\in[n],\, t\in\{0\}\cup[T]}$, from which the histories $\mathcal{C}_{g-1}$ are constructed; the prespecified supported grid of reported pairs $\mathcal{Q} := \mathcal{S} \subseteq \mathcal{I}_+$ and aggregation weights $\Omega_{g,t}$; a normalization $\nu \in \{\mathrm{HT}, \mathrm{Háj}\}$.

Output: Targeted estimators $\widehat{\tau}^{\mathrm{tar}}_{g,t}$, $(g,t) \in \mathcal{S}$, and the reported aggregates $\widehat{\Theta}_{\mathrm{tar}}$.

Step 1: initial nuisances. Obtain the control regressions $\widehat{m}^{\mathrm{ini}}_{g,k,0}$ and cascade ratios $\widehat{r}^{\mathrm{ini}}_{g,k}$ (both cross-fitted for $L \geq 2$ and full-sample for $L = 1$) of (3.2) as in Section 3, on the slices required by the reported target windows; form the row offsets (4.3), which equal the observed increment on the treated rows of each target cohort.

Step 2: auxiliary regression. Stack the regressions (4.4), giving each pair $(g,t) \in \mathcal{Q}$ its own copy of the rows (i,k) with $i \in [n]$ and $k \in [g,t]$, so that the system is block diagonal with one column per pair:

$$\Delta\mathbf{y} = \widehat{\mathbf{m}}^{\mathrm{ini}} + \widehat{\mathbf{H}}^{\mathrm{ini}}\,\boldsymbol{\epsilon} + \mathbf{u}, \tag{4.7}$$

and solve it by OLS without intercept, giving $\widehat{\boldsymbol{\epsilon}} = (\widehat{\epsilon}_{g,t})_{(g,t)\in\mathcal{Q}}$.

Step 3: update. For every $(g,t) \in \mathcal{Q}$, form the updated values (4.5) and the targeted control branches $\widehat{m}^{*}_{(g,t),k,0}$.

Step 4: estimate. For every reported $(g,t) \in \mathcal{S}$, compute $\widehat{\tau}^{\mathrm{tar}}_{g,t}$ by (4.6), and form each reported aggregate $\widehat{\Theta}_{\mathrm{tar}}$.

4.2 Theoretical Guarantees

The auxiliary regressions solve the granular empirical scores; an aggregate score is the weighted sum of those scores plus the unit-level weight score, whose empirical mean cancels.

Condition 5 (Empirical implementation)**.** *For every pair in $\mathcal{Q}$ and every reported aggregate over $\mathcal{S}$: (i) $\widehat{p}_g$, any Hájek normalizers $B_{n,g,k}$ used in the clever covariates, and $\sum_{(g,t)\in\mathcal{S}} \widehat{\omega}_{g,t}$ are nonzero; and (ii) the coefficient vector $\widehat{\boldsymbol{\epsilon}}$ used in the update (4.5) solves the normal equation*

$$\widehat{\mathbf{H}}^{\mathrm{ini}\top}\big(\Delta\mathbf{y} - \widehat{\mathbf{m}}^{\mathrm{ini}} - \widehat{\mathbf{H}}^{\mathrm{ini}}\widehat{\boldsymbol{\epsilon}}\big) = \mathbf{0}. \tag{4.8}$$

Item (i) is an event whose probability tends to one under Conditions 6 and 7 below; off it the estimators may be defined arbitrarily, which does not affect the asymptotic statements. The survival products $\prod_{s=g}^{k}\{1 - \widehat{\lambda}_{g,s}\}$ of the cascade need no such condition, because the upper projection in (3.2) keeps them positive. Item (ii) restricts the implementation, not the data. In the stacked system (4.7) each pair has its own copy of the rows, so the design is block diagonal, and on the event in (i) the column of (g,t) has the nonzero entry $1/\widehat{p}_g$ on the rows of cohort g with $k \in [g,t]$ (Appendix E.1). Equation (4.8) therefore has exactly one solution, pair by pair

$$\widehat{\epsilon}_{g,t} = -\frac{\sum_{k=g}^{t} \mathbb{P}_n\big[\mathbb{I}(G > k)\,\beta^{\mathrm{ct}}_{n,g,k}(\mathcal{C}_{g-1})\{\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{g-1})\}\big]}{\sum_{k=g}^{t} \mathbb{P}_n\big[\{\widehat{H}^{\mathrm{ini}}_{(g,t),k}(O)\}^2\big]},$$

in which the treated rows contribute nothing to the numerator because their initial residuals are zero; any exact least-squares solver returns it. Condition 5(ii) requires the coefficient used in the pairwise update (4.5) to satisfy (4.8). Penalized or early-stopped fits typically change this equation, as can a coefficient from a joint fit inserted into the pairwise update (4.5). Such a coefficient satisfies Condition 5(ii) exactly when it also solves the stated pairwise normal equation. Otherwise, the

empirical score in Theorem 4.1 equals n^{-1} times that pair's normal-equation residual rather than zero. Cohort-wise targeting instead evaluates every endpoint at the common joint update and solves its own joint normal equations (Appendix F.2); Figure F.1 contrasts the two designs.

With the divisors $B^{\nu}_{n,g,k}$ and the targeted control branches above, define the normalization-indexed empirical granular score

$$\widehat{\psi}^{\nu}_{g,t}(O) \;:=\; \sum_{k=g}^{t} \left\{ \frac{\mathbb{I}(G=g)}{\widehat{p}_g} - \frac{\mathbb{I}(G>k)\widehat{r}_{g,k}(\mathcal{C}_{g-1})}{B^{\nu}_{n,g,k}} \right\} \{\Delta Y_k \;-\; \widehat{m}^{*}_{(g,t),k,0}(\mathcal{C}_{g-1})\} \;-\; \frac{\mathbb{I}(G=g)}{\widehat{p}_g}\widehat{\tau}^{\mathrm{tar}}_{g,t}$$

and, with $\widehat{W}_{\mathcal{S}} = \sum_{\varsigma\in\mathcal{S}} \widehat{\omega}_{\varsigma}$ as defined after Lemma 3.1, the aggregate score

$$\widehat{\Psi}^{\nu}_{\Theta}(O) := \frac{1}{\widehat{W}_{\mathcal{S}}} \sum_{\varsigma\in\mathcal{S}} \left[\widehat{\omega}_{\varsigma}\widehat{\psi}^{\nu}_{\varsigma}(O) + \{\widehat{\tau}^{\mathrm{tar}}_{\varsigma} - \widehat{\Theta}_{\mathrm{tar}}\}\{\Omega_{\varsigma}(O) - \widehat{\omega}_{\varsigma}\} \right].$$

Theorem 4.1 (Normalization-Indexed Score Solution)**.** *Under Condition 5, for the normalization* $\nu \in \{\mathrm{HT}, \mathrm{Háj}\}$ *used in its clever covariate, the TMLE satisfies*

$$\mathbb{P}_n\big[\widehat{\psi}^{\nu}_{g,t}\big] = 0, \qquad (g,t) \in \mathcal{Q},$$

and, for each aggregate formed with the same normalization,

$$\mathbb{P}_n\big[\widehat{\Psi}^{\nu}_{\Theta}\big] = 0.$$

Under HT normalization the granular score is the canonical plug-in EIF score; under Hájek normalization it is the self-normalized score. The normalization affects the empirical equation, not the population EIF. The Hájek score need not solve the HT equation in finite samples; Proposition 4.1, together with Theorem 4.2(ii) and Corollary 4.1, gives the first-order equivalence of the resulting estimators. Appendix D.2 gives the score-solution proof.

Condition 6 (Target validity)**.** *Condition 3 holds for* $\mathcal{S}$*. The limits* $\overline{\eta}_{\mathcal{S}}$ *of the initial nuisances satisfy the integrability and envelope requirements of Theorem 3.1:* $\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{m}^2_{g,k,0}] < \infty$ *and* $0 \le \overline{r}_{g,k} \le M$ *on* $\{G>k\}$*.*

Condition 7 (Asymptotic regularity)**.** *For every pair* (g,t) *entering* $\mathcal{S}$*, with slice set* $[g,t]$*: (i)* T*,* L*, and* $|\mathcal{S}|$ *are fixed as* $n\to\infty$*; (ii)* $0 < V_{\Theta} < \infty$*,* $\max_{k\in[T]} \mathbb{E}[(\Delta Y_k)^2] < \infty$*, and, for every non-singleton aggregate,* $\max_{\varsigma\in\mathcal{S}} \mathbb{E}[\Omega^2_{\varsigma}] < \infty$*; (iii) Assumption 3 and the ratio truth-containment clause of Condition 4,* $r^0_{g,k} \le M$ *on* $\{G>k\}$*, hold on every slice* $k\in[g,t]$*; the fitted ratios are constructed as in (3.2); and there exist constants* $c_p, c_B > 0$ *such that* $\mathbb{P}(\widehat{p}_g \ge c_p) \to 1$ *for every* $g \in \mathcal{G}_{\mathcal{S}}$ *and, under Hájek normalization,* $\mathbb{P}\big(\min_{(g,t)\in\mathcal{S},\, k\in[g,t]} B_{n,g,k} \ge c_B\big) \to 1$*; (iv) the pair-level errors* $d_{m,g,t}$ *and* $d_{r,g,t}$ *defined in (3.9), applied to the initial cross-fitted nuisances, are* $o_p(1)$*, and* $\max_{(g,t)\in\mathcal{S}} \sqrt{n}\, d_{m,g,t}\, d_{r,g,t} = o_p(1)$*.*

For a fixed target cohort, $p_g > 0$ and the law of large numbers imply the floor on $\widehat{p}_g$. The Hájek control floor can require an additional condition when the ratio is misspecified.

Item (iv) takes the maxima over folds and slices separately in the two error factors. Targeting needs this pair-level product because each fluctuation is fitted on the full sample. Under factual initialization and the exact normal equation, Lemma D.2 in Appendix D.2 gives the rate $O_p(d_{m,g,t} + n^{-1/2})$: the ratio multiplies a conditionally centered control residual. Its contribution to the expansion then vanishes under the product rate, without a separate squared-ratio rate condition. Appendix E derives from the stated conditions the zero population auxiliary score and a positive limit for the denominator $\sum_{k=g}^{t} \mathbb{P}_n[\{\widehat{H}^{\mathrm{ini}}_{(g,t),k}(O)\}^2]$ of $\widehat{\epsilon}_{g,t}$.

Theorem 4.2 (Asymptotic Properties of the HT Targeted Estimator)**.** *Let Θ^0 be any generic aggregated parameter over an aggregation set $\mathcal{S}$, and let $\widehat{\Theta}^{\mathrm{HT}}_{\mathrm{tar}}$ be the corresponding TMLE constructed with the HT clever covariate and factual initialization (4.3). Suppose Conditions 5 and 6 hold.*

(i) ***Double robustness.*** *Suppose the out-of-fold initial nuisance collection converges, uniformly over the fixed folds, in the identified-support $L_2(P_0)$ norms of Theorem 3.1, with deterministic limits satisfying that theorem's integrability and envelope conditions. Also assume Condition 7(i), the moment clause $\max_{k\in[T]}\mathbb{E}[(\Delta Y_k)^2]<\infty$ of Condition 7(ii), and the empirical denominator floors of Condition 7(iii). Then $\widehat{\Theta}^{\mathrm{HT}}_{\mathrm{tar}}$ is consistent for Θ^0 in either of two cases, with convergence in the identified-support norms of Theorem 3.1:* (a) $\widehat{m}_{g,k,0,-\ell}\to m^0_{g,k,0}$ *for every $k\in[g,t]$ and every $(g,t)\in\mathcal{S}$; or* (b) $\widehat{r}_{g,k,-\ell}\to r^0_{g,k}$ *for every such slice.*

(ii) ***Asymptotic linearity and efficiency.*** *Suppose, in addition, that Condition 7(i)–(iv) holds. Then*

$$\sqrt{n}\big(\widehat{\Theta}^{\mathrm{HT}}_{\mathrm{tar}}-\Theta^0\big)=\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varphi^{\mathrm{eff}}\{\Theta^0;\eta^0_{\mathcal{S}}\}(O_i)+o_p(1)\rightsquigarrow\mathcal{N}(0,V_\Theta).$$

Under part (ii), the estimator attains the efficiency bound V_Θ of Theorem 3.2.

To compare the Hájek TMLE with the initial one-step estimator (3.6), define the Gram contribution and loading of slice k (Appendix D.1):

$$V_{n,g,k}:=\frac{1}{\widehat{p}_g}+\frac{\mathbb{P}_n[\mathbb{I}(G>k)(\widehat{r}^{\mathrm{ini}}_{g,k})^2]}{B^2_{n,g,k}},\qquad c_{n,g,k}:=\frac{1}{\widehat{p}_g}\left(1+\frac{\mathbb{P}_n[\mathbb{I}(G=g)\widehat{r}^{\mathrm{ini}}_{g,k}]}{B_{n,g,k}}\right).$$

Proposition 4.1 (Asymptotic Equivalence under Hájek Normalization)**.** *Fix $(g,t)\in\mathcal{S}$ and use Hájek normalization. Assume Condition 5, factual initialization, and the following clauses of Condition 7 for this pair: the fixed T and L of item (i); the moment clause of item (ii) on the active slices, $\max_{k\in[g,t]}\mathbb{E}[(\Delta Y_k)^2]<\infty$; from item (iii), Assumption 3, the capped-cascade construction of the fitted ratios (so that $0\le\widehat{r}^{\mathrm{ini}}_{g,k}\le M$), and the empirical denominator floors; and, from item (iv), $d_{m,g,t}=o_p(1)$, $d_{r,g,t}=o_p(1)$, and $\sqrt{n}\,d_{m,g,t}\,d_{r,g,t}=o_p(1)$. Then the TMLE is*

$$\widehat{\tau}^{\mathrm{tar}}_{g,t}=\widehat{\tau}_{\mathrm{tr},(g,t)}-\Gamma_{n,g,t}\,\widehat{\tau}_{\mathrm{ct},(g,t)},\qquad\Gamma_{n,g,t}:=\frac{\sum_{k=g}^{t}c_{n,g,k}}{\sum_{k=g}^{t}V_{n,g,k}},$$

with $\Gamma_{n,g,t}\xrightarrow{p}1$ and $1-\Gamma_{n,g,t}=O_p(n^{-1/2}+d_{r,g,t})$. Moreover, $\widehat{\tau}^{\mathrm{tar}}_{g,t}-\widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t}=o_p(n^{-1/2})$ and $\widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t}-\widehat{\tau}^{\mathrm{os,HT}}_{g,t}=o_p(n^{-1/2})$. Whenever the conditions of Corollary 3.2(ii) hold, the Hájek targeted estimator therefore inherits the canonical HT influence-function expansion.

The proposition does not separately impose the truth-containment clauses of Condition 4. Its conditions entail the ratio clause: the cap gives $\max\{r^0_{g,k}-M,0\}\le|\widehat{r}_{g,k,-\ell}-r^0_{g,k}|$ pointwise, so the constant $\|\mathbb{I}(G>k)\max\{r^0_{g,k}-M,0\}\|_2\le d_{r,g,t}=o_p(1)$ is zero, and $r^0_{g,k}\le M$ almost surely on $\{G>k\}$ for every $k\in[g,t]$; the hazard clause is not used. Nor does the proposition use $V_\Theta>0$ or the aggregation-weight moments.

The Hájek normalizer floor in Condition 7(iii) is

$$\mathbb{P}\left(\min_{(g,t)\in\mathcal{S},\,k\in[g,t]}B_{n,g,k}\ge c_B\right)\longrightarrow 1\qquad\text{for some } c_B>0.\tag{4.9}$$

Corollary 4.1 (Asymptotic Properties under Hájek Normalization)**.** *Construct the TMLE with the Hájek control divisor $B_{n,g,k}$. Assume the corresponding conditions of Theorem 4.2 and, in either double-robustness branch, the floor (4.9). Then the Hájek estimator is consistent in either branch of Theorem 4.2(i). If, in addition, Condition 7 holds, then*

$$\sqrt{n}\big(\widehat{\Theta}^{\text{Háj}}_{\text{tar}} - \Theta^0\big) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varphi^{\text{eff}}\{\Theta^0; \eta^0_{\mathcal{S}}\}(O_i) + o_p(1) \rightsquigarrow \mathcal{N}(0, V_\Theta).$$

For binary outcomes, the efficiency conclusions of Theorem 4.2(ii) and of this corollary also apply to the causal model under the strict prefix-feasibility proviso stated after Theorem 3.2, with all their other conditions retained.

In the ratio-correct branch the floor (4.9) is automatic, because $B_{n,g,k} \xrightarrow{p} p_g > 0$ uniformly over the finite slice sets; under outcome correctness with a misspecified ratio, $B_{n,g,k}$ tends to $\mathbb{E}[\mathbb{I}(G > k)\overline{r}_{g,k}]$, which is not pinned to p_g, and the floor is an additional condition. In the outcome-correct branch the zero limit of the fluctuation is derived from factual initialization and conditional centering, not assumed. HT and Hájek normalization therefore deliver the same efficient first-order limit under their respective conditions.

4.3 Variance Estimation and Pointwise Inference

For a granular or aggregate target, report $\widehat{\text{se}}^{\,2} = n^{-2}\sum_{i=1}^{n}\widehat{\varphi}_i^2$. Here $\widehat{\varphi}_i$ is the empirical score under the chosen normalization, evaluated at the initial nuisances for one-step estimation and at the targeted nuisances for TMLE. Initial fits are cross-fitted for $L \geq 2$ and full-sample for $L = 1$; targeting fluctuations are estimated on the full sample. Aggregate scores retain the unit-level weight contribution in (2.6). Each estimator solves its empirical score equation, so $\mathbb{P}_n[\widehat{\varphi}] = 0$.

Under the applicable root-n conditions for HT or Hájek one-step or targeted estimation, collected in Corollary C.2 in Appendix C.2, with $V_\Theta \in (0, \infty)$ and Condition 3(iii) for aggregates, $n\widehat{\text{se}}^{\,2} \xrightarrow{p} V_\Theta$. These conclusions also apply under the full-sample extension of Corollary C.1 (Appendix C.1). The interval $\widehat{\theta} \pm z_{1-\alpha/2}\widehat{\text{se}}$ then has asymptotic coverage $1-\alpha$ for one target at a time; double-robust consistency alone does not justify this variance formula.

5 Monte Carlo Evidence

5.1 Common Setup and Reporting Conventions

The simulations have two parts. Section 5.2 uses CompactBinary, a finite-support design with exactly computed population limits, to assess double-robust consistency and empirical interval calibration. Section 5.3 examines Static and Dynamic stress designs with Gaussian covariates under which uniform overlap fails, including a design in which the sliding-pool parallel-trends restriction holds but the comparators' fixed-pool restrictions fail.

Each design has $T = 5$ and adoption dates $G \in \{2, 3, 4, 5, \infty\}$: in every period $k = 2, \ldots, 5$, each unit still untreated adopts with probability λ_{ik}, and adoption is absorbing. As in Section 2.1, Z_i denotes a time-invariant covariate and X_{ik} a time-varying one.

Write $\tau_{g,t} := \text{ATT}(g, t)$. The two causal targets are

$$\begin{aligned} \text{ATT}(2,4) &= \tau_{2,4} = \mathbb{E}\{Y_4(2) - Y_4(\infty) \mid G = 2\}, \\ \text{ES}(1) &= \frac{p_2\tau_{2,3} + p_3\tau_{3,4} + p_4\tau_{4,5}}{p_2 + p_3 + p_4}, \qquad p_g = \mathbb{P}(G = g). \end{aligned} \tag{5.1}$$

Thus ES(1) averages the effects one period after adoption for cohorts 2, 3, and 4, weighted by their population shares. The estimators use the empirical shares $\widehat{p}_g$, so their influence-function standard errors include the aggregation-weight score in (2.6).

Each design and scenario or route is evaluated at $n \in \{100, 500, 5000\}$, with five-fold cross-fitting ($L = 5$) and with full-sample fitting ($L = 1$), requesting 2,000 replications per cell; $L = 1$ means that the nuisances are fitted and evaluated on the full sample. Within each design, scenario or route, and sample size, all procedures and both fitting settings use the same simulated panels.

We estimate ATT$(2, 4)$ and ES(1) with the one-step estimating-equation estimators EEq-HT and EEq-Hájek of Section 3; the pairwise TMLEs TMLE-pw-HT and TMLE-pw-Hájek of Section 4; and the doubly robust estimators of Callaway and Sant'Anna (2021) with never-treated and not-yet-treated comparison groups, DR-nt and DR-ny, which do not cross-fit and whose results coincide across the two fitting settings. Pooled summaries use the four proposed procedures. We feature EEq-Hájek and TMLE-pw-Hájek in the main tables; Appendices H–J report the remaining procedures and four further preregistered variants, with printed names mapped to archive identifiers in Appendix G.5. The comparators use different conditioning histories and maintained restrictions, detailed in Appendix G.3.

We report nominal 95% pointwise Wald intervals, using the influence-function standard errors of Section 4.3 for the proposed procedures and the `did` package's multiplier-bootstrap standard errors for the comparators (Appendix G.3). A procedure–target result is *complete* when it returns a finite estimate and a finite, positive standard error, with no error or warning in the prespecified fits or package calls; no fallback is used. The *completion share* is the fraction of the 2,000 requested replications that are complete. Bias, coverage, and root mean squared error (RMSE) are computed over complete replications and are therefore conditional on completion. Coverage comparisons are descriptive: we call empirical coverage *low* when it is more than two pointwise Monte Carlo standard errors (MCSEs; Appendix G) below 0.95, without multiplicity adjustment.

5.2 Validation with Exact Population Limits

CompactBinary is a finite-support design in which the population targets and estimator limits can be computed exactly from the specified law, allowing the two double-robustness branches to be examined separately. It has Z distributed uniformly on $\{-1, 0, 1\}$, $T = 5$, adoption dates $G \in \{2, 3, 4, 5, \infty\}$, binary outcomes, and frozen history $\mathcal{C}_{g-1} = Z$. Write $\mu_s(a, z) := \mathbb{P}\{Y_s(a) = 1 \mid Z = z\}$ and let $\lambda_k(z) := \mathbb{P}(G = k \mid G \geq k, Z = z)$ be the adoption hazard. For $k = 2, \ldots, 5$, $s = 0, \ldots, 5$, and $g = 2, \ldots, 5$ with $t = g, \ldots, 5$, the law is

$$\begin{aligned} \lambda_k(z) &= \operatorname{logit}^{-1}\{-2 + 0.1(k-2) + 0.35z + \chi_{\text{haz}} z^2\}, \\ \mu_s(\infty, z) &= \operatorname{logit}^{-1}\{-0.7 + 0.12s + 0.35z + \chi_{\text{out}}(s/5) z^2\}, \\ \operatorname{logit} \mu_t(g, z) - \operatorname{logit} \mu_t(\infty, z) &= 0.4 + 0.08(g-2) + 0.05(t-g) + 0.1z. \end{aligned} \tag{5.2}$$

For $s < g$, set $\mu_s(g, z) = \mu_s(\infty, z)$, so adoption has no anticipatory effect. Conditional on Z, the random draws that generate the potential-outcome paths are independent across periods and independent of the adoption draws. A single outcome uniform is shared across potential regimes within each period; the observed outcome $Y_s = Y_s(G)$ uses the realized adoption date. In CompactBinary routes A, C1, C2, and C12, every untreated prefix mean (2.10) lies strictly between zero and one, so Theorem 2.4(iii), under its stated identification and moment conditions, identifies the efficient influence functions of Section 2 with those of the binary causal model. Consistency and rate conditions for the nuisance learners remain separate requirements.

The proposed procedures fit the binary control levels and frozen-history hazards by logistic regressions linear in Z, and difference the fitted control-level probabilities to obtain the outcome

increment regression (Appendix G.2). Routes C1 and C2 add to the outcome process and adoption index, respectively, a quadratic term in Z that the corresponding working model omits; C12 adds both. Table I shows which working forms contain the true functions in each route.

Table I: CompactBinary routes for the proposed procedures. Correct means that the prespecified working form contains the true function; fitted-limit recovery and the further assumptions of each theorem are separate requirements. The hazard determines the inverse-risk ratio through (3.2).

Route	χ_{out}	χ_{haz}	Outcome working form	Hazard working form
A	0	0	correct	correct
C1	0.6	0	misspecified	correct
C2	0	0.6	correct	misspecified
C12	0.6	0.6	misspecified	misspecified

Route A assesses consistency and efficient pointwise inference under the fitted-nuisance consistency, rate, and regularity conditions of Sections 3–4; C1 and C2 assess double-robust consistency when the fitted ratio or outcome regression, respectively, recovers the truth. One correct working form alone does not validate the EIF standard error. Route C12 assesses population drift when both forms are misspecified. Corollary C.1 supplies the full-sample extension for this finite-support design, while retaining the other conditions.

For a causal target $\theta \in \{\text{ATT}(2,4), \text{ES}(1)\}$, write $\overline{\theta}_{\text{proc}}$ for the population limit of a procedure, when it exists, and define its drift by

$$\widehat{\theta}_{\text{proc}} \xrightarrow{p} \overline{\theta}_{\text{proc}}, \qquad \text{drift}_{\text{proc}} := \overline{\theta}_{\text{proc}} - \theta. \tag{5.3}$$

Drift is a fixed population quantity; reported Monte Carlo bias is a finite-sample average over complete replications. Under the conditions of Proposition F.1 in Appendix F.3, each proposed procedure's limit is a combination of treated and control residual means in which recovery of the true ratio (routes A and C1) or of the true outcome regression (routes A and C2) returns the causal target, whereas in route C12 neither branch applies and the limits may differ from the target.

For CompactBinary, we compute the causal targets and solve the population working-model and targeting equations. The resulting limits of all proposed procedures in routes A, C1, and C2 agree with the causal targets, so these routes form the validation panel. The comparators have small nonzero population drifts under the `did` package's own working models, so the same validation premise does not extend to them. Appendix H.4 reports the numerical checks.

Table II reports all four routes at $n = 100, 500, 5000$ for EEq-Hájek and TMLE-pw-Hájek with five-fold cross-fitting. At $n = 5000$, across the three validation routes, both targets, both cross-fitting settings, and the four pooled procedures, empirical coverage ranges from 0.947 to 0.958. At $n = 100$, the ATT$(2,4)$ intervals have low empirical coverage in routes A and C1; these shortfalls are absent by $n = 500$, and RMSE falls roughly as $n^{-1/2}$ (Appendix H.4). Coverage at every reported sample size, including route A, is an empirical finding.

Route C12. With both working forms misspecified, route C12 lies outside the double-robustness guarantees. The computed limits of the four pooled procedures exceed the causal targets. At $n = 5000$, both featured procedures have low empirical coverage, consistent with their persistent centering error (Table H.4; Appendix H.4).

Table II: CompactBinary: calibration and misspecification, with five-fold cross-fitting. Compl. is completion; SD is the Monte Carlo standard deviation, SE the mean estimated standard error, and Cov. coverage of nominal 95% Wald intervals. Bias, SD, SE, and coverage use complete replications.

		EEq-Hájek					TMLE-pw-Hájek				
Route	n	Compl.	Bias	SD	SE	Cov.	Compl.	Bias	SD	SE	Cov.
Panel A: ATT$(2,4) = 0.126$ (A), 0.118 (C1), 0.125 (C2), 0.116 (C12)											
A	100	0.967	−0.0036	0.2265	0.2150	0.918	0.964	−0.0041	0.2249	0.2148	0.919
	500	1.000	0.0027	0.0959	0.0965	0.942	1.000	0.0027	0.0959	0.0965	0.942
	5,000	1.000	−0.0005	0.0308	0.0305	0.949	1.000	−0.0005	0.0308	0.0305	0.949
C1	100	0.968	−0.0040	0.2232	0.2123	0.920	0.965	−0.0035	0.2229	0.2119	0.920
	500	1.000	−0.0013	0.0968	0.0952	0.941	1.000	−0.0012	0.0968	0.0952	0.941
	5,000	1.000	−0.0004	0.0301	0.0301	0.958	1.000	−0.0004	0.0301	0.0301	0.958
C2	100	0.990	0.0053	0.2038	0.1992	0.939	0.990	0.0053	0.2035	0.1989	0.939
	500	1.000	0.0000	0.0882	0.0868	0.944	1.000	0.0000	0.0882	0.0868	0.944
	5,000	1.000	0.0000	0.0270	0.0273	0.955	1.000	0.0000	0.0270	0.0273	0.955
C12	100	0.992	0.0085	0.2004	0.1969	0.940	0.992	0.0087	0.2002	0.1967	0.940
	500	1.000	0.0073	0.0873	0.0856	0.942	1.000	0.0073	0.0873	0.0856	0.942
	5,000	1.000	0.0146	0.0268	0.0268	0.914	1.000	0.0146	0.0268	0.0268	0.914
Panel B: ES$(1) = 0.131$ (A), 0.122 (C1), 0.129 (C2), 0.120 (C12)											
A	100	0.945	−0.0006	0.1371	0.1372	0.941	0.934	−0.0010	0.1367	0.1370	0.941
	500	1.000	−0.0008	0.0599	0.0602	0.954	1.000	−0.0008	0.0599	0.0602	0.954
	5,000	1.000	−0.0003	0.0189	0.0189	0.947	1.000	−0.0003	0.0189	0.0189	0.947
C1	100	0.947	−0.0013	0.1371	0.1350	0.942	0.936	−0.0005	0.1370	0.1349	0.942
	500	1.000	0.0015	0.0609	0.0594	0.941	1.000	0.0015	0.0609	0.0594	0.941
	5,000	1.000	−0.0010	0.0186	0.0187	0.949	1.000	−0.0010	0.0186	0.0187	0.949
C2	100	0.979	0.0037	0.1307	0.1303	0.946	0.979	0.0036	0.1304	0.1302	0.946
	500	1.000	0.0014	0.0574	0.0562	0.946	1.000	0.0014	0.0574	0.0562	0.946
	5,000	1.000	0.0000	0.0176	0.0176	0.953	1.000	0.0000	0.0176	0.0176	0.953
C12	100	0.979	0.0060	0.1283	0.1289	0.952	0.978	0.0063	0.1281	0.1288	0.952
	500	1.000	0.0069	0.0564	0.0556	0.944	1.000	0.0069	0.0564	0.0556	0.944
	5,000	1.000	0.0087	0.0174	0.0174	0.918	1.000	0.0087	0.0174	0.0174	0.918

Note: Bias is relative to the causal target. For the displayed procedures, the population limit equals the causal target in A, C1, and C2; in C12 it exceeds the target by 0.0130 for ATT$(2,4)$ and 0.0083 for ES(1), to four decimals (Section 5.2; Table H.4). Bias and coverage MCSEs are at most 0.0052 and 0.0063, respectively. Appendix H.4 adds RMSE, bias minus drift, full-sample fitting, and the other procedures.

5.3 Stress Designs and Results

In contrast to CompactBinary, the Static and Dynamic designs violate the uniform overlap bound of Assumption 3 in every scenario: their Gaussian covariates allow adoption hazards to approach one in the tails, so the true inverse-risk ratio is unbounded and the fixed cap prevents its recovery (Appendix G.2). These designs therefore lie outside the conditions of the consistency and inference theorems.

Each has continuous and binary outcomes under scenarios A, B, B+, C1, and C2. Scenario A is the baseline, with adoption intercept and slope $(\gamma_0, \gamma_1) = (-2.0, 1.0)$; B and B+ strengthen confounding, with $(-2.5, 1.5)$ and $(-3.0, 2.5)$. Scenarios C1 and C2 keep the values of A and add to the outcome process and to the adoption index, respectively, a quadratic term that the corresponding working model omits. Its coefficient is $\xi_{\text{out}} = 0.5$ in C1 and $\xi_{\text{trt}} = 0.5$ in C2; otherwise $\xi_{\text{out}} = \xi_{\text{trt}} = 0$. These perturbations induce or add to working-model misspecification, depending on the design (Table G.1). In the continuous designs treatment adds $\tau = 1$ to the outcome in every treated period, so $\text{ATT}(2, 4) = \text{ES}(1) = 1$. In the binary designs it adds $\tau = 1$ to the outcome logit, and the targets are probability-scale effects that differ across scenarios (Tables G.3 and G.4).

Static design. A time-invariant covariate $Z_i \sim \mathcal{N}(1, 1)$ drives adoption through the same hazard in every period, $\text{logit}\, \lambda_{ik} = \gamma_0 + \gamma_1 Z_i + \xi_{\text{trt}} Z_i^2$. With a unit effect $\alpha_i \sim \mathcal{N}(0, 1)$ independent of Z_i and $D_{ik} = \mathbb{I}(G_i \leq k)$, the outcome index is

$$\mu_{ik} = \alpha_i + 0.5k + 2Z_i + \tau D_{ik} + \xi_{\text{out}}\, c_k Z_i^2. \tag{5.4}$$

Continuous outcomes are $Y_{ik} = \mu_{ik} + \sigma_{ik} u_{ik}$ with $\sigma_{ik} = \exp(0.5 D_{ik} + 0.2|Z_i|)$, independent standard normal u_{ik}, and $c_k = k$, so that under C1 the untreated increment has mean $0.5 + 0.5 Z_i^2$; a quadratic term constant in k would cancel from the increments. Binary outcomes are independent across periods given (α_i, Z_i, G_i), with $\mathbb{P}(Y_{ik} = 1 \mid \alpha_i, Z_i, G_i) = \text{logit}^{-1}(\mu_{ik} - 2)$ and $c_k = 1$. The history is $\mathcal{C}_{g-1} = Z$.

Dynamic continuous design. With $Z_i \sim \mathcal{N}(0, 1)$ and a pre-panel outcome $Y_{i0} \sim \mathcal{N}(0, 1)$, untreated outcomes and adoption follow

$$\begin{aligned} Y_{ik}(\infty) &= Y_{i,k-1}(\infty) + 0.5 + Z_i + \xi_{\text{out}} Z_i^2 + u_{ik}, \\ \text{logit}\, \lambda_{ik} &= \gamma_0 + \gamma_1 Z_i + 0.1\, Y_{i,k-1}(\infty) + \xi_{\text{trt}} Z_i^2, \end{aligned} \tag{5.5}$$

with independent standard normal u_{ik} and $Y_{ik} = Y_{ik}(\infty) + \tau D_{ik}$. The history is $\mathcal{C}_{g-1} = (Z, Y_{g-1})$. Adoption responds to the lagged outcome, but u_{ik} is independent of everything dated $k-1$ or earlier, so $\mathbb{E}\{\Delta Y_k(\infty) \mid \mathcal{C}_{g-1}, G = g\} = \mathbb{E}\{\Delta Y_k(\infty) \mid \mathcal{C}_{g-1}, G > k\} = 0.5 + Z + \xi_{\text{out}} Z^2$ and Assumption 2 holds. The same equality holds given Z alone, so this design illustrates the sliding-pool restriction without establishing a need to condition on the lag.

Dynamic binary design. For a binary outcome, the conditional mean of ΔY_k depends on the level Y_{k-1}, so adoption that responded to the lagged outcome would make cohort membership informative about untreated increments for $k > g$ and violate Assumption 2. The binary design therefore routes selection through a serially correlated covariate that does not enter the outcome. With $Z_i \sim \mathcal{N}(0, 1)$, $X_{i0} \sim \mathcal{N}(0, 1)$, $X_{ik} = 0.5 X_{i,k-1} + 0.75^{1/2} \nu_{ik}$ for independent standard normal

ν_{ik}, and $Y_{i0}(\infty) \sim \text{Bernoulli}(0.4)$,

$$\mathbb{P}\{Y_{ik}(\infty) = 1 \mid Z_i, Y_{i,k-1}(\infty)\} = \text{logit}^{-1}\{-0.4 + Z_i + \xi_{\text{out}} Z_i^2 + 0.8\, Y_{i,k-1}(\infty)\},$$
$$\text{logit}\, \lambda_{ik} = \gamma_0 + \gamma_1 Z_i + 0.6 X_{i,k-1} + \xi_{\text{trt}} X_{i,k-1}^2. \tag{5.6}$$

Treatment adds $\tau = 1$ to the current-period logit of the treated path, whose lag is its own, so the effect accumulates; both potential paths use the same uniform draws. The history is $\mathcal{C}_{g-1} = (Z, X_{g-1})$. Given Z, the outcome paths are independent of X and G, so Assumption 2 holds for every reported target. The two Dynamic designs are therefore not nested: one selects on the lagged outcome and the other on a lagged covariate.

The proposed working models enter the frozen history linearly: logistic regressions for adoption hazards, linear regressions for continuous outcome increments, and differenced logistic level regressions for binary increments. The outcome form is correct in the continuous designs outside C1, but approximate in the Gaussian-covariate binary designs even in A. The Static hazard form is correct outside C2; in the Dynamic designs, future frozen-history hazards are approximate. C1 and C2 apply the same kinds of perturbation as the CompactBinary routes, but the overlap failure and ratio cap mean that these scenarios do not isolate the double-robustness branches under the stated theorems. Table G.1 gives the full status map and Appendix G.2 the fitting details. The Static and Dynamic designs have no C12 scenario; both-working-form misspecification is assessed by CompactBinary route C12 in Section 5.2. Appendix G.1 describes computation of the binary targets.

In the continuous Dynamic design, the comparators' fixed-pool parallel-trends restrictions fail while the sliding-pool restriction holds (Appendix G.3). At $n = 5000$, comparator bias is present already under scenario A, consistent with this identification failure (Tables G.3–G.4). Under B and B+, their bias ranges from 0.035 to 0.223; in the binary Dynamic design their bias is smaller relative to the target but still exceeds two pointwise MCSEs in absolute value. Low coverage in the continuous design accompanies substantial bias despite standard errors close to the Monte Carlo spread (Appendix G.4).

Table III reports the featured procedures and the comparators in the continuous and binary Dynamic designs under B and B+. Scenario B+ moves adoption probabilities further toward the overlap boundary. At $n = 100$ the comparators complete far less often than the featured procedures, so summaries there describe differently selected subsets of the panels and do not rank the procedures (Section 5.4 examines the failure patterns).

At $n = 5000$, under B+ the TMLE-pw-Hájek intervals have low empirical coverage in both Dynamic designs and for both targets, 0.838–0.916, despite small bias relative to the targets. The EEq-Hájek intervals cover 0.939–0.952 in the same cells and are within two MCSEs of 0.95 except in the continuous ATT$(2, 4)$ cell. The two procedures have nearly identical average standard errors, but TMLE-pw-Hájek has greater Monte Carlo variability, leaving its SE/SD below one. A milder version of this pattern appears under B in the continuous design (Appendix G.4).

In the Static designs under B+ (Appendix H.3), the pairwise intervals again have low empirical coverage at $n = 5000$, and so do the continuous-outcome EEq-Hájek intervals and the binary-outcome comparator intervals.

Scenario C1 of the continuous Static and Dynamic designs illustrates why working-form correctness must be distinguished from recovery of the fitted nuisance. At $n = 5000$ with five-fold cross-fitting, absolute biases of the featured procedures range from 0.039 to 0.432, and those of the comparators from 0.209 to 2.136. In Static C1 the hazard form is correct, but the cap keeps the fitted ratio from converging to the true ratio, so with the outcome regression misspecified neither branch applies. For EEq-HT and EEq-Hájek in Static C1, the computed asymptotic biases explain

Table III: Dynamic designs under B and B+; five-fold cross-fitting for the proposed procedures. Compl. is the completion share. Bias and coverage (Cov.) of nominal 95% Wald intervals, with MCSEs in parentheses, and SE/SD, the average estimated standard error over the Monte Carlo standard deviation, are at $n = 5000$. Tables I.2–I.3 give completion, bias, coverage, and RMSE at all sample sizes; Table G.2 gives SE/SD at $n = 5000$.

			Compl.		$n = 5{,}000$		
Design	Scen.	Procedure	$n = 100$	$n = 5{,}000$	Bias	Cov.	SE/SD
Panel A: ATT$(2, 4)$							
Continuous	B	EEq-Hájek	0.925	1.000	0.004 (0.002)	0.947 (0.005)	0.98
		TMLE-pw-Hájek	0.925	1.000	0.005 (0.003)	0.914 (0.006)	0.89
		DR-nt	0.233	1.000	0.195 (0.003)	0.654 (0.011)	1.00
		DR-ny	0.233	1.000	0.107 (0.003)	0.848 (0.008)	1.01
	B+	EEq-Hájek	0.645	1.000	−0.001 (0.003)	0.939 (0.005)	0.95
		TMLE-pw-Hájek	0.645	1.000	−0.001 (0.004)	0.838 (0.008)	0.72
		DR-nt	0.039	1.000	0.223 (0.004)	0.722 (0.010)	0.99
		DR-ny	0.039	1.000	0.129 (0.004)	0.880 (0.007)	0.97
Binary	B	EEq-Hájek	0.941	1.000	−0.0012 (0.0008)	0.958 (0.004)	1.01
		TMLE-pw-Hájek	0.941	1.000	−0.0013 (0.0008)	0.946 (0.005)	0.94
		DR-nt	0.150	1.000	−0.0091 (0.0008)	0.937 (0.005)	0.99
		DR-ny	0.150	1.000	−0.0067 (0.0007)	0.945 (0.005)	1.00
	B+	EEq-Hájek	0.692	1.000	−0.0010 (0.0011)	0.952 (0.005)	0.99
		TMLE-pw-Hájek	0.692	1.000	−0.0013 (0.0014)	0.866 (0.008)	0.76
		DR-nt	0.030	1.000	−0.0154 (0.0012)	0.946 (0.005)	1.01
		DR-ny	0.030	1.000	−0.0128 (0.0011)	0.947 (0.005)	0.99
Panel B: ES(1)							
Continuous	B	EEq-Hájek	0.786	1.000	0.002 (0.001)	0.946 (0.005)	0.99
		TMLE-pw-Hájek	0.772	1.000	0.002 (0.001)	0.939 (0.005)	0.96
		DR-nt	0.232	0.998	0.106 (0.002)	0.673 (0.011)	1.01
		DR-ny	0.232	1.000	0.035 (0.001)	0.907 (0.006)	1.01
	B+	EEq-Hájek	0.382	1.000	−0.001 (0.002)	0.941 (0.005)	0.97
		TMLE-pw-Hájek	0.352	1.000	−0.001 (0.002)	0.898 (0.007)	0.84
		DR-nt	0.039	0.997	0.131 (0.002)	0.723 (0.010)	0.99
		DR-ny	0.039	1.000	0.040 (0.002)	0.919 (0.006)	0.97
Binary	B	EEq-Hájek	0.791	1.000	−0.0005 (0.0005)	0.948 (0.005)	1.03
		TMLE-pw-Hájek	0.770	1.000	−0.0005 (0.0005)	0.947 (0.005)	1.02
		DR-nt	0.150	1.000	−0.0043 (0.0005)	0.946 (0.005)	1.01
		DR-ny	0.150	1.000	−0.0020 (0.0004)	0.944 (0.005)	1.02
	B+	EEq-Hájek	0.373	1.000	−0.0001 (0.0007)	0.946 (0.005)	0.98
		TMLE-pw-Hájek	0.343	1.000	−0.0001 (0.0008)	0.916 (0.006)	0.87
		DR-nt	0.029	1.000	−0.0084 (0.0007)	0.941 (0.005)	1.01
		DR-ny	0.029	1.000	−0.0044 (0.0006)	0.948 (0.005)	0.96

the persistent bias: full-sample Monte Carlo biases are close to these asymptotic biases, whereas five-fold biases remain below them (Appendix G.2).

The EEq-Hájek intervals also have low empirical coverage under C2 in the continuous designs: 0.907 and 0.911 for ATT$(2,4)$ and ES(1) in the Static design and 0.927 for ATT$(2,4)$ in the Dynamic design, with SE/SD between 0.85 and 0.91 (Tables G.3–G.4 and G.2).

5.4 Completion and Finite-Sample Failures

Table I.1 reports availability for every procedure. For the pooled procedures, unavailability occurs almost entirely at $n = 100$, is rare at $n = 500$, and is absent at $n = 5000$. The worst cells combine five-fold cross-fitting and strong confounding (B+): ATT$(2,4)$ in the continuous Dynamic design and ES(1) in the binary Dynamic design.

Most failures in these cells arise in fold-specific frozen-hazard fits, through warnings consistent with separation or because a training risk set contains no adoptions. Fitting on the full sample improves completion on the same panels. Appendix I reports the counts and failure diagnostics, and Table H.1 compares the two fitting settings.

Comparator unavailability largely reflects package warnings about small cohorts, which the no-fallback protocol discards before estimation. A few additional results are discarded because of warnings about simultaneous bands that are not used here, despite the package having computed a pointwise estimate and standard error; these results are recovered in the analytic-standard-error sensitivity check (Appendix H.5). Comparator availability therefore reflects the protocol's strictness as well as the design's difficulty, and it says nothing about accuracy in the excluded replications.

5.5 Summary

The CompactBinary population calculations corroborate double-robust consistency in routes A, C1, and C2; empirical coverage is close to nominal at the largest sample size, with the theoretical scope stated in Section 5.2. The stress designs show that small bias can coexist with low coverage, including for pairwise targeting under B+ and estimating-equation inference under C2. At $n = 5000$ in the continuous Dynamic design, the comparators' bias is present in every scenario, consistent with the failure of their fixed-pool restrictions there. Availability is a separate finite-sample concern, concentrated at $n = 100$; comparator availability also reflects the protocol's treatment of package warnings.

6 Discussion

This paper develops a history-adjusted framework for staggered difference-in-differences with time-varying covariates. By formulating conditional parallel trends relative to a sliding risk set ($G > k$) and anchoring the conditioning set to the pre-adoption history $\mathcal{C}_{g-1}$, the framework uses not-yet-treated units while preserving exact identification: apart from restrictions implied by the support of the outcome, it places no restriction on the observed data. This structure leads to efficient influence functions for granular ATT(g,t) parameters and prespecified aggregates, a cross-fitted doubly robust one-step estimator whose second-order remainder is controlled by the product of the control-regression and fitted inverse-risk-ratio estimation errors, and a pairwise targeted minimum loss-based estimator (TMLE) constructed on a fixed target grid, with reported pairs and summaries selected in advance. Our estimators accommodate pre-adoption histories of time-varying covariates and cross-fitted nuisance learners satisfying the stated consistency and rate conditions.

At a common target and conditioning history, the cohort-by-cohort restrictions of Chen et al. (2025) imply our pooled risk-set restriction, giving the weak efficiency-bound ordering of Proposition 2.1 at common laws where the target is pathwise differentiable in the pooled model. Their displayed benchmark conditions on baseline Z, whereas ours uses the cohort-indexed $\mathcal{C}_{g-1}$; without restrictions linking these histories, that ordering does not apply.

The simulations distinguish consistency from interval calibration. In CompactBinary routes A, C1, and C2, the known population limits corroborate double robustness, and empirical coverage is close to nominal at the largest sample size. Under strong confounding at $n = 100$, completion falls for every procedure: the comparators complete least often, and the proposed procedures' failures, mostly in fold-specific hazard fits and consistent with separation in small training risk sets, are far less frequent without cross-fitting. In the stress designs, bias small relative to the target can still coexist with low coverage: this occurs for the pairwise targeted intervals under B+ and in several cells outside B+, including C2, and for the one-step intervals in several continuous-outcome cells under B+ and C2. Standard errors below the Monte Carlo spread are consistent with variance underestimation, although other contributors remain possible (Appendices G.4 and H.3). These findings describe the studied designs; completion and interval calibration both matter when assessing the procedures. The simulations use low-dimensional working models and do not assess high-dimensional performance.

The framework also has causal limitations. Conditioning on $\mathcal{C}_{g-1}$ avoids adjustment for post-adoption covariates, but it cannot rule out unobserved factors that influence both adoption timing and future untreated trends. For example, Medicaid expansion may respond to latent state-level fiscal changes, and treatment decisions in chronic-disease management may respond to latent health deterioration. In such settings, the observed pre-adoption history may not suffice for conditional parallel trends.

A possible extension is to condition on counterfactual covariate histories, $\mathcal{C}_k(\infty)$. These histories follow the untreated trajectory but are unobserved for treated units, so the corresponding full-data EIF is not directly estimable. Identification from proxies would require additional assumptions: a proximal approach would need suitable bridge functions and completeness conditions (Miao et al., 2018), while a missing-at-random approach would need additional observed variables and its own identification conditions.

Finally, the current framework assumes balanced panel data, requiring units to be observed from the initial through the final period. This excludes repeated cross-sectional designs in which different units appear in each period. Extending the approach to such settings would require modeling the period-sampling mechanism together with staggered treatment adoption.

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A Notation

Table A.1 summarizes the principal symbols used in the main text; quantities used only in proofs are defined where they appear.

Conventions. Superscript 0 denotes the true law, hats denote estimates, ini denotes initial fits, $*$ denotes targeted fits, and $-\ell$ denotes training off fold ℓ (on all units when $L = 1$). Overlines on nuisance and procedure quantities denote deterministic probability limits; $\overline{a}$ (projection margin) and $\overline{g}$ (largest value of G) are instead fixed. The final subscript 0 in $m_{g,k,0}$ denotes the control branch. The indices i, g, k and ℓ refer to units, cohorts, periods and folds; t is calendar time or a target endpoint. G is the adoption cohort, T is the time horizon, and $\mathcal{T}$ denotes a tangent space. The norm $\|\cdot\|_2$ uses P_0; fitted functions are held fixed when their norms are computed.

Table A.1: Principal notation in the main text.

Symbol	Meaning
O_i; Y_{it}, X_{it}, Z_i; ΔY_k	Observed panel; outcome, time-varying and time-invariant covariates; increment $\Delta Y_k = Y_k - Y_{k-1}$ (Sections 2.1–2.2).
G_i, D_{it}; $Y_t(g)$, $X_t(g)$	Adoption cohort (∞ if never treated), treatment indicator, and potential outcome and covariates under adoption at g (see (1.1) and Section 2.1).
P_0, P, $\mathbb{P}_n$; $\|\cdot\|_2$	True and generic observed-data laws, empirical average, and $L_2(P_0)$ norm. Unsubscripted $\mathbb{E}$ and $\mathbb{P}$ refer to the compatible full-data law and agree with P_0 on observed quantities (Section 2.1).
p_g; $\mathcal{G}_+$, $\mathcal{I}_+$	Cohort probability $\mathbb{P}(G = g)$, positive-probability cohorts, and supported cohort–endpoint pairs (Section 2.1).
$\mathcal{S} = \mathcal{Q}$; $\mathcal{G}_\mathcal{S}$, $t_\mathcal{S}(g)$	Fixed target grid, its represented cohorts, and the largest endpoint for cohort g (Sections 2.1, 2.2, 2.4 and 4.1).
$\mathcal{C}_{g-1}$; a_g, κ_g	Frozen pre-adoption history, its prespecified cutoff, and its selection or summary map (see (2.1)).
$\tau_{g,t} = \mathrm{ATT}(g,t)$; Θ, $\mathrm{ES}(e)$	Granular effect, generic aggregate, and event-study effect at exposure length e (see (1.1) and (2.3)).
$\Omega_{g,t}$, $\omega_{g,t}$; $W_\mathcal{S}$, $\widehat{W}_\mathcal{S}$	Known unit-level weight, its expectation, total population weight mass $\sum_\mathcal{S} \omega_{g,t}$, and empirical weight mass (see (2.3) and the text after Lemma 3.1).
$\lambda_{g,s}$, $q_{g,k}$, e_g	Frozen-history adoption hazard, survival probability conditional on $G \geq g$, and entry probability $\mathbb{P}(G \geq g \mid \mathcal{C}_{g-1})$ (see (2.2) and the surrounding text).

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Table A.1 (continued)

Symbol	Meaning
$m_{g,k,0}$, $r_{g,k}$; $m_{g,k,G}$	Control increment regression, inverse-risk ratio, and treated branch of the composite regression (Section 3.1; see (4.3) for factual initialization).
η_g, $\eta_{\mathcal{S}}$; η_g^0, $\overline{\eta}_{\mathcal{S}}$	Nuisance collections for a cohort or target grid; true nuisance collection and deterministic limits of initial estimates (Section 3.1; text after Lemma 3.1; Theorem 3.1).
$\phi_{g,k}$, $\psi_{g,t}$; $\varphi^{\text{eff}}\{\Theta\}$	Uncentered increment signal, granular EIF, and aggregate EIF (see (2.4)–(2.6)).
$\mathcal{M}_{\text{obs}}(\mathcal{S})$; $\mathcal{M}_{\text{caus}}(\mathcal{S};\mathcal{Y})$	Observed-data model and its subset compatible with the causal assumptions and outcome support (Sections 2.3–2.4).
$\mathcal{Y}$, $\mu_{g,s}^P$	Known outcome support and identified untreated prefix mean; feasibility requires $\mu_{g,s}^P \in \text{conv}(\mathcal{Y})$ on cohort-g histories (see (2.10)–(2.11)).
$\mathcal{H}_g$; $\mathcal{M}_{\text{pool}}(\mathcal{H})$, $\mathcal{M}_{\text{cbc}}(\mathcal{H})$	Common pre-adoption history and the pooled and cohort-by-cohort models (Proposition 2.1).
$V_{\text{pool}}(\theta;P)$; $V_{\text{cbc}}(\theta;P)$	Scalar semiparametric variance bounds under the pooled and cohort-by-cohort models at law P (Proposition 2.1).
L, $\mathcal{I}_\ell$, $\mathcal{I}_\ell^c$; $\widehat{\eta}_{g,-\ell}$	Fold count, held-out fold, training complement, and nuisance fit; $L=1$ denotes full-sample fitting and evaluation (Section 3.2).
$\overline{a}$, M; $\widetilde{\lambda}$, $\widehat{\lambda}$, $\widehat{r}$	Fixed hazard projection margin and ratio cap; pre-clipping learner, projected hazard, and capped cascade ratio (see (3.2)).
$\widehat{\tau}_{g,t}^{\text{os}}$, $\widehat{\Theta}_{\text{os}}$; $\widehat{p}_g$, $\widehat{\omega}_{g,t}$	One-step granular and aggregate estimates; empirical cohort share and aggregation-weight mean (see (3.4)–(3.5)).
ν, $B_{n,g,k}^{\nu}$; $B_{n,g,k}$, $\overline{B}_{g,k}$	Normalization and control divisor: $\widehat{p}_g$ for HT, empirical control weight mass $B_{n,g,k}$ for Hájek; $\overline{B}_{g,k}$ is its limit (Section 3.2; Corollary 3.2(i); see (4.2)).
$\widehat{H}_{(g,t),k}^{\text{ini},\nu}$; $\widehat{\epsilon}_{g,t}$	Pairwise clever covariate and scalar least-squares fluctuation coefficient (see (4.2)–(4.5)).
$\widehat{m}_{g,(i,k)}^{\text{ini}}$	Initial unit–period row value; treated rows equal their observed increments (see (4.3)).
$\widehat{m}_{(g,t),(i,k)}^{*}$; $\widehat{m}_{(g,t),k,0}^{*}$	Targeted row value and control branch as a function of history (see (4.5) and the text before (4.6)).
$\widehat{\tau}_{g,t}^{\text{tar}}$, $\widehat{\Theta}_{\text{tar}}$	Pairwise targeted granular and aggregate estimates (see (4.6) and the following display).
$\widehat{\eta}_{g,t}^{*}$, $\widehat{\eta}_{\mathcal{S}}^{*}$	Targeted nuisance collections for a pair and for the grid (text after (4.6)).

Continued on next page

Table A.1 (continued)

Symbol	Meaning
$\widehat{\psi}^{\nu}_{g,t}$, $\widehat{\Psi}^{\nu}_{\Theta}$	Empirical granular and aggregate scores solved by targeting (Section 4.2).
$d_{m,g,t}$, $d_{r,g,t}$	Regression and ratio errors, each maximized separately over folds and the target's slices (see (3.9)).
V_{Θ}; $\widehat{\mathrm{se}}$	Aggregate EIF variance and estimated standard error (Theorem 3.2; Section 4.3).

Appendices: Proofs

Appendices B, C and D prove the results of Sections 2, 3 and 4, respectively. Appendix C also contains the variance-consistency lemma and the pointwise Wald intervals (Lemma C.3 and Corollary C.2), and Appendix D the limit and rate of the pooled fluctuation and the rate of the Hájek anchor (Lemmas D.1–D.3). Appendix E defines the population auxiliary score and Gram matrix and shows that the auxiliary OLS is well posed. Appendix F develops cohort-wise targeting, the two-fluctuation diagnostic, and their exact representations, with the proofs given there. As in Section 2.1, $\|f\|_2 = \{\mathbb{E}_{P_0}[f(O)^2]\}^{1/2}$ is the $L_2(P_0)$ norm.

B Proofs for Section 2

B.1 Regular Paths and Tangent Spaces

Regular-path convention. For tangent spaces, canonical gradients, and efficiency bounds, a regular path at an observed law P is a path $u \mapsto P_u$ in the model under consideration, with $P_{u=0} = P$, dominated by P and differentiable in quadratic mean at zero, whose likelihood ratio $L_u := \mathrm{d}P_u/\mathrm{d}P$ satisfies

$$a \le L_u \le b \qquad P\text{-almost surely}$$

uniformly for all sufficiently small $|u|$, for path-specific constants $0 < a \le 1 \le b < \infty$. The tangent space is the closed linear span in $L_2^0(P)$ of the scores of these paths. A scalar functional Ψ is pathwise differentiable at P relative to this class if there is a continuous linear map $\dot{\Psi}_P$ on the tangent space such that

$$\frac{\Psi(P_u) - \Psi(P)}{u} \longrightarrow \dot{\Psi}_P(h)$$

along every regular path with score h; see van der Vaart (1998, Section 25.3) for the tangent-space approach to pathwise differentiability. Pathwise differentiability, canonical gradients, and efficiency bounds throughout are relative to this regular-path class, including in the causal models and the common-history comparison. This convention restricts local paths, not the sets of model laws.

Dense scores and derivative representations. In the observed-data model, the bounded tilts of Lemma 2.1 remain admitted, with scores dense in $L_2^0(P)$. At strictly prefix-feasible laws, the localized bounded tilts in the proof of Theorem 2.4(iii) give the same density in the causal model. Thus the observed-data tangent space and, at strictly prefix-feasible laws, the causal tangent space are all of $L_2^0(P)$. The bounded-tilt calculations identify the candidate gradients. The proofs of Theorems 2.2 and 2.3 then verify their derivative representations along every admitted path, using overlap at P_0 and the stated second moments; the proof of Theorem 2.4(iii) transfers these representations to the causal model. Each displayed canonical gradient is therefore the unique gradient in $L_2^0(P)$ in the applicable model, and its variance is the efficiency bound relative to this path class.

Why the likelihood bounds are imposed. For unbounded outcomes, finite second moments at each law do not by themselves justify differentiation along every quadratic-mean differentiable path. For example, for $\Psi(P) = \mathbb{E}_P[Y]$, let $P = \mathcal{N}(0,1)$ and

$$P_u = (1-u^4)\mathcal{N}(0,1) + u^4\mathcal{N}(u^{-3},1), \qquad 0 < |u| < 1,$$

with $P_{u=0} = P$. Each law has finite second moment, and its squared Hellinger distance from P is $O(u^4)$, so the path is differentiable in quadratic mean with score zero. Nevertheless, $\Psi(P_u) = u$ has derivative one at zero, whereas a derivative linear in the score would be zero. The local likelihood bounds exclude such uncontrolled tail perturbations.

B.2 Aggregate-Score Bookkeeping

Remark B.1 (Mean and Vanishing of the EIF Correction Term)**.** For any weight pair with $\omega_{g,t} = \mathbb{E}[\Omega_{g,t}(O)]$, $\mathbb{E}[\Omega_{g,t}(O) - \omega_{g,t}\mathbb{I}(G = g)/\mathbb{P}(G = g)] = \omega_{g,t} - \omega_{g,t} = 0$. Since $\tau_{g,t} - \Theta$ is a constant, the term (2.8) has mean zero; together with the mean Θ of (2.7) and the mean $-\Theta$ of (2.9), this gives $\mathbb{E}[\varphi^{\text{eff}}\{\Theta\}] = 0$. The term vanishes identically in two leading schemes. (i) If $\Omega_{g,t}(O) = \mathbb{I}(G = g)$ and $\omega_{g,t} = \mathbb{P}(G = g)$, its bracket vanishes identically. (ii) If $\Omega_{g,t} = \omega_{g,t} = 1$ and $\mathcal{S} = \{(\tilde{g}, t) : t \geq \tilde{g}\}$, its bracket does not depend on t, and $\sum_{t\geq\tilde{g}}(\tau_{\tilde{g},t} - \Theta) = 0$ by (2.3). In both cases the EIF is the sum of (2.7) and (2.9).

Remark B.2 (Finite-Sample Cancellation of Weight Corrections)**.** With the empirical frequencies $\widehat{\omega}_{g,t} = \mathbb{P}_n[\Omega_{g,t}(O)]$ and $\widehat{p}_g = \mathbb{P}_n[\mathbb{I}(G = g)]$ substituted, the term (2.8) vanishes exactly in every sample,

$$\mathbb{P}_n\big[\Omega_{g,t}(O) - \widehat{\omega}_{g,t}\mathbb{I}(G = g)/\widehat{p}_g\big](\tau_{g,t} - \Theta) = (\widehat{\omega}_{g,t} - \widehat{\omega}_{g,t})(\tau_{g,t} - \Theta) = 0,$$

so neither the one-step equation for Θ nor the targeting step of Section 4 needs to account for it. The cancellation concerns the empirical mean only: the unit-level deviations $\Omega_{g,t}(O_i) - \widehat{\omega}_{g,t}\mathbb{I}(G_i = g)/\widehat{p}_g$ are generally nonzero, so their score contribution must be retained in influence-function variances and in covariances across (g, t) cells. The decomposition (2.7)–(2.9) and the form (2.6) coincide whenever $\widehat{\Theta}$ is the $\widehat{\omega}$-weighted average of the $\widehat{\tau}_{g,t}$, and (2.6) prices the sampling variation of $\widehat{\omega}_{g,t}$ without a separate delta-method step.

B.3 Proofs

Proof of Theorem 2.1. Fix $t \geq g$ and write $\mathcal{C} = \mathcal{C}_{g-1}$. By Assumption B-CON, $\mathrm{ATT}(g, t) = \mathbb{E}[Y_t \mid G = g] - \mathbb{E}[Y_t(\infty) \mid G = g]$. Assumption 1 with $(j, s) = (g, g - 1)$ and consistency give $\mathbb{E}[Y_{g-1}(\infty) \mid \mathcal{C}, G = g] = \mathbb{E}[Y_{g-1}(g) \mid \mathcal{C}, G = g] = \mathbb{E}[Y_{g-1} \mid \mathcal{C}, G = g]$, so, telescoping the untreated path from $g - 1$ to t,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \sum_{k=g}^{t} \mathbb{E}\big[\mathbb{E}[\Delta Y_k(\infty) \mid \mathcal{C}, G = g] \,\big|\, G = g\big].$$

Assumption 2 replaces $\{G = g\}$ by $\{G > k\}$ in each inner expectation. For each finite $j \in \mathcal{G}_+$ with $j > k$, Assumption 1 at $s = k$ and at $s = k-1$, differenced, and consistency give $\mathbb{E}[\Delta Y_k(\infty) \mid \mathcal{C}, G = j] = \mathbb{E}[\Delta Y_k(j) \mid \mathcal{C}, G = j] = \mathbb{E}[\Delta Y_k \mid \mathcal{C}, G = j]$; for $j = \infty$ this is consistency itself. Averaging over j with the weights $\mathbb{P}(G = j \mid \mathcal{C}, G > k)$ gives

$$\mathbb{E}[\Delta Y_k(\infty) \mid \mathcal{C}, G > k] = \mathbb{E}[\Delta Y_k \mid \mathcal{C}, G > k], \tag{B.1}$$

and substituting (B.1) into the telescoped expression gives the stated formula. ∎

Proof of Lemma 2.1. Fix a bounded $h \in L_2^0(P_0)$ and $\delta \in (0, 1)$, and for $|u|\|h\|_\infty \leq \delta$ let $\mathrm{d}P_u/\mathrm{d}P_0 = L_u := 1+uh$. Since $\mathbb{E}_{P_0}h = 0$ and $1-\delta \leq L_u \leq 1+\delta$, P_u is a probability law equivalent to P_0, and the path is differentiable in quadratic mean with score h. Equivalence preserves the reduced support exactly. The quantitative restrictions persist. First, $(1-\delta)p_g(P_0) \leq p_g(P_u) \leq (1+\delta)p_g(P_0)$.

Second, with $q^0_{g,k} := \mathbb{P}_{P_0}(G > k \mid \mathcal{C}_{g-1}, G \geq g)$ and $a_\delta := (1-\delta)/(1+\delta)$, conditional Bayes' rule gives $q^{P_u}_{g,k} \geq a_\delta q^0_{g,k}$ and $1-q^{P_u}_{g,k} \geq a_\delta(1-q^0_{g,k})$, so a margin ε_{P_0} for $q^0_{g,k}$ and $1-q^0_{g,k}$ becomes the margin $a_\delta\varepsilon_{P_0}$, uniformly over the finite set $\mathcal{S}$. Third, $\mathbb{E}_{P_u}[\zeta^2] = \mathbb{E}_{P_0}[L_u\zeta^2] \leq (1+\delta)\mathbb{E}_{P_0}[\zeta^2] < \infty$ for every variable ζ in Condition 2(i)–(ii), and likewise $\mathbb{E}_{P_u}[|Y_{g-1}|\,\mathbb{I}(G \geq g)] \leq (1+\delta)\mathbb{E}_{P_0}[|Y_{g-1}|\,\mathbb{I}(G \geq g)] < \infty$. Fourth, $W_{\mathcal{S}}(P_u) = W_{\mathcal{S}}(P_0) + u\,\mathbb{E}_{P_0}[\{\sum_{\varsigma\in\mathcal{S}}\Omega_\varsigma\}h]$, whose slope is finite because h is bounded, stays nonzero for small $|u|$. Every bounded centered h is thus a score of $\mathcal{M}_{\text{obs}}(\mathcal{S})$. These are dense in $L^0_2(P_0)$, so the tangent space is all of $L^0_2(P_0)$. ∎

Proof of Theorem 2.2. Fix a bounded $h \in L^0_2(P_0)$ and the tilt $\mathrm{d}P_u/\mathrm{d}P_0 = 1+uh$, $|u|\|h\|_\infty \leq \delta < 1$, of the proof of Lemma 2.1, and suppress the subscript 0 at the truth. Write $\mathcal{C} = \mathcal{C}_{g-1}$ and $\tau_{g,t}(P) = A(P)/p_g(P)$, with $A(P) = \mathbb{E}_P[\mathbb{I}(G=g)\sum_{k=g}^t\{\Delta Y_k - m_{k,P}(\mathcal{C})\}]$, $m_{k,P}(\mathcal{C}) := m^P_{g,k,0}(\mathcal{C}) = \mathbb{E}_P[\Delta Y_k \mid \mathcal{C}, G > k]$ and $p_g(P) = \mathbb{E}_P[\mathbb{I}(G=g)]$. By conditional Bayes' rule, $m_{k,P_u} - m_k = u\,\dot m_k/(1+u\mathbb{E}[h \mid \mathcal{C}, G>k])$ exactly, with $\dot m_k(\mathcal{C}) := \mathbb{E}[\{\Delta Y_k - m_k(\mathcal{C})\}h \mid \mathcal{C}, G>k]$ and a denominator in $[1-\delta, 1+\delta]$. With $r_{g,k}(\mathcal{C}) = \mathbb{P}(G=g \mid \mathcal{C})/\mathbb{P}(G>k \mid \mathcal{C})$ on $\{\mathbb{P}(G \geq g \mid \mathcal{C}) > 0\}$, the domain of Section 2.2, iterated expectations give $\mathbb{E}[\mathbb{I}(G=g)f(\mathcal{C})] = \mathbb{E}[\mathbb{I}(G>k)r_{g,k}(\mathcal{C})f(\mathcal{C})]$ for nonnegative f, hence whenever $\mathbb{I}(G=g)f(\mathcal{C})$ is integrable. As $|\dot m_k| \leq \|h\|_\infty\mathbb{E}[|\Delta Y_k - m_k(\mathcal{C})| \mid \mathcal{C}, G>k]$ and $r_{g,k} \leq \varepsilon^{-1}$, $\mathbb{I}(G=g)\dot m_k(\mathcal{C})$ is integrable, dominated convergence differentiates $A(P_u)$ under $\mathbb{E}_{P_0}$, and $\mathbb{E}[\mathbb{I}(G=g)\dot m_k(\mathcal{C})] = \mathbb{E}[\mathbb{I}(G>k)r_{g,k}(\mathcal{C})\{\Delta Y_k - m_k(\mathcal{C})\}h]$. Hence $\dot p_g = \mathbb{E}[\mathbb{I}(G=g)h]$, $\dot A = \mathbb{E}[\sum_{k=g}^t\{\mathbb{I}(G=g) - \mathbb{I}(G>k)r_{g,k}(\mathcal{C})\}\{\Delta Y_k - m_k(\mathcal{C})\}h]$, and, since $A(P_0) = p_g\tau_{g,t}$, differentiating A/p_g gives $\dot\tau_{g,t} = \mathbb{E}[\psi_{g,t}h]$ with

$$\psi_{g,t} = \sum_{k=g}^t \frac{\mathbb{I}(G=g) - \mathbb{I}(G>k)r_{g,k}(\mathcal{C})}{p_g}\{\Delta Y_k - m_k(\mathcal{C})\} - \frac{\mathbb{I}(G=g)}{p_g}\tau_{g,t}.$$

This function has mean zero and lies in $L_2(P_0)$: Assumption 3 gives $r_{g,k} \leq \varepsilon^{-1}$, conditional Jensen bounds the control regression, and $\mathbb{E}[\mathbb{I}(G=g)m_k(\mathcal{C})^2] = \mathbb{E}[\mathbb{I}(G>k)r_{g,k}(\mathcal{C})m_k(\mathcal{C})^2] \leq \varepsilon^{-1}\mathbb{E}[\mathbb{I}(G>k)(\Delta Y_k)^2]$.

General regular paths. Let P_u be any regular path at P_0 under the convention of Appendix B.1, with score $h \in L^0_2(P_0)$ and local bounds $a \leq L_u \leq b$. Quadratic-mean differentiability gives $(\sqrt{L_u} - 1)/u \to h/2$ in $L_2(P_0)$. Multiplication by the uniformly bounded $\sqrt{L_u} + 1$ yields

$$\frac{L_u - 1}{u} \longrightarrow h \quad \text{in } L_2(P_0).$$

Indeed, the quadratic-mean remainder still tends to zero in $L_2(P_0)$ after this multiplication, and $h(\sqrt{L_u} - 1) \to 0$ in L_2, by truncating the fixed $h \in L_2$ and using the uniform bound. Write $\mathbb{I}_g := \mathbb{I}(G=g)$ and $\mathbb{I}_{>k} := \mathbb{I}(G>k)$, and define

$$L_{g,u}(\mathcal{C}) := \mathbb{E}[L_u \mid \mathcal{C}, G=g], \qquad L_{>k,u}(\mathcal{C}) := \mathbb{E}[L_u \mid \mathcal{C}, G>k], \qquad \widetilde{L}_{k,u} := \frac{L_{g,u}}{L_{>k,u}} - 1.$$

Both conditional likelihood ratios lie between a and b on the common history support. Conditional Bayes' rule gives

$$m_{k,P_u} - m_k = \frac{\mathbb{E}[(\Delta Y_k - m_k)(L_u - 1) \mid \mathcal{C}, G>k]}{L_{>k,u}}.$$

Overlap gives $\varepsilon \leq r_{g,k} \leq \varepsilon^{-1}$. Conditional Jensen and the balancing identity therefore imply

$$\|\mathbb{I}_{>k}\widetilde{L}_{k,u}\|_2 \leq \frac{1+\varepsilon^{-1/2}}{a}\|L_u - 1\|_2 = O(|u|), \qquad |\widetilde{L}_{k,u}| \leq b/a + 1.$$

The conditional-regression identity and balancing give the exact numerator expansion

$$A(P_u) - A(P_0) = p_g \mathbb{E}\left[\sum_{k=g}^{t} \phi_{g,k}(L_u - 1)\right] - \sum_{k=g}^{t} \mathbb{E}[\mathbb{I}_{>k} r_{g,k} \widetilde{L}_{k,u}(\Delta Y_k - m_k)(L_u - 1)].$$

Each $\phi_{g,k}$ and $\mathbb{I}_{>k}(\Delta Y_k - m_k)$ lies in $L_2(P_0)$ by the moment verification above. For fixed $N > 0$,

$$\|\mathbb{I}_{>k}\widetilde{L}_{k,u}h\|_2 \le N\|\mathbb{I}_{>k}\widetilde{L}_{k,u}\|_2 + (b/a+1)\|\mathbb{I}_{>k}h\mathbb{I}(|h| > N)\|_2 \longrightarrow 0,$$

by first taking $u \to 0$ and then $N \to \infty$. Consequently the absolute value of the second sum divided by u is bounded by

$$\varepsilon^{-1}\sum_{k=g}^{t}\|\mathbb{I}_{>k}(\Delta Y_k - m_k)\|_2 \left\{(b/a+1)\left\|\frac{L_u - 1}{u} - h\right\|_2 + \|\mathbb{I}_{>k}\widetilde{L}_{k,u}h\|_2\right\} = o(1).$$

The first term divided by u converges to $p_g\mathbb{E}[\sum_{k=g}^{t}\phi_{g,k}h]$, and $(p_g(P_u) - p_g)/u \to \mathbb{E}[\mathbb{I}_g h]$. The quotient rule gives $\dot{\tau}_{g,t}(h) = \mathbb{E}[\psi_{g,t}h]$ along every admitted path. Lemma 2.1 gives the full tangent space, so the verified mean-zero L_2 representer is its unique canonical gradient. Writing $r_{g,k}(\mathcal{C}) = \mathbb{P}(G = g \mid \mathcal{C}, G \ge g)/\mathbb{P}(G > k \mid \mathcal{C}, G \ge g)$ gives the stated form. ■

Proof of Theorem 2.3. Let P_u be any regular path at P_0 under the convention of Appendix B.1, with score $h \in L_2^0(P_0)$. The proof of Theorem 2.2 gives $(L_u - 1)/u \to h$ in $L_2(P_0)$ and $\dot{\tau}_{g,t}(h) = \langle \psi_{g,t}, h\rangle_{P_0}$ along this path. Since $\Omega_{g,t} \in L_2(P_0)$,

$$\frac{\omega_{g,t}(P_u) - \omega_{g,t}}{u} = \mathbb{E}_{P_0}\left[\Omega_{g,t}\frac{L_u - 1}{u}\right] \longrightarrow \langle \Omega_{g,t} - \omega_{g,t}, h\rangle_{P_0}.$$

In particular $W_{\mathcal{S}}(P_u) \to W_{\mathcal{S}}(P_0) \neq 0$, so the quotient is defined for small $|u|$. With $W = \sum_{\mathcal{S}}\omega_{g,t}$, the quotient rule applied to $\Theta = W^{-1}\sum_{\mathcal{S}}\omega_{g,t}\tau_{g,t}$ gives

$$\begin{aligned}\dot{\Theta}(h) &= \frac{1}{W}\sum_{(g,t)\in\mathcal{S}}\{\omega_{g,t}\dot{\tau}_{g,t}(h) + (\tau_{g,t} - \Theta)\dot{\omega}_{g,t}(h)\} \\ &= \left\langle \frac{1}{W}\sum_{(g,t)\in\mathcal{S}}\{\omega_{g,t}\psi_{g,t} + (\tau_{g,t} - \Theta)(\Omega_{g,t} - \omega_{g,t})\},\ h\right\rangle_{P_0},\end{aligned}$$

whose representer is (2.6). As in the proof of Theorem 2.2, continuity and density make it the unique canonical gradient. ■

B.4 Outcome Support and the Causal Model

This subsection proves Theorem 2.4 and records its special cases. Throughout, $\mathcal{Y}$ is nonempty and closed, $P \in \mathcal{M}_{\text{obs}}(\mathcal{S})$ has $\mathcal{Y}$-valued outcomes, ε_P is its overlap margin, and $\mathcal{C} = \mathcal{C}_{g-1}$ when the cohort g is fixed. Write $\underline{y} := \inf\mathcal{Y}$ and $\overline{y} := \sup\mathcal{Y}$. Because $\mathcal{Y}$ is closed, each finite endpoint belongs to $\mathcal{Y}$, so $\operatorname{conv}(\mathcal{Y}) = \{x \in \mathbb{R} : \underline{y} \le x \le \overline{y}\}$. Let

$$\rho_{g,s}^{P}(c) := \min\{\mu_{g,s}^{P}(c) - \underline{y},\ \overline{y} - \mu_{g,s}^{P}(c)\},$$

with the conventions $x - (-\infty) = +\infty$ and $(+\infty) - x = +\infty$, be the distance of a prefix mean from the boundary of $\operatorname{conv}(\mathcal{Y})$. Strict prefix feasibility is $\rho_{g,s}^{P}(\mathcal{C}_{g-1}) > 0$ almost surely on $\{G = g\}$.

Lemma B.1 (Two-point kernel). *For $x \in \mathrm{conv}(\mathcal{Y})$, let $y^-(x) := \max\{y \in \mathcal{Y} : y \le x\}$ and $y^+(x) := \min\{y \in \mathcal{Y} : y \ge x\}$. Let $w(x) := \{x - y^-(x)\}/\{y^+(x) - y^-(x)\}$ when $y^+(x) > y^-(x)$, and $w(x) := 0$ otherwise. Define*

$$\varpi(x, v) := y^-(x) + \{y^+(x) - y^-(x)\}\,\mathbb{I}\{v \le w(x)\}, \qquad x \in \mathrm{conv}(\mathcal{Y}),\ 0 < v < 1.$$

Then ϖ is Borel, $\varpi(x, v) \in \mathcal{Y}$, $\int_0^1 \varpi(x, v)\,\mathrm{d}v = x$, and $\int_0^1 |\varpi(x, v)|\,\mathrm{d}v \le |x| + c_{\mathcal{Y}}$. The constant is $c_{\mathcal{Y}} := \max\{|y^-(0)|, y^+(0)\}$ if $0 \in \mathrm{conv}(\mathcal{Y}) \setminus \mathcal{Y}$, and $c_{\mathcal{Y}} := 0$ otherwise.

Proof. For $x \in \mathrm{conv}(\mathcal{Y})$ both sets are nonempty: $\mathcal{Y}$ has points on each side of x, or contains the finite endpoint. Both sets are closed, so the maximum and the minimum exist. Both $y^{\pm}$ are nondecreasing in x, hence Borel, and so are w and ϖ. The kernel takes the values $y^{\pm}(x) \in \mathcal{Y}$, and its mean is $y^- + (y^+ - y^-)w = x$.

If $y^-(x) \ge 0$ or $y^+(x) \le 0$, the kernel has a constant sign, so its absolute mean is $|x|$. Otherwise $y^-(x) < 0 < y^+(x)$. By the extremal definitions, no point of $\mathcal{Y}$ lies strictly between them, so the open interval they bound is the component of $\mathbb{R} \setminus \mathcal{Y}$ that contains 0. Hence $y^{\pm}(x) = y^{\pm}(0)$, and the absolute mean is at most $c_{\mathcal{Y}}$. ■

Proof of Theorem 2.4(i): necessity. Let Q be a full-data law with observed marginal P that satisfies the assumptions defining $\mathcal{M}_{\mathrm{caus}}(\mathcal{S}; \mathcal{Y})$, and fix a represented cohort g and $s \in [g, t_{\mathcal{S}}(g)]$. The proof of Theorem 2.1, applied to the target (g, s), gives $\mathbb{E}_Q[Y_s(\infty) \mid \mathcal{C}, G = g] = \mu^P_{g,s}(\mathcal{C})$ almost surely on $\{G = g\}$. That proof uses Assumption 1 at $(g, g-1)$, the risk-set identity (B.1) for $k \le s$, and Assumption 2 for $k \in [g, s]$. All of these are available because $(g, t_{\mathcal{S}}(g)) \in \mathcal{S}$ and $s \le t_{\mathcal{S}}(g)$. The variable $Y_s(\infty)$ is integrable and takes values in $\mathcal{Y} \subseteq \mathrm{conv}(\mathcal{Y})$, so its conditional mean lies in $\mathrm{conv}(\mathcal{Y})$ almost surely. ■

Proof of Theorem 2.4(i): sufficiency. Enlarge the probability space by variables $\upsilon_{j,s}$, $j \in [T]$, $s \in [j, T]$, that are uniform on $(0, 1)$, independent of each other and of O. Fix $y_0 \in \mathcal{Y}$, and construct one full-data vector as follows.

- *Covariates.* Set $X_s(g') := X_s$ for every regime $g' \in [T] \cup \{\infty\}$ and every s. No maintained assumption restricts the counterfactual covariates.
- *Never-treated units.* If $G = \infty$, set $Y_s(\infty) := Y_s$ for all s.
- *Pre-adoption periods.* If $G = j < \infty$, set $Y_s(\infty) := Y_s$ for $s < j$.
- *Target windows.* If $G = j < \infty$ and j is represented in $\mathcal{S}$, set $Y_s(\infty) := \varpi\big(\mu^P_{j,s}(\mathcal{C}_{j-1}), \upsilon_{j,s}\big)$ for $s \in [j, t_{\mathcal{S}}(j)]$. Use a Borel version of $\mu^P_{j,s}$, and replace its argument by y_0 on the null set where it leaves $\mathrm{conv}(\mathcal{Y})$.
- *Remaining dates.* At every date not covered above, set $Y_s(\infty) := y_0$. These levels enter none of the assumptions.
- *Factual and other regimes.* Set $Y_s(G) := Y_s$, and $Y_s(g') := Y_s(\infty)$ for $g' \ne G$.

Every potential outcome lies in $\mathcal{Y}$ by Lemma B.1, the observed marginal is P, and Assumptions B-STA and B-CON hold by construction. On $\{G = j\}$, $Y_s(j) = Y_s(\infty) = Y_s$ for every $s < j$, so Assumption 1 holds pointwise for every target and every cohort.

Fix a represented g and $k \in [g, t_{\mathcal{S}}(g)]$. Because $\upsilon_{g,s}$ is independent of O, Lemma B.1 gives $\mathbb{E}[Y_s(\infty) \mid \mathcal{C}, G = g] = \mu^P_{g,s}(\mathcal{C})$ for $s \in [g, t_{\mathcal{S}}(g)]$, while $Y_{g-1}(\infty) = Y_{g-1}$. Hence, with $\mu^P_{g,g-1}(\mathcal{C}) := \mathbb{E}_P[Y_{g-1} \mid \mathcal{C}, G = g]$,

$$\mathbb{E}[Y_k(\infty) - Y_{k-1}(\infty) \mid \mathcal{C}, G = g] = \mu^P_{g,k}(\mathcal{C}) - \mu^P_{g,k-1}(\mathcal{C}) = m^P_{g,k,0}(\mathcal{C}).$$

On $\{G > k\}$, both $k-1$ and k precede adoption, so the untreated levels are the observed ones and the right side of Assumption 2 is the same $m^P_{g,k,0}(\mathcal{C})$.

For integrability, Lemma B.1 gives

$$\mathbb{E}\big[|Y_s(\infty)|\,\mathbb{I}(G=g)\big] \le \mathbb{E}_P\big[\mathbb{I}(G=g)\,|\mu^P_{g,s}(\mathcal{C})|\big] + c_{\mathcal{Y}}.$$

The right side is finite, because $\mathbb{E}_P[|Y_{g-1}|\,\mathbb{I}(G \ge g)] < \infty$ and, with $r^P_{g,k} \le \varepsilon_P^{-1}$,

$$\mathbb{E}_P\big[\mathbb{I}(G=g)\,|m^P_{g,k,0}(\mathcal{C})|\big] = \mathbb{E}_P\big[\mathbb{I}(G>k)\,r^P_{g,k}(\mathcal{C})\,|m^P_{g,k,0}(\mathcal{C})|\big] \le \varepsilon_P^{-1}\mathbb{E}_P|\Delta Y_k|.$$

Every other level named in Condition 1 is either the constant y_0 or an observed level on $\{G \ge g\}$ at a date $s \in [g-1, t_{\mathcal{S}}(g)]$, for some represented g. The latter are integrable because $\mathbb{E}_P[|Y_{g-1}|\,\mathbb{I}(G \ge g)] < \infty$ and $\mathbb{E}_P[(\Delta Y_k)^2] < \infty$. ∎

Proof of Theorem 2.4(ii). If $\mathrm{conv}(\mathcal{Y}) = \mathbb{R}$, then $\mu^P_{g,s}(c) \in \mathrm{conv}(\mathcal{Y})$ for every c, and part (i) applies to every such law. ∎

Proof of Theorem 2.4(iii). Fix a bounded $h \in L^0_2(P_0)$, write $\varepsilon := \varepsilon_{P_0}$ for the law-specific overlap margin, and write $\mu_{g,s} := \mu^{P_0}_{g,s}$ and $m_k := m^0_{g,k,0}$. For the anchor, also define $\mu_{g,g-1}(c) := \mathbb{E}_{P_0}[Y_{g-1} \mid \mathcal{C} = c, G = g]$. For each represented g, define the $\mathcal{C}$-measurable functions

$$b_g(c) := \min_{s \in [g, t_{\mathcal{S}}(g)]} \rho^{P_0}_{g,s}(c),$$

$$v_g(c) := \mathbb{E}_{P_0}\big[|Y_{g-1} - \mu_{g,g-1}(c)| \;\big|\; \mathcal{C} = c, G = g\big] + \sum_{k=g}^{t_{\mathcal{S}}(g)} \mathbb{E}_{P_0}\big[|\Delta Y_k - m_k(c)| \;\big|\; \mathcal{C} = c, G > k\big].$$

By strict feasibility, $b_g(\mathcal{C}) > 0$ almost surely on $\{G = g\}$. By integrability, the anchor summand in v_g is finite under the target-cohort history law, and each increment summand is finite under its own risk-set history law. The overlap inequalities below transfer these finitely many null sets, as well as strict feasibility, to $\{G \ge g\}$. Thus $v_g(\mathcal{C}) < \infty$ and $b_g(\mathcal{C}) > 0$ almost surely there. For a $\mathcal{C}$-measurable event F, overlap gives

$$\mathbb{P}_{P_0}(F \cap \{G = g\}) = \mathbb{E}_{P_0}\big[\mathbb{I}(F)\mathbb{I}(G \ge g)(1 - q_{g,g})\big] \ge \varepsilon\,\mathbb{P}_{P_0}(F \cap \{G \ge g\}),$$

and in the same way $\mathbb{P}_{P_0}(F \cap \{G > k\}) = \mathbb{E}_{P_0}[\mathbb{I}(F)\mathbb{I}(G \ge g)q_{g,k}] \ge \varepsilon\,\mathbb{P}_{P_0}(F \cap \{G \ge g\})$; so null sets transfer.

For $N \ge 1$, let

$$A_N := \bigcap_g \big[\{G < g\} \cup \{b_g(\mathcal{C}_{g-1}) \ge 1/N,\ v_g(\mathcal{C}_{g-1}) \le N\}\big],$$

with the intersection over the represented cohorts. Let $h_N := \mathbb{I}(A_N)h - \mathbb{E}_{P_0}\{\mathbb{I}(A_N)h\}$ and $K_N := \|h_N\|_\infty$. The sets A_N increase and $\mathbb{P}_{P_0}(A_N) \to 1$, so $h_N \to h$ in $L_2(P_0)$. If $K_N = 0$, the zero score is immediate. Otherwise, for u with $\delta := |u|K_N \le 1/2$, $\mathrm{d}P_{u,N}/\mathrm{d}P_0 := 1 + uh_N$ defines a path that is differentiable in quadratic mean with score h_N. As in the proof of Lemma 2.1, the path stays in $\mathcal{M}_{\mathrm{obs}}(\mathcal{S})$ for small $|u|$. It is an equivalent law, so the support and $\mathcal{Y}$ are preserved. It also keeps the moments up to factors $1+\delta$, an overlap margin $a_\delta\varepsilon$, and $W_{\mathcal{S}}(P_{u,N}) \ne 0$, where $a_\delta := (1-\delta)/(1+\delta)$.

Histories outside the good set. Fix a represented g, and let B be $\{G = g\}$ or $\{G > k\}$, both contained in $\{G \ge g\}$. On $\{b_g(\mathcal{C}) < 1/N\} \cup \{v_g(\mathcal{C}) > N\}$, every observation with $G \ge g$ lies outside A_N, where h_N is constant. The tilt therefore leaves $\mathbb{E}[\cdot \mid \mathcal{C}, B]$ unchanged there, so $\mu^{P_{u,N}}_{g,s}(\mathcal{C}) = \mu_{g,s}(\mathcal{C}) \in \mathrm{conv}(\mathcal{Y})$.

Histories in the good set. For an integrable random variable ζ,

$$\mathbb{E}_{P_{u,N}}[\zeta \mid \mathcal{C}, B] - \mathbb{E}_{P_0}[\zeta \mid \mathcal{C}, B] = \frac{u \operatorname{cov}_{P_0}(\zeta, h_N \mid \mathcal{C}, B)}{1 + u\mathbb{E}_{P_0}[h_N \mid \mathcal{C}, B]},$$

and $|\operatorname{cov}_{P_0}(\zeta, h_N \mid \mathcal{C}, B)| \le 2K_N \mathbb{E}_{P_0}\big[|\zeta - \mathbb{E}_{P_0}[\zeta \mid \mathcal{C}, B]| \;\big|\; \mathcal{C}, B\big]$, because $|h_N - \mathbb{E}_{P_0}[h_N \mid \mathcal{C}, B]| \le 2K_N$. Apply this to $\zeta = Y_{g-1}$ with $B = \{G = g\}$ and to $\zeta = \Delta Y_k$ with $B = \{G > k\}$, and sum. On $\{v_g(\mathcal{C}) \le N\}$ this gives, uniformly over $s \in [g, t_{\mathcal{S}}(g)]$,

$$\big|\mu_{g,s}^{P_{u,N}}(\mathcal{C}) - \mu_{g,s}(\mathcal{C})\big| \le \frac{2|u|K_N\, v_g(\mathcal{C})}{1 - |u|K_N} \le 4|u|K_N\, N.$$

If also $4|u|K_N N < 1/N$, each prefix mean moves by less than $1/N \le b_g(\mathcal{C})$ and stays inside $\operatorname{conv}(\mathcal{Y})$.

Thus $P_{u,N}$ is prefix feasible for all small $|u|$, and part (i) places it in $\mathcal{M}_{\text{caus}}(\mathcal{S}; \mathcal{Y})$; each law is completed separately, so no measurability in u is needed. Every h_N is therefore a score of the causal model. Since $h_N \to h$ in $L_2(P_0)$ and bounded centered functions are dense in $L_2^0(P_0)$, the tangent space is $L_2^0(P_0)$.

Transfer of the gradients. The functional of Theorem 2.1 identifies the target at every law of $\mathcal{M}_{\text{caus}}(\mathcal{S}; \mathcal{Y})$. Every regular path in the causal model is a regular path in the observed-data model under the convention of Appendix B.1. Its derivative is therefore supplied by the general-path arguments in the proofs of Theorems 2.2 and 2.3. With a full tangent space, the representer is the unique canonical gradient. ■

Remark B.3 (Special cases and scope)**.** *Binary outcomes.* For $\mathcal{Y} = \{0, 1\}$ and $0 < x < 1$, $\varpi(x, v) = \mathbb{I}\{v \le x\}$. The construction then draws the post-adoption untreated levels $Y_s(\infty) = \mathbb{I}\{v_{g,s} \le \mu_{g,s}^P(\mathcal{C}_{g-1})\}$. The untreated increments are differences of these levels, not independent draws. Independence of the $v_{g,s}$ across s is only a witnessing coupling: the assumptions restrict marginal conditional means, not the joint law of the untreated path.

Finite supports. For a finite support, only its smallest and largest points enter (i). For example, a variable on $\{0, 2\}$ can have mean one, so interior support points play no role.

Counts and integers. For counts, $\mathcal{Y} = \{0, 1, 2, \ldots\}$, the restriction is $\mu_{g,s}^P \ge 0$. For $\mathcal{Y} = \mathbb{Z}$, it is vacuous, and the kernel randomizes between $\lfloor x \rfloor$ and $\lfloor x \rfloor + 1$.

Closedness. The theorem requires $\mathcal{Y}$ to be closed, which covers every finite support and every subset of $\mathbb{Z}$. For the set $\{1, 1/2, 1/3, \ldots\}$, which is not closed, the closed convex hull contains zero, but no variable with values in the set has mean zero, so (i) would fail with the closed hull.

Boundary laws. Strict feasibility can fail when $Y_{g-1} \in \mathcal{C}_{g-1}$ sits at an endpoint of $\operatorname{conv}(\mathcal{Y})$ on histories of positive probability and the cumulative risk-set trend vanishes there. This happens, for example, when the untreated binary outcome is absorbing. Such a law is prefix feasible, so Theorem 2.1 identifies the target, but part (iii) does not apply.

The anchor. No margin is needed at $s = g - 1$. When $Y_{g-1} \in \mathcal{C}_{g-1}$, the anchor $\mu_{g,g-1}^P = Y_{g-1}$ lies in $\mathcal{Y}$ and may sit on the boundary. A tilt cannot move it, and feasibility restricts only $s \ge g$.

The localization constant. For bounded $\mathcal{Y}$, $v_g \le (1 + 2T)(\overline{y} - \underline{y})$, so the second localization in part (iii) is vacuous.

Scope. Restrictions on the untreated paths beyond their support, such as absorbing states, monotone paths, or prohibited transitions, would impose further compatibility conditions and are not covered.

B.5 Common-History Nesting and Efficiency Bounds

For the fixed finite target collection $\mathcal{S}$, each $\mathcal{H}_g \subseteq \sigma(O)$ is generated by prespecified observed variables dated at most $g - 1$. We use the same reduced sample space, $\mathcal{Y}$-valued observed and po-

tential outcomes for the nonempty closed $\mathcal{Y}$ of Theorem 2.4, and applicable observed-data moment requirements of Condition 2(i)–(ii) in both models. For an aggregate, the weight functions $\Omega_{g,t}$ are the same prespecified functions of O and $W_{\mathcal{S}} \neq 0$. The history-specific identification domain imposes $p_g(P) > 0$, consistency, the analogues of Assumptions 1–3 with $\mathcal{H}_g$ in place of $\mathcal{C}_{g-1}$, and integrability of every potential-outcome level used in identification, as in Condition 1. In particular, for a law-specific $\varepsilon_P > 0$,

$$q_{g,k}^{P,\mathcal{H}} := \mathbb{P}_P(G > k \mid \mathcal{H}_g, G \geq g) \in (\varepsilon_P, 1 - \varepsilon_P), \qquad k \in [g, t_{\mathcal{S}}(g)],$$

almost surely on $\{G \geq g\}$. The cohort-by-cohort model replaces only the pooled parallel-trends restriction by the component-cohort restrictions displayed before Proposition 2.1; all other requirements are shared.

Proof of Proposition 2.1. *(a) Identification with the chosen history.* Let Q be a compatible pooled full-data law and P its observed marginal. Consistency identifies the factual outcome $Y_t(g) = Y_t$ on $\{G = g\}$. At the anchor, no anticipation at $(j, s) = (g, g-1)$, followed by consistency $Y_{g-1}(g) = Y_{g-1}$ on that event, gives

$$\mathbb{E}_Q[Y_{g-1}(\infty) \mid \mathcal{H}_g, G = g] = \mathbb{E}_P[Y_{g-1} \mid \mathcal{H}_g, G = g].$$

For a finite $j > k$ with $p_j(P) > 0$, no anticipation at $s = k$ and $s = k - 1$, differenced and followed by consistency, gives

$$\mathbb{E}_Q[\Delta Y_k(\infty) \mid \mathcal{H}_g, G = j] = \mathbb{E}_P[\Delta Y_k \mid \mathcal{H}_g, G = j].$$

For $j = \infty$ with $\mathbb{P}_P(G = \infty) > 0$, this follows directly from consistency. Absent comparison cohorts are omitted. Applying the conditional-mixture identity and null-set argument of (c) below, which use only integrability, to the untreated and observed increments with the weights $\varrho_{g,j|k}^{\mathcal{H}}$ of (b) therefore identifies the risk-set mean as

$$\mathbb{E}_Q[\Delta Y_k(\infty) \mid \mathcal{H}_g, G > k] = \mathbb{E}_P[\Delta Y_k \mid \mathcal{H}_g, G > k].$$

Telescoping the untreated path from $g-1$ to t, using the anchor equality and pooled parallel trends, now gives

$$\begin{aligned} \mathrm{ATT}(g,t) &= \Psi_{\mathcal{H},g,t}(P) \\ &:= \mathbb{E}_P[Y_t - Y_{g-1} \mid G = g] - \sum_{k=g}^{t} \mathbb{E}_P[\mathbb{E}_P[\Delta Y_k \mid \mathcal{H}_g, G > k] \,|\, G = g]\,. \end{aligned}$$

Integrability justifies the differences, telescoping, and iterated expectations. The history-specific overlap makes the history laws given $G = g$, $G \geq g$, and $G > k$ mutually absolutely continuous, so the control conditional means are defined on target-cohort histories. Their integrability within that cohort also follows from the bounded ratio

$$\frac{\mathbb{P}_P(G = g \mid \mathcal{H}_g, G \geq g)}{\mathbb{P}_P(G > k \mid \mathcal{H}_g, G \geq g)} \leq \varepsilon_P^{-1}.$$

This establishes identification for $\mathcal{H}_g$ itself, rather than invoking Theorem 2.1 with an unchanged conditioning history.

(b) Observed-data models and component positivity. Write $\mathcal{M}_{\text{pool}}(\mathcal{H})$ and $\mathcal{M}_{\text{cbc}}(\mathcal{H})$ for the sets of observed marginals of full-data laws satisfying the respective restrictions and the shared requirements above. For $\mathcal{J}_k := \{k+1, \ldots, T, \infty\}$, let

$$\varrho^{\mathcal{H}}_{g,j|k} := \mathbb{P}_P(G = j \mid \mathcal{H}_g, G > k), \qquad j \in \mathcal{J}_k.$$

A component-cohort equality is imposed only on histories where $\varrho^{\mathcal{H}}_{g,j|k} > 0$; no positivity is required for an absent component.

(c) Mixture identity and null sets. Conditional iterated expectations for the integrable untreated increment yield

$$\mathbb{E}_Q[\Delta Y_k(\infty) \mid \mathcal{H}_g, G > k] = \sum_{j \in \mathcal{J}_k} \varrho^{\mathcal{H}}_{g,j|k} \mathbb{E}_Q[\Delta Y_k(\infty) \mid \mathcal{H}_g, G = j],$$

with zero-weight terms omitted. If $\mathcal{E}_j \in \mathcal{H}_g$ is null for the history law within cohort j, then

$$\mathbb{E}_P[\mathbb{I}(G > k)\varrho^{\mathcal{H}}_{g,j|k}\mathbb{I}(\mathcal{E}_j)] = \mathbb{P}_P(G = j, \mathcal{E}_j) = 0.$$

Thus each weighted term is independent of the chosen conditional-mean version. The finite union of the events $\{\varrho^{\mathcal{H}}_{g,j|k} > 0\} \cap \mathcal{E}_j$ is null under the risk-set history law, so all positive-weight component means can be used simultaneously outside one null set. Component-cohort history laws need not have identical supports.

(d) Inclusion. Under the cohort-by-cohort restrictions every positive-weight component in (c) equals $\mathbb{E}_Q[\Delta Y_k(\infty) \mid \mathcal{H}_g, G = g]$. The weights sum to one, so the pooled restriction holds, first on risk-set histories and then on target-cohort histories by overlap. Every compatible cohort-by-cohort full-data law is consequently a pooled full-data law, and

$$\mathcal{M}_{\text{cbc}}(\mathcal{H}) \subseteq \mathcal{M}_{\text{pool}}(\mathcal{H}).$$

(e) Common identified functional. By (d), part (a) applies to both models and identifies every group-period target by the same $\Psi_{\mathcal{H},g,t}(P)$, regardless of which compatible full-data law generates P. A prespecified aggregate is the same functional

$$\Psi_{\mathcal{H},\Theta}(P) := \frac{\sum_{(g,t)\in\mathcal{S}} \mathbb{E}_P[\Omega_{g,t}]\, \Psi_{\mathcal{H},g,t}(P)}{\sum_{(g,t)\in\mathcal{S}} \mathbb{E}_P[\Omega_{g,t}]}$$

in the two models.

(f) Tangent spaces. Fix a common observed law P. For either model, define its tangent space as the closed linear span in $L^0_2(P)$ of scores of regular paths under the convention of Appendix B.1. By (d), every smaller-model regular path is a pooled-model regular path: the models share the observed-data experiment and dominating law, so its quadratic-mean differentiability and local likelihood-ratio bounds are unchanged. Taking linear spans and then closures therefore gives

$$\mathcal{T}_{\text{cbc}} \subseteq \mathcal{T}_{\text{pool}}.$$

(g) Canonical gradients. For either $\theta = \tau_{g,t}$ or $\theta = \Theta$, write $\Psi_{\mathcal{H}}$ for its common scalar functional in (e), and $\dot{\Psi}_{\mathcal{H},P}(h)$ for its pathwise derivative at P along a path with score h. Suppose it is pathwise differentiable in the pooled model, with canonical gradient $D_{\text{pool}} \in \mathcal{T}_{\text{pool}}$. Its pathwise derivative restricts to every smaller-model path. Define the orthogonal projection

$$D_{\text{cbc}} := \Pi_{\mathcal{T}_{\text{cbc}}} D_{\text{pool}}.$$

It belongs to $\mathcal{T}_{\mathrm{cbc}}$, and for every smaller-model score h,

$$\langle D_{\mathrm{cbc}}, h\rangle_P = \langle D_{\mathrm{pool}}, h\rangle_P = \dot{\Psi}_{\mathcal{H},P}(h),$$

because the projection residual is orthogonal to $\mathcal{T}_{\mathrm{cbc}}$. These verify both defining properties of the smaller-model canonical gradient. If another element of $\mathcal{T}_{\mathrm{cbc}}$ represented the derivative, their difference would be orthogonal to every score and hence to the closed linear span containing that difference; its squared norm would be zero. The gradient is therefore unique.

(h) Efficiency-bound order. The scalar bounds for θ are the squared norms of their canonical gradients. Pythagoras gives

$$V_{\mathrm{pool}}(\theta; P) - V_{\mathrm{cbc}}(\theta; P) = \mathbb{E}_P[(D_{\mathrm{pool}} - D_{\mathrm{cbc}})^2] \geq 0.$$

Applying this argument to $\tau_{g,t}$ and Θ gives the respective inequalities in the proposition.

(i) Equality. For the selected target θ, the bounds coincide if and only if the projection residual is zero, equivalently $D_{\mathrm{pool}} \in \mathcal{T}_{\mathrm{cbc}}$, which is already closed by definition.

(j) Link to the displayed EIF. The ordering above uses inclusion and pathwise differentiability and needs no strict prefix feasibility. When $\mathcal{H}_g = \sigma(\mathcal{C}_{g-1})$, consistency and the shared level-integrability clause give, for any compatible full-data law Q,

$$\mathbb{E}_P[|Y_{g-1}|\mathbb{I}(G \geq g)] = \sum_{j\in\{g,\ldots,T,\infty\}} \mathbb{E}_Q[|Y_{g-1}(j)|\mathbb{I}(G = j)] < \infty.$$

Together with the shared support, overlap and moment requirements, this gives exactly the domain defining $\mathcal{M}_{\mathrm{caus}}(\mathcal{S}; \mathcal{Y})$, so the pooled observed-data model is this causal model. At a strictly prefix-feasible law, the localized bounded-tilt paths constructed in the proof of Theorem 2.4(iii) belong to the pooled model, and their scores are dense in $L_2^0(P)$. Under Condition 1 and the applicable clauses (i)–(ii) of Condition 2, the proofs of Theorems 2.2 and 2.3 represent the derivative of the common identified functional on these paths by the corresponding displayed EIF. Density and uniqueness consequently identify D_{pool} of (g) with that EIF. For Θ, at the true law $P = P_0$ this gives $V_{\mathrm{pool}}(\Theta; P_0) = V_\Theta$. This EIF identification is not asserted at boundary laws or for a different history. ■

Over-identification and efficiency gains in other causal and data-combination models are studied by Chen and Xie (2025).

C Proofs for Section 3

C.1 Cross-Fitting and Full-Sample Fitting

This subsection details the L-fold cross-fitting of Section 3.2 (Chernozhukov et al., 2018). The folds are stratified by cohort: the index set $\mathcal{N}_j = \{i : G_i = j\}$ of each cohort j, of size n_j, is shuffled and split independently into L pieces $\mathcal{N}_{j,1}, \ldots, \mathcal{N}_{j,L}$ with sizes $n_{j,\ell}$ satisfying $\max_{j,\ell} |n_{j,\ell} - n_j/L| \leq 1$, and $\mathcal{I}_\ell := \bigcup_j \mathcal{N}_{j,\ell}$. Every training sample $\mathcal{I}_\ell^c$ thus contains each cohort in proportion to its share, up to rounding, so that no cohort with at least three units is absent from a training sample (separation and other degenerate fits remain possible); asymptotically, stratification is irrelevant. The number of folds $L \geq 2$ is fixed. Write $w_\ell := |\mathcal{I}_\ell|/n$, let $\mathbb{P}_{n,\ell}$ be the empirical measure of fold ℓ, so that $w_\ell \mathbb{P}_{n,\ell} f = n^{-1} \sum_{i\in\mathcal{I}_\ell} f(O_i)$, let $\ell(i)$ be the fold of unit i, and set $\mathbb{P}_n f_{\ell(\cdot)} := n^{-1} \sum_{i=1}^n f_{\ell(i)}(O_i)$; all arguments condition on the realized partition. For each positive-probability cohort j write $p_j := \mathbb{P}_{P_0}(G = j)$ and $P_{0,j} := P_0(\cdot \mid G = j)$ for the conditional law of O within cohort j.

Conditional on $(G_1, \ldots, G_n)$ and the training data $\{O_i : i \in \mathcal{I}_\ell^c\}$, the observations in the stratum $\mathcal{I}_{\ell,j} := \mathcal{I}_\ell \cap \mathcal{N}_j$ are iid from $P_{0,j}$, not from P_0, and independent of every function fitted on $\mathcal{I}_\ell^c$. Since $n_j/n = p_j + O_p(n^{-1/2})$, the balanced split gives $n_{j,\ell}/n = w_\ell p_j + O_p(n^{-1/2})$ and $\min_\ell n_{j,\ell} \to \infty$.

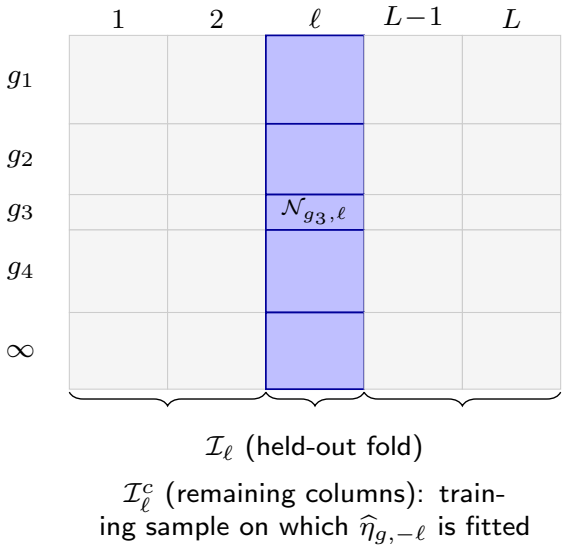


Figure C.1: Stratified L-fold cross-fitting. The index set $\mathcal{N}_j$ of each cohort j (rows; drawn for four adoption cohorts $g_1, \ldots, g_4$ and the never-treated group ∞) is shuffled and split independently into L pieces $\mathcal{N}_{j,1}, \ldots, \mathcal{N}_{j,L}$ (columns) of sizes within one of n_j/L, and fold ℓ pools one piece from every cohort, $\mathcal{I}_\ell = \bigcup_j \mathcal{N}_{j,\ell}$. Row heights are illustrative of unequal cohort sizes. Splitting within each cohort, rather than the pooled sample, keeps every cohort with at least three units represented in each training sample $\mathcal{I}_\ell^c$ in proportion to its share, up to rounding; a small cohort such as g_3 with $n_{g_3} \geq 3$ is therefore never absent from the sample on which $\widehat{\eta}_{g,-\ell}$ is fitted, although separation and other degenerate fits remain possible.

The proofs use one fact about cross-fitted averages. Let f_ℓ be a measurable function of O that may depend on $(G_1, \ldots, G_n)$ and on the training data of fold ℓ, let $\mathbb{E}_{P_0}[f_\ell] := \int f_\ell \, \mathrm{d}P_0$ and $\|f_\ell\|_2$ be computed with f_ℓ held fixed (that is, conditionally on the labels and the training data), and suppose that $\max_\ell \|f_\ell\|_2 < \infty$ with probability tending to one. Then

$$\frac{1}{\sqrt{n}} \sum_{\ell=1}^{L} \sum_{i \in \mathcal{I}_\ell} f_\ell(O_i) - \sqrt{n} \sum_{\ell=1}^{L} w_\ell \mathbb{E}_{P_0}[f_\ell] = O_p\Big(\max_\ell \|f_\ell\|_2 \Big),$$
$$\frac{1}{n} \sum_{\ell=1}^{L} \sum_{i \in \mathcal{I}_\ell} f_\ell(O_i)^2 = O_p\Big(\max_\ell \|f_\ell\|_2^2 \Big). \tag{C.1}$$

Indeed, conditionally on the labels and the training data, $n^{-1/2} \sum_{i \in \mathcal{I}_{\ell,j}} \{f_\ell(O_i) - \mathbb{E}_{P_0}[f_\ell \mid G = j]\}$ has mean zero and variance at most $(n_{j,\ell}/n)\mathbb{E}_{P_0}[f_\ell^2 \mid G = j] \leq \{n_{j,\ell}/(np_j)\}\|f_\ell\|_2^2$, so conditional Chebyshev makes it $O_p(\|f_\ell\|_2)$. The stratum means recombine: $\sum_j (n_{j,\ell}/n)\mathbb{E}_{P_0}[f_\ell \mid G = j] - w_\ell \mathbb{E}_{P_0}[f_\ell] = \sum_j (n_{j,\ell}/n - w_\ell p_j)\mathbb{E}_{P_0}[f_\ell \mid G = j] = O_p(n^{-1/2}\|f_\ell\|_2)$, because $|\mathbb{E}_{P_0}[f_\ell \mid G = j]| \leq p_j^{-1/2}\|f_\ell\|_2$. The second bound is conditional Markov, since the conditional mean of $\sum_{i \in \mathcal{I}_{\ell,j}} f_\ell(O_i)^2$ is $n_{j,\ell}\mathbb{E}_{P_0}[f_\ell^2 \mid G = j]$. Folds and strata are finitely many, and no Donsker condition is used. In particular, if $\|f_\ell - \overline{f}\|_2 \xrightarrow{p} 0$ for a fixed $\overline{f} \in L_2(P_0)$, then $n^{-1} \sum_\ell \sum_{i \in \mathcal{I}_\ell} f_\ell(O_i) \xrightarrow{p} \mathbb{E}_{P_0}[\overline{f}]$. Since

$w_\ell \to 1/L$, taking $f_{\ell'} = 0$ for $\ell' \neq \ell$ gives the bounds fold by fold: $\sqrt{n}\{\mathbb{P}_{n,\ell} f_\ell - \mathbb{E}_{P_0}[f_\ell]\} = O_p(\|f_\ell\|_2)$ and $\mathbb{P}_{n,\ell} f_\ell^2 = O_p(\|f_\ell\|_2^2)$.

Full-sample fitting on fixed finite support. The following bounds replace sample splitting when the observed-data law has fixed finite support.

Lemma C.1 (Finite-support empirical bounds)**.** *Let $O_1, \ldots, O_n$ be iid from a fixed law P_0 on a finite set $\mathcal{O}$, with $p_o = P_0(O = o) > 0$ for every $o \in \mathcal{O}$, and let $p_{n,o} = \mathbb{P}_n[\mathbb{I}(O = o)]$. For every realized function $f : \mathcal{O} \to \mathbb{R}$, including a function fitted on the entire sample,*

$$\sqrt{n}\,|(\mathbb{P}_n - P_0)f| \le \|f\|_2 A_n, \qquad A_n^2 := \sum_{o \in \mathcal{O}} \frac{n(p_{n,o} - p_o)^2}{p_o} = O_p(1),$$

$$\mathbb{P}_n f^2 \le \Lambda_n \|f\|_2^2, \qquad \Lambda_n := \max_{o \in \mathcal{O}} \frac{p_{n,o}}{p_o} = 1 + o_p(1).$$

The bounds hold simultaneously for all f; $P_0 f = \sum_o p_o f(o)$ integrates the realized function against a fresh P_0 argument.

Proof. Cauchy–Schwarz applied to $\sum_o (p_{n,o} - p_o) f(o)$ gives the first inequality, and $\mathbb{E}[A_n^2] = \sum_o (1 - p_o) = |\mathcal{O}| - 1$. The second follows by bounding each $p_{n,o}/p_o$ by its maximum; the law of large numbers and the fixed finite support give $\Lambda_n \xrightarrow{p} 1$. These are samplewise inequalities with common random bounds, so no independence between f and the sample is required. ■

Corollary C.1 (Full-sample extension)**.** *Suppose the observed-data law has the fixed finite support of Lemma C.1. Use the $L = 1$ convention of Section 3.2, with $\ell(i) = 1$ and $w_1 = 1$. Then the consistency and asymptotic-linearity conclusions of Theorems 3.1, 3.2, and 4.2, Corollaries 3.1, 3.2, and 4.1, and Proposition 4.1 continue to hold, as do the variance and pointwise-inference conclusions of Lemma C.3 and Corollary C.2. The same extension applies to the single-fluctuation cohort-wise conclusions of Theorem F.1. In each case replace the cross-fitting requirement by full-sample fitting and retain every other hypothesis of the invoked result, with maxima over folds evaluated at the sole full-sample fit.*

Proof. Lemma C.1 supplies both bounds in (C.1) for $L = 1$, and hence also empirical convergence to a fixed $L_2(P_0)$ limit and convergence of empirical squared differences. In transferring the proofs, every population expectation involving a fitted function holds that realized function fixed and integrates a fresh argument; training-conditional statements are replaced by these samplewise bounds, not by a conditional-independence claim about training observations. The balancing and ratio identities are identities for each realized history-measurable function, so their population remainder bounds are unchanged. The bounds apply to the fitted signal and residual differences, the normalizers, and the score and Gram entries used in the targeting proofs. The empirical normal equations, finite-dimensional fluctuation arguments, and HT–Hájek bridges are unchanged. Finally, the iid central limit theorem applies to the fixed limiting influence function, and the second bound gives the empirical squared-score convergence needed for the variance results. These are the uses of splitting in the cited proofs, including the cohort-wise argument. ■

This extension does not establish nuisance consistency or product rates, overlap or truth containment under the cap, estimator availability, positive denominator limits, or nondegenerate variance. Those requirements, and the strict prefix-feasibility qualification for causal-model efficiency, remain as in the invoked results. In particular, one correct nuisance yields consistency under the stated conditions, not automatically valid EIF-based inference. The law and its finite support are fixed;

the result makes no uniform claim over growing supports or vanishing atom probabilities. CompactBinary has an effective observation $(Z, Y_0, \ldots, Y_5, G)$ with at most $3 \cdot 2^6 \cdot 5 = 960$ atoms, so it satisfies this support requirement.

Remark C.1 (Full-sample fitting and Donsker conditions)**.** No additional Donsker condition is required in Corollary C.1: on fixed finite support, Lemma C.1 supplies both bounds in (C.1) simultaneously for all realized functions, including full-sample fits. Thus the empirical-process and empirical squared-norm controls follow directly from the support assumption.

Beyond finite support, a classical route uses stochastic equicontinuity. Write $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P_0)$. If $\widehat{f}_n$ and its fixed limit f_0 belong, with probability tending to one, to a fixed measurable P_0-Donsker class and $\|\widehat{f}_n - f_0\|_2 = o_p(1)$, then

$$\mathbb{G}_n(\widehat{f}_n - f_0) = o_p(1)$$

(van der Vaart, 1998, Chapter 19). This conclusion alone does not supply the stronger bound $\mathbb{G}_n(\widehat{f}_n - f_0) = O_p(\|\widehat{f}_n - f_0\|_2)$ used in (C.1), nor its companion relative bound for empirical squared norms.

A Donsker extension of all the corollary's conclusions would therefore require specifying suitable classes for the relevant fitted signals, residuals, normalizers, and targeting score and Gram entries, together with empirical squared-score convergence for variance estimation, for example through appropriate uniform laws of large numbers for squared differences. Each use of (C.1) would need justification with sufficient rates; a Donsker assumption on the nuisance regressions alone, without checking the induced classes, does not establish this transfer. Such an extension could cover some laws with infinite support, but is not established here. The other hypotheses retained by the corollary, including nuisance consistency and product rates, would still be required.

C.2 Proofs

Proof of Lemma 3.1. Write $\mathbb{I}_g = \mathbb{I}(G = g)$, $\mathbb{I}_{>k} = \mathbb{I}(G > k)$, $m^0 = m^0_{g,k,0}$, $r^0 = r^0_{g,k}$, $\widetilde{\phi} = \phi_{g,k}(O; \widetilde{m}, \widetilde{r})$ and $\phi^0 = \phi_{g,k}(O; m^0, r^0)$, and suppress the argument $\mathcal{C}_{g-1}$. Adding and subtracting,

$$p_g(\widetilde{\phi} - \phi^0) = (\mathbb{I}_g - \mathbb{I}_{>k} r^0)(m^0 - \widetilde{m}) - \mathbb{I}_{>k}(\widetilde{r} - r^0)(\Delta Y_k - m^0) + \mathbb{I}_{>k}(\widetilde{r} - r^0)(\widetilde{m} - m^0).$$

By the ratio identity (3.1), $\mathbb{E}[\mathbb{I}_{>k} r^0 \mid \mathcal{C}_{g-1}] = \pi_{>k|g-1} r^0 = \pi_{g|g-1} = \mathbb{E}[\mathbb{I}_g \mid \mathcal{C}_{g-1}]$. Hence, for every nonnegative $\mathcal{C}_{g-1}$-measurable f, and by $r^0 \le \varepsilon^{-1}$ (Assumption 3),

$$\mathbb{E}[\mathbb{I}_{>k} r^0 f] = \mathbb{E}[\mathbb{I}_g f] \le \varepsilon^{-1} \mathbb{E}[\mathbb{I}_{>k} f]. \tag{C.2}$$

With $f = \widetilde{m}^2$ and $f = (m^0)^2$, it makes every term integrable, given the risk-set moment of $\widetilde{m}$, $\max_k \mathbb{E}[(\Delta Y_k)^2] < \infty$, the envelopes $|\widetilde{r}| \le M$ and $r^0 \le \varepsilon^{-1}$, and conditional Jensen, $\mathbb{E}[\mathbb{I}_{>k}(m^0)^2] \le \mathbb{E}[\mathbb{I}_{>k}(\Delta Y_k)^2]$. Conditionally on $\mathcal{C}_{g-1}$, the first term has mean zero because $\mathbb{E}[\mathbb{I}_g - \mathbb{I}_{>k} r^0 \mid \mathcal{C}_{g-1}] = 0$, and the second because m^0 is the conditional mean of ΔY_k on the risk set; in each, the other factor is $\mathcal{C}_{g-1}$-measurable. Only the third term survives, which is the identity of the lemma, and Cauchy–Schwarz gives the bound. ■

Proof of Theorem 3.1. Fix $(g, t) \in \mathcal{S}$. The one-step estimator is $\widehat{\tau}^{\mathrm{os}}_{g,t} = \widehat{A}/\widehat{p}_g$, where $\widehat{A}$ is the average $n^{-1} \sum_{\ell=1}^{L} \sum_{i \in \mathcal{I}_\ell} \widehat{S}_\ell(O_i)$ of the cross-fitted signal

$$\widehat{S}_\ell := \sum_{k=g}^{t} \{\mathbb{I}(G = g) - \mathbb{I}(G > k)\widehat{r}_{g,k,-\ell}\}(\Delta Y_k - \widehat{m}_{g,k,0,-\ell}),$$

and $\overline{S}$ denotes the same signal at the limits $(\overline{m}_{g,k,0}, \overline{r}_{g,k})$. The difference $f_\ell := \widehat{S}_\ell - \overline{S}$ is $\sum_{k=g}^{t}(a_{k,\ell} - b_{k,\ell})$, where

$$a_{k,\ell} := \{\mathbb{I}(G=g) - \mathbb{I}(G>k)\widehat{r}_{g,k,-\ell}\}(\overline{m}_{g,k,0} - \widehat{m}_{g,k,0,-\ell}),$$
$$b_{k,\ell} := \mathbb{I}(G>k)(\widehat{r}_{g,k,-\ell} - \overline{r}_{g,k})U_k, \qquad U_k := \Delta Y_k - \overline{m}_{g,k,0}.$$

By (C.2) and the cap, $\|a_{k,\ell}\|_2 \le (\varepsilon^{-1/2} + M)\|\mathbb{I}(G>k)(\widehat{m}_{g,k,0,-\ell} - \overline{m}_{g,k,0})\|_2 = o_p(1)$. Also $\|b_{k,\ell}\|_2 \le 2M\|\mathbb{I}(G>k)U_k\|_2 < \infty$ and, by Cauchy–Schwarz, $|\mathbb{E}_{P_0}[b_{k,\ell}]| \le \|\mathbb{I}(G>k)(\widehat{r}_{g,k,-\ell} - \overline{r}_{g,k})\|_2 \|\mathbb{I}(G>k)U_k\|_2 = o_p(1)$. Hence $\|f_\ell\|_2 = O_p(1)$ and $\mathbb{E}_{P_0}[f_\ell] = o_p(1)$, and (C.1) and the law of large numbers for $\overline{S} \in L_2(P_0)$ give $\widehat{A} \xrightarrow{p} \mathbb{E}_{P_0}[\overline{S}]$. With $\widehat{p}_g \xrightarrow{p} p_g > 0$,

$$\widehat{\tau}^{\mathrm{os}}_{g,t} \xrightarrow{p} \frac{\mathbb{E}_{P_0}[\overline{S}]}{p_g} = \sum_{k=g}^{t} \mathbb{E}_{P_0}\big[\phi_{g,k}(O; \overline{m}_{g,k,0}, \overline{r}_{g,k})\big].$$

It remains to show that the right side is $\tau^0_{g,t}$ in either branch. In branch (a), $\overline{m}_{g,k,0} = m^0_{g,k,0}$, the conditional mean of ΔY_k given $(\mathcal{C}_{g-1}, G>k)$, on the risk set, and hence also on $\{G=g\}$ by (C.2) with $f = (\overline{m}_{g,k,0} - m^0_{g,k,0})^2$, so the control term has mean zero for any capped $\overline{r}_{g,k}$, and (B.1), Assumption 2 and consistency give

$$\begin{aligned}\mathbb{E}_{P_0}\big[\phi_{g,k}(O; m^0_{g,k,0}, \overline{r}_{g,k})\big] &= \mathbb{E}[\Delta Y_k \mid G=g] - \mathbb{E}\big[m^0_{g,k,0}(\mathcal{C}_{g-1}) \mid G=g\big] \\ &= \mathbb{E}\big[\Delta Y_k(g) - \Delta Y_k(\infty) \mid G=g\big]. \end{aligned} \tag{C.3}$$

By Theorem 2.1, the middle expressions sum over $k \in [g,t]$ to $\tau^0_{g,t}$; in particular this holds at the truth, $\overline{r}_{g,k} = r^0_{g,k}$. In branch (b), Lemma 3.1 applies with $\widetilde{m} = \overline{m}_{g,k,0}$ and $\widetilde{r} = \overline{r}_{g,k} = r^0_{g,k}$, by the moment and envelope conditions of the theorem. Its remainder contains the factor $\overline{r}_{g,k} - r^0_{g,k} = 0$, so $\mathbb{E}_{P_0}[\phi_{g,k}(O; \overline{m}_{g,k,0}, r^0_{g,k})] = \mathbb{E}_{P_0}[\phi_{g,k}(O; m^0_{g,k,0}, r^0_{g,k})]$, the value of branch (a) at the truth. In either branch, $\mathbb{E}_{P_0}[\psi_{g,t}(O; \overline{\eta}_g)] = \sum_k \mathbb{E}_{P_0}[\phi_{g,k}(O; \overline{m}_{g,k,0}, \overline{r}_{g,k})] - \tau^0_{g,t} = 0$.

For the aggregate, $\varphi^{\mathrm{eff}}\{\Theta^0; \overline{\eta}_{\mathcal{S}}\}$ is a fixed linear combination of the $\psi_{g,t}(O; \overline{\eta}_g)$ and of the weight scores $\Omega_{g,t}(O) - \omega^0_{g,t}$ (Theorem 2.3), all of which have mean zero. Moreover $\widehat{\Theta}_{\mathrm{os}}$ is the weighted average (3.5) of the granular estimators, $\widehat{\omega}_{g,t} \xrightarrow{p} \omega^0_{g,t}$ by the law of large numbers, and $W_{\mathcal{S}} \neq 0$ by Condition 3(ii), so $\widehat{\Theta}_{\mathrm{os}} \xrightarrow{p} \Theta^0$ in either branch. ∎

Proof of Theorem 3.2. *Step 1: exact ratio identity.* Fix $(g,t) \in \mathcal{S}$ and use the notation of the preceding proof. Let S^0 be the signal at the truth (m^0, r^0), $A^0 := \mathbb{E}_{P_0}[S^0] = p^0_g \tau^0_{g,t}$, $\delta_A := \widehat{A} - A^0$ and $\delta_p := \widehat{p}_g - p^0_g$. Elementary algebra gives the exact identity

$$\frac{\widehat{A}}{\widehat{p}_g} - \frac{A^0}{p^0_g} = \frac{\delta_A}{p^0_g} - \frac{A^0\delta_p}{(p^0_g)^2} - \frac{\delta_A\delta_p}{(p^0_g)^2} + \frac{\widehat{A}\delta_p^2}{(p^0_g)^2\widehat{p}_g}, \tag{C.4}$$

to whose last two terms no order is assigned until the numerator expansion

$$\sqrt{n}(\widehat{A} - A^0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\{S^0(O_i) - A^0\} + o_p(1) \tag{C.5}$$

is proved. With $f_\ell := \widehat{S}_\ell - S^0$, (C.1) reduces (C.5) to $\sqrt{n}\sum_\ell w_\ell \mathbb{E}_{P_0}[f_\ell] = o_p(1)$ (Step 2) and $\max_\ell \|f_\ell\|_2 = o_p(1)$ (Step 3). Since $\psi_{g,t}(O; \widehat{\eta}_{g,-\ell}) - \psi_{g,t}(O; \eta^0_g) = f_\ell / p^0_g$ at the true parameter values, these are the bias and empirical-process terms of the one-step estimator.

Step 2: bias, by Neyman orthogonality. Given $\mathcal{I}_\ell^c$, $(\widehat{m}_{g,k,0,-\ell}, \widehat{r}_{g,k,-\ell})$ is a fixed pair with $|\widehat{r}_{g,k,-\ell}| \le M$ by construction (3.2). Write $\delta_m := \widehat{m}_{g,k,0,-\ell} - m^0_{g,k,0}$ and $\delta_r := \widehat{r}_{g,k,-\ell} - r^0_{g,k}$. On the event that $\|\mathbb{I}(G > k)\delta_m\|_2 < \infty$ for every $k \in [g,t]$, which has probability tending to one by consistency, conditional Jensen gives $\mathbb{E}[\mathbb{I}(G > k)\widehat{m}^2_{g,k,0,-\ell} \mid \mathcal{I}_\ell^c] \le 2\|\mathbb{I}(G > k)\delta_m\|_2^2 + 2\mathbb{E}[(\Delta Y_k)^2] < \infty$, so the pair satisfies the conditions of Lemma 3.1. That lemma gives, exactly, $\mathbb{E}_{P_0}[f_\ell] = \sum_{k=g}^t \mathbb{E}_{P_0}[\mathbb{I}(G > k)\delta_r\delta_m \mid \mathcal{I}_\ell^c]$, and, by Cauchy–Schwarz, $\sqrt{n}\,|\mathbb{E}_{P_0}[f_\ell]| \le \sqrt{n}\sum_{k=g}^t \|\mathbb{I}(G > k)\delta_m\|_2\|\mathbb{I}(G > k)\delta_r\|_2 = o_p(1)$ by the product-rate condition.

Step 3: the empirical-process term. Write f_ℓ as the sum over k of $\{\mathbb{I}(G = g) - \mathbb{I}(G > k)\widehat{r}_{g,k,-\ell}\}(-\delta_m)$ and $-\mathbb{I}(G > k)\delta_r U_k$, with $U_k := \Delta Y_k - m^0_{g,k,0}$. As in the preceding proof, the first has norm at most $(\varepsilon^{-1/2} + M)\|\mathbb{I}(G > k)\delta_m\|_2 = o_p(1)$; no envelope for $\widehat{m}$ is needed, and no off-support value enters. For the second, the cap and Assumption 3 give $\delta_r^2 \le C_r := (M + \varepsilon^{-1})^2$. Since $L_2(P_0)$ convergence in probability does not give pointwise convergence of the random integrand, truncate: for every $N > 0$,

$$\mathbb{E}_{P_0}\big[\mathbb{I}(G > k)\delta_r^2 U_k^2 \,\big|\, \mathcal{I}_\ell^c\big] \le N\big\|\mathbb{I}(G > k)\delta_r\big\|_2^2 + C_r\,\mathbb{E}_{P_0}\big[\mathbb{I}(G > k)U_k^2\,\mathbb{I}(U_k^2 > N)\big].$$

The first term is $o_p(1)$ for each fixed N by consistency of the ratio in the risk-set norm; the second is a deterministic tail of the integrable $\mathbb{I}(G > k)U_k^2$, which vanishes as $N \to \infty$ by $\max_k \mathbb{E}_{P_0}[(\Delta Y_k)^2] < \infty$. Letting $n \to \infty$ and then $N \to \infty$ gives $\|f_\ell\|_2 = o_p(1)$ for every ℓ.

Step 4: ratio expansion and conclusion. Steps 2–3 prove (C.5), so $\delta_A = O_p(n^{-1/2})$; also $\delta_p = O_p(n^{-1/2})$ by the central limit theorem, and $\widehat{p}_g$ is bounded away from zero with probability tending to one. The two product terms in (C.4) are therefore $O_p(n^{-1})$, and substituting (C.5) and $A^0 = p_g^0\tau_{g,t}^0$ into its linear terms gives

$$\sqrt{n}(\widehat{\tau}^{\mathrm{os}}_{g,t} - \tau^0_{g,t}) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \frac{S^0(O_i) - \tau^0_{g,t}\mathbb{I}(G_i = g)}{p_g^0} + o_p(1) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{g,t}(O_i;\eta_g^0) + o_p(1),$$

which converges to $\mathcal{N}(0, V_{\tau_{g,t}})$, $V_{\tau_{g,t}} = \mathbb{E}_{P_0}[\psi^2_{g,t}(O;\eta_g^0)]$. For an aggregate, the exact identity

$$\widehat{\Theta}_{\mathrm{os}} - \Theta^0 = \frac{1}{\widehat{W}_{\mathcal{S}}}\sum_{\varsigma\in\mathcal{S}}\big[\widehat{\omega}_\varsigma(\widehat{\tau}^{\mathrm{os}}_\varsigma - \tau^0_\varsigma) + (\tau^0_\varsigma - \Theta^0)(\widehat{\omega}_\varsigma - \omega_\varsigma)\big]$$

uses only $\sum_\varsigma \omega_\varsigma(\tau^0_\varsigma - \Theta^0) = 0$. Replacing $\widehat{W}_{\mathcal{S}}$ and the leading $\widehat{\omega}_\varsigma$ by their limits costs only products of root-n errors, since $|W_{\mathcal{S}}| \ge c_W$ and $\max_\varsigma \mathbb{E}[\Omega_\varsigma^2] < \infty$ (Condition 3(iii)), so

$$\sqrt{n}(\widehat{\Theta}_{\mathrm{os}} - \Theta^0) = \frac{1}{W_{\mathcal{S}}}\sum_{\varsigma\in\mathcal{S}}\big[\omega_\varsigma\sqrt{n}(\widehat{\tau}^{\mathrm{os}}_\varsigma - \tau^0_\varsigma) + (\tau^0_\varsigma - \Theta^0)\sqrt{n}(\widehat{\omega}_\varsigma - \omega_\varsigma)\big] + o_p(1).$$

The granular expansions and $\sqrt{n}(\widehat{\omega}_\varsigma - \omega_\varsigma) = n^{-1/2}\sum_i\{\Omega_\varsigma(O_i) - \omega_\varsigma\}$ give the canonical gradient (2.6), and the central limit theorem gives the limit $\mathcal{N}(0, V_\Theta)$. ∎

Proof of Corollary 3.1. Fix a target cohort g and a slice k in a target window for g, and write $\mathcal{R}_{g,k}$ for the cascade functional, $\mathcal{R}_{g,k}(\boldsymbol{x}) := x_g/\prod_{s=g}^k(1 - x_s)$ for $\boldsymbol{x} = (x_g, \dots, x_k)^\top$. Then $r^0_{g,k} = \mathcal{R}_{g,k}(\boldsymbol{\lambda}^0_{g,[g:k]})$ by (3.1), and, by (3.2), $\widehat{r}_{g,k,-\ell} = \min\{\widehat{r}^{\mathrm{raw}}_{g,k,-\ell}, M\}$ with $\widehat{r}^{\mathrm{raw}}_{g,k,-\ell} = \mathcal{R}_{g,k}(\widehat{\boldsymbol{\lambda}}_{g,[g:k],-\ell})$ and $\widehat{\lambda}_{g,s,-\ell} = \Pi_{[0,1-\bar{a}]}(\widetilde{\lambda}_{g,s,-\ell})$.

Both parts rest on one bound, which carries the learners' errors to the fitted ratio:

$$\big\|\mathbb{I}(G > k)\{\widehat{r}_{g,k,-\ell} - r^0_{g,k}\}\big\|_2 \le J_{g,k}\,T\max_{s\in[g,k]}\big\|\mathbb{I}(G \ge s)\{\widetilde{\lambda}_{g,s,-\ell} - \lambda^0_{g,s}\}\big\|_2. \qquad (*)$$

Here $J_{g,k} := \sup_{\mathcal{B}_{g,k}} |\nabla \mathcal{R}_{g,k}|$, with $|\cdot|$ the Euclidean norm and $\mathcal{B}_{g,k} := [0, 1-\overline{a}]^{k-g+1}$, depends on neither n nor ℓ. Near $\mathcal{B}_{g,k}$ every factor $1-x_s$ is positive, and on $\mathcal{B}_{g,k}$ it is at least $\overline{a}$, so $\mathcal{R}_{g,k}$ is continuously differentiable there, with partial derivatives $1/\{(1-x_g)\prod_{s=g}^{k}(1-x_s)\}$ and $x_g/\{(1-x_j)\prod_{s=g}^{k}(1-x_s)\}$ for $j > g$, each at most $\overline{a}^{-(T+1)}$ on $\mathcal{B}_{g,k}$; hence $J_{g,k} \le \sqrt{T}\,\overline{a}^{-(T+1)}$. On $\{G > k\}$, which is contained in $\{G \ge s\}$ for $s \le k$, the hazard clause of Condition 4 puts $\boldsymbol{\lambda}^0_{g,[g:k]}$ in $\mathcal{B}_{g,k}$, where the 1-Lipschitz projection leaves it fixed (the first line of (3.7)), and $\widehat{\boldsymbol{\lambda}}_{g,[g:k],-\ell}$ lies in $\mathcal{B}_{g,k}$ by construction; the ratio clause $r^0_{g,k} \le M$ makes $r^0_{g,k}$ a fixed point of the 1-Lipschitz cap (the second line). So the cap, the mean value theorem on the convex box and the projection give, almost surely,

$$\mathbb{I}(G > k)\big|\widehat{r}_{g,k,-\ell} - r^0_{g,k}\big| \le J_{g,k} \sum_{s=g}^{k} \mathbb{I}(G \ge s)\big|\widetilde{\lambda}_{g,s,-\ell} - \lambda^0_{g,s}\big|,$$

each hazard error being evaluated only on the risk set on which $\lambda^0_{g,s}$ is identified and $\widetilde{\lambda}_{g,s,-\ell}$ is fitted; the bound is well defined on $\{e_g(\mathcal{C}_{g-1}) > 0\}$. Squaring, applying Cauchy–Schwarz to the at most T terms, and taking $\mathbb{E}_{P_0}[\cdot \mid \mathcal{I}^c_\ell]$ give $(*)$.

Step 1: double robustness. If the pre-clipping learners converge, $\|\mathbb{I}(G \ge s)\{\widetilde{\lambda}_{g,s,-\ell} - \lambda^0_{g,s}\}\|_2 \xrightarrow{p} 0$ uniformly over ℓ for every $g \in \mathcal{G}_{\mathcal{S}}$ and every s entering a target window for g, then $(*)$ gives $\max_\ell \|\mathbb{I}(G > k)\{\widehat{r}_{g,k,-\ell} - r^0_{g,k}\}\|_2 \xrightarrow{p} 0$ for every $(g,t) \in \mathcal{S}$ and $k \in [g,t]$. The ratio limit of Theorem 3.1 is therefore $r^0_{g,k}$ on $\{G > k\}$, which is at most M by the ratio clause, and the ratio-correct branch of that theorem applies. If instead $\overline{m}_{g,k,0} = m^0_{g,k,0}$, the outcome-correct branch applies whatever the ratio limit, without either clause.

Step 2: asymptotic efficiency. The corollary assumes the conditions of Theorem 3.1, and part (ii) the applicable moment and aggregation conditions of Theorem 3.2, which (3.8) does not imply; the fitted ratios are capped by construction. It remains to verify the consistency and product-rate conditions of that theorem on the target windows, in the identified-support norms in which its proof uses them. The outcome regressions are consistent by assumption, and the fitted ratios by $(*)$, as in Step 1; (3.8) alone would not give this, because a product can be $o_p(n^{-1/2})$ while one factor does not vanish. For the product rate, with $\delta_m := \widehat{m}_{g,k,0,-\ell} - m^0_{g,k,0}$ and $\delta_r := \widehat{r}_{g,k,-\ell} - r^0_{g,k}$, $(*)$ gives $\|\mathbb{I}(G > k)\delta_m\|_2 \|\mathbb{I}(G > k)\delta_r\|_2 \le J_{g,k} T \max_{s \in [g,k]} \|\mathbb{I}(G > k)\delta_m\|_2 \|\mathbb{I}(G \ge s)\{\widetilde{\lambda}_{g,s,-\ell} - \lambda^0_{g,s}\}\|_2$ for every ℓ, $(g,t) \in \mathcal{S}$ and $k \in [g,t]$. The indices are finitely many and the constants free of n, so the maximum of the left side is $o_p(n^{-1/2})$ by (3.8): this is the product-rate condition of Theorem 3.2. Every condition of that theorem holds, so the hazard-based one-step estimator is asymptotically linear with the canonical gradient and attains the bound V_Θ. ∎

Lemma C.2 (One-step normalizer and control-residual bounds)**.** *Fix (g,t) and use the fixed-fold cross-fitting protocol of Appendix C.1. Suppose $p_g > 0$, Assumption 3 holds on $k \in [g,t]$, $\max_{k \in [g,t]} \mathbb{E}_{P_0}[(\Delta Y_k)^2] < \infty$, and the out-of-fold ratios satisfy $0 \le \widehat{r}_{g,k,-\ell} \le M$ on $\{G > k\}$. Let $d_m := d_{m,g,t}$ and $d_r := d_{r,g,t}$ be defined by (3.9), and suppose $d_m, d_r = o_p(1)$. Then*

$$\max_{k \in [g,t]} |B_{n,g,k} - \widehat{p}_g| = O_p(n^{-1/2} + d_r), \qquad \max_{k \in [g,t]} |\widehat{\tau}_{\mathrm{ct},k}| = O_p(n^{-1/2} + d_m).$$

In particular, $B_{n,g,k} \xrightarrow{p} p_g$ uniformly over this fixed slice set, so these control-arm means are well defined with probability tending to one.

Proof. Write $\mathbb{I}_{>k} := \mathbb{I}(G > k)$, $\delta_{r,k,-\ell} := \widehat{r}_{g,k,-\ell} - r^0_{g,k}$, $\delta_{m,k,-\ell} := \widehat{m}_{g,k,0,-\ell} - m^0_{g,k,0}$, and $U_k := \Delta Y_k - m^0_{g,k,0}(\mathcal{C}_{g-1})$. All fitted functions below are evaluated out of fold. Assumption 3 bounds $r^0_{g,k}$ by ε^{-1} on the risk set, and (C.2) with $f = 1$ gives $\mathbb{E}_{P_0}[\mathbb{I}_{>k} r^0_{g,k}] = p_g$. The average of this bounded

true summand differs from p_g by $O_p(n^{-1/2})$. By (C.1), applied to $\mathbb{I}_{>k}\delta_{r,k,-\ell}$, its fitted perturbation has average $O_p(d_r)$: its $L_2(P_0)$ norm and the absolute value of its expectation are at most d_r. Hence $B_{n,g,k} - p_g = O_p(n^{-1/2} + d_r)$. Since $\widehat{p}_g - p_g = O_p(n^{-1/2})$, this proves the first bound. As $d_r = o_p(1)$ and $p_g > 0$, the fixed collection of normalizers is at least $p_g/2$ with probability tending to one.

Conditional Jensen gives $\mathbb{E}_{P_0}[\mathbb{I}_{>k}(m^0_{g,k,0})^2] \le \mathbb{E}_{P_0}[\mathbb{I}_{>k}(\Delta Y_k)^2] < \infty$, so $\mathbb{I}_{>k}U_k \in L_2(P_0)$. The numerator of the control-arm mean splits exactly as

$$\mathbb{P}_n[\mathbb{I}_{>k}\widehat{r}^{\,\mathrm{ini}}_{g,k}\{\Delta Y_k - \widehat{m}^{\,\mathrm{ini}}_{g,k,0}\}] = \mathbb{P}_n[\mathbb{I}_{>k}r^0_{g,k}U_k] - \mathbb{P}_n[\mathbb{I}_{>k}\widehat{r}^{\,\mathrm{ini}}_{g,k}\delta_{m,k,-\ell(\cdot)}] + \mathbb{P}_n[\mathbb{I}_{>k}\delta_{r,k,-\ell(\cdot)}U_k].$$

The first term is $O_p(n^{-1/2})$, because its summand is centered by the conditional-mean property of $m^0_{g,k,0}$ and has finite second moment. The second is $O_p(d_m)$ by the cap, Cauchy–Schwarz, and the second bound of (C.1). For the third, hold each trained function fixed and set $f_\ell := \mathbb{I}_{>k}\delta_{r,k,-\ell}U_k$. The ratio error is $\mathcal{C}_{g-1}$-measurable and $\mathbb{E}_{P_0}[\mathbb{I}_{>k}U_k \mid \mathcal{C}_{g-1}] = 0$, so $\mathbb{E}_{P_0}[f_\ell] = 0$. Also $\|f_\ell\|_2 \le (M+\varepsilon^{-1})\|\mathbb{I}_{>k}U_k\|_2$. The first bound of (C.1) therefore makes the third average $O_p(n^{-1/2})$. Division by $B_{n,g,k} \ge p_g/2$ proves the second bound. There are finitely many slices, so both bounds hold for their maxima. ∎

Proof of Corollary 3.2. *Consistency.* Fix $(g,t) \in \mathcal{S}$ and $k \in [g,t]$. The capped-ratio convergence and (C.1) give $B_{n,g,k} \xrightarrow{p} \overline{B}_{g,k}$, while $\widehat{p}_g \xrightarrow{p} p_g > 0$. To check the two numerators separately, write $\delta_{m,k,-\ell} := \widehat{m}_{g,k,0,-\ell} - \overline{m}_{g,k,0}$, $\delta_{r,k,-\ell} := \widehat{r}_{g,k,-\ell} - \overline{r}_{g,k}$, and $\overline{U}_k := \Delta Y_k - \overline{m}_{g,k,0}(\mathcal{C}_{g-1})$. The balancing identity (C.2) transfers the risk-set L_2 convergence of δ_m to $\{G = g\}$, so (C.1) gives

$$\mathbb{P}_n[\mathbb{I}(G=g)\{\Delta Y_k - \widehat{m}^{\,\mathrm{ini}}_{g,k,0}\}] \xrightarrow{p} \mathbb{E}_{P_0}[\mathbb{I}(G=g)\overline{U}_k].$$

The control-numerator perturbation is

$$-\mathbb{P}_n[\mathbb{I}(G>k)\widehat{r}^{\,\mathrm{ini}}_{g,k}\delta_{m,k,-\ell(\cdot)}] + \mathbb{P}_n[\mathbb{I}(G>k)\delta_{r,k,-\ell(\cdot)}\overline{U}_k].$$

Its first term is $o_p(1)$ by the cap and the second bound in (C.1). The functions in the second term have uniformly bounded L_2 norms, since $|\delta_{r,k,-\ell}| \le 2M$ and $\mathbb{I}(G>k)\overline{U}_k \in L_2(P_0)$; their expectations are $o_p(1)$ by Cauchy–Schwarz and ratio convergence. The first bound in (C.1) then makes this term $o_p(1)$. Consequently the control numerator converges to $\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}\overline{U}_k]$. All divisions are valid with probability tending to one under the positive limits in part (i), and

$$\widehat{\tau}^{\,\mathrm{os,Háj,\,ini}}_{g,t} \xrightarrow{p} \sum_{k=g}^{t}\left\{\frac{\mathbb{E}_{P_0}[\mathbb{I}(G=g)\overline{U}_k]}{p_g} - \frac{\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}\overline{U}_k]}{\overline{B}_{g,k}}\right\}.$$

In the outcome-correct branch, conditional centering makes every control numerator zero; the treated terms sum to $\tau^0_{g,t}$ by (C.3) and identification. In the ratio-correct branch, (C.2) gives $\overline{B}_{g,k} = p_g$, so the displayed limit is the HT limit proved to equal $\tau^0_{g,t}$ in Theorem 3.1. For an aggregate, $\widehat{\omega}_{g,t} \xrightarrow{p} \omega^0_{g,t}$ and the empirical weight mass converges to $W_{\mathcal{S}} \neq 0$, so the finite weighted average is consistent.

First-order equivalence. Under part (ii), Lemma C.2 applies to each pair. Exact subtraction of the two one-step estimators gives

$$\widehat{\tau}^{\,\mathrm{os,Háj,\,ini}}_{g,t} - \widehat{\tau}^{\,\mathrm{os,HT}}_{g,t} = \sum_{k=g}^{t}\widehat{\tau}_{\mathrm{ct},k}\frac{B_{n,g,k} - \widehat{p}_g}{\widehat{p}_g}.$$

Since $\widehat{p}_g \xrightarrow{p} p_g > 0$, the lemma bounds this difference by

$$O_p\{(n^{-1/2} + d_{m,g,t})(n^{-1/2} + d_{r,g,t})\} = o_p(n^{-1/2});$$

indeed, multiplication by $\sqrt{n}$ gives terms of orders $n^{-1/2}$, $d_{m,g,t}$, $d_{r,g,t}$, and $\sqrt{n}\, d_{m,g,t} d_{r,g,t}$, all tending to zero. The separate-maxima product condition also implies the product condition of Theorem 3.2, because every same-fold, same-slice product is bounded by $d_{m,g,t} d_{r,g,t}$. Thus that theorem supplies the canonical expansion of the HT one-step estimator.

Aggregation and efficiency. Here $\widehat{W}_{\mathcal{S}} = \sum_{\mathcal{S}} \widehat{\omega}_{g,t}$, as defined after Lemma 3.1. The exact identity

$$\widehat{\Theta}^{\text{Háj}}_{\text{os}} - \widehat{\Theta}^{\text{HT}}_{\text{os}} = \frac{1}{\widehat{W}_{\mathcal{S}}} \sum_{(g,t)\in\mathcal{S}} \widehat{\omega}_{g,t} \{\widehat{\tau}^{\text{os,Háj, ini}}_{g,t} - \widehat{\tau}^{\text{os,HT}}_{g,t}\} = o_p(n^{-1/2})$$

follows from the fixed finite index set, $\widehat{\omega}_{g,t} = O_p(1)$, and $\widehat{W}_{\mathcal{S}} \xrightarrow{p} W_{\mathcal{S}} \neq 0$. Hence the aggregate inherits the expansion of Theorem 3.2, whose canonical score is

$$\frac{1}{W_{\mathcal{S}}} \sum_{(g,t)\in\mathcal{S}} \left[\omega^0_{g,t} \psi_{g,t}(O; \eta^0_g) + (\tau^0_{g,t} - \Theta^0)\{\Omega_{g,t}(O) - \omega^0_{g,t}\}\right].$$

The empirical cancellation of the weight correction in the point equation does not remove this unit-level term from the expansion or its variance. The efficiency and central-limit conclusions follow from that theorem, and their causal-model interpretation at a strictly prefix-feasible law follows from Theorem 2.4. ■

Lemma C.3 (Empirical-score L_2 convergence implies variance consistency)**.** *Let $\theta^0 \in \{\tau^0_{g,t}, \Theta^0\}$ have efficient influence function $\psi_\theta(\cdot; \eta^0_\theta)$ with $V_\theta = \mathbb{E}_{P_0}[\psi^2_\theta] < \infty$, and let $\widehat{\psi}_{\theta,i}$ be any cross-fitted empirical score evaluation built from a consistent target/nuisance pair. If*

$$\mathbb{P}_n\left[(\widehat{\psi}_{\theta,i} - \psi_\theta(O_i; \eta^0_\theta))^2\right] = o_p(1), \tag{C.6}$$

then $\mathbb{P}_n[\widehat{\psi}^2_{\theta,i}] \xrightarrow{p} V_\theta$ and $(\mathbb{P}_n[\widehat{\psi}_{\theta,i}])^2 = o_p(1)$; consequently the plug-in variance estimator formed from $\{\widehat{\psi}_{\theta,i}\}$ is consistent for V_θ. Moreover, (C.6) holds in each of the following two cases.

(i) One-step: *under HT normalization, $\widehat{\psi}_{\theta,i} = \varphi^{\text{eff}}\{\widehat{\theta}_{\text{os}}; \widehat{\eta}_{\mathcal{S},-\ell(i)}\}(O_i)$, with $\widehat{\theta}_{\text{os}}$ the granular or aggregate one-step estimator matching θ^0, under the conditions of Theorem 3.2 and Condition 3(iii). The Hájek self-normalized analogue is included under the conditions of Corollary 3.2(ii), applied to the corresponding granular or aggregate target.*

(ii) Targeted: *under HT normalization, $\widehat{\psi}_{\theta,i} = \varphi^{\text{eff}}\{\widehat{\theta}_{\text{tar}}; \widehat{\eta}^*_{\mathcal{S},-\ell(i)}\}(O_i)$, with $\widehat{\theta}_{\text{tar}}$ the corresponding TMLE, under Theorem 4.2(ii) and Condition 3(iii). The Hájek self-normalized targeted-score analogue is included under the root-n part of Corollary 4.1, the same aggregate condition, and the normalizer floor (4.9), together with risk-set consistency of the ratio and $\max_k \mathbb{E}_{P_0}[(\Delta Y_k)^2] < \infty$; it is not reclassified as a population EIF.*

Proof. Given (C.6), expand $\mathbb{P}_n[\widehat{\psi}^2] = \mathbb{P}_n[\psi^2_\theta] + 2\mathbb{P}_n[\psi_\theta(\widehat{\psi} - \psi_\theta)] + \mathbb{P}_n[(\widehat{\psi} - \psi_\theta)^2]$: the first term converges to V_θ by the law of large numbers, the third is $o_p(1)$, and so, by Cauchy–Schwarz, is the second. Likewise $\mathbb{P}_n[\widehat{\psi}] = \mathbb{P}_n[\psi_\theta] + o_p(1) = o_p(1)$, since $\mathbb{E}[\psi_\theta] = 0$.

To verify (C.6), write $\widehat{\psi}_{\theta,i} - \psi_\theta(O_i; \eta^0_\theta) = \Delta_{i,\ell(i)} + R_i$, where $\Delta_{i,\ell}$ is the nuisance perturbation holding all scalar components at their true values, as defined after Lemma 3.1, and R_i collects the replacements of $(\tau^0_{g,t}, \Theta^0, p^0_g, \omega^0_{g,t}, W^0_{\mathcal{S}})$ by the corresponding procedure's estimates inside the score, with $\widehat{W}_{\mathcal{S}} = \sum_{\mathcal{S}} \widehat{\omega}_{g,t}$. In case (i), each granular component of $\Delta_{i,\ell}$ is $f_\ell(O_i)/p^0_g$, with $f_\ell := \widehat{S}_\ell - S^0$ the function of Step 1 of the proof of Theorem 3.2, whose norm is $o_p(1)$ by Step 3; the second

bound in (C.1) gives $\mathbb{P}_n[\Delta^2] = o_p(1)$. In case (ii), fix a component (g,t) and write $d_m = d_{m,g,t}$. The exact normal equation and factual initialization give, on an event whose probability tends to one,

$$\widehat{\epsilon}^{\nu}_{g,t} = -\frac{\sum_{k=g}^{t} \rho^{\nu}_{n,g,k} \widehat{\tau}_{\mathrm{ct},k}}{\sum_{k=g}^{t} \mathbb{P}_n[(\widehat{H}^{\mathrm{ini},\nu}_{(g,t),k})^2]}, \qquad \rho^{\mathrm{HT}}_{n,g,k} = \frac{B_{n,g,k}}{\widehat{p}_g}, \quad \rho^{\mathrm{Háj}}_{n,g,k} = 1.$$

The treated rows contribute $(t-g+1)/\widehat{p}_g$ to the denominator. By Lemma C.2, $\max_{k\in[g,t]} |\widehat{\tau}_{\mathrm{ct},k}| = O_p(n^{-1/2} + d_m)$ and $B_{n,g,k} \xrightarrow{p} p_g > 0$; hence

$$|\widehat{\epsilon}^{\nu}_{g,t}| \le \frac{\widehat{p}_g}{t-g+1} \sum_{k=g}^{t} \rho^{\nu}_{n,g,k} |\widehat{\tau}_{\mathrm{ct},k}| = O_p(n^{-1/2} + d_m) = o_p(1).$$

The targeted control branch differs from the initial one by $\widehat{\epsilon}^{\nu}_{g,t} \widehat{H}^{\mathrm{ini},\nu,\mathrm{ct}}_{(g,t),k,-\ell}$. The ratio cap and the positive denominator limits bound the latter factor by $O_p(1)$, so this perturbation is $o_p(1)$ in empirical L_2. Together with the case-(i) bound, this gives $\mathbb{P}_n[\Delta^2] = o_p(1)$ componentwise, and a reported aggregate is a fixed finite combination of components. Under Corollary 3.2(ii) for a Hájek one-step estimator, or the root-n conditions of Corollary 4.1 for a Hájek targeted estimator, $B_{n,g,k} \xrightarrow{p} p_g$ by (C.1) with $f_\ell = \mathbb{I}(G > k)\widehat{r}_{g,k,-\ell}$ and (C.2) with $f = 1$. At a common nuisance value $(\widetilde{m}, \widehat{r})$—the initial one, or the Hájek-targeted one—and a common centering estimate, the self-normalized and HT score evaluations of a component pair differ only in the control-arm divisor,

$$\widehat{\psi}^{\mathrm{Háj}}_i - \widehat{\psi}^{\mathrm{HT}}_i = \sum_{k=g}^{t} \Big(\frac{1}{\widehat{p}_g} - \frac{1}{B_{n,g,k}}\Big) \mathbb{I}(G_i > k)\, \widehat{r}_{g,k,-\ell(i)}(\mathcal{C}_{i,g-1}) \big\{\Delta Y_{i,k} - \widetilde{m}_{g,k,-\ell(i)}(\mathcal{C}_{i,g-1})\big\},$$

so $\mathbb{P}_n[(\widehat{\psi}^{\mathrm{Háj}} - \widehat{\psi}^{\mathrm{HT}})^2] \le (t-g+1) M^2 \max_k (1/\widehat{p}_g - 1/B_{n,g,k})^2 \sum_k \mathbb{P}_n[\mathbb{I}(G > k)(\Delta Y_k - \widetilde{m}_{g,k})^2] = o_p(1)$, the residual moment being $O_p(1)$ by the second bound in (C.1). At the Hájek-targeted nuisance the residual carries the additional term $\widehat{\epsilon}^{\mathrm{Háj}}_{g,t} \beta^{\mathrm{ct}}_{n,g,k}$, which is $o_p(1)$ by the same fluctuation bound and the ratio cap with $B_{n,g,k} \xrightarrow{p} p_g > 0$, so the case-(ii) argument applies to the HT evaluation at that nuisance. The centering estimates differ by $o_p(1)$, because both estimators are consistent. Each term of R_i is a product of bounded indicators, the capped ratio, weights with $\mathbb{E}[\Omega_\varsigma^2] < \infty$ (Condition 3(iii)), residual factors with empirical second moment $O_p(1)$ (by $\max_k \mathbb{E}[(\Delta Y_k)^2] < \infty$ and the second bound in (C.1)), and parameter differences that are $o_p(1)$ by Theorem 3.2, Corollary 3.2, Theorem 4.2 or Corollary 4.1 and the law of large numbers, so Cauchy–Schwarz gives $\mathbb{P}_n[R^2] = o_p(1)$. The weight score is retained in $\widehat{\psi}_\theta$ throughout, so the conclusion covers the full unit-level aggregate score. ■

Corollary C.2 (Pointwise Wald intervals)**.** *Let $\theta^0 \in \{\tau^0_{g,t}, \Theta^0\}$, with V_θ the efficiency bound V_Θ of Theorem 3.2 (for $\mathcal{S} = \{(g,t)\}$ when $\theta^0 = \tau^0_{g,t}$), let $\widehat{\theta}$ be one of the following cross-fitted estimators, and let $\widehat{\mathrm{se}}$ be formed from its empirical score: (a) the HT one-step estimator, under the conditions of Theorem 3.2 and Condition 3(iii); (b) the Hájek one-step estimator, under the conditions of Corollary 3.2(ii); (c) the HT TMLE, under the conditions of Theorem 4.2(ii); or (d) the Hájek TMLE, under the conditions of the root-n part of Corollary 4.1. If $V_\theta \in (0,\infty)$, then $n\widehat{\mathrm{se}}^2 \xrightarrow{p} V_\theta$ and, for every $\alpha \in (0,1)$, with $z_{1-\alpha/2}$ the standard normal quantile,*

$$\mathbb{P}\big(\widehat{\theta} - z_{1-\alpha/2}\,\widehat{\mathrm{se}} \le \theta^0 \le \widehat{\theta} + z_{1-\alpha/2}\,\widehat{\mathrm{se}}\big) \longrightarrow 1-\alpha.$$

Proof of Corollary C.2. Let $\widehat{\varphi}_i$ be the empirical score of the corollary and $\psi_\theta = \varphi^{\mathrm{eff}}\{\theta^0; \eta^0_\theta\}$, with $V_\theta = \mathbb{E}_{P_0}[\psi^2_\theta] \in (0,\infty)$. Lemma C.3 gives the first claim, $n\widehat{\mathrm{se}}^2 = \mathbb{P}_n[\widehat{\varphi}^2_i] \xrightarrow{p} V_\theta$: in case

(a) by its item (i) under HT normalization; in case (b) by its item (i) with the Hájek analogue, under the conditions of Corollary 3.2(ii); in case (c) by its item (ii) under HT normalization; and in case (d) by its item (ii) with the Hájek targeted-score analogue. The canonical expansion $\sqrt{n}(\widehat{\theta}-\theta^0) = n^{-1/2}\sum_{i=1}^n \psi_\theta(O_i) + o_p(1) \rightsquigarrow \mathcal{N}(0, V_\theta)$ holds by Theorem 3.2 in case (a); in case (b), by Corollary 3.2(ii); by Theorem 4.2(ii) in case (c); and by Corollary 4.1 in case (d). Because $V_\theta > 0$, $(n\widehat{\mathrm{se}}^2)^{-1/2} \xrightarrow{p} V_\theta^{-1/2}$, so $\widehat{\mathrm{se}} > 0$ with probability tending to one, and Slutsky's theorem gives $(\widehat{\theta}-\theta^0)/\widehat{\mathrm{se}} \rightsquigarrow \mathcal{N}(0,1)$. Since the standard normal distribution function Φ is continuous, the probability that $|\widehat{\theta}-\theta^0| \le z_{1-\alpha/2}\,\widehat{\mathrm{se}}$ converges to $2\Phi(z_{1-\alpha/2}) - 1 = 1-\alpha$. ■

D Proofs for Section 4

D.1 Notation for the Hájek Bridge

The clever covariate (4.2) divides its treated arm by $\widehat{p}_g = \mathbb{P}_n[\mathbb{I}(G=g)]$ and its control arm by $B^\nu_{n,g,k}$, which is $\widehat{p}_g$ under HT normalization and the Hájek self-normalizer $B_{n,g,k} := \mathbb{P}_n[\mathbb{I}(G>k)\widehat{r}^{\mathrm{ini}}_{g,k}(\mathcal{C}_{g-1})] = n^{-1}\sum_{i=1}^n \mathbb{I}(G_i>k)\,\widehat{r}_{g,k,-\ell(i)}(\mathcal{C}_{i,g-1})$ under Hájek normalization; here $\widehat{r}^{\mathrm{ini}}_{g,k}$ is the cross-fitted capped cascade ratio (3.2). Under consistency of $\widehat{r}^{\mathrm{ini}}_{g,k}$ and overlap, the balancing identity (C.2) gives $B_{n,g,k} \xrightarrow{p} p_g$. Hájek self-normalization is the default of Sant'Anna and Zhao (2020), of Callaway and Sant'Anna (2021), and of the `did` package.

With the treated-arm and control-arm weights $\beta^{\mathrm{tr}}_{n,g,k} = 1/\widehat{p}_g$ and $\beta^{\mathrm{ct}}_{n,g,k} = \widehat{r}^{\mathrm{ini}}_{g,k}/B^\nu_{n,g,k}$ of Section 4.1, the raw contrast of slice $k \ge g$ is $\widehat{H}^{\mathrm{ini}}_{g,k}(O) := \mathbb{I}(G=g)\beta^{\mathrm{tr}}_{n,g,k} - \mathbb{I}(G>k)\beta^{\mathrm{ct}}_{n,g,k}(\mathcal{C}_{g-1})$, and the clever covariate of the pair (g,t) is $\widehat{H}^{\mathrm{ini}}_{(g,t),k} := \mathbb{I}(g \le k \le t)\widehat{H}^{\mathrm{ini}}_{g,k}$.

The treated and control arms of slice k are disjoint, so its Gram contribution and loading are

$$V_{n,g,k} := \mathbb{P}_n\{(\widehat{H}^{\mathrm{ini}}_{g,k})^2\} = \widehat{p}_g(\beta^{\mathrm{tr}}_{n,g,k})^2 + \mathbb{P}_n\big[\mathbb{I}(G>k)(\beta^{\mathrm{ct}}_{n,g,k})^2\big],$$
$$c_{n,g,k} := \beta^{\mathrm{tr}}_{n,g,k} + \mathbb{P}_n\big[\mathbb{I}(G=g)\beta^{\mathrm{ct}}_{n,g,k}\big]/\widehat{p}_g.$$

Here the chosen normalization ν is fixed and suppressed: $V_{n,g,k}$ and $c_{n,g,k}$ denote $V^\nu_{n,g,k}$ and $c^\nu_{n,g,k}$, respectively, as written explicitly in Appendix F.2. Under Hájek normalization they are the expressions defined in Section 4.2. On the event of Condition 5(i), where they are defined, both are at least $1/\widehat{p}_g > 0$. The single column $\widehat{\mathbf{h}}^{\mathrm{ini}}_{(g,t)} \in \mathbb{R}^{n(t-g+1)}$ of the pair regression (4.4) has normalized Gram entry $n^{-1}\widehat{\mathbf{h}}^{\mathrm{ini}\top}_{(g,t)}\widehat{\mathbf{h}}^{\mathrm{ini}}_{(g,t)} = \sum_{k=g}^t V_{n,g,k}$. For a treated row of cohort g with $k \in [g,t]$, the updated row value and the targeted control branch of Section 4.1 satisfy $\widehat{m}^*_{(g,t),(i,k)} - \widehat{m}^*_{(g,t),k,0}(\mathcal{C}_{i,g-1}) = \Delta Y_{i,k} - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{i,g-1}) + \widehat{\epsilon}_{g,t}\{\beta^{\mathrm{tr}}_{n,g,k} + \beta^{\mathrm{ct}}_{n,g,k}(\mathcal{C}_{i,g-1})\}$, so averaging (4.6) over the treated cohort gives the loading form

$$\widehat{\tau}^{\mathrm{tar}}_{g,t} = \sum_{k=g}^t \frac{\mathbb{P}_n[\mathbb{I}(G=g)\{\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{g-1})\}]}{\widehat{p}_g} + \Big(\sum_{k=g}^t c_{n,g,k}\Big)\widehat{\epsilon}_{g,t},$$

whose first term is the untargeted value. We write $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},k}$ and $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},k}$ for the within-arm Hájek means $\widehat{\tau}_{\mathrm{tr},k}$ and $\widehat{\tau}_{\mathrm{ct},k}$ defined in Section 3.2, and $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},(g,t)}$ and $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)}$ for their sums over $k \in [g,t]$.

D.2 Proofs

Throughout, the targeting step uses the row offsets (4.3) and the updated row values (4.5); no treated-branch regression enters the efficient-score algebra.

Proof of Theorem 4.1. By Step 2 of Algorithm 1, $\widehat{\boldsymbol{\epsilon}}$ solves the normal equation (4.8), $\widehat{\mathbf{H}}^{\mathrm{ini}\top}(\Delta\mathbf{y} - \widehat{\mathbf{m}}^*) = \mathbf{0}$, where $\widehat{\mathbf{m}}^* = \widehat{\mathbf{m}}^{\mathrm{ini}} + \widehat{\mathbf{H}}^{\mathrm{ini}}\widehat{\boldsymbol{\epsilon}}$ stacks the updated outcomes, one block per target pair. The column of $\epsilon_{g,t}$ holds $\widehat{H}^{\mathrm{ini}}_{(g,t),k}$, which vanishes outside $k \in [g,t]$, so its coordinate, divided by n, is

$$\frac{1}{n}\sum_{i=1}^{n}\sum_{k=g}^{t}\widehat{H}^{\mathrm{ini}}_{(g,t),k}(O_i)\big\{\Delta Y_{i,k} - \widehat{m}^*_{(g,t),(i,k)}\big\} = 0. \tag{D.1}$$

Let ν be the normalization of the column. On the active control rows the updated row value equals the targeted control branch, so adding and subtracting that branch on every active row gives

$$\begin{aligned}\widehat{H}^{\mathrm{ini},\nu}_{(g,t),k}(O_i)\{\Delta Y_{i,k} - \widehat{m}^*_{(g,t),(i,k)}\} = \Big\{\frac{\mathbb{I}(G_i = g)}{\widehat{p}_g} - \frac{\mathbb{I}(G_i > k)\widehat{r}_{g,k}(\mathcal{C}_{i,g-1})}{B^{\nu}_{n,g,k}}\Big\}\{\Delta Y_{i,k} - \widehat{m}^*_{(g,t),k,0}(\mathcal{C}_{i,g-1})\}\\ - \frac{\mathbb{I}(G_i = g)}{\widehat{p}_g}\{\widehat{m}^*_{(g,t),(i,k)} - \widehat{m}^*_{(g,t),k,0}(\mathcal{C}_{i,g-1})\}.\end{aligned}$$

Summing over k and taking $\mathbb{P}_n$, the definition (4.6) and $\mathbb{P}_n[\mathbb{I}(G = g)/\widehat{p}_g] = 1$ turn the right side into $\mathbb{P}_n[\widehat{\psi}^{\nu}_{g,t}]$, so (D.1) is the granular score equation: the canonical plug-in EIF score when $\nu = \mathrm{HT}$, and the self-normalized score when $\nu = \mathrm{Háj}$. For the aggregate, $\mathbb{P}_n[\Omega_\varsigma - \widehat{\omega}_\varsigma] = 0$ for every $\varsigma \in \mathcal{S}$ gives $\mathbb{P}_n[\widehat{\Psi}^{\nu}_{\Theta}] = \widehat{W}_{\mathcal{S}}^{-1}\sum_{\varsigma\in\mathcal{S}}\widehat{\omega}_\varsigma\mathbb{P}_n[\widehat{\psi}^{\nu}_{\varsigma}] = 0$. ∎

Lemma D.1 (Probability limit of the pooled auxiliary fluctuation)**.** *Fix a pair $\mathfrak{b} = (g,t)$, whose auxiliary block has slice set $K_{\mathfrak{b}} = [g,t]$ and one fluctuation coefficient, and use factual initialization (4.3). Suppose that, uniformly over the fixed folds and slices in $K_{\mathfrak{b}}$, the initial control regressions and ratios converge in their identified-support $L_2(P_0)$ norms to deterministic limits $\overline{m}_{g,k,0}$ and $\overline{r}_{g,k}$; $\mathbb{E}_{P_0}[\mathbb{I}(G > k)\overline{m}_{g,k,0}(\mathcal{C}_{g-1})^2] < \infty$ for every $k \in K_{\mathfrak{b}}$; the empirical divisors converge to the corresponding positive population divisors; and the fixed-dimensional clause of Condition 7(i), the moment clause $\max_{k\in[T]}\mathbb{E}[(\Delta Y_k)^2] < \infty$ of item (ii), and the cap and empirical-floor clauses in item (iii) hold. Then*

$$\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} \xrightarrow{p} \overline{\boldsymbol{\epsilon}}_{\mathfrak{b}} := \mathbf{G}_{\mathfrak{b}}(\overline{r})^{-1}\mathbf{S}_{\mathfrak{b}}(\overline{m},\overline{r}),$$

with the score and Gram matrix defined in (E.1)–(E.2). In particular, if $\overline{m}_{g,k,0} = m^0_{g,k,0}$ on every active risk set, then $\mathbf{S}_{\mathfrak{b}}(m^0,\overline{r}) = \mathbf{0}$ and the unique limiting fluctuation is $\overline{\boldsymbol{\epsilon}}_{\mathfrak{b}} = \mathbf{0}$, whether or not $\overline{r} = r^0$.

Proof. Let $\widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell}$ collect the block's out-of-fold clever-covariate columns and $\widehat{U}_{\mathfrak{b},k,-\ell} := \mathbb{I}(G > k)\{\Delta Y_k - \widehat{m}_{g,k,0,-\ell}(\mathcal{C}_{g-1})\}$. Factual initialization makes the zero-fluctuation response vanish on the treated rows, so the pooled normal equation of the block is $\widehat{\mathbf{G}}_{\mathfrak{b}}\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} = \widehat{\mathbf{S}}_{\mathfrak{b}}$, with

$$\widehat{\mathbf{S}}_{\mathfrak{b}} := \sum_{\ell=1}^{L} w_\ell\mathbb{P}_{n,\ell}\sum_{k\in K_{\mathfrak{b}}}\widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell}\widehat{U}_{\mathfrak{b},k,-\ell}, \qquad \widehat{\mathbf{G}}_{\mathfrak{b}} := \sum_{\ell=1}^{L} w_\ell\mathbb{P}_{n,\ell}\sum_{k\in K_{\mathfrak{b}}}\widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell}\widehat{\mathbf{h}}^{\top}_{\mathfrak{b},k,-\ell}.$$

The divisors $\widehat{p}_g$ and, under Hájek normalization, $B_{n,g,k}$ are full-sample averages and so are not training-sample measurable. Write each arm of each column as its cross-fitted numerator, $\mathbb{I}(G = g)$ or $\mathbb{I}(G > k)\widehat{r}_{g,k,-\ell} \le M$, times the inverse of its divisor, $\widehat{p}_g$ or $B^{\nu}_{n,g,k}$. By the assumed divisor convergence and floors, replacing every empirical inverse divisor by its positive population limit changes each score and Gram entry by $o_p(1)$, and Cauchy–Schwarz controls the accompanying residual factor, whose limit is square integrable on the risk set by the assumed outcome moments and the stated square integrability of $\overline{m}_{g,k,0}$. The remaining functions of fold ℓ are fitted on its

training data, so (C.1), the assumed $L_2(P_0)$ convergence and Cauchy–Schwarz give $\widehat{\mathbf{S}}_{\mathfrak{b}} \xrightarrow{p} \mathbf{S}_{\mathfrak{b}}(\overline{m}, \overline{r})$ and $\widehat{\mathbf{G}}_{\mathfrak{b}} \xrightarrow{p} \mathbf{G}_{\mathfrak{b}}(\overline{r})$, jointly because L, $K_{\mathfrak{b}}$ and the block dimension are fixed. The treated-row bound following (E.2) makes $\mathbf{G}_{\mathfrak{b}}(\overline{r})$ positive definite, so inversion is continuous at the limit and $\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} \xrightarrow{p} \mathbf{G}_{\mathfrak{b}}(\overline{r})^{-1}\mathbf{S}_{\mathfrak{b}}(\overline{m}, \overline{r})$. If $\overline{m} = m^0$, every component of $\overline{\mathbf{h}}^{\nu}_{\mathfrak{b},k}$ is a function of $\mathcal{C}_{g-1}$ on $\{G > k\}$, and $\mathbb{E}\{\Delta Y_k - m^0_{g,k,0} \mid \mathcal{C}_{g-1}, G > k\} = 0$, so iterated expectations give $\mathbb{E}_{P_0}[\overline{\mathbf{h}}^{\nu}_{\mathfrak{b},k}\, \mathbb{I}(G > k)\{\Delta Y_k - m^0_{g,k,0}\}] = \mathbf{0}$ for every k. Summing over the slices gives $\mathbf{S}_{\mathfrak{b}}(m^0, \overline{r}) = \mathbf{0}$, and hence the final claim. ■

Lemma D.2 (Rate of the pooled auxiliary fluctuation)**.** *Under factual initialization (4.3) and Condition 7, consider the auxiliary OLS update of a pair $\mathfrak{b} = (g,t) \in \mathcal{S}$, whose single fluctuation coefficient is $\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} = \widehat{\epsilon}_{g,t}$. For fold ℓ and active period k, let $\delta_{m,\mathfrak{b},k,-\ell}$ be the initial control-outcome-regression error, extended by zero off the not-yet-treated control arm, and let $\delta_{r,\mathfrak{b},k,-\ell} = \mathbb{I}(G > k)\{\widehat{r}_{g,k,-\ell} - r^0_{g,k}\}$ be the inverse-risk-ratio error* on the risk set*, matching the norm used in Condition 7(iv). Set*

$$d_{m,\mathfrak{b}} := \max_{\ell,k} \|\delta_{m,\mathfrak{b},k,-\ell}\|_2, \qquad d_{r,\mathfrak{b}} := \max_{\ell,k} \|\delta_{r,\mathfrak{b},k,-\ell}\|_2.$$

These are the pair-level errors $d_{m,g,t}$ and $d_{r,g,t}$ of Condition 7(iv). Then

$$\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} = O_p\Big(d_{m,\mathfrak{b}} + n^{-1/2}\Big).$$

Writing $d_m := d_{m,\mathfrak{b}}$ and $d_r := d_{r,\mathfrak{b}}$ for the fixed pair, this is the rate $O_p(d_m + n^{-1/2})$ used below; the ratio error does not enter it, because under factual initialization the fitted ratio multiplies only control residuals, which are conditionally centered on the risk set. The lemma is stated for the fixed-dimensional regime of Condition 7(i). Centering follows from (E.1) at (m^0, r^0), and positivity follows from the treated rows after (E.2); neither is a separate primitive assumption. Under factual initialization the treated-arm rows contribute zero initial residual to the zero-fluctuation score $\widehat{\mathbf{S}}_{\mathfrak{b}}$, which is why the score bound involves only control-arm terms. Pooling the fluctuation over the full sample after cross-fitted initial estimation adds no unaccounted first-order term, as quantified in Step 2.3 of the proof of Theorem 4.2 through the exact factorization $\Delta^{\mathrm{ols}}_{i,\ell} = \widehat{\epsilon}_{g,t}\widetilde{F}_{g,\ell}(O_i)$.

Proof. Fix $\mathfrak{b} = (g,t)$. Let $\widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell}(O)$ be the active clever covariate of the pair at period k, $\mathbf{h}^0_{\mathfrak{b},k}(O)$ its true counterpart, $\Delta\mathbf{h}_{\mathfrak{b},k,-\ell} := \widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell} - \mathbf{h}^0_{\mathfrak{b},k}$, and $U_{\mathfrak{b},k} := \mathbb{I}(G > k)\{\Delta Y_k - m^0_{g,k,0}(\mathcal{C}_{g-1})\}$ the true control residual. Factual initialization makes the treated initial residual zero, so $U_{\mathfrak{b},k} - \delta_{m,\mathfrak{b},k,-\ell}$ is the initial residual on every row with a nonzero clever covariate, and the normal equation of the proof of Lemma D.1 reads $\widehat{\mathbf{G}}_{\mathfrak{b}}\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} = \widehat{\mathbf{S}}_{\mathfrak{b}}$, with $\widehat{\mathbf{S}}_{\mathfrak{b}} = \sum_\ell w_\ell \mathbb{P}_{n,\ell} \sum_k \widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell}\{U_{\mathfrak{b},k} - \delta_{m,\mathfrak{b},k,-\ell}\}$.

Step 1: the clever covariates. Under HT normalization, on the event of the floors of Condition 7(iii), $|\Delta\mathbf{h}_{\mathfrak{b},k,-\ell}|$ is at most a constant times $\mathbb{I}(G > k)|\widehat{r}_{g,k,-\ell} - r^0_{g,k}| + |\widehat{p}_g - p_g|$, and $|\widehat{p}_g - p_g| = O_p(n^{-1/2})$. Under Hájek normalization the normalizer also enters. Lemma C.2 applies under the present positive cohort mass, cap, overlap, moment, fixed-fold and risk-set consistency conditions and gives $B_{n,g,k} - \widehat{p}_g = O_p(n^{-1/2} + d_{r,\mathfrak{b}})$. Since $\widehat{p}_g - p_g = O_p(n^{-1/2})$, it also gives $B_{n,g,k} - p_g = O_p(n^{-1/2} + d_{r,\mathfrak{b}})$. With the denominators bounded below on that event, both normalizations give

$$\max_{\ell,k} \big\{\mathbb{P}_{n,\ell}\|\Delta\mathbf{h}_{\mathfrak{b},k,-\ell}\|^2\big\}^{1/2} = O_p\big(d_{r,\mathfrak{b}} + n^{-1/2}\big). \tag{D.2}$$

Step 2: the zero-fluctuation score. Decompose

$$\widehat{\mathbf{S}}_{\mathfrak{b}} = \mathbb{P}_n \sum_k \mathbf{h}^0_{\mathfrak{b},k} U_{\mathfrak{b},k} + \sum_\ell w_\ell \mathbb{P}_{n,\ell} \sum_k \Delta\mathbf{h}_{\mathfrak{b},k,-\ell} U_{\mathfrak{b},k} - \sum_\ell w_\ell \mathbb{P}_{n,\ell} \sum_k \widehat{\mathbf{h}}_{\mathfrak{b},k,-\ell}\delta_{m,\mathfrak{b},k,-\ell}.$$

The first term is centered by (E.1) at the true nuisances and has a finite second moment by Condition 7(ii), so it is $O_p(n^{-1/2})$. The second term is $O_p(n^{-1/2})$. Because $U_{\mathfrak{b},k} = 0$ on the treated rows, only the control arm of $\Delta\mathbf{h}_{\mathfrak{b},k,-\ell}$ enters, and there the full-sample divisor $B^{\nu}_{n,g,k}$ factors out:

$$\Delta\mathbf{h}_{\mathfrak{b},k,-\ell}U_{\mathfrak{b},k} = -\mathbb{I}(G > k)\Big\{\frac{\widehat{r}_{g,k,-\ell} - r^0_{g,k}}{B^{\nu}_{n,g,k}} + r^0_{g,k}\Big(\frac{1}{B^{\nu}_{n,g,k}} - \frac{1}{p_g}\Big)\Big\}U_{\mathfrak{b},k}.$$

The function $f_{k,\ell} := \mathbb{I}(G > k)(\widehat{r}_{g,k,-\ell} - r^0_{g,k})U_{\mathfrak{b},k}$ depends on the training data only through the out-of-fold ratio, a function of $\mathcal{C}_{g-1}$, and $\mathbb{E}_{P_0}[U_{\mathfrak{b},k} \mid \mathcal{C}_{g-1}, G > k] = 0$; hence $\mathbb{E}_{P_0}[f_{k,\ell}(O) \mid \mathcal{I}^c_\ell] = 0$, while $\|f_{k,\ell}\|_2 \le (M + \varepsilon^{-1})\|\mathbb{I}(G > k)U_{\mathfrak{b},k}\|_2$, so (C.1) makes $\sum_\ell w_\ell \mathbb{P}_{n,\ell} f_{k,\ell} = O_p(n^{-1/2})$, and the floors bound $1/B^{\nu}_{n,g,k}$. The remaining piece is $(1/B^{\nu}_{n,g,k} - 1/p_g)\,\mathbb{P}_n[\mathbb{I}(G > k)r^0_{g,k}U_{\mathfrak{b},k}] = O_p(n^{-1/2} + d_{r,\mathfrak{b}})\,O_p(n^{-1/2})$, by Step 1 and the central limit theorem. On the event of Step 1, the cap and the floors bound the clever covariates, and the second bound in (C.1) gives $\max_{\ell,k}\mathbb{P}_{n,\ell}(\delta^2_{m,\mathfrak{b},k,-\ell}) = O_p(d^2_{m,\mathfrak{b}})$, so the third is $O_p(d_{m,\mathfrak{b}})$. Hence $\|\widehat{\mathbf{S}}_{\mathfrak{b}}\| = O_p(d_{m,\mathfrak{b}} + n^{-1/2})$.

Step 3: the Gram matrix. Condition 7 supplies the conditions of Lemma D.1 with $\overline{m} = m^0$, whose risk-set second moment is finite by conditional Jensen, and $\overline{r} = r^0$, the divisors converging by Step 1, so its proof gives $\widehat{\mathbf{G}}_{\mathfrak{b}} \xrightarrow{p} \mathbf{G}_{\mathfrak{b}}(r^0)$, which the treated-row bound following (E.2) makes nonsingular. Hence $\|\widehat{\mathbf{G}}^{-1}_{\mathfrak{b}}\|_{\mathrm{op}} = O_p(1)$ and $\widehat{\boldsymbol{\epsilon}}_{\mathfrak{b}} = \widehat{\mathbf{G}}^{-1}_{\mathfrak{b}}\widehat{\mathbf{S}}_{\mathfrak{b}} = O_p(d_{m,\mathfrak{b}} + n^{-1/2})$. There are finitely many pairs in $\mathcal{S}$, so the same order holds for the fluctuation vector of the stated collection. ∎

Proof of Theorem 4.2. Fix a granular target (g,t), with auxiliary block $\mathfrak{b} = (g,t)$ and coefficient $\widehat{\epsilon}_{g,t}$, and write $d_m := d_{m,\mathfrak{b}}$ and $d_r := d_{r,\mathfrak{b}}$ for the pair-level errors of Condition 7(iv); $\mathcal{S}$ is finite by Condition 7(i), so uniformity over it follows by taking maxima of the $o_p(1)$ terms below. By (4.5) and the control branch $\widehat{H}^{\mathrm{ini,ct}}_{(g,t),k}$ of the clever covariate of Section 4.1,

$$\widehat{m}^{*}_{(g,t),k,0,-\ell;\widehat{\boldsymbol{\epsilon}}}(\mathcal{C}_{g-1}) = \widehat{m}^{\mathrm{ini}}_{g,k,0,-\ell}(\mathcal{C}_{g-1}) + \widehat{\epsilon}_{g,t}\,\widehat{H}^{\mathrm{ini,ct}}_{(g,t),k,-\ell}(\mathcal{C}_{g-1}), \tag{D.3}$$

where $-\ell$ marks the cross-fitted first-stage evaluations, and the semicolon records that $\widehat{\epsilon}_{g,t}$ is fitted on the full sample, not refit within fold ℓ. We give the canonical HT expansion.

Step 1: double robustness. Under HT normalization, $\widehat{H}^{\mathrm{ini,ct}}_{(g,t),k,-\ell} = -\widehat{r}^{\mathrm{ini}}_{g,k,-\ell}/\widehat{p}_g$ for $k \in [g,t]$, and the increment signal is linear in the control regression, so Theorem 4.1 and (D.3) give the exact decomposition

$$\widehat{\tau}^{\mathrm{tar}}_{g,t} = \widehat{\tau}^{\mathrm{os}}_{g,t} + \frac{\widehat{\epsilon}_{g,t}}{\widehat{p}^2_g}\sum_{k=g}^{t}\mathbb{P}_n\Big[\mathbb{I}(G = g)\,\widehat{r}^{\mathrm{ini}}_{g,k} - \mathbb{I}(G > k)\big(\widehat{r}^{\mathrm{ini}}_{g,k}\big)^2\Big],$$

where $\widehat{\tau}^{\mathrm{os}}_{g,t}$ is the HT one-step estimator at the initial nuisances and each average uses every unit's out-of-fold ratio. In either branch, Theorem 3.1 gives $\widehat{\tau}^{\mathrm{os}}_{g,t} \xrightarrow{p} \tau^0_{g,t}$. The deterministic first-stage limits and divisor floors of part (i), with Lemma D.1, give a finite deterministic limit $\overline{\epsilon}_{g,t}$, so $\widehat{\epsilon}_{g,t} = O_p(1)$; and $\widehat{p}_g \xrightarrow{p} p_g > 0$. By the cap, (C.1) and (C.2), each average converges in probability to $\mathbb{E}_{P_0}[\mathbb{I}(G = g)\overline{r}_{g,k} - \mathbb{I}(G > k)\overline{r}^2_{g,k}]$. If $\overline{r}_{g,k} = r^0_{g,k}$, this limit is zero by (C.2) with $f = r^0_{g,k}$, whatever $\overline{\epsilon}_{g,t}$ is. If instead $\overline{m}_{g,k,0} = m^0_{g,k,0}$, Lemma D.1 gives $\overline{\epsilon}_{g,t} = 0$: conditional centering makes the population score zero for any admissible ratio limit, and the treated rows make the limiting Gram positive. In both branches the second term vanishes in probability, and $\widehat{\tau}^{\mathrm{tar}}_{g,t} \xrightarrow{p} \tau^0_{g,t}$. An aggregate is consistent because $\mathcal{S}$ is fixed and finite and its weight denominator is bounded away from zero by Condition 6.

Step 2: asymptotic normality under HT normalization. For part (ii), both initial nuisance estimators are consistent, so (E.1) is centered at the true nuisances, the derived Gram limit is positive, and Lemma D.2 gives $|\widehat{\epsilon}_{g,t}| = O_p(d_m + n^{-1/2})$.

Step 2.1: ratio identity. Let $\widehat{S}^*_\ell$ be the one-step signal $\widehat{S}_\ell$ of the proof of Theorem 3.1 with $\widehat{m}_{g,k,0,-\ell}$ replaced by the targeted branch (D.3), and $\widehat{A}^*$ the corresponding average of $\widehat{S}^*_{\ell(i)}(O_i)$. By Theorem 4.1, $\widehat{\tau}^{\mathrm{tar}}_{g,t} = \widehat{A}^*/\widehat{p}_g$, so the exact identity (C.4) holds with $\widehat{A}^*$ in place of $\widehat{A}$, and Step 4 of the proof of Theorem 3.2 applies once (C.5) is established for $\widehat{A}^*$. With S^0 the signal at the truth, write $\{\widehat{S}^*_\ell(O_i) - S^0(O_i)\}/p^0_g = \Delta^0_{i,\ell} + \Delta^{\mathrm{ols}}_{i,\ell}$, where $\Delta^0_{i,\ell} := \{\widehat{S}_\ell(O_i) - S^0(O_i)\}/p^0_g$ is the perturbation from the cross-fitted initial nuisances and $\Delta^{\mathrm{ols}}_{i,\ell} := \{\widehat{S}^*_\ell(O_i) - \widehat{S}_\ell(O_i)\}/p^0_g$ that from the pooled fluctuation. It suffices that $n^{-1/2}\sum_\ell \sum_{i\in\mathcal{I}_\ell} \Delta^0_{i,\ell}$ and $n^{-1/2}\sum_\ell \sum_{i\in\mathcal{I}_\ell} \Delta^{\mathrm{ols}}_{i,\ell}$ are $o_p(1)$.

Step 2.2: the initial-nuisance term. Here $\Delta^0_{i,\ell} = f_\ell(O_i)/p^0_g$, where $f_\ell := \widehat{S}_\ell - S^0$ is the function of the proof of Theorem 3.2 and depends only on the out-of-fold initial estimator. Condition 7(iv) implies the consistency and product-rate conditions of that theorem, and Conditions 6 and 7(ii)–(iii) its remaining ones: the cap gives the envelope of Lemma 3.1, and that lemma's risk-set moment condition for $\widehat{m}^{\mathrm{ini}}_{g,k,0,-\ell}$ holds with probability tending to one because $d_m = o_p(1)$. Steps 2–3 of that proof therefore give $\sqrt{n}\max_\ell |\mathbb{E}_{P_0}[f_\ell]| = o_p(1)$, the drift being a sum of same-fold, same-slice products of risk-set errors, and $\max_\ell \|f_\ell\|_2 = o_p(1)$, so (C.1) gives $n^{-1/2}\sum_\ell \sum_{i\in\mathcal{I}_\ell} \Delta^0_{i,\ell} = o_p(1)$, without a Donsker condition. No conditional-expectation argument is applied to the pooled update, which is fitted on the full sample and is not $\mathcal{I}^c_\ell$-measurable; its whole contribution, mean and fluctuation together, is handled in Step 2.3.

Step 2.3: the pooled-fluctuation term. The signal is linear in the control regression and the update (D.3) is linear in $\widehat{\epsilon}_{g,t}$, so the difference factors exactly as $\Delta^{\mathrm{ols}}_{i,\ell} = \widehat{\epsilon}_{g,t}\widetilde{F}_{g,\ell}(O_i)$, with the score derivative

$$\widetilde{F}_{g,\ell}(O) := -\sum_{k=g}^{t} \frac{1}{p^0_g}\{\mathbb{I}(G=g) - \widehat{r}^{\mathrm{ini}}_{g,k,-\ell}\,\mathbb{I}(G>k)\}\,\widehat{H}^{\mathrm{ini,ct}}_{(g,t),k,-\ell}(\mathcal{C}_{g-1}),$$

and $n^{-1/2}\sum_\ell \sum_{i\in\mathcal{I}_\ell} \Delta^{\mathrm{ols}}_{i,\ell} = (\sqrt{n}\,\widehat{\epsilon}_{g,t})\,\mathbb{P}_n\{\widetilde{F}_{g,\ell(\cdot)}\}$. Under HT normalization $\widehat{H}^{\mathrm{ini,ct}}_{(g,t),k,-\ell} = -\widehat{r}_{g,k,-\ell}/\widehat{p}_g$, so, with $\delta_{r,k,-\ell} := \widehat{r}_{g,k,-\ell} - r^0_{g,k}$ and $\zeta_{k,-\ell} := \mathbb{I}(G=g)\widehat{r}_{g,k,-\ell} - \mathbb{I}(G>k)\widehat{r}^2_{g,k,-\ell}$, we have $\widetilde{F}_{g,\ell} = \sum_{k=g}^t \zeta_{k,-\ell}/(p^0_g\widehat{p}_g)$, and the balancing identity gives, conditionally on the training data,

$$\mathbb{E}_{P_0}[\zeta_{k,-\ell}] = \mathbb{E}_{P_0}\big[\mathbb{I}(G>k)\{r^0_{g,k}\widehat{r}_{g,k,-\ell} - \widehat{r}^2_{g,k,-\ell}\}\big] = -\mathbb{E}_{P_0}\big[\mathbb{I}(G>k)\{r^0_{g,k}\delta_{r,k,-\ell} + \delta^2_{r,k,-\ell}\}\big].$$

This gives the exact identity

$$\mathbb{P}_n\{\widetilde{F}_{g,\ell(\cdot)}\} = \sum_{\ell=1}^{L} w_\ell \sum_{k=g}^{t} \frac{1}{p^0_g\widehat{p}_g}\Big[\{\mathbb{P}_{n,\ell}\zeta_{k,-\ell} - \mathbb{E}_{P_0}[\zeta_{k,-\ell}]\} - \mathbb{E}_{P_0}\{\mathbb{I}(G>k)r^0_{g,k}\delta_{r,k,-\ell}\} - \mathbb{E}_{P_0}\{\mathbb{I}(G>k)\delta^2_{r,k,-\ell}\}\Big]. \tag{D.4}$$

The capped ratios, the floor on $\widehat{p}_g$ and (C.1) make the centered term $O_p(n^{-1/2})$; Cauchy–Schwarz and bounded $r^0_{g,k}$ make the next $O_p(d_r)$; and the last is $O_p(d_r^2)$. Thus $\mathbb{P}_n\{\widetilde{F}_{g,\ell(\cdot)}\} = O_p(n^{-1/2} + d_r)$, and, with the rate of $\widehat{\epsilon}_{g,t}$, $\sqrt{n}\,\widehat{\epsilon}_{g,t}\,\mathbb{P}_n\{\widetilde{F}_{g,\ell(\cdot)}\} = O_p(\sqrt{n}(d_m + n^{-1/2})(d_r + n^{-1/2})) = O_p(\sqrt{n}\,d_m d_r + d_m + d_r + n^{-1/2}) = o_p(1)$, because $\sqrt{n}\,d_m d_r = o_p(1)$ and $d_m, d_r = o_p(1)$ by Condition 7(iv). The pair-level product, with the maxima over folds and slices taken separately in each factor, is what this step uses.

Step 2.4: conclusion. Steps 2.2–2.3 establish (C.5) for $\widehat{A}^*$, and Step 4 of the proof of Theorem 3.2 gives $\sqrt{n}(\widehat{\tau}^{\mathrm{tar}}_{g,t} - \tau^0_{g,t}) = n^{-1/2}\sum_{i=1}^n \psi_{g,t}(O_i;\eta^0_g) + o_p(1) \rightsquigarrow \mathcal{N}(0, V_{\tau_{g,t}})$. For an aggregate, the exact identity and expansion of that Step 4, with $\widehat{\tau}^{\mathrm{tar}}_\varsigma$ in place of $\widehat{\tau}^{\mathrm{os}}_\varsigma$, give the stated expansion, including the unit-level empirical weight score: the exact cancellation of Theorem 4.1 concerns only the point equation and does not remove this term. ■

Lemma D.3 (Rate of the Hájek anchor). *Fix a pair $\mathfrak{b} = (g,t) \in \mathcal{S}$ with slice set $K_{\mathfrak{b}} = [g,t]$, and use Hájek normalization. Let $d_m := d_{m,g,t}$ and $d_r := d_{r,g,t}$ be the pair-level errors of Condition 7(iv), both gated by $\mathbb{I}(G > k)$. Under the clauses of Condition 7 listed in Proposition 4.1, uniformly over this fixed slice set,*

$$\max_{k \in K_{\mathfrak{b}}} |B_{n,g,k} - \widehat{p}_g| = O_p(n^{-1/2} + d_r), \qquad \max_{k \in K_{\mathfrak{b}}} |\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},k}| = O_p(n^{-1/2} + d_m),$$

and

$$\max_{k \in K_{\mathfrak{b}}} \left| 1 - \frac{c_{n,g,k}}{V_{n,g,k}} \right| = O_p(n^{-1/2} + d_r).$$

Consequently,

$$(1 - \Gamma_{n,g,t})\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)} = O_p\Big\{ (n^{-1/2} + d_r)(n^{-1/2} + d_m) \Big\} = o_p(n^{-1/2}).$$

Proof. Lemma C.2 gives the first two bounds: the conditions here imply that lemma's fixed-fold protocol, positive cohort mass, overlap, cap, moment and risk-set consistency requirements. Its normalizer bound also gives the needed floor on $B_{n,g,k}$. Summing the control means over the fixed slice set gives $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)} = O_p(n^{-1/2} + d_m)$.

For the multiplier, write $Q_1 := \mathbb{P}_n[\mathbb{I}(G = g)\widehat{r}^{\mathrm{ini}}_{g,k}]$ and $Q_2 := \mathbb{P}_n[\mathbb{I}(G > k)(\widehat{r}^{\mathrm{ini}}_{g,k})^2]$, so that $V_{n,g,k} - c_{n,g,k} = (\widehat{p}_g Q_2 - B_{n,g,k} Q_1)/(\widehat{p}_g B^2_{n,g,k})$. By (C.2) with $f = r^0_{g,k}$, $\mathbb{E}[\mathbb{I}(G > k)(r^0_{g,k})^2] = \mathbb{E}[\mathbb{I}(G = g) r^0_{g,k}]$: the population Gram contribution and loading coincide. Hence $\mathbb{P}_n[\mathbb{I}(G > k)(r^0_{g,k})^2 - \mathbb{I}(G = g) r^0_{g,k}] = O_p(n^{-1/2})$, its summand being bounded and centered. Replacing $r^0_{g,k}$ by the out-of-fold ratio changes Q_2 by $O_p(d_r)$, because $(\widehat{r}^{\mathrm{ini}}_{g,k})^2 - (r^0_{g,k})^2 = (\widehat{r}^{\mathrm{ini}}_{g,k} - r^0_{g,k})(\widehat{r}^{\mathrm{ini}}_{g,k} + r^0_{g,k})$ with second factor at most $M + \varepsilon^{-1}$, and Q_1 by $O_p(d_r)$, because (C.2) gives $\|\mathbb{I}(G = g)(\widehat{r}^{\mathrm{ini}}_{g,k,-\ell} - r^0_{g,k})\|_2 \le \varepsilon^{-1/2} d_r$; both bounds use the second bound in (C.1) and Cauchy–Schwarz. So $Q_2 - Q_1 = O_p(n^{-1/2} + d_r)$, and $\widehat{p}_g Q_2 - B_{n,g,k} Q_1 = \widehat{p}_g(Q_2 - Q_1) - (B_{n,g,k} - \widehat{p}_g) Q_1 = O_p(n^{-1/2} + d_r)$ by the bound on $B_{n,g,k} - \widehat{p}_g$. With the floors on $\widehat{p}_g$ and $B_{n,g,k}$, $V_{n,g,k} - c_{n,g,k} = O_p(n^{-1/2} + d_r)$, and, since $V_{n,g,k} \ge 1/\widehat{p}_g$, $\max_k |1 - c_{n,g,k}/V_{n,g,k}| = O_p(n^{-1/2} + d_r)$. The window is fixed and $\sum_k V_{n,g,k} \ge 1/\widehat{p}_g$, so $1 - \Gamma_{n,g,t} = \sum_k (V_{n,g,k} - c_{n,g,k}) / \sum_k V_{n,g,k} = O_p(n^{-1/2} + d_r)$, and multiplication gives the final display. ∎

Proof of Proposition 4.1. Under Hájek normalization the raw slice contrast is $\widehat{H}^{\mathrm{ini}}_{g,k}(O) = \mathbb{I}(G = g)/\widehat{p}_g - \mathbb{I}(G > k)\widehat{r}^{\mathrm{ini}}_{g,k}(\mathcal{C}_{g-1})/B_{n,g,k}$. Factual initialization (4.3) makes the treated residual vanish on $\{G = g\}$, so for each slice $\mathbb{P}_n[\widehat{H}^{\mathrm{ini}}_{g,k}(\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k})] = -\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},k}$, and the one-step estimator at the initial nuisance is $\widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t} = \widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},(g,t)} - \widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)}$. The first-order condition of the pair's OLS pools the active slices with one coefficient on the same raw contrasts, so, by the Gram entry of Appendix D.1, $0 = -\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)} - \widehat{\epsilon}_{g,t} \sum_{k=g}^{t} V_{n,g,k}$. The loading form of Appendix D.1, whose untargeted term is $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},(g,t)}$ under factual initialization, then gives

$$\widehat{\tau}^{\mathrm{tar}}_{g,t} = \widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},(g,t)} + \Big(\textstyle\sum_{k=g}^{t} c_{n,g,k} \Big) \widehat{\epsilon}_{g,t} = \widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},(g,t)} - \Gamma_{n,g,t} \widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)},$$

with $\Gamma_{n,g,t} := \sum_{k=g}^{t} c_{n,g,k} / \sum_{k=g}^{t} V_{n,g,k}$. Lemma D.3, whose conditions are those of the proposition, gives $1 - \Gamma_{n,g,t} = O_p(n^{-1/2} + d_r)$, which is $o_p(1)$ because $d_r = o_p(1)$, and $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)} = O_p(n^{-1/2} + d_m)$. So $\Gamma_{n,g,t} \xrightarrow{p} 1$, and the targeted and the self-normalized one-step estimators differ by $(1 - \Gamma_{n,g,t})\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)} = o_p(n^{-1/2})$. The Hájek–HT one-step bridge is supplied independently by Lemma C.2 and the exact subtraction used in the proof of Corollary 3.2(ii). Whenever the remaining conditions of that corollary hold, its canonical expansion therefore transfers to the targeted estimator. ∎

Proof of Corollary 4.1. *Consistency.* Under Hájek normalization the control branch of the clever covariate is $-\widehat{r}^{\mathrm{ini}}_{g,k,-\ell}/B_{n,g,k}$ for $k \in [g,t]$, so, as in Step 1 of the proof of Theorem 4.2,

$$\widehat{\tau}^{\mathrm{tar}}_{g,t} = \widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t} + \widehat{\epsilon}_{g,t} \sum_{k=g}^{t} \left\{ \frac{\mathbb{P}_n[\mathbb{I}(G=g)\widehat{r}^{\mathrm{ini}}_{g,k}]}{\widehat{p}_g B_{n,g,k}} - \frac{\mathbb{P}_n[\mathbb{I}(G>k)(\widehat{r}^{\mathrm{ini}}_{g,k})^2]}{B^2_{n,g,k}} \right\},$$

with $\widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t} = \widehat{\tau}^{\mathrm{ini}}_{\mathrm{tr},(g,t)} - \widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},(g,t)}$. The floor (4.9) and Lemma D.1 give $\widehat{\epsilon}_{g,t} = O_p(1)$, and the averages and $B_{n,g,k}$ converge as in that Step 1. In the ratio-correct branch, (C.2) with $f=1$ gives $B_{n,g,k} \xrightarrow{p} p_g$, and with $f = r^0_{g,k}$ it makes the braces tend to zero; moreover $\widehat{\tau}^{\mathrm{os,Háj,\,ini}}_{g,t} - \widehat{\tau}^{\mathrm{os,HT}}_{g,t} = \sum_{k=g}^{t} \widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},k}(B_{n,g,k} - \widehat{p}_g)/\widehat{p}_g \xrightarrow{p} 0$, each $\widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},k}$ being $O_p(1)$ by the cap, the moment clause, the risk-set convergence of the initial regressions to square-integrable limits, the second bound in (C.1) and the floor, and Theorem 3.1 gives $\widehat{\tau}^{\mathrm{os,HT}}_{g,t} \xrightarrow{p} \tau^0_{g,t}$. In the outcome-correct branch, Lemma D.1 gives $\widehat{\epsilon}_{g,t} \xrightarrow{p} 0$, the braces being $O_p(1)$ by the cap and the floor; the treated-arm anchors converge to the values $\mathbb{E}[\Delta Y_k(g) - \Delta Y_k(\infty) \mid G=g]$ of (C.3), which sum to $\tau^0_{g,t}$, by the balancing identity and branch (a) of the proof of Theorem 3.1; and each control-arm anchor converges to $\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}\{\Delta Y_k - m^0_{g,k,0}\}]/\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}] = 0$ by conditional centering, its denominator being at least c_B. Every granular Hájek target is thus consistent in either branch, and so is an aggregate, because $\mathcal{S}$ is fixed and finite and its weight denominator is bounded away from zero by Condition 6.

Root-n expansion. Conditions 5 and 7 and factual initialization supply the conditions of Lemma D.3 and Proposition 4.1, so $\widehat{\tau}^{\mathrm{tar,Háj}}_{\varsigma} = \widehat{\tau}^{\mathrm{os,Háj,ini}}_{\varsigma} + o_p(n^{-1/2}) = \widehat{\tau}^{\mathrm{os,HT}}_{\varsigma} + o_p(n^{-1/2})$ for every granular $\varsigma \in \mathcal{S}$. Theorem 3.2 supplies the canonical expansion of the HT one-step estimator, so each Hájek granular targeted estimator has that expansion. The fixed-dimensional aggregate expansion of Step 4 of the proof of Theorem 3.2 carries the finite sum of $o_p(n^{-1/2})$ bridges and retains the unit-level weight score, which gives the stated aggregate expansion and its central limit theorem. Combined with Theorem 4.2, it also yields the root-n equivalence of the two targeted normalizations. ∎

E Pooled Fluctuation and Well-Posedness

In the proofs of Appendix D, the initial control regressions and the ratio numerators of the clever covariates are evaluated out of fold; the divisors $\widehat{p}_g$ and $B_{n,g,k}$ are full-sample averages, replaced by their limits in the proof of Lemma D.1, with $\widehat{p}_g$ factored out as a scalar in Step 2.3 of the proof of Theorem 4.2; and the coefficient $\widehat{\boldsymbol{\epsilon}}$ is estimated once, by the pooled normal equation (4.8), and carries no fold index. The coefficient is handled directly, through its finite-dimensional normal equation in Lemmas D.1 and D.2 and the exact factorization $\Delta^{\mathrm{ols}}_{i,\ell} = \widehat{\epsilon}_{g,t}\widetilde{F}_{g,\ell}(O_i)$ of that Step 2.3, with no conditional empirical-process argument.

The zero population score and the positive Gram used by the proofs are consequences of Condition 7 and factual initialization, not additional primitive clauses. Fix a pair $\mathfrak{b} = (g,t)$, with slice set $K_{\mathfrak{b}} = [g,t]$; let $\overline{m}_{g,k,0}$ and $\overline{r}_{g,k}$ be deterministic nuisance limits, and $\overline{\mathbf{h}}^{\nu}_{\mathfrak{b},k}(O)$ the limiting clever-covariate columns of the block, with control divisor $\overline{B}^{\mathrm{HT}}_{g,k} = p_g$ or $\overline{B}^{\mathrm{Háj}}_{g,k} = \mathbb{E}_{P_0}\{\mathbb{I}(G>k)\overline{r}_{g,k}\}$.

Under factual initialization, the population zero-fluctuation score and Gram matrix are

$$\mathbf{S}_{\mathfrak{b}}(\overline{m},\overline{r}) := \mathbb{E}_{P_0}\Big[\sum_{k\in K_{\mathfrak{b}}} \overline{\mathbf{h}}^{\nu}_{\mathfrak{b},k}(O)\,\mathbb{I}(G>k)\{\Delta Y_k - \overline{m}_{g,k,0}(\mathcal{C}_{g-1})\}\Big], \tag{E.1}$$

$$\mathbf{G}_{\mathfrak{b}}(\overline{r}) := \mathbb{E}_{P_0}\Big[\sum_{k\in K_{\mathfrak{b}}} \overline{\mathbf{h}}^{\nu}_{\mathfrak{b},k}(O)\,\overline{\mathbf{h}}^{\nu\top}_{\mathfrak{b},k}(O)\Big]. \tag{E.2}$$

If $\overline{m} = m^0$, iterated expectations give $\mathbf{S}_{\mathfrak{b}}(m^0, \overline{r}) = \mathbf{0}$ for every admissible ratio limit: each control residual is conditionally centered on its risk set, and the treated residual is zero by factual initialization. The treated rows alone give $\mathbf{G}_{\mathfrak{b}}(\overline{r}) \geq \mathbb{E}_{P_0}[\sum_{k \in K_{\mathfrak{b}}} \mathbb{I}(G = g)/p_g^2] = (t - g + 1)/p_g > 0$, whatever the ratio limit, and cross-fitted laws of large numbers give convergence of the empirical Gram to (E.2).

Remark E.1 (Interpreting Condition 7)**.** *(a) Identified-support norms.* Both errors in Condition 7(iv) are gated by $\mathbb{I}(G > k)$. The control regression and inverse-risk ratio are identified on that risk set, and the clever covariate uses them only there. In Corollary 3.1, the errors of the pre-clipping learners are correspondingly measured on $\{G \geq s\}$.

(b) The normalizer floor. The floor for $B_{n,g,k}$ in (iii) is implied when the ratio is consistent, since $\mathbb{E}[\mathbb{I}(G > k) r^0_{g,k}] = p_g$ gives $B_{n,g,k} \xrightarrow{p} p_g > 0$; it must be maintained directly only when the ratio is misspecified. The merely nonzero denominators of Condition 5 are a finite-sample requirement and do not deliver this floor.

E.1 Finite-Sample Well-Posedness of the Auxiliary OLS

The pairwise auxiliary OLS needs no additional finite-sample rank condition. On the event of Condition 5(i), every empirical denominator of the clever covariate (4.2) is nonzero, including, under Hájek normalization, the control-arm normalizer, so the design is defined. A unit with $G_i = g$ has $D_{ik} = 1$ for every $k \in [g, t]$, so its control-arm term switches off and its clever covariate reduces to $\widehat{p}_g^{-1}$, free of the fitted ratio under either normalization. The column of the pair (g, t) is therefore nonzero, the block-diagonal design, with one column per pair, has full column rank, and the auxiliary least squares (4.4) has a unique minimizer. Uniqueness thus rests on the treated arm; under Hájek normalization the fitted ratios, through the normalizer, govern whether the design exists, not the uniqueness of its solution. If $\widehat{p}_g = 0$, the targeting update is undefined rather than collinear.

F Cohort-Wise Targeting

F.1 Definitions

Cohort-wise targeting (TMLE-cw-HT and TMLE-cw-Hájek) replaces the pairwise regressions (4.4) by one auxiliary regression for each cohort g, with the $T - g + 1$ clever covariates $\widehat{H}^{\mathrm{ini}}_{(g,t'),k}$, $t' \in [g, T]$, as joint regressors. With $\widehat{\epsilon}^{\mathrm{cw}}_{g,t'}$ the jointly fitted coefficients, every target (g, t) of the cohort is evaluated at the common update $\widehat{m}^{\mathrm{ini}}_{g,k} + \sum_{t'=g}^{T} \widehat{H}^{\mathrm{ini}}_{(g,t'),k} \widehat{\epsilon}^{\mathrm{cw}}_{g,t'}$ in place of (4.5), and its branches enter (4.6).

Two-fluctuation targeting (TMLE-2eps-pw-Hájek and TMLE-2eps-cw-Hájek), reported in Appendix H.1 as the two-fluctuation diagnostic, uses Hájek normalization and gives the treated component $\mathbb{I}(g \leq k \leq t')\,\mathbb{I}(G = g)/\widehat{p}_g$ and the control component $-\mathbb{I}(g \leq k \leq t')\,\mathbb{I}(G > k)\,\widehat{r}^{\mathrm{ini}}_{g,k}/B_{n,g,k}$ of each clever covariate separate fluctuation coefficients, with $t' = t$ under the pairwise and $t' \in [g, T]$ under the cohort-wise architecture.

Appendix F.5 establishes the required Gram positivity and the unique auxiliary least-squares solution, then proves that the treated coefficients are zero under factual initialization.

The constructions below use factual initialization (4.3) and exact auxiliary least squares without intercept. This appendix gives their finite-sample representations and the asymptotic guarantees for the single-fluctuation cohort-wise estimator. Appendix G.5 maps the printed procedure names to the replication identifiers.

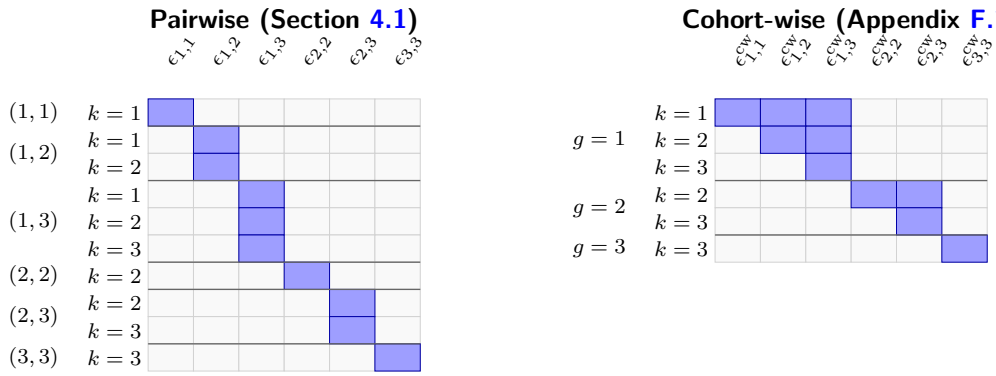


Figure F.1: Row structure of the two auxiliary-regression designs for $T = 3$ with every pair supported, $\mathcal{Q} = \{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$. Each displayed row stands for the n unit rows (i,k) of one slice k. A shaded cell marks the rows of a block on which that column's clever covariate can be nonzero; a column's regressor is identically zero outside its own block, and within a shaded slice only the units of cohort g and of the risk set $\{G > k\}$ can have a nonzero entry. **Left:** the pairwise design of Algorithm 1, Step 2, gives each pair its own block of rows $k \in [g,t]$ and its own column, so the stacked system (4.7) is block diagonal and each $\widehat{\epsilon}_{g,t}$ solves the scalar normal equation of (4.4). **Right:** the cohort-wise design of Appendix F.1 gives each cohort one block of rows $k \in [g,T]$ whose $T-g+1$ regressors are the clever covariates $\widehat{H}^{\mathrm{ini}}_{(g,t'),k}$, $t' \in [g,T]$, with coefficients $\epsilon^{\mathrm{cw}}_{g,t'}$; the supports are nested within a block because slice k enters every column with $t' \geq k$.

F.2 Slice Decomposition, Availability, and Score Solution

Fix a target cohort g and a normalization $\nu \in \{\mathrm{HT}, \mathrm{Háj}\}$. The full cohort construction uses every slice $k \in [g,T]$, including slices after a reported endpoint. Write $\widehat{r}^{\mathrm{ini}}_{g,k}$ and $\widehat{m}^{\mathrm{ini}}_{g,k,0}$ for the initial cross-fitted nuisances, with their out-of-fold evaluation understood, and put

$$h^{\nu}_{g,k}(O) := \frac{\mathbb{I}(G=g)}{\widehat{p}_g} - \frac{\mathbb{I}(G>k)\widehat{r}^{\mathrm{ini}}_{g,k}(\mathcal{C}_{g-1})}{B^{\nu}_{n,g,k}}.$$

Here $B^{\mathrm{HT}}_{n,g,k} = \widehat{p}_g$ and $B^{\mathrm{Háj}}_{n,g,k} = B_{n,g,k}$ as in (4.2). Define the initial treated and control residual means and the slice Gram contribution and loading by

$$\begin{aligned}
\widehat{\tau}_{\mathrm{tr},k} &:= \frac{\mathbb{P}_n[\mathbb{I}(G=g)\{\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{g-1})\}]}{\widehat{p}_g}, \\
\widehat{\tau}^{\nu}_{\mathrm{ct},k} &:= \frac{\mathbb{P}_n[\mathbb{I}(G>k)\widehat{r}^{\mathrm{ini}}_{g,k}(\mathcal{C}_{g-1})\{\Delta Y_k - \widehat{m}^{\mathrm{ini}}_{g,k,0}(\mathcal{C}_{g-1})\}]}{B^{\nu}_{n,g,k}}, \\
V^{\nu}_{n,g,k} &:= \mathbb{P}_n[(h^{\nu}_{g,k})^2] = \frac{1}{\widehat{p}_g} + \frac{\mathbb{P}_n[\mathbb{I}(G>k)(\widehat{r}^{\mathrm{ini}}_{g,k})^2]}{(B^{\nu}_{n,g,k})^2}, \\
c^{\nu}_{n,g,k} &:= \frac{1}{\widehat{p}_g} + \frac{\mathbb{P}_n[\mathbb{I}(G=g)\widehat{r}^{\mathrm{ini}}_{g,k}]}{\widehat{p}_g B^{\nu}_{n,g,k}}.
\end{aligned}$$

The Hájek definitions agree with those in Section 4.2. For the chosen normalization, $h^{\nu}_{g,k}$ is the raw slice contrast $\widehat{H}^{\mathrm{ini}}_{g,k}$ of Appendix D.1; under Hájek normalization, $\widehat{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k} = \widehat{\tau}_{\mathrm{ct},k} = \widehat{\tau}^{\mathrm{ini}}_{\mathrm{ct},k}$. The loading is the change in the empirical treated contrast produced by one unit of the slice fluctuation.

Let $\mathcal{A}_n^\nu$ denote the full-design availability event: for every represented target cohort, all required hazard fits through T and all control-regression fits on $[g,T]$ produce finite predictions at every history used by the construction, the initial row offsets are finite, $\widehat{p}_g > 0$, and $B_{n,g,k}^\nu > 0$ for every $k \in [g,T]$. The full cohort asymptotic results below require

$$\mathbb{P}(\mathcal{A}_n^\nu) \longrightarrow 1. \tag{F.1}$$

This is an availability requirement, separate from consistency or rates on the slices of a reported target. Because the slice-T control regression is fitted on $\{G > T\}$, full-grid availability requires positive population mass in the terminal untreated group under either normalization. Under Hájek normalization, $B_{n,g,T} > 0$ requires this mass as well. Overlap on a reported window ending before T does not by itself supply this requirement.

Lemma F.1 (Cohort slice decomposition and empirical score solution)**.** *On $\mathcal{A}_n^\nu$, the exact cohort-wise auxiliary least-squares problem has a unique solution. Order the endpoint and slice indices as $g, g+1, \ldots, T$, and let A be the endpoint-by-slice matrix $A_{t',k} = \mathbb{I}(k \le t')$. If H_g^ν is the endpoint-column design, then*

$$(H_g^\nu)^\top H_g^\nu = A \,\mathrm{diag}\{nV_{n,g,k}^\nu : k \in [g,T]\}\, A^\top. \tag{F.2}$$

Its endpoint coefficients and the equivalent slice coefficients satisfy

$$\widehat{\boldsymbol{\gamma}}_g^\nu = A^\top \widehat{\boldsymbol{\epsilon}}_g^{\mathrm{cw},\nu}, \qquad \widehat{\gamma}_{g,k}^\nu = \sum_{t'=k}^{T} \widehat{\epsilon}_{g,t'}^{\mathrm{cw},\nu} = -\frac{\widehat{\tau}_{\mathrm{ct},k}^\nu}{V_{n,g,k}^\nu}. \tag{F.3}$$

At the resulting common update, every slice empirical score and every endpoint empirical score is zero. For every reported aggregate whose empirical weight mass is nonzero, the full aggregate empirical score, including the aggregation-weight term, also has empirical mean zero.

Proof. Let Z_g^ν have one column per slice. Its column k equals $h_{g,k}^\nu(O_i)$ on row (i,k) and is zero on every other slice. Rows on which all columns are zero may be omitted. The columns have disjoint row supports, and consequently

$$(Z_g^\nu)^\top Z_g^\nu = \mathrm{diag}\{nV_{n,g,k}^\nu : k \in [g,T]\}.$$

An endpoint column sums the slice columns through that endpoint, so $H_g^\nu = Z_g^\nu A^\top$. This proves (F.2) with the stated matrix orientation. The matrix A is unit lower triangular and hence invertible. Also $V_{n,g,k}^\nu \ge \widehat{p}_g^{-1} > 0$, because the treated rows contribute $\mathbb{P}_n[\mathbb{I}(G=g)/\widehat{p}_g^2] = \widehat{p}_g^{-1}$. Thus both designs have full column rank and the exact least-squares solution is unique.

For candidate (unhatted) cohort-wise endpoint and slice coefficients $\boldsymbol{\epsilon}_g^{\mathrm{cw}}$ and $\boldsymbol{\gamma}_g$ at the fixed normalization ν, the transformation $\boldsymbol{\gamma}_g = A^\top \boldsymbol{\epsilon}_g^{\mathrm{cw}}$ is one-to-one; explicitly, $\epsilon_{g,t'}^{\mathrm{cw}} = \gamma_{g,t'} - \gamma_{g,t'+1}$ with the boundary convention $\gamma_{g,T+1} = 0$. Their fitted values appear in (F.3). Because the slice columns have disjoint row supports, least squares is equivalent to separate univariate regressions on those columns. If $R_{g,k}(O_i) = \Delta Y_{i,k} - \widehat{m}_{g,(i,k)}^{\mathrm{ini}}$, factual initialization makes $R_{g,k} = 0$ on $\{G_i = g\}$ and gives

$$\mathbb{P}_n[h_{g,k}^\nu R_{g,k}] = -\widehat{\tau}_{\mathrm{ct},k}^\nu.$$

Division by $V_{n,g,k}^\nu$ proves (F.3). In particular, each $\widehat{\gamma}_{g,k}^\nu$ depends only on its own slice, although the full cohort construction must be available through T.

The common updated row value and control branch on slice k are

$$\widehat{m}_{g,(i,k)}^{*,\mathrm{cw},\nu} = \widehat{m}_{g,(i,k)}^{\mathrm{ini}} + h_{g,k}^\nu(O_i)\widehat{\gamma}_{g,k}^\nu, \qquad \widehat{m}_{g,k,0}^{*,\mathrm{cw},\nu}(c) = \widehat{m}_{g,k,0}^{\mathrm{ini}}(c) - \frac{\widehat{r}_{g,k}^{\mathrm{ini}}(c)}{B_{n,g,k}^\nu}\widehat{\gamma}_{g,k}^\nu.$$

Set

$$\widehat{a}^{\nu}_{g,k} := \frac{1}{n}\sum_{i=1}^{n} \frac{\mathbb{I}(G_i = g)}{\widehat{p}_g}\{\widehat{m}^{*,\mathrm{cw},\nu}_{g,(i,k)} - \widehat{m}^{*,\mathrm{cw},\nu}_{g,k,0}(\mathcal{C}_{i,g-1})\}.$$

Direct substitution yields $\widehat{a}^{\nu}_{g,k} = \widehat{\tau}_{\mathrm{tr},k} + c^{\nu}_{n,g,k}\widehat{\gamma}^{\nu}_{g,k}$. The empirical score for this slice, with $\widehat{e}^{\nu}_{g,k}(O) := \Delta Y_k - \widehat{m}^{*,\mathrm{cw},\nu}_{g,k,0}(\mathcal{C}_{g-1})$, is

$$\widehat{\psi}^{\mathrm{sl},\nu}_{g,k}(O) := \frac{\mathbb{I}(G = g)}{\widehat{p}_g}\{\widehat{e}^{\nu}_{g,k}(O) - \widehat{a}^{\nu}_{g,k}\} - \frac{\mathbb{I}(G > k)\widehat{r}^{\mathrm{ini}}_{g,k}}{B^{\nu}_{n,g,k}}\widehat{e}^{\nu}_{g,k}(O).$$

Add and subtract the updated row value on treated rows. The empirical mean of the resulting treated contrast is zero by the definition of $\widehat{a}^{\nu}_{g,k}$ and $\mathbb{P}_n[\mathbb{I}(G = g)/\widehat{p}_g] = 1$. The remaining mean is

$$\mathbb{P}_n[\widehat{\psi}^{\mathrm{sl},\nu}_{g,k}] = \mathbb{P}_n[h^{\nu}_{g,k}\{R_{g,k} - h^{\nu}_{g,k}\widehat{\gamma}^{\nu}_{g,k}\}] = 0$$

by the slice normal equation. Define the cohort-wise endpoint score and estimate at the common update by

$$\widehat{\psi}^{\mathrm{cw},\nu}_{g,t}(O) := \sum_{k=g}^{t} \widehat{\psi}^{\mathrm{sl},\nu}_{g,k}(O),$$

$$\widehat{\tau}^{\mathrm{cw},\nu}_{g,t} := \sum_{k=g}^{t} \widehat{a}^{\nu}_{g,k}.$$

Summing the slice equations gives $\mathbb{P}_n[\widehat{\psi}^{\mathrm{cw},\nu}_{g,t}] = 0$. For the supported reported targets, these are the canonical empirical EIF scores under HT normalization and the self-normalized empirical scores under Hájek normalization.

Finally let $\widehat{W}_{\mathcal{S}} = \sum_{\mathcal{S}} \widehat{\omega}_{g,t} \neq 0$ and $\widehat{\Theta}^{\nu}_{\mathrm{cw}} = \widehat{W}_{\mathcal{S}}^{-1}\sum_{\mathcal{S}} \widehat{\omega}_{g,t}\widehat{\tau}^{\mathrm{cw},\nu}_{g,t}$. The aggregate empirical score is

$$\widehat{\Psi}^{\mathrm{cw},\nu}_{\Theta}(O) = \frac{1}{\widehat{W}_{\mathcal{S}}}\sum_{(g,t)\in\mathcal{S}} \left[\widehat{\omega}_{g,t}\widehat{\psi}^{\mathrm{cw},\nu}_{g,t}(O) + (\widehat{\tau}^{\mathrm{cw},\nu}_{g,t} - \widehat{\Theta}^{\nu}_{\mathrm{cw}})\{\Omega_{g,t}(O) - \widehat{\omega}_{g,t}\}\right].$$

Every endpoint score has mean zero, and $\mathbb{P}_n[\Omega_{g,t} - \widehat{\omega}_{g,t}] = 0$ exactly. Hence $\mathbb{P}_n[\widehat{\Psi}^{\mathrm{cw},\nu}_{\Theta}] = 0$. ■

The lemma concerns a finite mathematical design and its exact normal-equation solution. The production implementation also checks learner and solver diagnostics; a nonzero mathematical Gram alone does not guarantee software completion. The implementation uses the equivalent disjoint slice basis, fitting $\widehat{\gamma}^{\nu}_{g}$ directly.

F.3 Exact Representations and Population Limits

For a common initial cross-fitted nuisance collection, use the treated and normalization-indexed control residual means, $c^{\nu}_{n,g,k}$ and $V^{\nu}_{n,g,k}$ of Appendix F.2. Write $\widehat{\tau}^{\mathrm{os},\nu}_{g,t}$ for (3.4) under HT normalization and (3.6) under Hájek normalization, suppressing its initial-fit superscript. Write $\widehat{\tau}^{\mathrm{pw},\nu}_{g,t}$ for the pairwise TMLE of Section 4 and $\widehat{\tau}^{\mathrm{cw},\nu}_{g,t}$ for the common-update cohort-wise estimator. The superscripts $2\mathrm{eps,pw}$ and $2\mathrm{eps,cw}$ refer to the Hájek two-fluctuation diagnostics of Appendix F.5; their control-only quantities are

$$V'_{n,g,k} = \frac{\mathbb{P}_n[\mathbb{I}(G > k)(\widehat{r}^{\mathrm{ini}}_{g,k})^2]}{B^2_{n,g,k}}, \qquad c'_{n,g,k} = \frac{\mathbb{P}_n[\mathbb{I}(G = g)\widehat{r}^{\mathrm{ini}}_{g,k}]}{\widehat{p}_g B_{n,g,k}}.$$

These omit the treated contribution $\widehat{p}_g^{-1}$ in $(V^{\mathrm{Háj}}_{n,g,k}, c^{\mathrm{Háj}}_{n,g,k})$.

Proposition F.1 (Exact representations and population limits)**.** *Fix (g,t) and put $K_{g,t}=[g,t]$.* (i) Exact representations. *For one-step estimation, suppose the required initial functions and divisors on $K_{g,t}$ are finite and the divisors are positive. For targeting, additionally use factual initialization (4.3) and exact auxiliary least squares without intercept. The pairwise construction must be available on its active window, and the cohort-wise construction on its full design $[g,T]$, as in $\mathcal{A}_n^\nu$. The two-fluctuation identities are stated on the corresponding availability event, on which the control Gram denominators are positive; their uniqueness is proved in Appendix F.5. On the corresponding availability events, define*

$$\Gamma^{\nu}_{n,g,t}:=\frac{\sum_{k=g}^{t}c^{\nu}_{n,g,k}}{\sum_{k=g}^{t}V^{\nu}_{n,g,k}},\qquad \Gamma'_{n,g,t}:=\frac{\sum_{k=g}^{t}c'_{n,g,k}}{\sum_{k=g}^{t}V'_{n,g,k}}.$$

Then

$$\begin{aligned}
\widehat{\tau}^{\mathrm{os},\nu}_{g,t}&=\sum_{k=g}^{t}(\widehat{\tau}_{\mathrm{tr},k}-\widehat{\tau}^{\nu}_{\mathrm{ct},k}),\\
\widehat{\tau}^{\mathrm{pw},\nu}_{g,t}&=\sum_{k=g}^{t}\widehat{\tau}_{\mathrm{tr},k}-\Gamma^{\nu}_{n,g,t}\sum_{k=g}^{t}\widehat{\tau}^{\nu}_{\mathrm{ct},k},\\
\widehat{\tau}^{\mathrm{cw},\nu}_{g,t}&=\sum_{k=g}^{t}\left\{\widehat{\tau}_{\mathrm{tr},k}-\frac{c^{\nu}_{n,g,k}}{V^{\nu}_{n,g,k}}\widehat{\tau}^{\nu}_{\mathrm{ct},k}\right\},\\
\widehat{\tau}^{\mathrm{2eps,pw}}_{g,t}&=\sum_{k=g}^{t}\widehat{\tau}_{\mathrm{tr},k}-\Gamma'_{n,g,t}\sum_{k=g}^{t}\widehat{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k},\\
\widehat{\tau}^{\mathrm{2eps,cw}}_{g,t}&=\sum_{k=g}^{t}\left\{\widehat{\tau}_{\mathrm{tr},k}-\frac{c'_{n,g,k}}{V'_{n,g,k}}\widehat{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k}\right\}.
\end{aligned}\tag{F.4}$$

The first three lines each apply under $\nu\in\{\mathrm{HT},\mathrm{Háj}\}$, and the last two only under Hájek normalization: the display covers all eight proposed procedures. For any one of these procedures, indexed by proc*, let $\widehat{\tau}^{\mathrm{proc}}_{g,t}$ denote its displayed granular estimate, including its normalization, and define*

$$\widehat{\Theta}_{\mathrm{proc}}:=\frac{\sum_{(g,t)\in\mathcal{S}}\widehat{\omega}_{g,t}\widehat{\tau}^{\mathrm{proc}}_{g,t}}{\sum_{(g,t)\in\mathcal{S}}\widehat{\omega}_{g,t}}.$$

For $\mathrm{proc}=(\mathrm{cw},\nu)$ this is the $\widehat{\Theta}^{\nu}_{\mathrm{cw}}$ used in Appendix F.4.

(ii) Population limits. *Suppose the applicable availability event has probability tending to one, $p_g>0$, and T, the fold count and the reported collection are fixed. On every active slice $k\in K_{g,t}$, suppose the initial functions converge, uniformly over folds, to deterministic functions $\overline{m}_{g,k,0}$ and $\overline{r}_{g,k}$ in $L_2(P_0)$ on both $\{G=g\}$ and $\{G>k\}$, with $0\le\widehat{r}^{\mathrm{ini}}_{g,k}\le M$ and*

$$\mathbb{E}_{P_0}\left[(\mathbb{I}(G=g)+\mathbb{I}(G>k))\{(\Delta Y_k)^2+\overline{m}_{g,k,0}(\mathcal{C}_{g-1})^2\}\right]<\infty.$$

Use only quantities with positive denominators, with the requirements below imposed for the proce-

dure under consideration. Put $\overline{U}_{g,k} := \Delta Y_k - \overline{m}_{g,k,0}(\mathcal{C}_{g-1})$ *and define*

$$\begin{aligned}
\overline{B}^{\mathrm{HT}}_{g,k} &:= p_g, & \overline{B}^{\mathrm{Háj}}_{g,k} &:= \mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}],\\
\overline{\tau}_{\mathrm{tr},k} &:= \frac{\mathbb{E}_{P_0}[\mathbb{I}(G=g)\overline{U}_{g,k}]}{p_g}, & \overline{\tau}^{\nu}_{\mathrm{ct},k} &:= \frac{\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}_{g,k}\overline{U}_{g,k}]}{\overline{B}^{\nu}_{g,k}},\\
\overline{V}^{\nu}_{g,k} &:= \frac{1}{p_g} + \frac{\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}^2_{g,k}]}{(\overline{B}^{\nu}_{g,k})^2}, & \overline{c}^{\nu}_{g,k} &:= \frac{1}{p_g} + \frac{\mathbb{E}_{P_0}[\mathbb{I}(G=g)\overline{r}_{g,k}]}{p_g\overline{B}^{\nu}_{g,k}},\\
\overline{V}'_{g,k} &:= \frac{\mathbb{E}_{P_0}[\mathbb{I}(G>k)\overline{r}^2_{g,k}]}{(\overline{B}^{\mathrm{Háj}}_{g,k})^2}, & \overline{c}'_{g,k} &:= \frac{\mathbb{E}_{P_0}[\mathbb{I}(G=g)\overline{r}_{g,k}]}{p_g\overline{B}^{\mathrm{Háj}}_{g,k}}.
\end{aligned}$$

For a Hájek procedure require $\overline{B}^{\mathrm{Háj}}_{g,k} > 0$ *on every active slice. For a two-fluctuation diagnostic, this positivity also implies* $\overline{V}'_{g,k} \geq 1/\mathbb{P}(G>k) > 0$ *by Cauchy–Schwarz, so both the cohort-wise control Gram denominators and their pairwise sums are positive. Define*

$$\overline{\Gamma}^{\nu}_{g,t} := \frac{\sum_{k=g}^{t}\overline{c}^{\nu}_{g,k}}{\sum_{k=g}^{t}\overline{V}^{\nu}_{g,k}}, \qquad \overline{\Gamma}'_{g,t} := \frac{\sum_{k=g}^{t}\overline{c}'_{g,k}}{\sum_{k=g}^{t}\overline{V}'_{g,k}}.$$

The probability limits are

$$\begin{aligned}
\overline{\tau}^{\mathrm{os},\nu}_{g,t} &= \sum_{k=g}^{t}(\overline{\tau}_{\mathrm{tr},k} - \overline{\tau}^{\nu}_{\mathrm{ct},k}),\\
\overline{\tau}^{\mathrm{pw},\nu}_{g,t} &= \sum_{k=g}^{t}\overline{\tau}_{\mathrm{tr},k} - \overline{\Gamma}^{\nu}_{g,t}\sum_{k=g}^{t}\overline{\tau}^{\nu}_{\mathrm{ct},k},\\
\overline{\tau}^{\mathrm{cw},\nu}_{g,t} &= \sum_{k=g}^{t}\left\{\overline{\tau}_{\mathrm{tr},k} - \frac{\overline{c}^{\nu}_{g,k}}{\overline{V}^{\nu}_{g,k}}\overline{\tau}^{\nu}_{\mathrm{ct},k}\right\},\\
\overline{\tau}^{\mathrm{2eps,pw}}_{g,t} &= \sum_{k=g}^{t}\overline{\tau}_{\mathrm{tr},k} - \overline{\Gamma}'_{g,t}\sum_{k=g}^{t}\overline{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k},\\
\overline{\tau}^{\mathrm{2eps,cw}}_{g,t} &= \sum_{k=g}^{t}\left\{\overline{\tau}_{\mathrm{tr},k} - \frac{\overline{c}'_{g,k}}{\overline{V}'_{g,k}}\overline{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k}\right\}.
\end{aligned} \tag{F.5}$$

For any one of these procedures, if the conditions above hold for every $(g,t)\in\mathcal{S}$, $\widehat{\omega}_{g,t} \xrightarrow{p} \omega^0_{g,t}$ *for every such pair, and* $W_{\mathcal{S}} = \sum_{(g,t)\in\mathcal{S}}\omega^0_{g,t} \neq 0$, *its aggregate probability limit is*

$$\overline{\Theta}_{\mathrm{proc}} = \frac{1}{W_{\mathcal{S}}}\sum_{(g,t)\in\mathcal{S}}\omega^0_{g,t}\overline{\tau}^{\mathrm{proc}}_{g,t}.$$

Proof. The first line of (F.4) is the normalization-indexed one-step definition. Under factual initialization, the pairwise normal equation gives

$$\widehat{\epsilon}^{\mathrm{pw},\nu}_{g,t} = -\frac{\sum_{k=g}^{t}\widehat{\tau}^{\nu}_{\mathrm{ct},k}}{\sum_{k=g}^{t}V^{\nu}_{n,g,k}}.$$

A slice fluctuation e_k changes its empirical treated contrast by $c^{\nu}_{n,g,k}e_k$: the treated row gains $e_k/\widehat{p}_g$ and the control branch evaluated at a treated history loses $\widehat{r}^{\mathrm{ini}}_{g,k}e_k/B^{\nu}_{n,g,k}$. Therefore its contrast equals $\widehat{\tau}_{\mathrm{tr},k} + c^{\nu}_{n,g,k}e_k$. The common pairwise coefficient proves the second line; substituting the slice coefficient $\widehat{\gamma}^{\nu}_{g,k} = -\widehat{\tau}^{\nu}_{\mathrm{ct},k}/V^{\nu}_{n,g,k}$ of Lemma F.1 proves the third. Proposition F.2 first establishes uniqueness and then the zero treated coefficients under factual initialization. Its control coefficients are $-\sum_k \widehat{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k}/\sum_k V'_{n,g,k}$ under pairwise targeting and $-\widehat{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k}/V'_{n,g,k}$ under cohort-wise targeting; their contrast loading is $c'_{n,g,k}$. This proves the final two lines.

For part (ii), the law of large numbers gives $\widehat{p}_g \xrightarrow{p} p_g$. Fixed-fold cross-fitted averaging (C.1), the stated L_2 convergence and the ratio cap give convergence of every moment defining the residual means and $(V^{\nu}, c^{\nu}, V', c')$. For example, the perturbation of a control residual numerator is

$$-\mathbb{P}_n[\mathbb{I}(G > k)\widehat{r}^{\mathrm{ini}}_{g,k}(\widehat{m}^{\mathrm{ini}}_{g,k,0} - \overline{m}_{g,k,0})] + \mathbb{P}_n[\mathbb{I}(G > k)(\widehat{r}^{\mathrm{ini}}_{g,k} - \overline{r}_{g,k})\overline{U}_{g,k}],$$

which is $o_p(1)$ by the cap, Cauchy–Schwarz and (C.1). The treated numerator is handled by the convergence on $\{G = g\}$; bounded ratios similarly give convergence of their first and second moments. The positive limiting divisors permit continuous mapping in every line of (F.4), proving (F.5) and the finite aggregate formula. All these limits use only active-window moments. Full-design availability for a cohort-wise construction is still required, separately, and is not implied by consistency on $[g, t]$. ■

The support conditions in part (ii) make the required limit moments explicit without imposing identification overlap on a stress design. When Assumption 3 holds, the balancing identity (C.2) transfers the identified-risk-set L_2 convergence of Theorem 3.1 to the treated support, so those conditions also yield the moment convergence used here.

Implications for working-model routes. Under the identification assumptions and the applicable availability and positive-denominator conditions above, either nuisance-correctness branch collapses every applicable population representation to $\tau^0_{g,t}$. If $\overline{m}_{g,k,0} = m^0_{g,k,0}$ on every active slice, conditional centering gives $\overline{\tau}^{\nu}_{\mathrm{ct},k} = 0$, and the treated terms sum to the causal effect. If $\overline{r}_{g,k} = r^0_{g,k}$, the balancing identity gives $\overline{B}^{\nu}_{g,k} = p_g$, $\overline{c}^{\nu}_{g,k} = \overline{V}^{\nu}_{g,k}$ and, for the defined two-fluctuation diagnostics, $\overline{c}'_{g,k} = \overline{V}'_{g,k} > 0$. The multipliers are then one and the limits reduce to the consistent one-step functional. These are implications under the stated conditions. Nuisance correctness on a reported window does not by itself give full-cohort availability, nor, in the outcome-correct branch, the positivity $\overline{B}^{\mathrm{Háj}}_{g,k} > 0$ on which the Hájek procedures and the diagnostics' Gram denominators depend.

In CompactBinary, the working forms are correct for both nuisances in route A, for the ratio in C1, and for the outcome regression in C2 (Section 5.2). When the fitted limits recover these correctly specified functions and the conditions above hold, the proposed population limits equal the causal targets in these routes. Route C12 lies outside both correctness branches, and the displayed formulas describe its procedure-specific working-model limits. They do not require the C12 limits to differ from one another or from a causal target. The identity of a population limit with a causal target does not by itself supply a first-order expansion or validate a Wald interval; the applicable asymptotic conditions are stated separately.

F.4 Cohort-Wise Asymptotic Guarantees

Theorem F.1 (Asymptotic properties of single-fluctuation cohort-wise targeting)**.** *Fix a supported finite reported set $\mathcal{S}$ and $\nu \in \{\mathrm{HT}, \mathrm{Háj}\}$. Use the full cohort-wise construction of Appendix F.2,*

with factual initialization and its exact joint least-squares solution, and assume $\mathbb{P}(\mathcal{A}_n^\nu) \to 1$ *for the full-cohort availability event* (F.1). *Define the estimators arbitrarily off this event; for an aggregate, also define it arbitrarily when its empirical weight mass is zero. The sample, horizon, target set and fold count follow the fixed-dimensional cross-fitting regime of Appendix C.1. Every nuisance convergence or rate condition below is imposed only on reported windows* $k \in [g,t]$, $(g,t) \in \mathcal{S}$.

(i) ***Double-robust consistency.*** *Suppose the conditions of Theorem 3.1 hold for* $\mathcal{S}$, *with either its outcome-correct or its ratio-correct branch on the reported windows. Under Hájek normalization, in the outcome-correct branch additionally require* $\overline{B}_{g,k} = \mathbb{E}_{P_0}[\mathbb{I}(G > k)\overline{r}_{g,k}] > 0$ *for every reported slice. Then every reported* $\widehat{\tau}_{g,t}^{\mathrm{cw},\nu}$ *and every aggregate formed with the empirical weights of* (3.5) *is consistent for its target. Under ratio correctness, the Hájek positivity requirement follows from* $\overline{B}_{g,k} = p_g > 0$.

(ii) ***Equivalence to the same-normalization one-step estimator.*** *Suppose* $p_g > 0$, *Assumption 3,* $\max_{k\in[g,t]} \mathbb{E}_{P_0}[(\Delta Y_k)^2] < \infty$, *and* $0 \le \widehat{r}_{g,k,-\ell} \le M$ *hold for every reported pair. Let the separately maximized errors* $d_{m,g,t}$ *and* $d_{r,g,t}$ *be those in* (3.9). *Require both to be* $o_p(1)$ *and*

$$\max_{(g,t)\in\mathcal{S}} \sqrt{n}\, d_{m,g,t} d_{r,g,t} = o_p(1).$$

Then

$$\max_{(g,t)\in\mathcal{S}} \left|\widehat{\tau}_{g,t}^{\mathrm{cw},\nu} - \widehat{\tau}_{g,t}^{\mathrm{os},\nu}\right| = o_p(n^{-1/2}).$$

For an aggregate, if additionally $\mathbb{E}_{P_0}|\Omega_{g,t}| < \infty$ *and* $W_{\mathcal{S}} \neq 0$ *as in Condition 3(ii), the aggregate cohort-wise and one-step estimators with the same normalization also differ by* $o_p(n^{-1/2})$. *This assertion does not require positive influence-function variance or a correctness condition on unreported later slices.*

(iii) ***Canonical expansion, efficiency and inference.*** *In addition to part (ii), suppose the applicable one-step conditions of Theorem 3.2 under HT normalization, or Corollary 3.2(ii) under Hájek normalization, hold for the reported targets. For an aggregate, include Condition 3(iii). Then*

$$\sqrt{n}(\widehat{\Theta}_{\mathrm{cw}}^{\nu} - \Theta^0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi^{\mathrm{eff}}\{\Theta^0; \eta_{\mathcal{S}}^0\}(O_i) + o_p(1) \rightsquigarrow \mathcal{N}(0, V_\Theta).$$

The unit-level canonical gradient includes the aggregation-weight score in (2.6). *The estimator attains the observed-data efficiency bound* V_Θ, *and also the causal-model bound at a strictly prefix-feasible law as in Theorem 2.4(iii). Let* $\widehat{\varphi}_i^{\mathrm{cw},\nu} := \widehat{\Psi}_\Theta^{\mathrm{cw},\nu}(O_i)$ *be the empirical score of Lemma F.1, evaluated at the common cohort-wise update and including its weight score. Then*

$$\mathbb{P}_n[(\widehat{\varphi}_i^{\mathrm{cw},\nu})^2] \xrightarrow{p} V_\Theta.$$

With $V_\Theta > 0$, *the standard error* $\widehat{\mathrm{se}}_{\mathrm{cw},\nu}^2 := n^{-1}\mathbb{P}_n[(\widehat{\varphi}_i^{\mathrm{cw},\nu})^2]$ *gives*

$$\frac{\widehat{\Theta}_{\mathrm{cw}}^{\nu} - \Theta^0}{\widehat{\mathrm{se}}_{\mathrm{cw},\nu}} \rightsquigarrow \mathcal{N}(0,1).$$

Each assertion includes granular targets by taking $\mathcal{S} = \{(g,t)\}$ *with* $\Omega_{g,t} \equiv 1$.

Proof. Work on $\mathcal{A}_n^\nu$, whose complement has probability tending to zero, and fix $(g,t) \in \mathcal{S}$. By Lemma F.1 and Proposition F.1,

$$\widehat{\tau}_{g,t}^{\mathrm{cw},\nu} = \sum_{k=g}^{t} \left\{ \widehat{\tau}_{\mathrm{tr},k} - \frac{c_{n,g,k}^\nu}{V_{n,g,k}^\nu} \widehat{\tau}_{\mathrm{ct},k}^\nu \right\}.$$

Thus, with $b_{n,g,k}^\nu := c_{n,g,k}^\nu / V_{n,g,k}^\nu$,

$$\widehat{\tau}_{g,t}^{\mathrm{cw},\nu} - \widehat{\tau}_{g,t}^{\mathrm{os},\nu} = \sum_{k=g}^{t} (1 - b_{n,g,k}^\nu) \widehat{\tau}_{\mathrm{ct},k}^\nu. \tag{F.6}$$

This identity uses only slices in $[g,t]$.

(i) Consistency. The nuisance convergence conditions give $B_{n,g,k}^\nu \xrightarrow{p} \overline{B}_{g,k}^\nu$, with $\overline{B}_{g,k}^{\mathrm{HT}} = p_g$ and $\overline{B}_{g,k}^{\mathrm{Háj}} = \overline{B}_{g,k}$. The assumed or automatic positive limits and the cap give

$$0 \le b_{n,g,k}^\nu \le 1 + \frac{M \widehat{p}_g}{B_{n,g,k}^\nu} = O_p(1),$$

because $V_{n,g,k}^\nu \ge 1/\widehat{p}_g$ and $\mathbb{P}_n[\mathbb{I}(G=g)\widehat{r}_{g,k}^{\mathrm{ini}}] \le M\widehat{p}_g$. In the outcome-correct branch, the numerator convergence argument in the proof of Corollary 3.2(i) and conditional centering give $\widehat{\tau}_{\mathrm{ct},k}^\nu \xrightarrow{p} 0$ under either normalization. Hence the difference in (F.6) is $o_p(1)$.

In the ratio-correct branch, set

$$Q_{1,k} := \mathbb{P}_n[\mathbb{I}(G=g)\widehat{r}_{g,k}^{\mathrm{ini}}], \qquad Q_{2,k} := \mathbb{P}_n[\mathbb{I}(G>k)(\widehat{r}_{g,k}^{\mathrm{ini}})^2].$$

Cross-fitted L_2 convergence, the cap, and (C.2) imply

$$Q_{1,k} \xrightarrow{p} \mathbb{E}_{P_0}[\mathbb{I}(G=g) r_{g,k}^0] = \mathbb{E}_{P_0}[\mathbb{I}(G>k)(r_{g,k}^0)^2] \xleftarrow{p} Q_{2,k}.$$

Also $B_{n,g,k}^\nu \xrightarrow{p} p_g$ under both normalizations. Thus $c_{n,g,k}^\nu$ and $V_{n,g,k}^\nu$ have the same finite, strictly positive limit, so $b_{n,g,k}^\nu \xrightarrow{p} 1$. The control means are $O_p(1)$ by the separate numerator convergence proved for the one-step estimator, so (F.6) again is $o_p(1)$. The one-step estimator is consistent by Theorem 3.1 under HT or Corollary 3.2(i) under Hájek normalization. Therefore the cohort-wise estimator is consistent. The weight law of large numbers and $W_{\mathcal{S}} \neq 0$ extend this conclusion to every fixed aggregate.

(ii) Local rates and the first-order bridge. Write $d_m = d_{m,g,t}$ and $d_r = d_{r,g,t}$ for this reported pair. Let $Q_{1,k}^0$ and $Q_{2,k}^0$ denote the same averages as $Q_{1,k}$ and $Q_{2,k}$ with the fitted ratio replaced by $r_{g,k}^0$. By (C.2) with $f = r_{g,k}^0$, $Q_{2,k}^0 - Q_{1,k}^0 = O_p(n^{-1/2})$: the difference has a bounded, centered summand. With $\delta_{r,k,-\ell} := \widehat{r}_{g,k,-\ell} - r_{g,k}^0$, (C.2) gives

$$\|\mathbb{I}(G=g)\delta_{r,k,-\ell}\|_2 \le \varepsilon^{-1/2} d_r.$$

The second bound of (C.1) and Cauchy–Schwarz therefore yield $Q_{1,k} - Q_{1,k}^0 = O_p(d_r)$. Since $(\widehat{r}_{g,k,-\ell})^2 - (r_{g,k}^0)^2 = \delta_{r,k,-\ell}(\widehat{r}_{g,k,-\ell} + r_{g,k}^0)$ and the second factor is at most $M + \varepsilon^{-1}$ on the risk set, the same argument gives $Q_{2,k} - Q_{2,k}^0 = O_p(d_r)$. Consequently, uniformly over the fixed reported window,

$$Q_{2,k} - Q_{1,k} = O_p(n^{-1/2} + d_r).$$

Under HT normalization, $B^{\nu}_{n,g,k} - \widehat{p}_g = 0$ exactly. Under Hájek normalization, Lemma C.2 gives $B^{\nu}_{n,g,k} - \widehat{p}_g = O_p(n^{-1/2} + d_r)$. The exact identity

$$c^{\nu}_{n,g,k} - V^{\nu}_{n,g,k} = \frac{B^{\nu}_{n,g,k} Q_{1,k} - \widehat{p}_g Q_{2,k}}{\widehat{p}_g (B^{\nu}_{n,g,k})^2} = \frac{(B^{\nu}_{n,g,k} - \widehat{p}_g) Q_{1,k} - \widehat{p}_g (Q_{2,k} - Q_{1,k})}{\widehat{p}_g (B^{\nu}_{n,g,k})^2}$$

and the limits $\widehat{p}_g, B^{\nu}_{n,g,k} \xrightarrow{p} p_g > 0$ give

$$\max_{k \in [g,t]} |1 - b^{\nu}_{n,g,k}| = O_p(n^{-1/2} + d_r).$$

Here division by $V^{\nu}_{n,g,k}$ is harmless since $V^{\nu}_{n,g,k} \geq 1/\widehat{p}_g$.

On $\{\min_{k \in [g,t]} B_{n,g,k} > 0\}$, whose probability tends to one by Lemma C.2, that lemma gives

$$\max_k |\widehat{\tau}^{\text{Háj}}_{\text{ct},k}| = O_p(n^{-1/2} + d_m).$$

Its numerator is the same under both normalizations, so

$$\widehat{\tau}^{\text{HT}}_{\text{ct},k} = (B_{n,g,k}/\widehat{p}_g) \widehat{\tau}^{\text{Háj}}_{\text{ct},k}.$$

Since $B_{n,g,k}/\widehat{p}_g \xrightarrow{p} 1$, the HT control residual mean has the same rate. Substituting these two bounds in (F.6) yields

$$\widehat{\tau}^{\text{cw},\nu}_{g,t} - \widehat{\tau}^{\text{os},\nu}_{g,t} = O_p\{(n^{-1/2} + d_m)(n^{-1/2} + d_r)\} = o_p(n^{-1/2}).$$

Multiplication by $\sqrt{n}$ leaves terms of orders $n^{-1/2}$, d_m, d_r and $\sqrt{n}\, d_m d_r$, all tending to zero. The reported pairs are finitely many, so the assertion holds for their maximum. For an aggregate, the exact weighted-average difference is

$$\frac{1}{\sum_{\mathcal{S}} \widehat{\omega}_{g,t}} \sum_{(g,t) \in \mathcal{S}} \widehat{\omega}_{g,t} \{\widehat{\tau}^{\text{cw},\nu}_{g,t} - \widehat{\tau}^{\text{os},\nu}_{g,t}\}.$$

The weight assumptions make the coefficients $O_p(1)$ and the denominator converge to $W_{\mathcal{S}} \neq 0$, so this too is $o_p(n^{-1/2})$.

(iii) Canonical expansion and variance consistency. Part (ii) transfers the corresponding one-step expansion to the cohort-wise estimator. The one-step source is Theorem 3.2 for HT and Corollary 3.2(ii) for Hájek normalization. In particular, its canonical gradient is

$$\frac{1}{W_{\mathcal{S}}} \sum_{(g,t) \in \mathcal{S}} \left[\omega^0_{g,t} \psi_{g,t}(O; \eta^0_g) + (\tau^0_{g,t} - \Theta^0) \{\Omega_{g,t}(O) - \omega^0_{g,t}\} \right].$$

The empirical cancellation of the weight correction from the point equation does not remove this unit-level score or its contribution to V_Θ.

To verify variance consistency without invoking a pairwise targeting theorem, let $\widehat{\gamma}^{\nu}_{g,k}$ be the slice coefficient of Lemma F.1. Its exact formula and the control-residual rate in part (ii) give

$$\max_{k \in [g,t]} |\widehat{\gamma}^{\nu}_{g,k}| = \max_{k \in [g,t]} \frac{|\widehat{\tau}^{\nu}_{\text{ct},k}|}{V^{\nu}_{n,g,k}} = O_p(n^{-1/2} + d_m) = o_p(1).$$

On reported slices the common targeted control branch is $\widehat{m}^{\text{ini}}_{g,k,0} - \widehat{\gamma}^{\nu}_{g,k} \widehat{r}^{\text{ini}}_{g,k} / B^{\nu}_{n,g,k}$. Its difference from the initial branch is bounded in absolute value by $O_p(\max_{k \in [g,t]} |\widehat{\gamma}^{\nu}_{g,k}|) = o_p(1)$ on the target

and control history supports, by the ratio cap and the positive reported-slice divisor limits. Thus its effect on each granular empirical score is $o_p(1)$ in empirical L_2; the treated and control score loadings are also bounded by the cap and these positive divisor limits. The initial one-step scores have the empirical L_2 convergence verified in Lemma C.3(i). Consistency of the point estimates and empirical weights handles their scalar replacements inside the score. For the aggregate-weight terms, the empirical second moments $\mathbb{P}_n[\Omega_{g,t}^2]$ are $O_p(1)$ by Condition 3(iii); they multiply scalar parameter differences that are $o_p(1)$. The fixed finite combination therefore satisfies

$$\mathbb{P}_n\left[\left\{\widehat{\varphi}_i^{\mathrm{cw},\nu} - \varphi^{\mathrm{eff}}\{\Theta^0; \eta_{\mathcal{S}}^0\}(O_i)\right\}^2\right] = o_p(1).$$

This is the generic premise of Lemma C.3, which yields the stated variance consistency. With $V_\Theta > 0$, Slutsky's theorem gives studentized normality. The observed-data efficiency claim follows from the canonical expansion; its causal-model interpretation requires the strict prefix feasibility of Theorem 2.4(iii).

Finally, the exact slice decomposition shows why no accuracy or rate condition after a reported endpoint is used. Coefficients in the original cumulative basis may depend on later slices, but the update on a reported slice uses only its own coefficient $\widehat{\gamma}_{g,k}^{\nu}$. Full-cohort availability remains a separate requirement: the entire joint regression must exist to define this construction. ■

F.5 The Two-Fluctuation Diagnostic

The reported two-fluctuation diagnostic uses Hájek normalization. Its separate treated and control columns on slice k are

$$h_{g,k}^{\mathrm{tr}}(O) = \frac{\mathbb{I}(G=g)}{\widehat{p}_g}, \qquad h_{g,k}^{\mathrm{ct}}(O) = -\frac{\mathbb{I}(G>k)\widehat{r}_{g,k}^{\mathrm{ini}}}{B_{n,g,k}}.$$

Define its control-only Gram contribution and loading by

$$V'_{n,g,k} := \frac{\mathbb{P}_n[\mathbb{I}(G>k)(\widehat{r}_{g,k}^{\mathrm{ini}})^2]}{B_{n,g,k}^2}, \qquad c'_{n,g,k} := \frac{\mathbb{P}_n[\mathbb{I}(G=g)\widehat{r}_{g,k}^{\mathrm{ini}}]}{\widehat{p}_g B_{n,g,k}}.$$

These omit the treated contribution $\widehat{p}_g^{-1}$ present in $V_{n,g,k}^{\mathrm{Háj}}$ and $c_{n,g,k}^{\mathrm{Háj}}$.

Proposition F.2 (Unique solution and representations of the two-fluctuation diagnostic)**.** *On $\mathcal{A}_n^{\mathrm{Háj}}$ for the cohort-wise architecture, or on the corresponding availability event restricted to $[g,t]$ for an isolated pairwise fit, the two-fluctuation least-squares solution is unique. Under factual initialization, all treated fluctuation coefficients are zero. The control coefficients are*

$$\widehat{\epsilon}_{g,t}^{\mathrm{2eps,pw,ct}} = -\frac{\sum_{k=g}^{t}\widehat{\tau}_{\mathrm{ct},k}^{\mathrm{Háj}}}{\sum_{k=g}^{t}V'_{n,g,k}}, \qquad \widehat{\gamma}_{g,k}^{\mathrm{2eps,cw,ct}} = -\frac{\widehat{\tau}_{\mathrm{ct},k}^{\mathrm{Háj}}}{V'_{n,g,k}}.$$

Consequently

$$\widehat{\tau}_{g,t}^{\mathrm{2eps,pw}} = \sum_{k=g}^{t}\widehat{\tau}_{\mathrm{tr},k} - \frac{\sum_{k=g}^{t}c'_{n,g,k}}{\sum_{k=g}^{t}V'_{n,g,k}}\sum_{k=g}^{t}\widehat{\tau}_{\mathrm{ct},k}^{\mathrm{Háj}},$$

$$\widehat{\tau}_{g,t}^{\mathrm{2eps,cw}} = \sum_{k=g}^{t}\left\{\widehat{\tau}_{\mathrm{tr},k} - \frac{c'_{n,g,k}}{V'_{n,g,k}}\widehat{\tau}_{\mathrm{ct},k}^{\mathrm{Háj}}\right\}.$$

Proof. First establish positivity without using factual initialization. Availability gives $\widehat{p}_g > 0$ and $B_{n,g,k} > 0$. Since $\widehat{r}^{\mathrm{ini}}_{g,k} \geq 0$, the latter implies that at least one control row has a positive ratio. Thus the control column is nonzero and $V'_{n,g,k} > 0$. More explicitly, with $q_{n,k} := \mathbb{P}_n[\mathbb{I}(G > k)] \in (0, 1]$, Cauchy–Schwarz gives

$$B^2_{n,g,k} \leq q_{n,k}\, \mathbb{P}_n[\mathbb{I}(G > k)(\widehat{r}^{\mathrm{ini}}_{g,k})^2], \qquad V'_{n,g,k} \geq q^{-1}_{n,k} > 0.$$

The treated column is also nonzero, with squared norm $n/\widehat{p}_g$. Treated and control columns have disjoint row supports. For a pairwise fit, its Gram is therefore

$$n \operatorname{diag}\left\{ \frac{t-g+1}{\widehat{p}_g}, \sum_{k=g}^{t} V'_{n,g,k} \right\},$$

which is positive definite. In the cohort slice basis its Gram is diagonal, with entries $n/\widehat{p}_g$ and $nV'_{n,g,k}$ for each slice $k \in [g, T]$. The endpoint basis is obtained by the invertible triangular map of Lemma F.1 applied to each arm. Its Gram is also positive definite. Both exact least-squares solutions are consequently unique.

Now impose factual initialization. Every treated row has initial residual zero, so the treated normal-equation numerator is zero. Since its Gram is positive, its unique coefficient is zero, under either architecture. The control numerator on slice k is $-\widehat{\tau}^{\mathrm{Háj}}_{\mathrm{ct},k}$. The control normal equations therefore give the displayed coefficients. Write $\widehat{\epsilon}^{\mathrm{ct}}$ for $\widehat{\epsilon}^{\mathrm{2eps,pw,ct}}_{g,t}$ in the pairwise construction and for $\widehat{\gamma}^{\mathrm{2eps,cw,ct}}_{g,k}$ in the cohort-wise construction. The treated row value remains $\Delta Y_{i,k}$, whereas the targeted control branch changes by $-\widehat{r}^{\mathrm{ini}}_{g,k}\widehat{\epsilon}^{\mathrm{ct}}/B_{n,g,k}$. Substitution into the treated empirical contrast gives a loading $c'_{n,g,k}$ and proves both representations. ∎

The single-fluctuation Hájek loading ratio is the following exact weighted average:

$$\frac{c^{\mathrm{Háj}}_{n,g,k}}{V^{\mathrm{Háj}}_{n,g,k}} = \frac{\widehat{p}_g^{-1} \cdot 1 + V'_{n,g,k}\{c'_{n,g,k}/V'_{n,g,k}\}}{\widehat{p}_g^{-1} + V'_{n,g,k}}.$$

The identity follows from $c^{\mathrm{Háj}}_{n,g,k} = \widehat{p}_g^{-1} + c'_{n,g,k}$ and $V^{\mathrm{Háj}}_{n,g,k} = \widehat{p}_g^{-1} + V'_{n,g,k}$. It describes the finite-sample multipliers. It does not order bias, variance, coverage, or mean squared error.

F.6 A Qualified Design-Quadratic Comparison

Remark F.1 (Unscaled design quadratics)**.** For a fixed window $[g, t]$, suppress the indices of $c_k = c^{\nu}_{n,g,k}$ and $V_k = V^{\nu}_{n,g,k} > 0$, and define the unscaled quadratics

$$Q_{\mathrm{cw}} := \sum_{k=g}^{t} \frac{c_k^2}{V_k}, \qquad Q_{\mathrm{pw}} := \frac{(\sum_{k=g}^{t} c_k)^2}{\sum_{k=g}^{t} V_k}.$$

If $\Gamma = (\sum_k c_k)/(\sum_k V_k)$, then

$$Q_{\mathrm{cw}} - Q_{\mathrm{pw}} = \sum_{k=g}^{t} V_k \left(\frac{c_k}{V_k} - \Gamma \right)^2 \geq 0.$$

Indeed, expanding the squares gives $\sum_k c_k^2/V_k - 2\Gamma \sum_k c_k + \Gamma^2 \sum_k V_k$, which is the displayed difference. Equality holds if and only if c_k/V_k is constant over the window. The same algebra applies to the control-only pair (c'_k, V'_k) in the two-fluctuation diagnostic.

These quadratics have a covariance interpretation only under an artificial working model. Hold the auxiliary design and loadings fixed, and give a hypothetical auxiliary residual vector the working covariance $\sigma^2 I$ across the stacked rows (i,k). The working variances of the linear fluctuation contributions are then $(\sigma^2/n)Q_{\mathrm{cw}}$ and $(\sigma^2/n)Q_{\mathrm{pw}}$; the factor σ^2/n is not part of the definitions of Q_{cw} and Q_{pw}. Equal marginal residual variances alone do not supply the working covariance $\sigma^2 I$. Stacked outcomes from the same unit can be correlated, and factual initialization fixes the treated row residuals at zero.

The working calculation omits within-unit correlation and the variability of the initial treated residual mean, estimated nuisances, normalizers, and loadings, together with their covariance with the fluctuation. Thus these are design quadratics, rather than variances of the complete estimators. Under ratio consistency and the applicable moment and denominator conditions, $c_k/V_k \xrightarrow{p} 1$ on the fixed window and the displayed difference is $o_p(1)$; the same conclusion holds for c'_k/V'_k. For the single-fluctuation estimators, this is consistent with their common canonical first-order limits under the applicable conditions of Theorem 4.2(ii) and the root-n part of Corollary 4.1 for the pairwise estimators, and Theorem F.1(iii) for the cohort-wise estimators.

Appendices: Simulation Designs and Complete Results

Appendices G–J document the Monte Carlo study of Section 5. Appendix G documents numerical truth computation, working models, and comparators for the designs in Section 5, summarizes all scenarios at $n = 5000$, and defines the procedure identifiers; Appendix H reports the comparisons of normalizations and architectures, the setting without cross-fitting, the Static designs under the stress scenarios B and B+, every CompactBinary cell, and a refit of the comparators with analytic standard errors; Appendix I reports the completion and stress-scenario tables; Appendix J reports every cell of the Static and Dynamic designs. All eight preregistered proposed variants are reported here; Section 5 pools four of them. All eight variants use the printed procedure names described in Appendix G.5, which maps them to stable replication-archive identifiers; Appendix F.1 defines the four further variants.

G Simulation Designs and Large-Sample Summary

This section documents numerical truth computation, working models, and comparators for the Static and Dynamic designs specified in Section 5.3 (Appendices G.1–G.3); summarizes all five scenarios at $n = 5000$ (Appendix G.4); and maps the printed procedure names to the archive identifiers (Appendix G.5); Appendix F.1 defines the four further preregistered variants. Reporting conventions are those of Section 5.1. With R_c complete replications, the MCSE of the bias is the standard deviation of the estimates divided by $R_c^{1/2}$, that of a coverage rate $\widehat{c}$ is $\{\widehat{c}(1-\widehat{c})/R_c\}^{1/2}$, and that of the RMSE is the delta-method value $\mathrm{sd}(e^2)/(2\,\mathrm{RMSE}\,R_c^{1/2})$, where e is the estimation error. All random numbers derive from the master seed 20260530 through L'Ecuyer-CMRG streams keyed by design coordinates and replication index, so any cell can be regenerated in isolation and no result depends on the order of computation.

G.1 Numerical Truth Computation

The causal targets are defined in (5.1), and the Static and Dynamic designs are specified in Section 5.3.[1] In the binary Static and Dynamic designs the targets are probability-scale integrals over the Gaussian covariates and, in the Static design, the unit effect; they are computed by Gauss–Hermite quadrature and confirmed by an independent adaptive Gauss–Kronrod integration. The CompactBinary targets are finite sums (Section 5.2).

G.2 Working Models, Overlap, and the Ratio Cap

All proposed procedures share the same prespecified working models, fitted on the training folds (on all units when $L = 1$) and evaluated at the frozen history $\mathcal{C}_{g-1}$; every regressor enters linearly with an intercept. For each target cohort g and each required period s (through the largest reported endpoint for the one-step and pairwise targeted estimators, and through T for cohort-wise targeting), the frozen-history hazard $\lambda_{g,s}$ is fitted by a logistic regression of $\mathbb{I}(G = s)$ on $\mathcal{C}_{g-1}$ in the risk set $\{G \geq s\}$. The inverse-risk ratio is assembled from these hazards by (3.2) in every design, the Static design included; a direct logistic fit of $\mathbb{I}(G = g)$ against $\mathbb{I}(G > k)$ is computed only as a numerical cross-check. The logistic fits are projected onto $[0, 1 - \overline{a}]$ with $\overline{a} = 0.001$, which gives the fitted hazards, and the assembled ratio is capped at $M = 10$ as in (3.2). The control outcome regression for period k is fitted on $\{G > k\}$: a linear regression of ΔY_k on $\mathcal{C}_{g-1}$ for continuous

[1] The replication code names the time-invariant covariate `X` and the time-varying one `Z`; the mapping is one to one.

Table G.1: Status of the working models by design. *Correct*: the working model contains the true function. *Misspecified*: the scenario adds a quadratic term that the working model omits. *Approximate*: the true function lies outside the working model in every scenario, because it averages over variables that are not in $\mathcal{C}_{g-1}$—the unit effect or the lagged outcome for binary levels, and the history realized after $g-1$ for hazards at $s > g$. At $s = g$ the Dynamic hazards are correct except under C2. The last column refers to the uniform overlap bound of Assumption 3 and the truth-containment clause of Condition 4 with $M = 10$.

Design	Control outcome regression	Frozen-history hazard	Assumption 3 and $r^0_{g,k} \le M$
Static, continuous	correct; misspecified in C1	correct; misspecified in C2	fail
Dynamic, continuous	correct; misspecified in C1	approximate for $s > g$; misspecified in C2	fail
Static, binary	approximate; C1 adds Z^2	correct; misspecified in C2	fail
Dynamic, binary	approximate; C1 adds Z^2	approximate for $s > g$; misspecified in C2	fail
CompactBinary	correct in A, C2	correct in A, C1	hold

outcomes and, for binary outcomes, logistic regressions of Y_k and of Y_{k-1} on $\mathcal{C}_{g-1}$ whose fitted probabilities are differenced.

Treated-arm initialization. Algorithm 1 sets the offsets of the treated rows to their observed increments (4.3). The implementation instead initializes the treated arm with a fold-specific regression on $\mathcal{C}_{g-1}$ fitted to the target cohort—of the increment for continuous outcomes, and of the level by quasi-binomial logistic regression, differenced across periods, for binary outcomes—shifted additively (and clamped to $[0, 1]$ for binary levels) so that its average over the cohort equals the observed average. For binary levels, the shift is chosen so that the average after clamping equals the observed average. Calibration covers every level needed by the construction, from $g-1$ through T for cohort-wise targeting and through the largest reported endpoint for pairwise targeting, and makes the sum of the treated residuals zero on each slice. Because each treated clever-covariate component is constant on that slice, these sums leave the right-hand side of every pairwise or cohort-wise auxiliary normal equation unchanged. In exact arithmetic, the fluctuation coefficients, point estimates, and unit-level influence-function values therefore agree with those under (4.3) for both architectures and for the single- and two-fluctuation specifications. The implementation computes both versions for all six targeted variants and requires them to agree to within 10^{-11} plus 10^{-10} times the largest absolute value among the compared quantities in every accepted replication. The fitted treated function serves the range diagnostics for binary outcomes. It is nonetheless a further fit that can fail, and its failures make a targeted procedure's replication unavailable; they account for 3 percent of the unavailable replications of the eight proposed variants, and for 3 percent of those of the four procedures pooled in Section 5.4.

Overlap and the ratio cap. Table G.1 records the status of the working models. Two features of the Static and Dynamic designs matter for the theory. First, their covariates are Gaussian and every adoption index increases without bound in a covariate, so the adoption hazards approach one, and the survival probabilities zero, in the tails. The uniform overlap bound of Assumption 3 therefore holds for no $\varepsilon > 0$, and the true ratio $r^0_{g,k}$ is unbounded, so the truth-containment clause $r^0_{g,k} \le M$ of Condition 4 fails. The cap is correspondingly active: the share of complete replications in which at least one fitted ratio reaches M is one in every cell at $n = 5000$ and at least 62 percent in every cell at $n = 100$. In CompactBinary, by contrast, the true ratio lies below M, and a

fitted ratio reached M in a single replication (route C2, $n = 100$, five-fold cross-fitting, cohort-wise procedures). Second, in the binary designs the outcome working models are only approximations even in scenario A, and in the Dynamic designs so are the hazards at $s > g$.

Consequently, no Static or Dynamic design satisfies the conditions of Theorem 3.1: overlap fails in all of them, no fitted ratio converges to $r^0_{g,k}$, and the outcome regression is correct only in the continuous designs outside C1. In the continuous designs scenario C1 therefore does not isolate the outcome-regression departure. In the Static design the hazard model is correct under C1, but the logistic fits are projected onto $[0, 1-\overline{a}]$ before the ratio is assembled and capped, so the fitted ratio converges to the projected-then-capped limit

$$\overline{r}_{g,k} = \min\Big\{\lambda^{\Pi}_{g,g} \Big/ \prod_{s=g}^{k}(1-\lambda^{\Pi}_{g,s}),\ M\Big\}, \qquad \lambda^{\Pi}_{g,s} := \min\{\lambda^{0}_{g,s},\, 1-\overline{a}\}.$$

The projection binds only where a true hazard exceeds $1-\overline{a} = 0.999$, which in this design means $Z > 2+\log 999 \approx 8.907$, an event whose marginal covariate probability is about 1.3×10^{-15}. Because the hazard is the same in every period, the assembled ratio there is at least $0.999/0.001 = 999 > M$ with or without the projection, so the cap sets the limit to M either way, and $\overline{r}_{g,k} = \min(r^0_{g,k}, M)$ exactly. The cap binds with positive probability, so $\overline{r}_{g,k}$ differs from $r^0_{g,k}$ on an event of positive probability, and the product remainders from the identity of Lemma 3.1, which does not use the overlap margin when its expectations exist, are nonzero at these limits once summed over each target window and, for ES(1), weighted by cohort shares, as the computed HT biases below show. We computed the probability limits of the two estimating-equation estimators in this design by numerical integration, with the ratio at this limit and the outcome regression at its population least-squares fit. The resulting asymptotic biases—probability limit minus target—are 0.461 (HT) and 0.396 (Hájek) for ATT$(2,4)$ and 0.071 and 0.064 for ES(1). Without cross-fitting, the Monte Carlo biases at $n = 5000$ are 0.459, 0.395, 0.073, and 0.066, each within one MCSE of the corresponding asymptotic bias. With five-fold cross-fitting they are 0.449, 0.387, 0.065, and 0.059, 2.0–2.8 MCSEs below the corresponding asymptotic bias. In the continuous Dynamic design, C1 misspecifies the outcome regression while the hazard model is itself approximate. In both designs, therefore, C1 is a joint departure from both branches rather than a test of one of them.

G.3 Comparators

DR-nt and DR-ny are the doubly robust estimators of Callaway and Sant'Anna (2021) with never-treated and not-yet-treated comparison groups. In each replication the `did` package is called as

```
att_gt(yname = "Y", tname = "time", idname = "id", gname = "G",
       xformla = <formula>, data = <panel>, panel = TRUE,
       allow_unbalanced_panel = FALSE, control_group = <group>,
       anticipation = 0, weightsname = NULL, alp = 0.05,
       bstrap = TRUE, cband = TRUE, biters = 1000,
       clustervars = NULL, est_method = "dr",
       base_period = "varying", faster_mode = TRUE,
       print_details = FALSE, pl = FALSE, cores = 1)
```

with `<group>` equal to `"nevertreated"` for DR-nt and `"notyettreated"` for DR-ny. The never-treated cohort is coded $G = 0$ for the package. ES(1) is read at event time one from the aggregation `aggte(type = "dynamic")`, whose remaining arguments are left at their defaults, so its inference settings are inherited from the `att_gt` object; the aggregation weights cohorts by their empirical shares, as our own ES(1) estimators do. As for the proposed procedures, any warning raised by

the package, including one about the band, makes the fit unavailable (Section 5.1 and Appendix I). Appendix H.5 refits the comparators with analytic standard errors and without the band in the cells where these warnings occur. In the code's naming, where `X` is Z_i and `lag_Y` and `lag_Z` are the previous period's outcome and time-varying covariate, `<formula>` is `~ X` in the Static and CompactBinary designs, `~ X + lag_Y` in the Dynamic continuous design, and `~ X + lag_Z` in the Dynamic binary design. For post-treatment cells `did` evaluates covariates in the base period $g-1$, so in the Dynamic designs the comparators condition on (Z, Y_{g-2}) and (Z, X_{g-2}), one period earlier than the frozen history $\mathcal{C}_{g-1}$. Its doubly robust estimator combines a linear regression for the comparison group's outcome change with a logistic propensity score, both linear in these covariates.

Pointwise standard errors. Every interval, for every procedure, uses the pointwise critical value 1.96; the critical value of the package's simultaneous band is not used. The standard errors follow the same influence-function principle, each computed from the procedure's own estimated score, but not the same formula. For the comparators, the `did` package (version 2.3.0) draws $n^{-1}\sum_i \xi_i \widehat{\psi}_i^{\text{did}}$ with independent Rademacher multipliers ξ_i applied to the comparator's estimated influence function $\widehat{\psi}_i^{\text{did}}$, holding its nuisance fits fixed, and reports the interquartile range of 1,000 such draws divided by $z_{0.75} - z_{0.25}$. Conditionally on the data the draws have variance $n^{-2}\sum_i(\widehat{\psi}_i^{\text{did}})^2$, the analogue of the squared standard error we report for the proposed procedures, but the interquartile-range scale need not coincide with its square root in finite samples and also carries bootstrap simulation error, so the two constructions need not give the same standard error. At $n = 5000$ the comparators' ratio of average estimated standard error to Monte Carlo standard deviation lies between 0.94 and 1.03 in every cell of Table G.2. In the twelve cells that contain the band warnings of Section 5.4, refitting the comparators with the package's analytic standard errors, the construction used for the proposed procedures, lowers rather than raises their empirical coverage (Appendix H.5).

The comparators' endpoint estimands use a long-difference restriction against the fixed pool $\{G = \infty\}$ or $\{G > t\}$ (Section 2.2); imposing equality for each increment is stronger, as is the maintained not-yet-treated assumption of Callaway and Sant'Anna (2021). In the continuous Dynamic design, adoption responds to the lagged outcome, so units that remain untreated through period $t > g$ have been selected on outcome innovations realized after $g-1$. Selection on these intermediate innovations changes the fixed pool's mean long difference. The not-yet-treated long-difference restriction therefore fails for every target with $t > g$, including both reported targets, and the never-treated restriction fails as well, whereas the sliding-risk-set restriction of Assumption 2 holds. In that design the comparators' parallel-trends restrictions fail, so their causal consistency guarantees do not apply even when their working models are correct. In the other designs adoption depends only on covariates, and given Z the outcome paths are independent of adoption, so the comparators' restrictions hold. Their outcome regression is correct, however, only in the continuous Static design outside C1, and their propensity-score model is misspecified for every reported target in every design. In the Static designs, for example, the hazard is the same in every period, so given Z the log-odds of cohort g against $\{G = \infty\}$ is $\operatorname{logit}\lambda_{ik} + (T-g)\log\{1+\exp(\operatorname{logit}\lambda_{ik})\}$, and against $\{G > t\}$ it is the same expression with $t-g$ in place of $T-g$; as $T-g$ and $t-g$ are positive for every pair in a reported target, neither log-odds is linear in Z. Consequently, the continuous Static design outside C1 is the only setting here in which one of the comparators' working models, their outcome regression, is correctly specified. Even there their double-robust consistency guarantee is not formally available: it also maintains overlap, which the Gaussian covariate defeats as it defeats Assumption 3 for the proposed procedures (Appendix G.2). Their centering there is consistent with, but not guaranteed by, the outcome-regression branch. In every other setting, the guarantee

Table G.2: Ratio of the average estimated standard error to the Monte Carlo standard deviation of the estimates, over complete replications, $n = 5000$, five-fold cross-fitting for the proposed procedures (EEq-Háj. and TMLE-pw-Háj. abbreviate EEq-Hájek and TMLE-pw-Hájek). A ratio below one means that the estimated standard errors are smaller on average than the Monte Carlo standard deviation; the ratio is a descriptive summary and says nothing about centering. With 2,000 replications the relative Monte Carlo error of the standard deviation is about $(2 \times 2000)^{-1/2} \approx 0.016$ for approximately normal estimates.

		ATT(2, 4)				ES(1)			
Design	Scen.	EEq-Háj.	TMLE-pw-Háj.	DR-nt	DR-ny	EEq-Háj.	TMLE-pw-Háj.	DR-nt	DR-ny
Static continuous	A	0.99	0.95	1.01	1.01	0.99	0.98	1.00	0.98
	B	0.97	0.85	0.99	1.00	0.97	0.92	0.96	1.00
	B+	0.89	0.68	1.01	1.01	0.91	0.77	0.94	0.97
	C1	1.22	1.04	1.01	1.00	1.17	1.07	1.01	0.99
	C2	0.85	0.64	1.02	1.02	0.89	0.72	1.03	1.00
Dynamic continuous	A	0.97	0.95	0.97	0.98	0.98	0.97	0.98	0.98
	B	0.98	0.89	1.00	1.01	0.99	0.96	1.01	1.01
	B+	0.95	0.72	0.99	0.97	0.97	0.84	0.99	0.97
	C1	1.33	1.09	1.00	1.00	1.25	1.11	0.98	0.98
	C2	0.91	0.77	1.01	1.02	0.97	0.92	1.01	1.02
Static binary	A	1.04	1.02	0.99	1.00	1.00	1.00	0.97	0.99
	B	1.05	0.93	1.00	0.98	1.02	0.98	0.99	0.99
	B+	1.05	0.75	0.96	0.97	1.02	0.85	0.96	0.99
	C1	1.05	1.04	1.01	1.02	1.04	1.04	1.00	1.02
	C2	1.13	0.78	0.99	1.00	1.11	0.89	0.98	1.02
Dynamic binary	A	0.98	0.97	0.99	0.99	0.99	0.98	0.98	0.99
	B	1.01	0.94	0.99	1.00	1.03	1.02	1.01	1.02
	B+	0.99	0.76	1.01	0.99	0.98	0.87	1.01	0.96
	C1	1.00	0.99	0.99	0.99	1.01	1.01	1.00	1.00
	C2	1.00	0.97	1.01	1.00	1.02	1.01	1.02	1.02

does not apply, whether or not overlap holds, because neither working model is correctly specified.

G.4 Large-Sample Summary

Tables G.3 and G.4 report bias and coverage at $n = 5000$, with five-fold cross-fitting, for the two featured procedures and the two comparators of Section 5. At this sample size every proposed replication is complete, and each comparator completes in at least 0.988 of replications in every cell. Table G.2 reports the ratio of the average estimated standard error to the Monte Carlo standard deviation of the estimates.

Outside continuous C1 the biases of the proposed procedures are small relative to the targets, though not always within Monte Carlo error. The absolute bias of EEq-Hájek is at most 0.006 in the continuous designs, where the targets equal one, and at most 0.0016, or 3.6 percent of the target, in the binary designs; for TMLE-pw-Hájek the bounds are 0.007 and 0.0042 (8.5 percent). In the 20 binary cells the bias exceeds two pointwise MCSEs in 4 for EEq-Hájek and in 6 for TMLE-pw-Hájek, by up to 5.0 MCSEs; in these designs neither branch of Theorem 3.1 holds exactly (Table G.1). Under binary C1 no proposed procedure has absolute bias above 0.0005.

Dynamic B and B+ diagnostics. At $n = 5000$ with five-fold cross-fitting, under B+ the TMLE-pw-Hájek intervals have low empirical coverage in both designs and for both targets, 0.838

to 0.916, although the estimator's bias is below 0.0014 in absolute value, whereas the EEq-Hájek intervals cover 0.939–0.952 in the same four cells and are within two MCSEs of 0.95 except in the continuous ATT$(2, 4)$ cell, approximately 2.1 MCSEs below. The two procedures share a standard-error construction, and their average standard errors differ by less than one percent in these cells, but the Monte Carlo standard deviation of TMLE-pw-Hájek is 13 to 32 percent larger, so its SE/SD is 0.72–0.87, against 0.95–0.99 for EEq-Hájek. The pattern appears, milder, under B in the continuous design (ATT$(2, 4)$: coverage 0.914, SE/SD 0.89).

The comparators are centered away from the target in the continuous design (bias 0.035–0.223), consistent with the failure of their comparison-group restrictions; in the binary design their bias is smaller in magnitude, at most 12 percent of the target, but still 4.9 to 13.0 MCSEs from zero. In the continuous design their SE/SD ratios in Table III range from 0.97 to 1.01. Thus, low coverage in the continuous design accompanies substantial bias despite standard errors close to the Monte Carlo spread.

Empirical coverage differs more. Outside continuous C1, the empirical coverage of EEq-Hájek is 0.907–0.949 in the continuous designs and 0.946–0.975 in the binary designs; that of TMLE-pw-Hájek is 0.794–0.942 and 0.851–0.957. In 20 of these cells TMLE-pw-Hájek has low empirical coverage, more than two pointwise MCSEs below 0.95: there its empirical coverage is 0.794–0.939, its absolute bias is at most 0.12 Monte Carlo standard deviations, and its standard-error ratio in Table G.2 is 0.64–0.96; in every one of them EEq-Hájek has the higher ratio (0.85–1.13) and the higher empirical coverage (0.907–0.975). Centering alone therefore does not account for the low empirical coverage of the targeted intervals in these cells. The pattern is consistent with standard errors that understate the estimator's sampling variability, but these summaries do not exclude other contributors, such as nonnormality, heavy tails, or dependence between an estimate and its estimated standard error. Both procedures use the same standard-error construction; the comparison is descriptive and does not rank them.

The comparators' results are consistent with Appendix G.3. In the continuous Static design outside C1 their absolute bias is at most 0.005 and their coverage 0.939–0.959. In the continuous Dynamic design they are centered away from the targets in every scenario, with bias 0.027–0.223 and coverage 0.567–0.919 outside C1. In the Static binary design their bias is negative in every cell, from -0.166 to -0.002, and their empirical coverage falls as low as 0.062. In the Dynamic binary design their bias is negative in every cell, at most 11.7 percent of the target, and more than two pointwise MCSEs from zero in 18 of the 20 cells; their empirical coverage is 0.937–0.954. Under continuous C1 the featured procedures and the comparators are all biased, the proposed ones by 0.039–0.432 and the comparators by 0.209–2.136, the comparators more so in every cell.

G.5 Procedure Identifiers and Preregistered Variants

The tables and text print EEq-HT and EEq-Hájek for the one-step estimators of Section 3, TMLE-pw-HT and TMLE-pw-Hájek for the pairwise TMLEs of Section 4, and TMLE-cw-HT and TMLE-cw-Hájek for the single-fluctuation cohort-wise variants. The replication archive retains stable algorithm identifiers and uses the ASCII spelling Hajek: its one-step identifiers are EEq-HT and EEq-Hajek; its pairwise single-fluctuation identifiers are TMLE-1eps-pw-HT and TMLE-1eps-pw-Hajek; and its cohort-wise single-fluctuation identifiers are TMLE-1eps-cw-HT and TMLE-1eps-cw-Hajek. The printed two-fluctuation names TMLE-2eps-pw-Hájek and TMLE-2eps-cw-Hájek keep the 2eps prefix, and their archive identifiers use Hajek. In an archive identifier, 1eps and 2eps give the number of fluctuation coefficients per clever covariate, and pw and cw indicate pairwise or cohort-wise targeting. Narrow tables abbreviate the Hájek suffix to Háj. The four further preregistered variants, defined in Appendix F.1, are reported in Appendices H–J under the

Table G.3: Static and Dynamic designs at $n = 5000$: ATT$(2, 4)$. Five-fold cross-fitting for the proposed procedures. Target is the causal target, which equals one in the continuous designs. Bias and coverage (Cov.) of nominal 95% Wald intervals are computed over complete replications; MCSEs in parentheses. Every proposed replication is complete at this sample size, and each comparator completes in at least 0.988 of replications. The Static B and B+ cells at all sample sizes are in Appendix H.3, and the Dynamic ones in Appendix I.

			EEq-Hájek		TMLE-pw-Hájek		DR-nt		DR-ny	
Design	Scen.	Target	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.
Static continuous	A	1	−0.001	0.949	−0.001	0.941	0.002	0.952	0.004	0.951
			(0.003)	(0.005)	(0.003)	(0.005)	(0.003)	(0.005)	(0.003)	(0.005)
	B	1	0.002	0.940	0.003	0.905	0.005	0.944	−0.003	0.959
			(0.004)	(0.005)	(0.004)	(0.007)	(0.004)	(0.005)	(0.004)	(0.004)
	B+	1	0.005	0.918	0.007	0.812	0.000	0.949	−0.002	0.947
			(0.005)	(0.006)	(0.006)	(0.009)	(0.005)	(0.005)	(0.005)	(0.005)
	C1	1	0.387	0.621	0.243	0.706	1.321	0.149	0.846	0.242
			(0.004)	(0.011)	(0.004)	(0.010)	(0.008)	(0.008)	(0.007)	(0.010)
	C2	1	−0.003	0.907	−0.004	0.794	−0.003	0.955	−0.002	0.957
			(0.004)	(0.006)	(0.005)	(0.009)	(0.004)	(0.005)	(0.003)	(0.005)
Dynamic continuous	A	1	−0.001	0.940	−0.001	0.935	0.182	0.574	0.090	0.828
			(0.002)	(0.005)	(0.002)	(0.006)	(0.002)	(0.011)	(0.002)	(0.008)
	B	1	0.004	0.947	0.005	0.914	0.195	0.654	0.107	0.848
			(0.002)	(0.005)	(0.003)	(0.006)	(0.003)	(0.011)	(0.003)	(0.008)
	B+	1	−0.001	0.939	−0.001	0.838	0.223	0.722	0.129	0.880
			(0.003)	(0.005)	(0.004)	(0.008)	(0.004)	(0.010)	(0.004)	(0.007)
	C1	1	0.432	0.478	0.173	0.762	2.136	0.002	1.457	0.012
			(0.004)	(0.011)	(0.003)	(0.010)	(0.005)	(0.001)	(0.005)	(0.002)
	C2	1	−0.002	0.927	−0.002	0.869	0.168	0.567	0.089	0.835
			(0.002)	(0.006)	(0.002)	(0.008)	(0.002)	(0.011)	(0.002)	(0.008)
Static binary	A	0.058	−0.0005	0.959	−0.0006	0.956	−0.0330	0.761	−0.0181	0.860
			(0.0004)	(0.004)	(0.0004)	(0.005)	(0.0006)	(0.010)	(0.0005)	(0.008)
	B	0.048	0.0000	0.963	−0.0004	0.934	−0.0681	0.508	−0.0406	0.696
			(0.0005)	(0.004)	(0.0005)	(0.006)	(0.0008)	(0.011)	(0.0006)	(0.010)
	B+	0.044	−0.0016	0.959	−0.0037	0.851	−0.1522	0.285	−0.1126	0.387
			(0.0008)	(0.004)	(0.0011)	(0.008)	(0.0015)	(0.010)	(0.0012)	(0.011)
	C1	0.043	−0.0003	0.959	−0.0002	0.957	−0.0189	0.827	−0.0081	0.922
			(0.0003)	(0.004)	(0.0003)	(0.005)	(0.0004)	(0.008)	(0.0003)	(0.006)
	C2	0.051	−0.0014	0.968	−0.0041	0.878	−0.1657	0.062	−0.1318	0.070
			(0.0006)	(0.004)	(0.0008)	(0.007)	(0.0011)	(0.005)	(0.0009)	(0.006)
Dynamic binary	A	0.175	0.0000	0.947	0.0000	0.942	−0.0032	0.948	−0.0024	0.942
			(0.0007)	(0.005)	(0.0007)	(0.005)	(0.0007)	(0.005)	(0.0006)	(0.005)
	B	0.154	−0.0012	0.958	−0.0013	0.946	−0.0091	0.937	−0.0067	0.945
			(0.0008)	(0.004)	(0.0008)	(0.005)	(0.0008)	(0.005)	(0.0007)	(0.005)
	B+	0.133	−0.0010	0.952	−0.0013	0.866	−0.0154	0.946	−0.0128	0.947
			(0.0011)	(0.005)	(0.0014)	(0.008)	(0.0012)	(0.005)	(0.0011)	(0.005)
	C1	0.149	0.0003	0.949	0.0003	0.948	−0.0038	0.954	−0.0027	0.952
			(0.0005)	(0.005)	(0.0005)	(0.005)	(0.0006)	(0.005)	(0.0005)	(0.005)
	C2	0.185	0.0013	0.950	0.0014	0.945	−0.0033	0.948	−0.0025	0.949
			(0.0006)	(0.005)	(0.0007)	(0.005)	(0.0007)	(0.005)	(0.0006)	(0.005)

Table G.4: Static and Dynamic designs at $n = 5000$: ES(1). Conventions as in Table G.3.

			EEq-Hájek		TMLE-pw-Hájek		DR-nt		DR-ny	
Design	Scen.	Target	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.
Static continuous	A	1	−0.005	0.945	−0.005	0.942	−0.004	0.951	−0.003	0.940
			(0.002)	(0.005)	(0.002)	(0.005)	(0.002)	(0.005)	(0.002)	(0.005)
	B	1	0.001	0.946	0.001	0.935	0.002	0.943	0.002	0.949
			(0.002)	(0.005)	(0.002)	(0.006)	(0.003)	(0.005)	(0.002)	(0.005)
	B+	1	0.002	0.928	0.003	0.870	0.000	0.939	0.002	0.945
			(0.003)	(0.006)	(0.003)	(0.008)	(0.004)	(0.005)	(0.004)	(0.005)
	C1	1	0.059	0.941	0.040	0.940	0.583	0.197	0.209	0.503
			(0.002)	(0.005)	(0.002)	(0.005)	(0.004)	(0.009)	(0.003)	(0.011)
	C2	1	−0.001	0.911	−0.002	0.839	−0.002	0.957	−0.003	0.951
			(0.002)	(0.006)	(0.003)	(0.008)	(0.002)	(0.005)	(0.002)	(0.005)
Dynamic continuous	A	1	0.000	0.940	0.000	0.940	0.099	0.590	0.028	0.907
			(0.001)	(0.005)	(0.001)	(0.005)	(0.001)	(0.011)	(0.001)	(0.006)
	B	1	0.002	0.946	0.002	0.939	0.106	0.673	0.035	0.907
			(0.001)	(0.005)	(0.001)	(0.005)	(0.002)	(0.011)	(0.001)	(0.006)
	B+	1	−0.001	0.941	−0.001	0.898	0.131	0.723	0.040	0.919
			(0.002)	(0.005)	(0.002)	(0.007)	(0.002)	(0.010)	(0.002)	(0.006)
	C1	1	0.083	0.878	0.039	0.925	0.920	0.006	0.409	0.034
			(0.002)	(0.007)	(0.001)	(0.006)	(0.003)	(0.002)	(0.002)	(0.004)
	C2	1	−0.001	0.942	−0.002	0.926	0.094	0.567	0.027	0.917
			(0.001)	(0.005)	(0.001)	(0.006)	(0.001)	(0.011)	(0.001)	(0.006)
Static binary	A	0.073	−0.0001	0.951	−0.0002	0.949	−0.0165	0.808	−0.0045	0.922
			(0.0003)	(0.005)	(0.0003)	(0.005)	(0.0003)	(0.009)	(0.0003)	(0.006)
	B	0.066	0.0001	0.959	0.0000	0.946	−0.0353	0.611	−0.0100	0.891
			(0.0003)	(0.004)	(0.0003)	(0.005)	(0.0005)	(0.011)	(0.0003)	(0.007)
	B+	0.066	−0.0006	0.956	−0.0015	0.903	−0.0851	0.467	−0.0362	0.756
			(0.0005)	(0.005)	(0.0006)	(0.007)	(0.0010)	(0.011)	(0.0006)	(0.010)
	C1	0.056	0.0001	0.957	0.0001	0.957	−0.0101	0.866	−0.0016	0.947
			(0.0002)	(0.005)	(0.0002)	(0.005)	(0.0002)	(0.008)	(0.0002)	(0.005)
	C2	0.075	−0.0006	0.975	−0.0017	0.915	−0.0911	0.188	−0.0539	0.278
			(0.0004)	(0.003)	(0.0005)	(0.006)	(0.0007)	(0.009)	(0.0005)	(0.010)
Dynamic binary	A	0.184	−0.0004	0.946	−0.0004	0.946	−0.0020	0.948	−0.0010	0.946
			(0.0004)	(0.005)	(0.0004)	(0.005)	(0.0004)	(0.005)	(0.0004)	(0.005)
	B	0.168	−0.0005	0.948	−0.0005	0.947	−0.0043	0.946	−0.0020	0.944
			(0.0005)	(0.005)	(0.0005)	(0.005)	(0.0005)	(0.005)	(0.0004)	(0.005)
	B+	0.152	−0.0001	0.946	−0.0001	0.916	−0.0084	0.941	−0.0044	0.948
			(0.0007)	(0.005)	(0.0008)	(0.006)	(0.0007)	(0.005)	(0.0006)	(0.005)
	C1	0.162	0.0000	0.950	0.0000	0.949	−0.0019	0.952	−0.0007	0.951
			(0.0004)	(0.005)	(0.0004)	(0.005)	(0.0003)	(0.005)	(0.0003)	(0.005)
	C2	0.193	0.0009	0.953	0.0009	0.951	−0.0014	0.954	−0.0003	0.952
			(0.0004)	(0.005)	(0.0004)	(0.005)	(0.0004)	(0.005)	(0.0004)	(0.005)

printed names described here.

H Additional Simulation Results: Normalization, Architecture, Cross-Fitting, Overlap, and Comparator Inference

This section compares the proposed variants with one another (Appendix H.1), reports the setting without cross-fitting (Appendix H.2) and the Static designs under B and B+ (Appendix H.3), reports CompactBinary in full (Appendix H.4), and refits the comparators with analytic standard errors (Appendix H.5).

H.1 Normalization, Architecture, and the Two-Fluctuation Diagnostic

The comparisons in this subsection use the 40 cells of the Static and Dynamic designs at a given sample size (four designs, five scenarios, two targets) with five-fold cross-fitting. They describe coverage differences larger than 0.01 in absolute value, about two pointwise MCSEs of a single coverage rate near 0.95. The threshold is descriptive: the procedures are evaluated on the same panels, so the Monte Carlo error of a difference is generally smaller than the two MCSEs combined, and we attach no test to it. Every underlying cell is in Appendix J.

Normalization. At $n = 500$ and $n = 5000$ no HT variant covers more often than its Hájek counterpart by more than 0.01. At $n = 5000$ HT coverage is lower by more than 0.01 in 16 of the 40 cells for EEq, in 18 for the pairwise targeted estimator, and in 20 for the cohort-wise one; outside continuous C1 the largest deficit for EEq is 0.089, in continuous Static C2. At $n = 100$ the comparison is mixed: HT coverage is higher by more than 0.01 in 7 cells for EEq, 16 for the pairwise and 19 for the cohort-wise targeted estimator, and lower by more than 0.01 in 10, 7, and 6. Corollary 3.2(ii) asserts root-n equivalence of the HT and Hájek one-step estimators, and Proposition 4.1 together with Theorem 4.2(ii) gives the same equivalence for the pairwise targeted estimators, under conditions that include L_2-consistency of the fitted ratio for $r^0_{g,k}$. The corresponding cohort-wise bridges and their conditions are stated in Appendix F.4; together with Corollary 3.2(ii), they give the same equivalence for the cohort-wise estimators. Because $r^0_{g,k}$ exceeds M with positive probability on $\{G > k\}$ in every Static and Dynamic design (Appendix G.2), the capped ratio cannot converge to it, so these results do not bear on these differences. Normalization does not affect completion: in every cell each HT variant completes exactly as often as its Hájek counterpart, consistent with failures in the nuisance fits that the two share.

Architecture. Appendix F.2 gives the common-update decomposition, and Appendix F.3 gives the exact endpoint representations. Theorem F.1(ii) ties each one-fluctuation cohort-wise estimator to the one-step estimator with the same normalization; Proposition 4.1 gives the pairwise Hájek link, and the proof of Theorem 4.2(ii) gives the pairwise HT link. Each link requires ratio consistency, which the cap prevents in these Static and Dynamic designs. The three variants available under both architectures are one fluctuation with Hájek or HT normalization and the Hájek two-fluctuation diagnostic. For all three, cohort-wise targeting never covers more often than pairwise targeting by more than 0.01, at any sample size. For the one-fluctuation Hájek estimator it covers less often by more than 0.01 in 29, 15, and 3 of the 40 cells at $n = 100$, 500, and 5000 (25, 10, and 11 under HT normalization, and 29, 15, and 6 for the diagnostic). At $n = 5000$ its coverage difference ranges from -0.066 to 0.004, and its RMSE is 0.997–1.095 times the pairwise RMSE. Cohort-wise targeting, which requires hazard fits through period T rather than through t, also completes no more often than pairwise targeting in any cell, for every variant, and in the two worst cells of Section 5.4 its one-fluctuation Hájek estimator completes in 0.399 and 0.282 of replications, against 0.645 and 0.343 under pairwise targeting.

Two fluctuations. Appendix F.5 gives the diagnostic's control-only representations and Gram conditions. At each sample size there are 80 comparisons of the Hájek two-fluctuation diagnostic with the one-fluctuation estimator of the same architecture (40 cells, two architectures). The diagnostic covers less often by more than 0.01 in 65, 48, and 30 of them at $n = 100$, 500, and 5000. It covers more often by more than 0.01 only at $n = 5000$, in 6 comparisons, all in continuous C1, where every featured procedure and comparator is biased (Appendix G.4) and the two-fluctuation estimates are less so.

H.2 No Cross-Fitting

The full-sample setting ($L = 1$) is covered for CompactBinary by Corollary C.1, under the other conditions of the invoked result. For the Gaussian-covariate stress designs it remains an implementation ablation: the finite-support extension does not apply, and uniform overlap fails in either fitting setting. Table H.1 reports completion at $n = 100$ in both settings for EEq-Hájek and TMLE-pw-Hájek in every design, CompactBinary included. For every proposed procedure and every cell, completion without cross-fitting is at least as high as with it. At $n = 100$ it is higher by up to 0.457 for the two procedures in the table and by up to 0.481 for the cohort-wise procedures, consistent with the failure mechanism described in Section 5.4. At $n = 5000$ the two settings give nearly identical results outside continuous C1: for the two procedures, coverage differs by at most 0.005 and bias by at most 0.0003.

H.3 Static Designs under Confounding and Strong Confounding

Tables H.2 and H.3 report the Static designs under B and B+ in the format of Tables I.2 and I.3. Availability follows the Dynamic pattern. At $n = 100$ the featured procedures complete in 0.923–0.989 of replications under B and 0.687–0.900 under B+, and the comparators in 0.577 and 0.188–0.189; from $n = 500$ on, every reported procedure completes in at least 0.998 of replications.

At $n = 5000$ the two outcome families differ. With continuous outcomes the comparators are centered, consistent with their correctly specified outcome regression (Appendix G.3), and cover 0.939–0.959, whereas both proposed procedures have low empirical coverage under B+: EEq-Hájek covers 0.918 and 0.928 for ATT$(2, 4)$ and ES(1), and TMLE-pw-Hájek 0.812 and 0.870. With binary outcomes the comparators exhibit bias in these simulations, consistent with misspecification of both working models there, and their empirical coverage in the B and B+ cells is 0.285–0.891. EEq-Hájek covers 0.956–0.963 in these cells, and TMLE-pw-Hájek covers 0.851 and 0.903 under B+. Wherever a proposed procedure has low empirical coverage in these cells, its standard-error ratio in Table G.2 is below one.

Table H.1: Completion at $n = 100$ with five-fold cross-fitting ($L = 5$) and without cross-fitting ($L = 1$, nuisances fitted on the full sample), on the same simulated panels. Completion is the share of the 2,000 requested replications that returned a finite estimate and a finite, positive standard error. CompactBinary routes are defined in Section 5.2.

		ATT(2, 4)				ES(1)			
		EEq-Hájek		TMLE-pw-Hájek		EEq-Hájek		TMLE-pw-Hájek	
Design	Scen.	$L = 5$	$L = 1$	$L = 5$	$L = 1$	$L = 5$	$L = 1$	$L = 5$	$L = 1$
Static continuous	A	0.997	1.000	0.997	1.000	0.982	0.997	0.981	0.997
	B	0.989	0.998	0.989	0.998	0.937	0.991	0.936	0.991
	B+	0.900	0.984	0.900	0.984	0.703	0.936	0.697	0.936
	C1	0.999	1.000	0.999	1.000	0.984	0.999	0.984	0.999
	C2	0.977	0.997	0.977	0.997	0.891	0.981	0.890	0.981
Dynamic continuous	A	0.986	0.999	0.986	0.999	0.934	0.994	0.929	0.994
	B	0.925	0.990	0.925	0.990	0.786	0.967	0.772	0.967
	B+	0.645	0.931	0.645	0.931	0.382	0.809	0.352	0.809
	C1	0.989	0.999	0.989	0.999	0.932	0.993	0.928	0.993
	C2	0.986	0.998	0.986	0.998	0.945	0.995	0.937	0.995
Static binary	A	0.988	0.999	0.988	0.999	0.936	0.991	0.936	0.991
	B	0.983	0.998	0.983	0.998	0.925	0.990	0.923	0.990
	B+	0.895	0.983	0.895	0.983	0.691	0.934	0.687	0.932
	C1	0.983	1.000	0.983	1.000	0.935	0.994	0.935	0.994
	C2	0.955	0.996	0.955	0.996	0.810	0.974	0.809	0.973
Dynamic binary	A	0.993	1.000	0.993	1.000	0.953	0.995	0.945	0.994
	B	0.941	0.993	0.941	0.993	0.791	0.972	0.770	0.958
	B+	0.692	0.938	0.692	0.938	0.373	0.818	0.343	0.777
	C1	0.987	1.000	0.987	1.000	0.952	0.996	0.942	0.992
	C2	0.995	1.000	0.995	1.000	0.953	0.995	0.951	0.995
CompactBinary	A	0.967	0.996	0.964	0.996	0.945	0.994	0.934	0.994
	C1	0.968	0.997	0.965	0.997	0.947	0.996	0.936	0.996
	C2	0.990	0.998	0.990	0.998	0.979	0.997	0.979	0.997
	C12	0.992	1.000	0.992	1.000	0.979	0.999	0.978	0.999

Table H.2: Stress scenarios B and B+ in the continuous Static design; five-fold cross-fitting for the proposed procedures. Bias, coverage (Cov.), and RMSE are conditional on completion (Compl.), with MCSEs in parentheses, and do not rank procedures whose completion differs.

			ATT(2, 4)				ES(1)			
Design	n	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	100	EEq-Hájek	0.989	−0.006	0.930	1.249	0.937	0.011	0.932	0.779
				(0.028)	(0.006)	(0.022)		(0.018)	(0.006)	(0.013)
		TMLE-pw-Hájek	0.989	0.002	0.826	1.485	0.936	0.011	0.883	0.820
				(0.033)	(0.009)	(0.029)		(0.019)	(0.007)	(0.014)
		DR-nt	0.577	−0.004	0.871	1.129	0.577	0.006	0.884	0.656
				(0.033)	(0.010)	(0.026)		(0.019)	(0.009)	(0.015)
		DR-ny	0.577	−0.037	0.879	1.022	0.577	0.006	0.914	0.604
				(0.030)	(0.010)	(0.026)		(0.018)	(0.008)	(0.014)
	500	EEq-Hájek	1.000	−0.019	0.931	0.534	1.000	−0.007	0.944	0.326
				(0.012)	(0.006)	(0.009)		(0.007)	(0.005)	(0.005)
		TMLE-pw-Hájek	1.000	−0.021	0.881	0.605	1.000	−0.007	0.918	0.344
				(0.014)	(0.007)	(0.010)		(0.008)	(0.006)	(0.005)
		DR-nt	1.000	−0.002	0.916	0.575	1.000	−0.001	0.940	0.331
				(0.013)	(0.006)	(0.015)		(0.007)	(0.005)	(0.007)
		DR-ny	1.000	−0.009	0.922	0.532	1.000	0.001	0.933	0.302
				(0.012)	(0.006)	(0.011)		(0.007)	(0.006)	(0.006)
	5,000	EEq-Hájek	1.000	0.002	0.940	0.164	1.000	0.001	0.946	0.102
				(0.004)	(0.005)	(0.003)		(0.002)	(0.005)	(0.002)
		TMLE-pw-Hájek	1.000	0.003	0.905	0.185	1.000	0.001	0.935	0.107
				(0.004)	(0.007)	(0.003)		(0.002)	(0.006)	(0.002)
		DR-nt	1.000	0.005	0.944	0.200	0.999	0.002	0.943	0.119
				(0.004)	(0.005)	(0.004)		(0.003)	(0.005)	(0.002)
		DR-ny	1.000	−0.003	0.959	0.177	1.000	0.002	0.949	0.099
				(0.004)	(0.004)	(0.003)		(0.002)	(0.005)	(0.002)
B+	100	EEq-Hájek	0.900	0.018	0.904	1.472	0.703	−0.005	0.900	0.973
				(0.035)	(0.007)	(0.026)		(0.026)	(0.008)	(0.020)
		TMLE-pw-Hájek	0.900	0.029	0.697	2.181	0.697	−0.003	0.754	1.222
				(0.051)	(0.011)	(0.063)		(0.033)	(0.012)	(0.027)
		DR-nt	0.189	0.001	0.767	1.402	0.188	−0.014	0.806	0.936
				(0.072)	(0.022)	(0.053)		(0.048)	(0.020)	(0.037)
		DR-ny	0.189	0.032	0.809	1.154	0.188	−0.007	0.875	0.730
				(0.059)	(0.020)	(0.047)		(0.038)	(0.017)	(0.034)
	500	EEq-Hájek	1.000	0.009	0.921	0.640	1.000	0.007	0.925	0.419
				(0.014)	(0.006)	(0.010)		(0.009)	(0.006)	(0.006)
		TMLE-pw-Hájek	1.000	0.006	0.792	0.846	1.000	0.005	0.853	0.499
				(0.019)	(0.009)	(0.014)		(0.011)	(0.008)	(0.008)
		DR-nt	1.000	0.014	0.894	0.704	0.999	0.010	0.888	0.482
				(0.016)	(0.007)	(0.013)		(0.011)	(0.007)	(0.009)
		DR-ny	1.000	0.008	0.906	0.628	1.000	0.010	0.920	0.438
				(0.014)	(0.007)	(0.011)		(0.010)	(0.006)	(0.009)
	5,000	EEq-Hájek	1.000	0.005	0.918	0.202	1.000	0.002	0.928	0.133
				(0.005)	(0.006)	(0.003)		(0.003)	(0.006)	(0.002)
		TMLE-pw-Hájek	1.000	0.007	0.812	0.262	1.000	0.003	0.870	0.156
				(0.006)	(0.009)	(0.004)		(0.003)	(0.008)	(0.003)
		DR-nt	1.000	0.000	0.949	0.245	0.998	0.000	0.939	0.180
				(0.005)	(0.005)	(0.004)		(0.004)	(0.005)	(0.003)
		DR-ny	1.000	−0.002	0.947	0.234	1.000	0.002	0.945	0.162
				(0.005)	(0.005)	(0.004)		(0.004)	(0.005)	(0.003)

Table H.3: Stress scenarios B and B+ in the binary Static design. Conventions as in Table H.2.

			ATT(2, 4)				ES(1)			
Design	n	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	100	EEq-Hájek	0.983	0.0061	0.966	0.1635	0.925	0.0078	0.965	0.1079
				(0.0037)	(0.004)	(0.0038)		(0.0025)	(0.004)	(0.0023)
		TMLE-pw-Hájek	0.983	0.0043	0.916	0.2020	0.923	0.0070	0.943	0.1179
				(0.0046)	(0.006)	(0.0063)		(0.0027)	(0.005)	(0.0026)
		DR-nt	0.577	−0.0744	0.909	0.2596	0.577	−0.0398	0.923	0.1548
				(0.0073)	(0.008)	(0.0065)		(0.0044)	(0.008)	(0.0041)
		DR-ny	0.577	−0.0396	0.942	0.1894	0.577	−0.0071	0.956	0.1020
				(0.0055)	(0.007)	(0.0053)		(0.0030)	(0.006)	(0.0028)
	500	EEq-Hájek	1.000	−0.0036	0.964	0.0660	1.000	−0.0028	0.962	0.0447
				(0.0015)	(0.004)	(0.0011)		(0.0010)	(0.004)	(0.0007)
		TMLE-pw-Hájek	1.000	−0.0043	0.934	0.0753	1.000	−0.0031	0.950	0.0466
				(0.0017)	(0.006)	(0.0013)		(0.0010)	(0.005)	(0.0008)
		DR-nt	1.000	−0.0767	0.896	0.1368	1.000	−0.0412	0.913	0.0799
				(0.0025)	(0.007)	(0.0024)		(0.0015)	(0.006)	(0.0015)
		DR-ny	1.000	−0.0463	0.932	0.0999	1.000	−0.0145	0.950	0.0460
				(0.0020)	(0.006)	(0.0019)		(0.0010)	(0.005)	(0.0008)
	5,000	EEq-Hájek	1.000	0.0000	0.963	0.0203	1.000	0.0001	0.959	0.0140
				(0.0005)	(0.004)	(0.0003)		(0.0003)	(0.004)	(0.0002)
		TMLE-pw-Hájek	1.000	−0.0004	0.934	0.0230	1.000	0.0000	0.946	0.0145
				(0.0005)	(0.006)	(0.0004)		(0.0003)	(0.005)	(0.0002)
		DR-nt	1.000	−0.0681	0.508	0.0767	1.000	−0.0353	0.611	0.0414
				(0.0008)	(0.011)	(0.0008)		(0.0005)	(0.011)	(0.0005)
		DR-ny	1.000	−0.0406	0.696	0.0493	1.000	−0.0100	0.891	0.0173
				(0.0006)	(0.010)	(0.0006)		(0.0003)	(0.007)	(0.0003)
B+	100	EEq-Hájek	0.895	−0.0102	0.978	0.2836	0.691	−0.0015	0.967	0.1839
				(0.0067)	(0.003)	(0.0061)		(0.0050)	(0.005)	(0.0041)
		TMLE-pw-Hájek	0.895	−0.0188	0.770	0.4557	0.687	−0.0058	0.852	0.2527
				(0.0108)	(0.010)	(0.0116)		(0.0068)	(0.010)	(0.0070)
		DR-nt	0.188	−0.1899	0.811	0.4451	0.188	−0.1245	0.824	0.3032
				(0.0208)	(0.020)	(0.0172)		(0.0143)	(0.020)	(0.0116)
		DR-ny	0.189	−0.1067	0.878	0.3090	0.188	−0.0348	0.912	0.1797
				(0.0150)	(0.017)	(0.0128)		(0.0091)	(0.015)	(0.0072)
	500	EEq-Hájek	1.000	−0.0037	0.972	0.1098	1.000	0.0002	0.966	0.0735
				(0.0025)	(0.004)	(0.0019)		(0.0016)	(0.004)	(0.0012)
		TMLE-pw-Hájek	1.000	−0.0067	0.852	0.1566	1.000	−0.0006	0.910	0.0904
				(0.0035)	(0.008)	(0.0028)		(0.0020)	(0.006)	(0.0014)
		DR-nt	1.000	−0.1622	0.814	0.2451	1.000	−0.0871	0.858	0.1564
				(0.0041)	(0.009)	(0.0039)		(0.0029)	(0.008)	(0.0026)
		DR-ny	1.000	−0.1239	0.859	0.2002	1.000	−0.0421	0.924	0.0971
				(0.0035)	(0.008)	(0.0033)		(0.0020)	(0.006)	(0.0018)
	5,000	EEq-Hájek	1.000	−0.0016	0.959	0.0347	1.000	−0.0006	0.956	0.0235
				(0.0008)	(0.004)	(0.0005)		(0.0005)	(0.005)	(0.0004)
		TMLE-pw-Hájek	1.000	−0.0037	0.851	0.0484	1.000	−0.0015	0.903	0.0284
				(0.0011)	(0.008)	(0.0008)		(0.0006)	(0.007)	(0.0004)
		DR-nt	1.000	−0.1522	0.285	0.1655	1.000	−0.0851	0.467	0.0968
				(0.0015)	(0.010)	(0.0014)		(0.0010)	(0.011)	(0.0010)
		DR-ny	1.000	−0.1126	0.387	0.1247	1.000	−0.0362	0.756	0.0465
				(0.0012)	(0.011)	(0.0012)		(0.0006)	(0.010)	(0.0006)

H.4 CompactBinary: Design, Route C12, and Complete Results

Design. The CompactBinary law is displayed in (5.2), and Table I gives the proposed procedures' working-model routes A, C1, C2, and C12. In these four routes the effects vary across (g, t), so the aggregation-weight score in (2.6) is not identically zero; it is retained in the standard errors for ES(1).

Population limits. For CompactBinary routes A, C1, C2, and C12, we compute both targets exactly (a separate enumeration that shares no code with the simulation reproduces them to machine precision) and solve every population working-model and targeting equation. The resulting limits of all proposed procedures in CompactBinary routes A, C1, and C2 coincide with the causal targets, with discrepancies below 10^{-15}, so these routes form the validation panel. The comparators' limits in these same routes, set by the `did` package's own working models, differ from the targets (the largest discrepancy is 4.8×10^{-4}), so the panel's premise does not extend to them.

Finite-sample calibration. At $n = 5000$, across the three validation routes A, C1, and C2, both targets, both cross-fitting settings, and the four pooled procedures (48 cells), the largest ratio of absolute bias to its MCSE is 2.46, and coverage ranges from 0.947 to 0.958. The coverage values at every reported sample size, including route A, are empirical findings. In the three validation routes of Table II, at $n = 100$ completion ranges from 0.934 to 0.990, and the ATT$(2, 4)$ intervals have low empirical coverage in routes A and C1 (0.918–0.920); both coverage shortfalls are absent by $n = 500$, and the RMSE falls roughly as $n^{-1/2}$. The lower $n = 100$ coverage in A and C1, where the expected cohort-2 count is 12.25 (compared with 17.55 in C2 and C12), is consistent with limited treated-cohort information; this comparison does not isolate its cause.

In route C12, each of the four pooled procedures converges to a method-specific pseudo-limit that exceeds the causal target by 0.0130 for ATT$(2, 4)$ and 0.0083 for ES(1) (to four decimals). Table H.4 reports this route, in which both proposed working models are misspecified, for the two procedures featured in Table II. Bias is measured from the causal target, and Bias − drift measures the discrepancy from the procedure's own population limit. The drift, defined in the caption, is computed on the exact law and is a fixed number, so the MCSE of that discrepancy equals the MCSE of the bias. At $n = 5000$ the coverage of both featured procedures, 0.914 and 0.918, is consistent with a centering shift equal to the observed discrepancy from the causal target, 0.0146 and 0.0087: treating the estimator as normal with the observed mean and Monte Carlo standard deviation, and the interval half-width as 1.96 times the average estimated standard error, gives 0.916 and 0.922; with the population drift alone in place of the observed discrepancy, the same calculation gives 0.923 and 0.925. The estimates exceed the pseudo-limits by 0.0016 and 0.0004, or 2.6 and 1.1 MCSEs; we report these residuals without multiplicity adjustment and draw no conclusion from them.

Tables H.5–H.12 report every procedure in every route, for both targets and both cross-fitting settings. Validation in the sense of Section 5.2 applies only to the proposed procedures in routes A, C1, and C2. There, with five-fold cross-fitting, the RMSE of EEq-Hájek and TMLE-pw-Hájek falls by factors of 2.2–2.4 from $n = 100$ to $n = 500$ and 3.1–3.3 from $n = 500$ to $n = 5000$, close to the factors $5^{1/2} \approx 2.2$ and $10^{1/2} \approx 3.2$ used here as a descriptive root-n benchmark. The theoretical scope of the routes is described in Section 5.2.

Table H.4: CompactBinary route C12, five-fold cross-fitting. Drift is the population pseudo-limit minus the causal target; Bias − drift is computed before rounding. MCSEs are in parentheses; other conventions are as in Table II. The MCSE of the RMSE is defined in Appendix G.

n	Procedure	Compl.	Bias	Drift	Bias − drift	Cov.	RMSE
Panel A: ATT$(2,4) = 0.116$							
100	EEq-Hájek	0.992	0.0085	0.0130	−0.0045	0.940	0.2005
			(0.0045)		(0.0045)	(0.005)	(0.0031)
	TMLE-pw-Hájek	0.992	0.0087	0.0130	−0.0043	0.940	0.2003
			(0.0045)		(0.0045)	(0.005)	(0.0031)
500	EEq-Hájek	1.000	0.0073	0.0130	−0.0057	0.942	0.0876
			(0.0020)		(0.0020)	(0.005)	(0.0014)
	TMLE-pw-Hájek	1.000	0.0073	0.0130	−0.0057	0.942	0.0876
			(0.0020)		(0.0020)	(0.005)	(0.0014)
5,000	EEq-Hájek	1.000	0.0146	0.0130	0.0016	0.914	0.0305
			(0.0006)		(0.0006)	(0.006)	(0.0005)
	TMLE-pw-Hájek	1.000	0.0146	0.0130	0.0016	0.914	0.0305
			(0.0006)		(0.0006)	(0.006)	(0.0005)
Panel B: ES$(1) = 0.120$							
100	EEq-Hájek	0.979	0.0060	0.0083	−0.0022	0.952	0.1284
			(0.0029)		(0.0029)	(0.005)	(0.0021)
	TMLE-pw-Hájek	0.978	0.0063	0.0083	−0.0020	0.952	0.1282
			(0.0029)		(0.0029)	(0.005)	(0.0021)
500	EEq-Hájek	1.000	0.0069	0.0083	−0.0013	0.944	0.0568
			(0.0013)		(0.0013)	(0.005)	(0.0009)
	TMLE-pw-Hájek	1.000	0.0069	0.0083	−0.0013	0.944	0.0568
			(0.0013)		(0.0013)	(0.005)	(0.0009)
5,000	EEq-Hájek	1.000	0.0087	0.0083	0.0004	0.918	0.0194
			(0.0004)		(0.0004)	(0.006)	(0.0003)
	TMLE-pw-Hájek	1.000	0.0087	0.0083	0.0004	0.918	0.0194
			(0.0004)		(0.0004)	(0.006)	(0.0003)

Table H.5: CompactBinary route A (both proposed working models correct): all procedures, with ATT$(2,4) = 0.126$. $L = 5$ is five-fold cross-fitting; $L = 1$ fits the nuisances on the full sample. Compl. is the completion share of the 2,000 requested replications; bias, coverage (Cov.) of nominal 95% Wald intervals, and RMSE are computed over complete replications, with MCSEs in parentheses. The comparators do not cross-fit; their results coincide in the two settings and are listed once, under $L = 5$.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.967	−0.0036 (0.0052)	0.919 (0.006)	0.2267 (0.0038)
		EEq-Hájek	0.967	−0.0036 (0.0052)	0.918 (0.006)	0.2265 (0.0038)
		TMLE-cw-HT	0.942	−0.0035 (0.0052)	0.920 (0.006)	0.2258 (0.0037)
		TMLE-cw-Hájek	0.942	−0.0035 (0.0052)	0.919 (0.006)	0.2256 (0.0037)
		TMLE-pw-HT	0.964	−0.0041 (0.0051)	0.920 (0.006)	0.2250 (0.0036)
		TMLE-pw-Hájek	0.964	−0.0041 (0.0051)	0.919 (0.006)	0.2249 (0.0036)
		TMLE-2eps-cw-Hájek	0.942	−0.0036 (0.0052)	0.921 (0.006)	0.2251 (0.0036)
		TMLE-2eps-pw-Hájek	0.964	−0.0042 (0.0051)	0.921 (0.006)	0.2244 (0.0036)
		DR-nt	0.891	−0.0027 (0.0055)	0.926 (0.006)	0.2310 (0.0041)
		DR-ny	0.891	−0.0028 (0.0053)	0.923 (0.006)	0.2255 (0.0040)
100	1	EEq-HT	0.996	−0.0035 (0.0051)	0.916 (0.006)	0.2266 (0.0038)
		EEq-Hájek	0.996	−0.0035 (0.0051)	0.916 (0.006)	0.2267 (0.0038)
		TMLE-cw-HT	0.994	−0.0033 (0.0051)	0.915 (0.006)	0.2268 (0.0038)
		TMLE-cw-Hájek	0.994	−0.0033 (0.0051)	0.915 (0.006)	0.2268 (0.0038)
		TMLE-pw-HT	0.996	−0.0035 (0.0051)	0.916 (0.006)	0.2267 (0.0038)
		TMLE-pw-Hájek	0.996	−0.0035 (0.0051)	0.916 (0.006)	0.2267 (0.0038)
		TMLE-2eps-cw-Hájek	0.994	−0.0033 (0.0051)	0.916 (0.006)	0.2269 (0.0038)
		TMLE-2eps-pw-Hájek	0.996	−0.0035 (0.0051)	0.916 (0.006)	0.2267 (0.0038)
500	5	EEq-HT	1.000	0.0027 (0.0021)	0.943 (0.005)	0.0959 (0.0015)
		EEq-Hájek	1.000	0.0027 (0.0021)	0.942 (0.005)	0.0959 (0.0015)
		TMLE-cw-HT	1.000	0.0027 (0.0021)	0.943 (0.005)	0.0959 (0.0015)
		TMLE-cw-Hájek	1.000	0.0027 (0.0021)	0.942 (0.005)	0.0959 (0.0015)
		TMLE-pw-HT	1.000	0.0027 (0.0021)	0.943 (0.005)	0.0959 (0.0015)
		TMLE-pw-Hájek	1.000	0.0027 (0.0021)	0.942 (0.005)	0.0959 (0.0015)
		TMLE-2eps-cw-Hájek	1.000	0.0027 (0.0021)	0.942 (0.005)	0.0959 (0.0015)
		TMLE-2eps-pw-Hájek	1.000	0.0027 (0.0021)	0.942 (0.005)	0.0959 (0.0015)
		DR-nt	1.000	0.0027 (0.0022)	0.944 (0.005)	0.0981 (0.0015)
		DR-ny	1.000	0.0029 (0.0022)	0.945 (0.005)	0.0965 (0.0015)
500	1	EEq-HT	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		EEq-Hájek	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		TMLE-cw-HT	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		TMLE-cw-Hájek	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		TMLE-pw-HT	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		TMLE-pw-Hájek	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		TMLE-2eps-cw-Hájek	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
		TMLE-2eps-pw-Hájek	1.000	0.0030 (0.0022)	0.941 (0.005)	0.0962 (0.0015)
5,000	5	EEq-HT	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		EEq-Hájek	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		TMLE-cw-HT	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		TMLE-cw-Hájek	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		TMLE-pw-HT	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		TMLE-pw-Hájek	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		TMLE-2eps-cw-Hájek	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		TMLE-2eps-pw-Hájek	1.000	−0.0005 (0.0007)	0.949 (0.005)	0.0308 (0.0005)
		DR-nt	1.000	−0.0004 (0.0007)	0.947 (0.005)	0.0313 (0.0005)
		DR-ny	1.000	−0.0004 (0.0007)	0.949 (0.005)	0.0307 (0.0005)
5,000	1	EEq-HT	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		EEq-Hájek	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		TMLE-cw-HT	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		TMLE-cw-Hájek	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		TMLE-pw-HT	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		TMLE-pw-Hájek	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		TMLE-2eps-cw-Hájek	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)
		TMLE-2eps-pw-Hájek	1.000	−0.0005 (0.0007)	0.948 (0.005)	0.0308 (0.0005)

Table H.6: CompactBinary route A (both proposed working models correct): all procedures, with ES(1) = 0.131. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.945	−0.0006 (0.0032)	0.941 (0.005)	0.1372 (0.0022)
		EEq-Hájek	0.945	−0.0006 (0.0032)	0.941 (0.005)	0.1371 (0.0022)
		TMLE-cw-HT	0.933	−0.0010 (0.0032)	0.941 (0.005)	0.1368 (0.0022)
		TMLE-cw-Hájek	0.933	−0.0010 (0.0032)	0.941 (0.005)	0.1367 (0.0022)
		TMLE-pw-HT	0.934	−0.0010 (0.0032)	0.941 (0.005)	0.1367 (0.0022)
		TMLE-pw-Hájek	0.934	−0.0010 (0.0032)	0.941 (0.005)	0.1366 (0.0022)
		TMLE-2eps-cw-Hájek	0.933	−0.0010 (0.0032)	0.940 (0.005)	0.1362 (0.0022)
		TMLE-2eps-pw-Hájek	0.934	−0.0010 (0.0032)	0.940 (0.005)	0.1362 (0.0022)
		DR-nt	0.891	−0.0019 (0.0030)	0.945 (0.005)	0.1272 (0.0021)
		DR-ny	0.891	−0.0013 (0.0031)	0.943 (0.006)	0.1295 (0.0021)
100	1	EEq-HT	0.994	−0.0005 (0.0030)	0.942 (0.005)	0.1351 (0.0021)
		EEq-Hájek	0.994	−0.0005 (0.0030)	0.942 (0.005)	0.1351 (0.0021)
		TMLE-cw-HT	0.993	−0.0005 (0.0030)	0.942 (0.005)	0.1352 (0.0021)
		TMLE-cw-Hájek	0.993	−0.0005 (0.0030)	0.942 (0.005)	0.1352 (0.0021)
		TMLE-pw-HT	0.994	−0.0005 (0.0030)	0.942 (0.005)	0.1351 (0.0021)
		TMLE-pw-Hájek	0.994	−0.0005 (0.0030)	0.942 (0.005)	0.1351 (0.0021)
		TMLE-2eps-cw-Hájek	0.993	−0.0005 (0.0030)	0.941 (0.005)	0.1352 (0.0021)
		TMLE-2eps-pw-Hájek	0.994	−0.0005 (0.0030)	0.941 (0.005)	0.1351 (0.0021)
500	5	EEq-HT	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		EEq-Hájek	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		TMLE-cw-HT	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		TMLE-cw-Hájek	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		TMLE-pw-HT	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		TMLE-pw-Hájek	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		TMLE-2eps-cw-Hájek	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		TMLE-2eps-pw-Hájek	1.000	−0.0008 (0.0013)	0.954 (0.005)	0.0599 (0.0010)
		DR-nt	1.000	−0.0006 (0.0013)	0.959 (0.004)	0.0563 (0.0009)
		DR-ny	1.000	−0.0009 (0.0013)	0.953 (0.005)	0.0570 (0.0009)
500	1	EEq-HT	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		EEq-Hájek	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		TMLE-cw-HT	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		TMLE-cw-Hájek	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		TMLE-pw-HT	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		TMLE-pw-Hájek	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		TMLE-2eps-cw-Hájek	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
		TMLE-2eps-pw-Hájek	1.000	−0.0007 (0.0013)	0.954 (0.005)	0.0597 (0.0010)
5,000	5	EEq-HT	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		EEq-Hájek	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		TMLE-cw-HT	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		TMLE-cw-Hájek	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		TMLE-pw-HT	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		TMLE-pw-Hájek	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	−0.0003 (0.0004)	0.947 (0.005)	0.0189 (0.0003)
		DR-nt	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0177 (0.0003)
		DR-ny	1.000	−0.0003 (0.0004)	0.949 (0.005)	0.0180 (0.0003)
5,000	1	EEq-HT	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		EEq-Hájek	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		TMLE-cw-HT	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		TMLE-cw-Hájek	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		TMLE-pw-HT	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		TMLE-pw-Hájek	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0189 (0.0003)

Table H.7: CompactBinary route C1 (proposed outcome working model misspecified): all procedures, with $\mathrm{ATT}(2,4) = 0.118$. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.968	−0.0043 (0.0051)	0.922 (0.006)	0.2234 (0.0035)
		EEq-Hájek	0.968	−0.0040 (0.0051)	0.920 (0.006)	0.2232 (0.0035)
		TMLE-cw-HT	0.945	−0.0039 (0.0052)	0.921 (0.006)	0.2238 (0.0035)
		TMLE-cw-Hájek	0.945	−0.0037 (0.0051)	0.920 (0.006)	0.2237 (0.0035)
		TMLE-pw-HT	0.965	−0.0036 (0.0051)	0.922 (0.006)	0.2230 (0.0035)
		TMLE-pw-Hájek	0.965	−0.0035 (0.0051)	0.920 (0.006)	0.2229 (0.0035)
		TMLE-2eps-cw-Hájek	0.945	−0.0029 (0.0051)	0.921 (0.006)	0.2230 (0.0035)
		TMLE-2eps-pw-Hájek	0.965	−0.0027 (0.0051)	0.921 (0.006)	0.2223 (0.0035)
		DR-nt	0.882	0.0012 (0.0054)	0.929 (0.006)	0.2276 (0.0038)
		DR-ny	0.882	0.0017 (0.0053)	0.918 (0.007)	0.2217 (0.0037)
100	1	EEq-HT	0.997	−0.0003 (0.0050)	0.917 (0.006)	0.2210 (0.0035)
		EEq-Hájek	0.997	−0.0003 (0.0050)	0.917 (0.006)	0.2210 (0.0035)
		TMLE-cw-HT	0.996	−0.0002 (0.0050)	0.917 (0.006)	0.2211 (0.0035)
		TMLE-cw-Hájek	0.996	−0.0002 (0.0050)	0.917 (0.006)	0.2211 (0.0035)
		TMLE-pw-HT	0.997	−0.0003 (0.0050)	0.917 (0.006)	0.2210 (0.0035)
		TMLE-pw-Hájek	0.997	−0.0003 (0.0049)	0.917 (0.006)	0.2210 (0.0035)
		TMLE-2eps-cw-Hájek	0.996	−0.0001 (0.0050)	0.917 (0.006)	0.2210 (0.0035)
		TMLE-2eps-pw-Hájek	0.997	−0.0002 (0.0049)	0.917 (0.006)	0.2209 (0.0035)
500	5	EEq-HT	1.000	−0.0013 (0.0022)	0.941 (0.005)	0.0968 (0.0016)
		EEq-Hájek	1.000	−0.0013 (0.0022)	0.941 (0.005)	0.0968 (0.0016)
		TMLE-cw-HT	1.000	−0.0012 (0.0022)	0.941 (0.005)	0.0968 (0.0016)
		TMLE-cw-Hájek	1.000	−0.0012 (0.0022)	0.941 (0.005)	0.0968 (0.0016)
		TMLE-pw-HT	1.000	−0.0013 (0.0022)	0.941 (0.005)	0.0968 (0.0016)
		TMLE-pw-Hájek	1.000	−0.0012 (0.0022)	0.941 (0.005)	0.0968 (0.0016)
		TMLE-2eps-cw-Hájek	1.000	−0.0012 (0.0022)	0.940 (0.005)	0.0968 (0.0016)
		TMLE-2eps-pw-Hájek	1.000	−0.0012 (0.0022)	0.940 (0.005)	0.0968 (0.0016)
		DR-nt	1.000	−0.0021 (0.0022)	0.950 (0.005)	0.0984 (0.0016)
		DR-ny	1.000	−0.0022 (0.0022)	0.945 (0.005)	0.0966 (0.0016)
500	1	EEq-HT	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		EEq-Hájek	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		TMLE-cw-HT	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		TMLE-cw-Hájek	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		TMLE-pw-HT	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		TMLE-pw-Hájek	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		TMLE-2eps-cw-Hájek	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
		TMLE-2eps-pw-Hájek	1.000	−0.0011 (0.0022)	0.941 (0.005)	0.0967 (0.0016)
5,000	5	EEq-HT	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		EEq-Hájek	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		TMLE-cw-HT	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		TMLE-cw-Hájek	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		TMLE-pw-HT	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		TMLE-pw-Hájek	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		TMLE-2eps-cw-Hájek	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		TMLE-2eps-pw-Hájek	1.000	−0.0004 (0.0007)	0.958 (0.005)	0.0301 (0.0004)
		DR-nt	1.000	0.0006 (0.0007)	0.953 (0.005)	0.0302 (0.0005)
		DR-ny	1.000	0.0005 (0.0007)	0.959 (0.004)	0.0298 (0.0004)
5,000	1	EEq-HT	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		EEq-Hájek	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		TMLE-cw-HT	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		TMLE-cw-Hájek	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		TMLE-pw-HT	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		TMLE-pw-Hájek	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		TMLE-2eps-cw-Hájek	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)
		TMLE-2eps-pw-Hájek	1.000	−0.0004 (0.0007)	0.957 (0.005)	0.0301 (0.0004)

Table H.8: CompactBinary route C1 (proposed outcome working model misspecified): all procedures, with ES(1) = 0.122. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.947	−0.0015 (0.0032)	0.943 (0.005)	0.1372 (0.0022)
		EEq-Hájek	0.947	−0.0013 (0.0032)	0.942 (0.005)	0.1371 (0.0022)
		TMLE-cw-HT	0.936	−0.0006 (0.0032)	0.944 (0.005)	0.1370 (0.0022)
		TMLE-cw-Hájek	0.936	−0.0005 (0.0032)	0.942 (0.005)	0.1369 (0.0022)
		TMLE-pw-HT	0.936	−0.0006 (0.0032)	0.944 (0.005)	0.1371 (0.0022)
		TMLE-pw-Hájek	0.936	−0.0005 (0.0032)	0.942 (0.005)	0.1369 (0.0022)
		TMLE-2eps-cw-Hájek	0.936	0.0000 (0.0032)	0.945 (0.005)	0.1365 (0.0022)
		TMLE-2eps-pw-Hájek	0.936	0.0000 (0.0032)	0.945 (0.005)	0.1366 (0.0022)
		DR-nt	0.882	0.0011 (0.0030)	0.944 (0.005)	0.1275 (0.0021)
		DR-ny	0.882	0.0011 (0.0031)	0.943 (0.006)	0.1282 (0.0021)
100	1	EEq-HT	0.996	0.0009 (0.0030)	0.945 (0.005)	0.1357 (0.0021)
		EEq-Hájek	0.996	0.0010 (0.0030)	0.945 (0.005)	0.1357 (0.0021)
		TMLE-cw-HT	0.996	0.0009 (0.0030)	0.945 (0.005)	0.1357 (0.0021)
		TMLE-cw-Hájek	0.996	0.0010 (0.0030)	0.945 (0.005)	0.1357 (0.0021)
		TMLE-pw-HT	0.996	0.0009 (0.0030)	0.945 (0.005)	0.1357 (0.0021)
		TMLE-pw-Hájek	0.996	0.0010 (0.0030)	0.945 (0.005)	0.1357 (0.0021)
		TMLE-2eps-cw-Hájek	0.996	0.0010 (0.0030)	0.944 (0.005)	0.1357 (0.0021)
		TMLE-2eps-pw-Hájek	0.996	0.0010 (0.0030)	0.944 (0.005)	0.1357 (0.0021)
500	5	EEq-HT	1.000	0.0015 (0.0014)	0.941 (0.005)	0.0609 (0.0010)
		EEq-Hájek	1.000	0.0015 (0.0014)	0.941 (0.005)	0.0609 (0.0010)
		TMLE-cw-HT	1.000	0.0015 (0.0014)	0.941 (0.005)	0.0609 (0.0010)
		TMLE-cw-Hájek	1.000	0.0015 (0.0014)	0.941 (0.005)	0.0609 (0.0010)
		TMLE-pw-HT	1.000	0.0015 (0.0014)	0.941 (0.005)	0.0609 (0.0010)
		TMLE-pw-Hájek	1.000	0.0015 (0.0014)	0.941 (0.005)	0.0609 (0.0010)
		TMLE-2eps-cw-Hájek	1.000	0.0016 (0.0014)	0.940 (0.005)	0.0609 (0.0010)
		TMLE-2eps-pw-Hájek	1.000	0.0016 (0.0014)	0.940 (0.005)	0.0609 (0.0010)
		DR-nt	1.000	0.0010 (0.0013)	0.934 (0.006)	0.0571 (0.0010)
		DR-ny	1.000	0.0013 (0.0013)	0.940 (0.005)	0.0580 (0.0010)
500	1	EEq-HT	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
		EEq-Hájek	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
		TMLE-cw-HT	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
		TMLE-cw-Hájek	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
		TMLE-pw-HT	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
		TMLE-pw-Hájek	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
		TMLE-2eps-cw-Hájek	1.000	0.0017 (0.0014)	0.939 (0.005)	0.0610 (0.0010)
		TMLE-2eps-pw-Hájek	1.000	0.0017 (0.0014)	0.940 (0.005)	0.0610 (0.0010)
5,000	5	EEq-HT	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		EEq-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-cw-HT	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-cw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-pw-HT	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-pw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		DR-nt	1.000	−0.0005 (0.0004)	0.952 (0.005)	0.0173 (0.0003)
		DR-ny	1.000	−0.0008 (0.0004)	0.951 (0.005)	0.0177 (0.0003)
5,000	1	EEq-HT	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		EEq-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-cw-HT	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-cw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-pw-HT	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-pw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	−0.0010 (0.0004)	0.949 (0.005)	0.0186 (0.0003)

Table H.9: CompactBinary route C2 (proposed hazard working model misspecified): all procedures, with $\mathrm{ATT}(2,4) = 0.125$. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.990	0.0053 (0.0046)	0.939 (0.005)	0.2041 (0.0032)
		EEq-Hájek	0.990	0.0053 (0.0046)	0.939 (0.005)	0.2038 (0.0032)
		TMLE-cw-HT	0.979	0.0063 (0.0046)	0.940 (0.005)	0.2032 (0.0032)
		TMLE-cw-Hájek	0.979	0.0063 (0.0046)	0.939 (0.005)	0.2031 (0.0032)
		TMLE-pw-HT	0.990	0.0053 (0.0046)	0.939 (0.005)	0.2037 (0.0032)
		TMLE-pw-Hájek	0.990	0.0053 (0.0046)	0.939 (0.005)	0.2035 (0.0032)
		TMLE-2eps-cw-Hájek	0.979	0.0063 (0.0046)	0.940 (0.005)	0.2025 (0.0032)
		TMLE-2eps-pw-Hájek	0.990	0.0053 (0.0046)	0.939 (0.005)	0.2029 (0.0032)
		DR-nt	0.960	0.0032 (0.0049)	0.939 (0.005)	0.2124 (0.0036)
		DR-ny	0.960	0.0033 (0.0045)	0.942 (0.005)	0.1975 (0.0032)
100	1	EEq-HT	0.998	0.0054 (0.0045)	0.936 (0.005)	0.1995 (0.0031)
		EEq-Hájek	0.998	0.0055 (0.0045)	0.936 (0.005)	0.1995 (0.0031)
		TMLE-cw-HT	0.997	0.0057 (0.0045)	0.936 (0.005)	0.1996 (0.0031)
		TMLE-cw-Hájek	0.997	0.0057 (0.0045)	0.936 (0.005)	0.1996 (0.0031)
		TMLE-pw-HT	0.998	0.0055 (0.0045)	0.936 (0.005)	0.1995 (0.0031)
		TMLE-pw-Hájek	0.998	0.0055 (0.0045)	0.936 (0.005)	0.1996 (0.0031)
		TMLE-2eps-cw-Hájek	0.997	0.0057 (0.0045)	0.935 (0.006)	0.1998 (0.0031)
		TMLE-2eps-pw-Hájek	0.998	0.0055 (0.0045)	0.935 (0.006)	0.1997 (0.0031)
500	5	EEq-HT	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		EEq-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		TMLE-cw-HT	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		TMLE-cw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		TMLE-pw-HT	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		TMLE-pw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		TMLE-2eps-cw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		TMLE-2eps-pw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0882 (0.0014)
		DR-nt	1.000	0.0005 (0.0021)	0.944 (0.005)	0.0929 (0.0015)
		DR-ny	1.000	0.0006 (0.0020)	0.947 (0.005)	0.0873 (0.0014)
500	1	EEq-HT	1.000	0.0000 (0.0020)	0.943 (0.005)	0.0879 (0.0014)
		EEq-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0879 (0.0014)
		TMLE-cw-HT	1.000	0.0000 (0.0020)	0.943 (0.005)	0.0879 (0.0014)
		TMLE-cw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0879 (0.0014)
		TMLE-pw-HT	1.000	0.0000 (0.0020)	0.943 (0.005)	0.0879 (0.0014)
		TMLE-pw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0879 (0.0014)
		TMLE-2eps-cw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0879 (0.0014)
		TMLE-2eps-pw-Hájek	1.000	0.0000 (0.0020)	0.944 (0.005)	0.0879 (0.0014)
5,000	5	EEq-HT	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		EEq-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-cw-HT	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		TMLE-cw-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-pw-HT	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		TMLE-pw-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-2eps-cw-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-2eps-pw-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		DR-nt	1.000	−0.0004 (0.0006)	0.955 (0.005)	0.0281 (0.0004)
		DR-ny	1.000	−0.0007 (0.0006)	0.956 (0.005)	0.0272 (0.0004)
5,000	1	EEq-HT	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		EEq-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-cw-HT	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		TMLE-cw-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-pw-HT	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		TMLE-pw-Hájek	1.000	0.0000 (0.0006)	0.955 (0.005)	0.0270 (0.0004)
		TMLE-2eps-cw-Hájek	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)
		TMLE-2eps-pw-Hájek	1.000	0.0000 (0.0006)	0.954 (0.005)	0.0270 (0.0004)

Table H.10: CompactBinary route C2 (proposed hazard working model misspecified): all procedures, with ES(1) = 0.129. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.979	0.0038 (0.0030)	0.947 (0.005)	0.1315 (0.0022)
		EEq-Hájek	0.979	0.0037 (0.0030)	0.946 (0.005)	0.1307 (0.0021)
		TMLE-cw-HT	0.979	0.0036 (0.0029)	0.947 (0.005)	0.1305 (0.0021)
		TMLE-cw-Hájek	0.979	0.0036 (0.0029)	0.946 (0.005)	0.1304 (0.0021)
		TMLE-pw-HT	0.979	0.0036 (0.0030)	0.947 (0.005)	0.1305 (0.0021)
		TMLE-pw-Hájek	0.979	0.0036 (0.0029)	0.946 (0.005)	0.1304 (0.0021)
		TMLE-2eps-cw-Hájek	0.979	0.0036 (0.0029)	0.947 (0.005)	0.1299 (0.0021)
		TMLE-2eps-pw-Hájek	0.979	0.0035 (0.0029)	0.947 (0.005)	0.1299 (0.0021)
		DR-nt	0.960	0.0008 (0.0027)	0.947 (0.005)	0.1194 (0.0020)
		DR-ny	0.960	0.0024 (0.0027)	0.948 (0.005)	0.1186 (0.0020)
100	1	EEq-HT	0.997	0.0020 (0.0029)	0.940 (0.005)	0.1276 (0.0020)
		EEq-Hájek	0.997	0.0021 (0.0029)	0.941 (0.005)	0.1276 (0.0020)
		TMLE-cw-HT	0.997	0.0021 (0.0029)	0.940 (0.005)	0.1276 (0.0020)
		TMLE-cw-Hájek	0.997	0.0021 (0.0029)	0.941 (0.005)	0.1276 (0.0020)
		TMLE-pw-HT	0.997	0.0021 (0.0029)	0.940 (0.005)	0.1276 (0.0020)
		TMLE-pw-Hájek	0.997	0.0021 (0.0029)	0.941 (0.005)	0.1276 (0.0020)
		TMLE-2eps-cw-Hájek	0.997	0.0021 (0.0029)	0.941 (0.005)	0.1276 (0.0020)
		TMLE-2eps-pw-Hájek	0.997	0.0021 (0.0029)	0.941 (0.005)	0.1276 (0.0020)
500	5	EEq-HT	1.000	0.0014 (0.0013)	0.947 (0.005)	0.0574 (0.0010)
		EEq-Hájek	1.000	0.0014 (0.0013)	0.946 (0.005)	0.0574 (0.0010)
		TMLE-cw-HT	1.000	0.0014 (0.0013)	0.947 (0.005)	0.0574 (0.0010)
		TMLE-cw-Hájek	1.000	0.0014 (0.0013)	0.946 (0.005)	0.0574 (0.0010)
		TMLE-pw-HT	1.000	0.0014 (0.0013)	0.947 (0.005)	0.0574 (0.0010)
		TMLE-pw-Hájek	1.000	0.0014 (0.0013)	0.946 (0.005)	0.0574 (0.0010)
		TMLE-2eps-cw-Hájek	1.000	0.0014 (0.0013)	0.947 (0.005)	0.0574 (0.0010)
		TMLE-2eps-pw-Hájek	1.000	0.0014 (0.0013)	0.947 (0.005)	0.0574 (0.0010)
		DR-nt	1.000	0.0013 (0.0012)	0.947 (0.005)	0.0532 (0.0009)
		DR-ny	1.000	0.0012 (0.0012)	0.947 (0.005)	0.0534 (0.0009)
500	1	EEq-HT	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		EEq-Hájek	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		TMLE-cw-HT	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		TMLE-cw-Hájek	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		TMLE-pw-HT	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		TMLE-pw-Hájek	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		TMLE-2eps-cw-Hájek	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
		TMLE-2eps-pw-Hájek	1.000	0.0015 (0.0013)	0.946 (0.005)	0.0570 (0.0009)
5,000	5	EEq-HT	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		EEq-Hájek	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		TMLE-cw-HT	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		TMLE-cw-Hájek	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		TMLE-pw-HT	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		TMLE-pw-Hájek	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	0.0000 (0.0004)	0.953 (0.005)	0.0176 (0.0003)
		DR-nt	1.000	−0.0003 (0.0004)	0.952 (0.005)	0.0164 (0.0003)
		DR-ny	1.000	−0.0003 (0.0004)	0.948 (0.005)	0.0165 (0.0003)
5,000	1	EEq-HT	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		EEq-Hájek	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		TMLE-cw-HT	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		TMLE-cw-Hájek	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		TMLE-pw-HT	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		TMLE-pw-Hájek	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	0.0000 (0.0004)	0.954 (0.005)	0.0176 (0.0003)

Table H.11: CompactBinary route C12 (both proposed working models misspecified): all procedures, with $\mathrm{ATT}(2,4) = 0.116$. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.992	0.0084 (0.0045)	0.944 (0.005)	0.2008 (0.0031)
		EEq-Hájek	0.992	0.0085 (0.0045)	0.940 (0.005)	0.2005 (0.0031)
		TMLE-cw-HT	0.979	0.0085 (0.0045)	0.943 (0.005)	0.2009 (0.0031)
		TMLE-cw-Hájek	0.979	0.0086 (0.0045)	0.940 (0.005)	0.2007 (0.0031)
		TMLE-pw-HT	0.992	0.0086 (0.0045)	0.943 (0.005)	0.2005 (0.0031)
		TMLE-pw-Hájek	0.992	0.0087 (0.0045)	0.940 (0.005)	0.2003 (0.0031)
		TMLE-2eps-cw-Hájek	0.979	0.0090 (0.0045)	0.939 (0.005)	0.2002 (0.0031)
		TMLE-2eps-pw-Hájek	0.992	0.0091 (0.0045)	0.939 (0.005)	0.1998 (0.0030)
		DR-nt	0.966	0.0151 (0.0048)	0.943 (0.005)	0.2095 (0.0033)
		DR-ny	0.966	0.0146 (0.0045)	0.939 (0.005)	0.1983 (0.0031)
100	1	EEq-HT	1.000	0.0113 (0.0044)	0.935 (0.005)	0.1969 (0.0030)
		EEq-Hájek	1.000	0.0112 (0.0044)	0.936 (0.005)	0.1970 (0.0030)
		TMLE-cw-HT	0.999	0.0112 (0.0044)	0.935 (0.006)	0.1970 (0.0030)
		TMLE-cw-Hájek	0.999	0.0112 (0.0044)	0.936 (0.005)	0.1970 (0.0030)
		TMLE-pw-HT	1.000	0.0112 (0.0044)	0.935 (0.005)	0.1970 (0.0030)
		TMLE-pw-Hájek	1.000	0.0112 (0.0044)	0.937 (0.005)	0.1970 (0.0030)
		TMLE-2eps-cw-Hájek	0.999	0.0110 (0.0044)	0.936 (0.005)	0.1970 (0.0030)
		TMLE-2eps-pw-Hájek	1.000	0.0110 (0.0044)	0.936 (0.005)	0.1970 (0.0030)
500	5	EEq-HT	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		EEq-Hájek	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		TMLE-cw-HT	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		TMLE-cw-Hájek	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		TMLE-pw-HT	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		TMLE-pw-Hájek	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		TMLE-2eps-cw-Hájek	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		TMLE-2eps-pw-Hájek	1.000	0.0073 (0.0020)	0.942 (0.005)	0.0876 (0.0014)
		DR-nt	1.000	0.0140 (0.0020)	0.942 (0.005)	0.0920 (0.0015)
		DR-ny	1.000	0.0101 (0.0019)	0.943 (0.005)	0.0876 (0.0014)
500	1	EEq-HT	1.000	0.0077 (0.0019)	0.941 (0.005)	0.0873 (0.0014)
		EEq-Hájek	1.000	0.0076 (0.0019)	0.942 (0.005)	0.0873 (0.0014)
		TMLE-cw-HT	1.000	0.0076 (0.0019)	0.941 (0.005)	0.0873 (0.0014)
		TMLE-cw-Hájek	1.000	0.0076 (0.0019)	0.942 (0.005)	0.0873 (0.0014)
		TMLE-pw-HT	1.000	0.0076 (0.0019)	0.941 (0.005)	0.0873 (0.0014)
		TMLE-pw-Hájek	1.000	0.0076 (0.0019)	0.942 (0.005)	0.0873 (0.0014)
		TMLE-2eps-cw-Hájek	1.000	0.0075 (0.0019)	0.943 (0.005)	0.0873 (0.0014)
		TMLE-2eps-pw-Hájek	1.000	0.0075 (0.0019)	0.943 (0.005)	0.0873 (0.0014)
5,000	5	EEq-HT	1.000	0.0146 (0.0006)	0.913 (0.006)	0.0306 (0.0005)
		EEq-Hájek	1.000	0.0146 (0.0006)	0.914 (0.006)	0.0305 (0.0005)
		TMLE-cw-HT	1.000	0.0146 (0.0006)	0.913 (0.006)	0.0305 (0.0005)
		TMLE-cw-Hájek	1.000	0.0146 (0.0006)	0.914 (0.006)	0.0305 (0.0005)
		TMLE-pw-HT	1.000	0.0146 (0.0006)	0.913 (0.006)	0.0305 (0.0005)
		TMLE-pw-Hájek	1.000	0.0146 (0.0006)	0.914 (0.006)	0.0305 (0.0005)
		TMLE-2eps-cw-Hájek	1.000	0.0145 (0.0006)	0.914 (0.006)	0.0305 (0.0005)
		TMLE-2eps-pw-Hájek	1.000	0.0145 (0.0006)	0.914 (0.006)	0.0305 (0.0005)
		DR-nt	1.000	0.0196 (0.0006)	0.886 (0.007)	0.0341 (0.0005)
		DR-ny	1.000	0.0166 (0.0006)	0.902 (0.007)	0.0315 (0.0005)
5,000	1	EEq-HT	1.000	0.0146 (0.0006)	0.912 (0.006)	0.0306 (0.0005)
		EEq-Hájek	1.000	0.0146 (0.0006)	0.913 (0.006)	0.0306 (0.0005)
		TMLE-cw-HT	1.000	0.0146 (0.0006)	0.912 (0.006)	0.0306 (0.0005)
		TMLE-cw-Hájek	1.000	0.0146 (0.0006)	0.913 (0.006)	0.0306 (0.0005)
		TMLE-pw-HT	1.000	0.0146 (0.0006)	0.912 (0.006)	0.0306 (0.0005)
		TMLE-pw-Hájek	1.000	0.0146 (0.0006)	0.913 (0.006)	0.0306 (0.0005)
		TMLE-2eps-cw-Hájek	1.000	0.0145 (0.0006)	0.914 (0.006)	0.0305 (0.0005)
		TMLE-2eps-pw-Hájek	1.000	0.0145 (0.0006)	0.914 (0.006)	0.0305 (0.0005)

Table H.12: CompactBinary route C12 (both proposed working models misspecified): all procedures, with ES(1) = 0.120. Conventions as in Table H.5.

n	L	Procedure	Compl.	Bias (MCSE)	Cov. (MCSE)	RMSE (MCSE)
100	5	EEq-HT	0.979	0.0058 (0.0029)	0.954 (0.005)	0.1286 (0.0021)
		EEq-Hájek	0.979	0.0060 (0.0029)	0.952 (0.005)	0.1284 (0.0021)
		TMLE-cw-HT	0.978	0.0062 (0.0029)	0.953 (0.005)	0.1283 (0.0021)
		TMLE-cw-Hájek	0.978	0.0063 (0.0029)	0.952 (0.005)	0.1282 (0.0021)
		TMLE-pw-HT	0.978	0.0062 (0.0029)	0.953 (0.005)	0.1283 (0.0021)
		TMLE-pw-Hájek	0.978	0.0063 (0.0029)	0.952 (0.005)	0.1282 (0.0021)
		TMLE-2eps-cw-Hájek	0.978	0.0066 (0.0029)	0.954 (0.005)	0.1279 (0.0021)
		TMLE-2eps-pw-Hájek	0.978	0.0067 (0.0029)	0.953 (0.005)	0.1279 (0.0021)
		DR-nt	0.966	0.0087 (0.0027)	0.943 (0.005)	0.1203 (0.0020)
		DR-ny	0.966	0.0070 (0.0027)	0.939 (0.005)	0.1184 (0.0020)
100	1	EEq-HT	0.999	0.0072 (0.0028)	0.945 (0.005)	0.1263 (0.0021)
		EEq-Hájek	0.999	0.0071 (0.0028)	0.946 (0.005)	0.1264 (0.0021)
		TMLE-cw-HT	0.999	0.0071 (0.0028)	0.945 (0.005)	0.1264 (0.0021)
		TMLE-cw-Hájek	0.999	0.0071 (0.0028)	0.946 (0.005)	0.1264 (0.0021)
		TMLE-pw-HT	0.999	0.0071 (0.0028)	0.945 (0.005)	0.1264 (0.0021)
		TMLE-pw-Hájek	0.999	0.0071 (0.0028)	0.946 (0.005)	0.1264 (0.0021)
		TMLE-2eps-cw-Hájek	0.999	0.0070 (0.0028)	0.945 (0.005)	0.1264 (0.0021)
		TMLE-2eps-pw-Hájek	0.999	0.0070 (0.0028)	0.945 (0.005)	0.1264 (0.0021)
500	5	EEq-HT	1.000	0.0069 (0.0013)	0.944 (0.005)	0.0568 (0.0009)
		EEq-Hájek	1.000	0.0069 (0.0013)	0.944 (0.005)	0.0568 (0.0009)
		TMLE-cw-HT	1.000	0.0069 (0.0013)	0.945 (0.005)	0.0568 (0.0009)
		TMLE-cw-Hájek	1.000	0.0069 (0.0013)	0.944 (0.005)	0.0568 (0.0009)
		TMLE-pw-HT	1.000	0.0069 (0.0013)	0.945 (0.005)	0.0568 (0.0009)
		TMLE-pw-Hájek	1.000	0.0069 (0.0013)	0.944 (0.005)	0.0568 (0.0009)
		TMLE-2eps-cw-Hájek	1.000	0.0069 (0.0013)	0.944 (0.005)	0.0568 (0.0009)
		TMLE-2eps-pw-Hájek	1.000	0.0069 (0.0013)	0.944 (0.005)	0.0568 (0.0009)
		DR-nt	1.000	0.0097 (0.0012)	0.941 (0.005)	0.0528 (0.0009)
		DR-ny	1.000	0.0078 (0.0012)	0.942 (0.005)	0.0530 (0.0008)
500	1	EEq-HT	1.000	0.0071 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		EEq-Hájek	1.000	0.0071 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		TMLE-cw-HT	1.000	0.0071 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		TMLE-cw-Hájek	1.000	0.0070 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		TMLE-pw-HT	1.000	0.0071 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		TMLE-pw-Hájek	1.000	0.0070 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		TMLE-2eps-cw-Hájek	1.000	0.0070 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
		TMLE-2eps-pw-Hájek	1.000	0.0070 (0.0013)	0.943 (0.005)	0.0566 (0.0009)
5,000	5	EEq-HT	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		EEq-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-cw-HT	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-cw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-pw-HT	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-pw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		DR-nt	1.000	0.0113 (0.0004)	0.893 (0.007)	0.0195 (0.0003)
		DR-ny	1.000	0.0092 (0.0004)	0.910 (0.006)	0.0186 (0.0003)
5,000	1	EEq-HT	1.000	0.0087 (0.0004)	0.917 (0.006)	0.0194 (0.0003)
		EEq-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-cw-HT	1.000	0.0087 (0.0004)	0.917 (0.006)	0.0194 (0.0003)
		TMLE-cw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-pw-HT	1.000	0.0087 (0.0004)	0.917 (0.006)	0.0194 (0.0003)
		TMLE-pw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-2eps-cw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)
		TMLE-2eps-pw-Hájek	1.000	0.0087 (0.0004)	0.918 (0.006)	0.0194 (0.0003)

H.5 Comparators with Analytic Standard Errors

This subsection reports a sensitivity check recorded for this version; the production results above remain the primary results. It refits DR-nt and DR-ny in the twelve design cells that contain the 94 band-warning fits of Section 5.4: the Static and Dynamic continuous designs under scenarios B, B+, and C1, at $n = 500$ and $n = 5000$, with all 2,000 replications of each cell. The panels are regenerated from the production random-number streams. Of the 96,000 refitted comparator results, the 95,906 that have a finite production estimate agree with it to within 10^{-12} (the largest absolute difference is 5.0×10^{-13}); the other 94 are the band-warning fits, for which the production archives hold no finite estimate or standard error. This agreement is consistent with the regenerated panels being the production panels, but it does not prove it, and it says nothing about the standard errors. The only change to the call of Appendix G.3 is `bstrap = FALSE` and `cband = FALSE`. The package then reports the analytic standard error $n^{-1}(\sum_i \widehat{\psi}_i^2)^{1/2}$ computed from the same estimated influence function $\widehat{\psi}_i$, which for ES(1) includes the term for the estimated aggregation weights; this is the construction we use for the proposed procedures. Intervals remain ± 1.96 standard errors, every other argument is unchanged, and any package warning still makes a fit unavailable. The Linux refits used R 4.3.3; the production run and local spot check used R 4.5.2. All three used `did` 2.3.0, `DRDID` 1.2.3, and `BMisc` 1.4.8. As a spot check, 24 panels, replication 1 and the first band-warning replication of each cell, were regenerated and refitted under that local installation: each regenerated panel's digest equals its archived production digest, and all 96 local results, 12 of them band-warning fits, return a finite estimate and a positive standard error; the estimates and standard errors agree with the Linux refits to within 2×10^{-13} and 2×10^{-14}, respectively. This covers 24 of the 24,000 panels; it is a spot check, not a rerun of all refits in the production environment.

Table H.13 reports the results. Every refit returns a finite estimate and a positive standard error, including the 94 fits that the production protocol lost to the band warning; 81 of these 94 intervals cover the target. The analytic standard errors are usually smaller: their median ratio to the production standard errors lies between 0.95 and 0.99 across the 48 combinations of cell, procedure, and target, and the analytic standard error is the smaller in 72 percent of the 95,906 paired results. The two constructions differ in scale: the production standard error is the interquartile range of 1,000 multiplier draws divided by $z_{0.75} - z_{0.25}$ (Appendix G.3; `R/mboot.R` in `did` 2.3.0), whereas the analytic standard error is the draws' conditional root-mean-square scale; the two need not coincide in finite samples, and the bootstrap value also carries the simulation error of a finite number of draws. The check records the direction of the difference in these cells, not its mechanism. On the replications that both protocols complete, empirical coverage with analytic standard errors is lower than the production coverage in every one of the 48 combinations, by 0.05 to 3.5 percentage points, and at $n = 5000$ the comparators' ratio of average estimated standard error to Monte Carlo standard deviation falls from 0.94–1.01 to 0.88–1.00. Over each protocol's own complete replications, all 2,000 for the refits, analytic coverage is lower by 0.05 to 2.85 percentage points. The two comparisons differ only for DR-nt ES(1), the twelve combinations with band warnings: the warning exclusions affect completion, and through the set of completed replications they change the reported coverage; restoring the excluded fits changes analytic coverage by less than 1.0 percentage point. In these twelve cells, replacing the production bootstrap standard errors with analytic standard errors lowers comparator coverage in every paired comparison, so the inference choice does not explain a comparator coverage disadvantage in these cells. The cells were selected because they contain band warnings: the check does not extend to the other designs or to other comparator failure mechanisms, and its differences are Monte Carlo differences, not a population ordering of the procedures.

Table H.13: Comparators with the package's multiplier-bootstrap standard errors (Prod., the production results) and with its analytic standard errors (Anal., `bstrap = FALSE`, `cband = FALSE`) on panels regenerated from the production random-number streams, continuous designs, 2,000 replications per cell. Cov. is the empirical coverage of the nominal 95% interval ±1.96 standard errors over the replications that both protocols complete, so that Prod. and Anal. are paired; these are the production protocol's complete replications, because every refit is complete. Ratio is the median, over the same replications, of the analytic standard error divided by the production standard error. Warn. is the number of DR-nt fits that the production protocol lost to the simultaneous-band warning; the refit recovers them, and both coverage columns exclude them (the text reports their effect). This is a sensitivity check; the production results are primary.

			DR-nt				DR-ny		
Design	Scen.	n	Prod. Cov.	Anal. Cov.	Ratio	Warn.	Prod. Cov.	Anal. Cov.	Ratio
ATT$(2, 4)$									
Static continuous	B	500	0.916	0.906	0.97	0	0.922	0.908	0.97
	B	5,000	0.944	0.940	0.97	0	0.959	0.955	0.98
	B+	500	0.894	0.875	0.96	0	0.906	0.886	0.96
	B+	5,000	0.949	0.940	0.96	0	0.947	0.937	0.96
	C1	500	0.430	0.403	0.95	0	0.595	0.571	0.96
	C1	5,000	0.149	0.120	0.96	0	0.242	0.218	0.96
Dynamic continuous	B	500	0.898	0.891	0.98	0	0.922	0.917	0.98
	B	5,000	0.654	0.645	0.99	0	0.848	0.839	0.99
	B+	500	0.880	0.869	0.97	0	0.907	0.897	0.97
	B+	5,000	0.722	0.702	0.98	0	0.880	0.858	0.98
	C1	500	0.102	0.087	0.98	0	0.280	0.265	0.98
	C1	5,000	0.002	0.001	0.98	0	0.012	0.007	0.98
ES(1)									
Static continuous	B	500	0.940	0.928	0.97	1	0.933	0.928	0.98
	B	5,000	0.943	0.933	0.98	3	0.949	0.946	0.98
	B+	500	0.888	0.873	0.96	2	0.920	0.902	0.96
	B+	5,000	0.939	0.931	0.96	4	0.945	0.933	0.97
	C1	500	0.520	0.488	0.95	19	0.806	0.797	0.97
	C1	5,000	0.197	0.162	0.96	23	0.503	0.488	0.98
Dynamic continuous	B	500	0.900	0.890	0.98	1	0.935	0.932	0.99
	B	5,000	0.673	0.663	0.99	5	0.907	0.907	0.99
	B+	500	0.879	0.868	0.97	2	0.919	0.913	0.98
	B+	5,000	0.723	0.701	0.98	7	0.919	0.914	0.98
	C1	500	0.169	0.141	0.97	12	0.503	0.494	0.98
	C1	5,000	0.006	0.002	0.98	15	0.034	0.024	0.99

I Completion and Stress-Scenario Tables

Table I.1 reports how often each of the ten procedures returned a reportable result, and Tables I.2 and I.3 report the stress scenarios B and B+ in the two Dynamic designs; Table III summarizes them at $n = 100$ and $n = 5000$. Counts refer to procedure-and-target results across replications. For the four pooled procedures (EEq-HT, EEq-Hájek, TMLE-pw-HT and TMLE-pw-Hájek), 45,410 of 2,304,000 requested results (1.97%; 3.29% with five-fold cross-fitting and 0.65% without) are unavailable in 390 of 1,152 procedure-by-target cells, and the smallest cell completion is 0.343. Over all eight proposed variants, 107,960 of 4,608,000 requested results (2.34%) are unavailable. For the comparators, 87,428 of 576,000 requested results (15.18%) are unavailable, and the smallest cell completion is 0.029.

For the pooled procedures unavailability is a small-sample phenomenon: no replication fails at $n = 5000$, 18 fail at $n = 500$, all with five-fold cross-fitting, and the remaining 45,392 fail at $n = 100$. The worst cells combine $n = 100$, five-fold cross-fitting, and the strong-confounding scenario B+: EEq-Hájek and TMLE-pw-Hájek complete in 0.645 of replications for ATT$(2, 4)$ in the continuous Dynamic design, and in 0.373 and 0.343 for ES(1) in the binary Dynamic design.

Pooling both targets and the four procedures, at least 97 percent of the unavailable procedure-replications in these two design cells fail in a fold-specific frozen-hazard logistic fit, either through a fatal warning (fitted probabilities numerically 0 or 1, with or without nonconvergence) or because the fold's training risk set contains no adoption in that period; the no-fallback rule treats both as fatal. These failures are consistent with complete or quasi-complete separation under the steep covariate–adoption gradient of B+. With $n = 100$ and $L = 5$, each training sample contains 80 units, with fewer in each period's risk set.

On the same panels, fitting the nuisances on the full sample ($L = 1$) raises completion in these cells to 0.931 for ATT$(2, 4)$ and to 0.818 and 0.777 for ES(1).

Across all cells, 92 percent of the unavailable replications of the four pooled procedures fail in frozen-hazard fits (85 percent through a fatal warning and 7 percent through a training risk set without adoptions), 5 percent in the outcome-level logistic regressions of the binary designs, 3 percent in treated-arm fits, and 0.3 percent because a target cohort is absent from the simulated sample; outcome-level failures account for most unavailable replications in some Static binary cells.

Of the 87,428 unavailable comparator results, 87,300 (99.9%) carry the `did` package's preprocessing warning that some cohorts contain very few units. This warning depends on realized cohort sizes rather than the reported target and, at $n = 100$, occurs in every design, in 3.4 to 97 percent of comparator fits. It is issued before any estimate is computed, and the no-fallback rule discards the fit at that point, so whether a finite pointwise estimate and standard error would have followed is not recorded. Another 94, all DR-nt fits of ES(1) at $n \geq 500$, are fits in which the package had already computed the pointwise estimate and standard error and then warned that its simultaneous band—which no reported interval uses—was narrower than the pointwise interval. These results are unavailable only because the protocol treats every warning as fatal; they account for the 12 additional incomplete DR-nt cells in Panel B of Table I.1, and all are recovered with `bstrap = FALSE` and `cband = FALSE` (Appendix H.5). The remaining 34 fail for other reasons.

Table I.1: Completion and availability over all design cells. Completion is the share of requested replications that returned a finite estimate and a finite, positive standard error, pooled over cells, and Unavailable is the number that did not. For a proposed procedure the pool is 144 cells (72 combinations of design, scenario or route, and sample size, each in two cross-fitting settings) of 2,000 replications each; the comparators do not use cross-fitting, their results in the two settings are identical, and each of their 72 combinations is counted once. Incomplete cells counts cells with completion below one; Minimum is the smallest cell completion, and the last column gives the design cell in which it occurs. These are availability comparisons under a common no-fallback protocol, not performance rankings.

Procedure	Completion	Unavailable	Incomplete cells	Minimum	Minimum attained in
Panel A: ATT$(2,4)$					
EEq-HT	0.990	2,992	47/144	0.645	Dynamic continuous B+, $n = 100$, $L = 5$
EEq-Hájek	0.990	2,992	47/144	0.645	Dynamic continuous B+, $n = 100$, $L = 5$
TMLE-cw-HT	0.974	7,518	50/144	0.399	Dynamic continuous B+, $n = 100$, $L = 5$
TMLE-cw-Hájek	0.974	7,518	50/144	0.399	Dynamic continuous B+, $n = 100$, $L = 5$
TMLE-pw-HT	0.990	3,003	47/144	0.645	Dynamic continuous B+, $n = 100$, $L = 5$
TMLE-pw-Hájek	0.990	3,003	47/144	0.645	Dynamic continuous B+, $n = 100$, $L = 5$
TMLE-2eps-cw-Hájek	0.974	7,518	50/144	0.399	Dynamic continuous B+, $n = 100$, $L = 5$
TMLE-2eps-pw-Hájek	0.990	3,003	47/144	0.645	Dynamic continuous B+, $n = 100$, $L = 5$
DR-nt	0.848	21,831	24/72	0.030	Dynamic binary B+, $n = 100$
DR-ny	0.848	21,830	24/72	0.030	Dynamic binary B+, $n = 100$
Panel B: ES(1)					
EEq-HT	0.972	8,118	50/144	0.373	Dynamic binary B+, $n = 100$, $L = 5$
EEq-Hájek	0.972	8,118	50/144	0.373	Dynamic binary B+, $n = 100$, $L = 5$
TMLE-cw-HT	0.967	9,467	51/144	0.282	Dynamic binary B+, $n = 100$, $L = 5$
TMLE-cw-Hájek	0.967	9,467	51/144	0.282	Dynamic binary B+, $n = 100$, $L = 5$
TMLE-pw-HT	0.970	8,592	51/144	0.343	Dynamic binary B+, $n = 100$, $L = 5$
TMLE-pw-Hájek	0.970	8,592	51/144	0.343	Dynamic binary B+, $n = 100$, $L = 5$
TMLE-2eps-cw-Hájek	0.967	9,467	51/144	0.282	Dynamic binary B+, $n = 100$, $L = 5$
TMLE-2eps-pw-Hájek	0.970	8,592	51/144	0.343	Dynamic binary B+, $n = 100$, $L = 5$
DR-nt	0.848	21,931	36/72	0.029	Dynamic binary B+, $n = 100$
DR-ny	0.848	21,836	24/72	0.029	Dynamic binary B+, $n = 100$

Table I.2: Stress scenarios B and B+ in the continuous Dynamic design; five-fold cross-fitting for the proposed procedures. Bias, coverage (Cov.) of nominal 95% Wald intervals, and RMSE are conditional on completion (Compl.); MCSEs in parentheses. Where completion differs across procedures, the summaries describe different subsets of panels and do not rank the procedures. Standard errors as in Section 5.1; Static-design counterparts are in Appendix H.3.

			ATT(2, 4)				ES(1)			
Design	n	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	100	EEq-Hájek	0.925	0.014	0.937	0.880	0.786	0.011	0.938	0.487
				(0.020)	(0.006)	(0.016)		(0.012)	(0.006)	(0.010)
		TMLE-pw-Hájek	0.925	0.012	0.857	1.007	0.772	0.008	0.894	0.480
				(0.023)	(0.008)	(0.026)		(0.012)	(0.008)	(0.011)
		DR-nt	0.233	0.109	0.890	0.887	0.232	0.075	0.892	0.466
				(0.041)	(0.014)	(0.036)		(0.021)	(0.014)	(0.017)
		DR-ny	0.233	0.042	0.901	0.791	0.232	0.015	0.929	0.380
				(0.037)	(0.014)	(0.033)		(0.018)	(0.012)	(0.013)
	500	EEq-Hájek	1.000	−0.001	0.943	0.348	1.000	0.000	0.942	0.194
				(0.008)	(0.005)	(0.006)		(0.004)	(0.005)	(0.003)
		TMLE-pw-Hájek	1.000	−0.002	0.900	0.378	1.000	0.000	0.926	0.195
				(0.008)	(0.007)	(0.006)		(0.004)	(0.006)	(0.003)
		DR-nt	1.000	0.193	0.898	0.439	1.000	0.106	0.900	0.242
				(0.009)	(0.007)	(0.007)		(0.005)	(0.007)	(0.004)
		DR-ny	1.000	0.101	0.922	0.379	1.000	0.031	0.935	0.187
				(0.008)	(0.006)	(0.007)		(0.004)	(0.006)	(0.003)
	5,000	EEq-Hájek	1.000	0.004	0.947	0.109	1.000	0.002	0.946	0.059
				(0.002)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
		TMLE-pw-Hájek	1.000	0.005	0.914	0.120	1.000	0.002	0.939	0.060
				(0.003)	(0.006)	(0.002)		(0.001)	(0.005)	(0.001)
		DR-nt	1.000	0.195	0.654	0.233	0.998	0.106	0.673	0.127
				(0.003)	(0.011)	(0.003)		(0.002)	(0.011)	(0.001)
		DR-ny	1.000	0.107	0.848	0.158	1.000	0.035	0.907	0.067
				(0.003)	(0.008)	(0.002)		(0.001)	(0.006)	(0.001)
B+	100	EEq-Hájek	0.645	−0.005	0.937	1.015	0.382	−0.016	0.928	0.599
				(0.028)	(0.007)	(0.021)		(0.022)	(0.009)	(0.016)
		TMLE-pw-Hájek	0.645	0.015	0.714	1.545	0.352	−0.017	0.808	0.666
				(0.043)	(0.013)	(0.054)		(0.025)	(0.015)	(0.021)
		DR-nt	0.039	−0.014	0.821	1.061	0.039	0.037	0.821	0.593
				(0.121)	(0.043)	(0.087)		(0.067)	(0.043)	(0.046)
		DR-ny	0.039	0.097	0.821	0.945	0.039	−0.031	0.936	0.454
				(0.107)	(0.043)	(0.074)		(0.052)	(0.028)	(0.036)
	500	EEq-Hájek	1.000	−0.002	0.939	0.434	0.999	−0.003	0.936	0.252
				(0.010)	(0.005)	(0.007)		(0.006)	(0.005)	(0.004)
		TMLE-pw-Hájek	1.000	−0.008	0.805	0.583	0.999	−0.004	0.858	0.288
				(0.013)	(0.009)	(0.010)		(0.006)	(0.008)	(0.005)
		DR-nt	1.000	0.223	0.880	0.549	0.999	0.127	0.879	0.318
				(0.011)	(0.007)	(0.011)		(0.007)	(0.007)	(0.006)
		DR-ny	1.000	0.136	0.907	0.487	1.000	0.039	0.919	0.257
				(0.010)	(0.006)	(0.011)		(0.006)	(0.006)	(0.005)
	5,000	EEq-Hájek	1.000	−0.001	0.939	0.135	1.000	−0.001	0.941	0.076
				(0.003)	(0.005)	(0.002)		(0.002)	(0.005)	(0.001)
		TMLE-pw-Hájek	1.000	−0.001	0.838	0.178	1.000	−0.001	0.898	0.088
				(0.004)	(0.008)	(0.003)		(0.002)	(0.007)	(0.001)
		DR-nt	1.000	0.223	0.722	0.283	0.997	0.131	0.723	0.164
				(0.004)	(0.010)	(0.004)		(0.002)	(0.010)	(0.002)
		DR-ny	1.000	0.129	0.880	0.213	1.000	0.040	0.919	0.096
				(0.004)	(0.007)	(0.004)		(0.002)	(0.006)	(0.002)

Table I.3: Stress scenarios B and B+ in the binary Dynamic design, conditional on completion. Conventions as in Table I.2.

			ATT(2, 4)				ES(1)			
Design	n	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	100	EEq-Hájek	0.941	0.0052	0.953	0.2955	0.791	0.0035	0.953	0.1841
				(0.0068)	(0.005)	(0.0055)		(0.0046)	(0.005)	(0.0036)
		TMLE-pw-Hájek	0.941	0.0051	0.882	0.3288	0.770	0.0055	0.919	0.1808
				(0.0076)	(0.007)	(0.0076)		(0.0046)	(0.007)	(0.0039)
		DR-nt	0.150	0.0179	0.913	0.2783	0.150	−0.0026	0.920	0.1567
				(0.0161)	(0.016)	(0.0134)		(0.0091)	(0.016)	(0.0062)
		DR-ny	0.150	−0.0038	0.926	0.2555	0.150	−0.0030	0.923	0.1382
				(0.0148)	(0.015)	(0.0119)		(0.0080)	(0.015)	(0.0055)
	500	EEq-Hájek	1.000	0.0042	0.950	0.1166	1.000	0.0008	0.947	0.0719
				(0.0026)	(0.005)	(0.0020)		(0.0016)	(0.005)	(0.0012)
		TMLE-pw-Hájek	1.000	0.0042	0.918	0.1240	1.000	0.0006	0.934	0.0722
				(0.0028)	(0.006)	(0.0022)		(0.0016)	(0.006)	(0.0012)
		DR-nt	1.000	−0.0053	0.940	0.1172	1.000	−0.0038	0.941	0.0653
				(0.0026)	(0.005)	(0.0020)		(0.0015)	(0.005)	(0.0011)
		DR-ny	1.000	−0.0035	0.935	0.1071	1.000	−0.0014	0.945	0.0607
				(0.0024)	(0.006)	(0.0018)		(0.0014)	(0.005)	(0.0010)
	5,000	EEq-Hájek	1.000	−0.0012	0.958	0.0352	1.000	−0.0005	0.948	0.0210
				(0.0008)	(0.004)	(0.0006)		(0.0005)	(0.005)	(0.0003)
		TMLE-pw-Hájek	1.000	−0.0013	0.946	0.0374	1.000	−0.0005	0.947	0.0212
				(0.0008)	(0.005)	(0.0006)		(0.0005)	(0.005)	(0.0003)
		DR-nt	1.000	−0.0091	0.937	0.0375	1.000	−0.0043	0.946	0.0208
				(0.0008)	(0.005)	(0.0006)		(0.0005)	(0.005)	(0.0003)
		DR-ny	1.000	−0.0067	0.945	0.0330	1.000	−0.0020	0.944	0.0185
				(0.0007)	(0.005)	(0.0005)		(0.0004)	(0.005)	(0.0003)
B+	100	EEq-Hájek	0.692	−0.0014	0.966	0.3844	0.373	−0.0022	0.954	0.2319
				(0.0103)	(0.005)	(0.0087)		(0.0085)	(0.008)	(0.0076)
		TMLE-pw-Hájek	0.692	0.0048	0.788	0.5711	0.343	−0.0019	0.872	0.2631
				(0.0154)	(0.011)	(0.0278)		(0.0101)	(0.013)	(0.0095)
		DR-nt	0.030	−0.0060	0.950	0.2735	0.029	−0.0004	0.931	0.1694
				(0.0356)	(0.028)	(0.0277)		(0.0224)	(0.033)	(0.0120)
		DR-ny	0.030	−0.0002	0.983	0.2536	0.029	−0.0125	0.845	0.2013
				(0.0330)	(0.017)	(0.0240)		(0.0266)	(0.048)	(0.0214)
	500	EEq-Hájek	1.000	−0.0051	0.960	0.1542	1.000	−0.0003	0.956	0.0966
				(0.0034)	(0.004)	(0.0025)		(0.0022)	(0.005)	(0.0016)
		TMLE-pw-Hájek	1.000	−0.0066	0.848	0.2021	1.000	0.0009	0.896	0.1103
				(0.0045)	(0.008)	(0.0035)		(0.0025)	(0.007)	(0.0019)
		DR-nt	1.000	−0.0192	0.932	0.1648	1.000	−0.0091	0.921	0.1018
				(0.0037)	(0.006)	(0.0034)		(0.0023)	(0.006)	(0.0020)
		DR-ny	1.000	−0.0208	0.937	0.1473	1.000	−0.0070	0.932	0.0871
				(0.0033)	(0.005)	(0.0028)		(0.0019)	(0.006)	(0.0015)
	5,000	EEq-Hájek	1.000	−0.0010	0.952	0.0493	1.000	−0.0001	0.946	0.0302
				(0.0011)	(0.005)	(0.0008)		(0.0007)	(0.005)	(0.0005)
		TMLE-pw-Hájek	1.000	−0.0013	0.866	0.0637	1.000	−0.0001	0.916	0.0342
				(0.0014)	(0.008)	(0.0010)		(0.0008)	(0.006)	(0.0005)
		DR-nt	1.000	−0.0154	0.946	0.0553	1.000	−0.0084	0.941	0.0325
				(0.0012)	(0.005)	(0.0010)		(0.0007)	(0.005)	(0.0006)
		DR-ny	1.000	−0.0128	0.947	0.0502	1.000	−0.0044	0.948	0.0286
				(0.0011)	(0.005)	(0.0009)		(0.0006)	(0.005)	(0.0005)

J Complete Simulation Tables

The tables of this section report every cell of the Static and Dynamic designs: the continuous and binary outcome versions of each design, scenarios A, B, B+, C1, and C2, sample sizes $n \in \{100, 500, 5{,}000\}$, and both cross-fitting settings, for all ten procedures and both targets, ATT$(2,4)$ and ES(1). The designs are specified in Section 5.3, and the working models and comparator specification are in Appendix G; the CompactBinary results are reported in full in Appendix H.4.

Conventions. Compl. is the share of the 2,000 requested replications in which the procedure returned a finite estimate and a finite, positive standard error. Bias, coverage (Cov.) of the nominal 95% Wald interval, and root mean squared error (RMSE) are computed over those complete replications, so they are conditional on completion. Monte Carlo standard errors (MCSEs), in parentheses below the estimates, are pointwise and unadjusted for multiplicity, and every comparison of empirical coverage is descriptive. In the continuous designs both targets equal one; in the binary designs they are probability-scale effects that vary with the scenario and are listed in each table's caption. Bias and RMSE are reported to three decimals in the continuous designs and four in the binary designs. With five-fold cross-fitting ($L = 5$) the tables list the eight proposed procedures and the two comparators; without cross-fitting ($L = 1$) they list the proposed procedures only, because the comparators do not cross-fit and their $L = 1$ results equal their $L = 5$ results. Procedures: EEq-HT and EEq-Hájek are the one-step estimating-equation estimators of Section 3; TMLE-pw-HT and TMLE-pw-Hájek are the pairwise TMLEs of Section 4; the preregistered variants appear as TMLE-cw-{HT,Hájek} for cohort-wise targeting and TMLE-2eps-{cw,pw}-Hájek for the two-fluctuation diagnostic, with archive identifiers mapped in Appendix G.5; DR-nt and DR-ny are the `did` doubly robust estimators with never-treated and not-yet-treated comparison groups.

Design	ATT$(2,4)$	ES(1)
Static, continuous	Tables J.1, J.2	Tables J.3, J.4
Dynamic, continuous	Tables J.5, J.6	Tables J.7, J.8
Static, binary	Tables J.9, J.10	Tables J.11, J.12
Dynamic, binary	Tables J.13, J.14	Tables J.15, J.16

Table J.1: Static design, continuous outcome: ATT$(2,4) = 1$; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.997	0.029	0.926	1.169	1.000	0.005	0.943	0.426	1.000	−0.001	0.945	0.133
			(0.026)	(0.006)	(0.021)		(0.010)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)
	EEq-Hájek	0.997	0.027	0.922	1.118	1.000	0.005	0.945	0.432	1.000	−0.001	0.949	0.136
			(0.025)	(0.006)	(0.019)		(0.010)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-cw-HT	0.982	0.032	0.874	1.088	1.000	0.005	0.920	0.442	1.000	−0.001	0.930	0.141
			(0.025)	(0.007)	(0.019)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.982	0.036	0.868	1.116	1.000	0.005	0.924	0.446	1.000	−0.001	0.939	0.142
			(0.025)	(0.008)	(0.021)		(0.010)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-pw-HT	0.997	0.033	0.889	1.128	1.000	0.005	0.922	0.443	1.000	−0.002	0.933	0.140
			(0.025)	(0.007)	(0.020)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	TMLE-pw-Hájek	0.997	0.035	0.886	1.142	1.000	0.005	0.925	0.447	1.000	−0.001	0.941	0.141
			(0.026)	(0.007)	(0.020)		(0.010)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-2eps-cw-Hájek	0.982	0.041	0.852	1.150	1.000	0.006	0.919	0.454	1.000	−0.001	0.934	0.145
			(0.026)	(0.008)	(0.024)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	TMLE-2eps-pw-Hájek	0.997	0.040	0.867	1.172	1.000	0.005	0.919	0.456	1.000	−0.001	0.935	0.144
			(0.026)	(0.008)	(0.022)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	DR-nt	0.809	−0.012	0.865	1.037	1.000	−0.008	0.937	0.450	1.000	0.002	0.952	0.156
			(0.026)	(0.008)	(0.020)		(0.010)	(0.005)	(0.007)		(0.003)	(0.005)	(0.003)
	DR-ny	0.809	0.023	0.894	0.903	1.000	−0.011	0.949	0.396	1.000	0.004	0.951	0.133
			(0.022)	(0.008)	(0.018)		(0.009)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)

Table J.1, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.989	−0.007	0.915	1.201	1.000	−0.017	0.909	0.500	1.000	0.001	0.924	0.151
			(0.027)	(0.006)	(0.020)		(0.011)	(0.006)	(0.008)		(0.003)	(0.006)	(0.002)
	EEq-Hájek	0.989	−0.006	0.930	1.249	1.000	−0.019	0.931	0.534	1.000	0.002	0.940	0.164
			(0.028)	(0.006)	(0.022)		(0.012)	(0.006)	(0.009)		(0.004)	(0.005)	(0.003)
	TMLE-cw-HT	0.937	−0.002	0.806	1.324	1.000	−0.020	0.849	0.578	1.000	0.003	0.856	0.180
			(0.031)	(0.009)	(0.024)		(0.013)	(0.008)	(0.010)		(0.004)	(0.008)	(0.003)
	TMLE-cw-Hájek	0.937	0.003	0.793	1.471	1.000	−0.021	0.870	0.604	1.000	0.003	0.898	0.187
			(0.034)	(0.009)	(0.031)		(0.014)	(0.008)	(0.010)		(0.004)	(0.007)	(0.003)
	TMLE-pw-HT	0.989	0.002	0.825	1.358	1.000	−0.020	0.853	0.580	1.000	0.002	0.867	0.177
			(0.031)	(0.009)	(0.024)		(0.013)	(0.008)	(0.009)		(0.004)	(0.008)	(0.003)
	TMLE-pw-Hájek	0.989	0.002	0.826	1.485	1.000	−0.021	0.881	0.605	1.000	0.003	0.905	0.185
			(0.033)	(0.009)	(0.029)		(0.014)	(0.007)	(0.010)		(0.004)	(0.007)	(0.003)
	TMLE-2eps-cw-Hájek	0.937	0.005	0.757	1.591	1.000	−0.021	0.860	0.625	1.000	0.004	0.884	0.193
			(0.037)	(0.010)	(0.037)		(0.014)	(0.008)	(0.011)		(0.004)	(0.007)	(0.003)
	TMLE-2eps-pw-Hájek	0.989	0.004	0.794	1.579	1.000	−0.022	0.866	0.627	1.000	0.003	0.889	0.192
			(0.036)	(0.009)	(0.032)		(0.014)	(0.008)	(0.010)		(0.004)	(0.007)	(0.003)
	DR-nt	0.577	−0.004	0.871	1.129	1.000	−0.002	0.916	0.575	1.000	0.005	0.944	0.200
			(0.033)	(0.010)	(0.026)		(0.013)	(0.006)	(0.015)		(0.004)	(0.005)	(0.004)
	DR-ny	0.577	−0.037	0.879	1.022	1.000	−0.009	0.922	0.532	1.000	−0.003	0.959	0.177
			(0.030)	(0.010)	(0.026)		(0.012)	(0.006)	(0.011)		(0.004)	(0.004)	(0.003)
B+	EEq-HT	0.900	0.013	0.837	1.209	1.000	0.010	0.833	0.514	1.000	0.004	0.837	0.163
			(0.029)	(0.009)	(0.021)		(0.011)	(0.008)	(0.008)		(0.004)	(0.008)	(0.003)
	EEq-Hájek	0.900	0.018	0.904	1.472	1.000	0.009	0.921	0.640	1.000	0.005	0.918	0.202
			(0.035)	(0.007)	(0.026)		(0.014)	(0.006)	(0.010)		(0.005)	(0.006)	(0.003)
	TMLE-cw-HT	0.703	−0.005	0.632	1.609	1.000	0.006	0.656	0.723	1.000	0.006	0.673	0.229
			(0.043)	(0.013)	(0.032)		(0.016)	(0.011)	(0.011)		(0.005)	(0.010)	(0.004)
	TMLE-cw-Hájek	0.703	−0.034	0.628	2.211	1.000	0.002	0.778	0.854	1.000	0.007	0.812	0.263
			(0.059)	(0.013)	(0.070)		(0.019)	(0.009)	(0.014)		(0.006)	(0.009)	(0.004)
	TMLE-pw-HT	0.900	0.010	0.653	1.680	1.000	0.008	0.655	0.726	1.000	0.005	0.674	0.229
			(0.040)	(0.011)	(0.030)		(0.016)	(0.011)	(0.011)		(0.005)	(0.010)	(0.004)
	TMLE-pw-Hájek	0.900	0.029	0.697	2.181	1.000	0.006	0.792	0.846	1.000	0.007	0.812	0.262
			(0.051)	(0.011)	(0.063)		(0.019)	(0.009)	(0.014)		(0.006)	(0.009)	(0.004)
	TMLE-2eps-cw-Hájek	0.703	−0.042	0.599	2.407	1.000	0.001	0.762	0.883	1.000	0.008	0.799	0.271
			(0.064)	(0.013)	(0.084)		(0.020)	(0.010)	(0.015)		(0.006)	(0.009)	(0.004)
	TMLE-2eps-pw-Hájek	0.900	0.030	0.676	2.304	1.000	0.006	0.776	0.876	1.000	0.007	0.798	0.270
			(0.054)	(0.011)	(0.068)		(0.020)	(0.009)	(0.014)		(0.006)	(0.009)	(0.004)
	DR-nt	0.189	0.001	0.767	1.402	1.000	0.014	0.894	0.704	1.000	0.000	0.949	0.245
			(0.072)	(0.022)	(0.053)		(0.016)	(0.007)	(0.013)		(0.005)	(0.005)	(0.004)
	DR-ny	0.189	0.032	0.809	1.154	1.000	0.008	0.906	0.628	1.000	−0.002	0.947	0.234
			(0.059)	(0.020)	(0.047)		(0.014)	(0.007)	(0.011)		(0.005)	(0.005)	(0.004)
C1	EEq-HT	0.999	0.144	0.923	1.642	1.000	0.394	0.889	0.732	1.000	0.449	0.442	0.487
			(0.037)	(0.006)	(0.031)		(0.014)	(0.007)	(0.011)		(0.004)	(0.011)	(0.004)
	EEq-Hájek	0.999	0.249	0.926	1.483	1.000	0.355	0.920	0.689	1.000	0.387	0.621	0.428
			(0.033)	(0.006)	(0.025)		(0.013)	(0.006)	(0.011)		(0.004)	(0.011)	(0.004)
	TMLE-cw-HT	0.984	0.333	0.894	1.285	1.000	0.288	0.884	0.602	1.000	0.284	0.586	0.329
			(0.028)	(0.007)	(0.021)		(0.012)	(0.007)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-cw-Hájek	0.984	0.294	0.893	1.274	1.000	0.263	0.900	0.586	1.000	0.260	0.671	0.307
			(0.028)	(0.007)	(0.021)		(0.012)	(0.007)	(0.009)		(0.004)	(0.011)	(0.003)
	TMLE-pw-HT	0.999	0.252	0.905	1.312	1.000	0.255	0.893	0.588	1.000	0.264	0.636	0.311
			(0.029)	(0.007)	(0.022)		(0.012)	(0.007)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-pw-Hájek	0.999	0.201	0.911	1.315	1.000	0.233	0.907	0.574	1.000	0.243	0.706	0.292
			(0.029)	(0.006)	(0.024)		(0.012)	(0.007)	(0.009)		(0.004)	(0.010)	(0.003)
	TMLE-2eps-cw-Hájek	0.984	0.272	0.867	1.235	1.000	0.225	0.890	0.554	1.000	0.215	0.702	0.265
			(0.027)	(0.008)	(0.021)		(0.011)	(0.007)	(0.009)		(0.003)	(0.010)	(0.003)
	TMLE-2eps-pw-Hájek	0.999	0.150	0.883	1.306	1.000	0.170	0.905	0.540	1.000	0.174	0.787	0.234
			(0.029)	(0.007)	(0.029)		(0.011)	(0.007)	(0.009)		(0.003)	(0.009)	(0.003)
	DR-nt	0.802	1.842	0.594	2.549	1.000	1.430	0.430	1.772	1.000	1.321	0.149	1.372
			(0.044)	(0.012)	(0.036)		(0.023)	(0.011)	(0.020)		(0.008)	(0.008)	(0.007)
	DR-ny	0.802	1.183	0.713	1.965	1.000	0.940	0.595	1.275	1.000	0.846	0.242	0.906
			(0.039)	(0.011)	(0.031)		(0.019)	(0.011)	(0.015)		(0.007)	(0.010)	(0.006)

Table J.1, continued

Scen.	Procedure	$n = 100$ Compl.	Bias	Cov.	RMSE	$n = 500$ Compl.	Bias	Cov.	RMSE	$n = 5{,}000$ Compl.	Bias	Cov.	RMSE
C2	EEq-HT	0.977	0.007	0.811	1.107	1.000	0.013	0.802	0.478	1.000	−0.003	0.819	0.148
			(0.025)	(0.009)	(0.019)		(0.011)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	EEq-Hájek	0.977	0.011	0.886	1.270	1.000	0.011	0.900	0.549	1.000	−0.003	0.907	0.169
			(0.029)	(0.007)	(0.022)		(0.012)	(0.007)	(0.009)		(0.004)	(0.006)	(0.003)
	TMLE-cw-HT	0.891	0.030	0.637	1.427	1.000	0.011	0.641	0.642	1.000	−0.004	0.674	0.198
			(0.034)	(0.011)	(0.026)		(0.014)	(0.011)	(0.010)		(0.004)	(0.010)	(0.003)
	TMLE-cw-Hájek	0.891	0.034	0.680	1.735	1.000	0.010	0.762	0.731	1.000	−0.005	0.792	0.221
			(0.041)	(0.011)	(0.039)		(0.016)	(0.010)	(0.012)		(0.005)	(0.009)	(0.003)
	TMLE-pw-HT	0.977	0.017	0.652	1.484	1.000	0.014	0.650	0.641	1.000	−0.004	0.671	0.200
			(0.034)	(0.011)	(0.028)		(0.014)	(0.011)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-pw-Hájek	0.977	0.030	0.718	1.776	1.000	0.011	0.774	0.727	1.000	−0.004	0.794	0.222
			(0.040)	(0.010)	(0.041)		(0.016)	(0.009)	(0.012)		(0.005)	(0.009)	(0.003)
	TMLE-2eps-cw-Hájek	0.891	0.035	0.643	1.856	1.000	0.010	0.733	0.768	1.000	−0.005	0.772	0.232
			(0.044)	(0.011)	(0.044)		(0.017)	(0.010)	(0.013)		(0.005)	(0.009)	(0.004)
	TMLE-2eps-pw-Hájek	0.977	0.033	0.684	1.897	1.000	0.012	0.749	0.765	1.000	−0.005	0.768	0.233
			(0.043)	(0.011)	(0.046)		(0.017)	(0.010)	(0.013)		(0.005)	(0.009)	(0.004)
	DR-nt	0.451	0.041	0.847	1.206	1.000	0.009	0.931	0.539	1.000	−0.003	0.955	0.164
			(0.040)	(0.012)	(0.029)		(0.012)	(0.006)	(0.009)		(0.004)	(0.005)	(0.003)
	DR-ny	0.451	−0.026	0.872	1.046	1.000	0.002	0.934	0.485	1.000	−0.002	0.957	0.147
			(0.035)	(0.011)	(0.028)		(0.011)	(0.006)	(0.008)		(0.003)	(0.005)	(0.002)

Table J.2: Static design, continuous outcome: $\text{ATT}(2,4) = 1$; no cross-fitting (proposed procedures).

Scen.	Procedure	$n = 100$ Compl.	Bias	Cov.	RMSE	$n = 500$ Compl.	Bias	Cov.	RMSE	$n = 5{,}000$ Compl.	Bias	Cov.	RMSE
A	EEq-HT	1.000	0.020	0.900	0.986	1.000	0.004	0.940	0.412	1.000	−0.001	0.945	0.133
			(0.022)	(0.007)	(0.017)		(0.009)	(0.005)	(0.006)		(0.003)	(0.005)	(0.002)
	EEq-Hájek	1.000	0.020	0.909	0.990	1.000	0.004	0.947	0.419	1.000	−0.001	0.950	0.135
			(0.022)	(0.006)	(0.017)		(0.009)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-cw-HT	0.997	0.024	0.853	1.015	1.000	0.004	0.914	0.432	1.000	−0.001	0.930	0.141
			(0.023)	(0.008)	(0.017)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.997	0.028	0.852	1.050	1.000	0.004	0.924	0.438	1.000	−0.001	0.940	0.142
			(0.024)	(0.008)	(0.019)		(0.010)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-pw-HT	1.000	0.023	0.855	1.037	1.000	0.004	0.918	0.433	1.000	−0.001	0.934	0.139
			(0.023)	(0.008)	(0.018)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	TMLE-pw-Hájek	1.000	0.025	0.867	1.063	1.000	0.004	0.926	0.438	1.000	−0.001	0.943	0.141
			(0.024)	(0.008)	(0.018)		(0.010)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-2eps-cw-Hájek	0.997	0.033	0.831	1.106	1.000	0.005	0.916	0.447	1.000	−0.001	0.934	0.144
			(0.025)	(0.008)	(0.023)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	TMLE-2eps-pw-Hájek	1.000	0.029	0.842	1.116	1.000	0.004	0.917	0.449	1.000	−0.001	0.935	0.143
			(0.025)	(0.008)	(0.020)		(0.010)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
B	EEq-HT	0.998	−0.014	0.896	1.049	1.000	−0.016	0.909	0.488	1.000	0.001	0.925	0.150
			(0.023)	(0.007)	(0.018)		(0.011)	(0.006)	(0.008)		(0.003)	(0.006)	(0.002)
	EEq-Hájek	0.998	−0.011	0.918	1.114	1.000	−0.017	0.929	0.523	1.000	0.002	0.939	0.163
			(0.025)	(0.006)	(0.019)		(0.012)	(0.006)	(0.009)		(0.004)	(0.005)	(0.003)
	TMLE-cw-HT	0.991	−0.010	0.784	1.216	1.000	−0.019	0.839	0.569	1.000	0.003	0.854	0.179
			(0.027)	(0.009)	(0.022)		(0.013)	(0.008)	(0.009)		(0.004)	(0.008)	(0.003)
	TMLE-cw-Hájek	0.991	−0.010	0.782	1.388	1.000	−0.020	0.865	0.597	1.000	0.003	0.898	0.187
			(0.031)	(0.009)	(0.028)		(0.013)	(0.008)	(0.010)		(0.004)	(0.007)	(0.003)
	TMLE-pw-HT	0.998	−0.004	0.790	1.255	1.000	−0.018	0.841	0.570	1.000	0.001	0.865	0.177
			(0.028)	(0.009)	(0.021)		(0.013)	(0.008)	(0.009)		(0.004)	(0.008)	(0.003)
	TMLE-pw-Hájek	0.998	−0.002	0.805	1.400	1.000	−0.019	0.877	0.597	1.000	0.002	0.903	0.185
			(0.031)	(0.009)	(0.027)		(0.013)	(0.007)	(0.010)		(0.004)	(0.007)	(0.003)
	TMLE-2eps-cw-Hájek	0.991	−0.009	0.740	1.530	1.000	−0.021	0.853	0.619	1.000	0.003	0.886	0.193
			(0.034)	(0.010)	(0.034)		(0.014)	(0.008)	(0.010)		(0.004)	(0.007)	(0.003)
	TMLE-2eps-pw-Hájek	0.998	0.000	0.772	1.522	1.000	−0.020	0.861	0.621	1.000	0.002	0.890	0.192
			(0.034)	(0.009)	(0.031)		(0.014)	(0.008)	(0.010)		(0.004)	(0.007)	(0.003)

Table J.2, continued

Scen.	Procedure	$n = 100$				$n = 500$				$n = 5{,}000$			
		Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B+	EEq-HT	0.984	−0.001	0.790	1.132	1.000	0.009	0.824	0.507	1.000	0.003	0.836	0.163
			(0.026)	(0.009)	(0.018)		(0.011)	(0.009)	(0.008)		(0.004)	(0.008)	(0.003)
	EEq-Hájek	0.984	0.001	0.881	1.347	1.000	0.008	0.917	0.628	1.000	0.005	0.917	0.202
			(0.030)	(0.007)	(0.023)		(0.014)	(0.006)	(0.010)		(0.005)	(0.006)	(0.003)
	TMLE-cw-HT	0.936	−0.008	0.619	1.490	1.000	0.006	0.644	0.713	1.000	0.006	0.672	0.229
			(0.034)	(0.011)	(0.026)		(0.016)	(0.011)	(0.011)		(0.005)	(0.011)	(0.004)
	TMLE-cw-Hájek	0.936	−0.045	0.605	2.190	1.000	0.002	0.773	0.851	1.000	0.007	0.812	0.264
			(0.051)	(0.011)	(0.069)		(0.019)	(0.009)	(0.015)		(0.006)	(0.009)	(0.004)
	TMLE-pw-HT	0.984	−0.005	0.605	1.595	1.000	0.008	0.651	0.716	1.000	0.005	0.675	0.229
			(0.036)	(0.011)	(0.027)		(0.016)	(0.011)	(0.011)		(0.005)	(0.010)	(0.004)
	TMLE-pw-Hájek	0.984	−0.006	0.646	2.144	1.000	0.006	0.783	0.841	1.000	0.007	0.807	0.262
			(0.048)	(0.011)	(0.053)		(0.019)	(0.009)	(0.014)		(0.006)	(0.009)	(0.004)
	TMLE-2eps-cw-Hájek	0.936	−0.056	0.573	2.423	1.000	0.001	0.747	0.881	1.000	0.007	0.796	0.271
			(0.056)	(0.011)	(0.084)		(0.020)	(0.010)	(0.016)		(0.006)	(0.009)	(0.004)
	TMLE-2eps-pw-Hájek	0.984	−0.007	0.629	2.292	1.000	0.005	0.765	0.872	1.000	0.007	0.794	0.270
			(0.052)	(0.011)	(0.058)		(0.019)	(0.009)	(0.015)		(0.006)	(0.009)	(0.004)
C1	EEq-HT	1.000	0.638	0.878	1.487	1.000	0.492	0.854	0.771	1.000	0.459	0.421	0.496
			(0.030)	(0.007)	(0.024)		(0.013)	(0.008)	(0.011)		(0.004)	(0.011)	(0.004)
	EEq-Hájek	1.000	0.608	0.899	1.413	1.000	0.435	0.897	0.720	1.000	0.395	0.591	0.436
			(0.029)	(0.007)	(0.022)		(0.013)	(0.007)	(0.011)		(0.004)	(0.011)	(0.004)
	TMLE-cw-HT	0.999	0.502	0.864	1.258	1.000	0.331	0.861	0.617	1.000	0.289	0.571	0.332
			(0.026)	(0.008)	(0.021)		(0.012)	(0.008)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-cw-Hájek	0.999	0.431	0.878	1.228	1.000	0.298	0.886	0.596	1.000	0.264	0.657	0.310
			(0.026)	(0.007)	(0.020)		(0.012)	(0.007)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-pw-HT	1.000	0.438	0.873	1.259	1.000	0.300	0.877	0.601	1.000	0.269	0.619	0.314
			(0.026)	(0.007)	(0.021)		(0.012)	(0.007)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-pw-Hájek	1.000	0.358	0.890	1.243	1.000	0.269	0.894	0.582	1.000	0.247	0.697	0.295
			(0.027)	(0.007)	(0.022)		(0.012)	(0.007)	(0.010)		(0.004)	(0.010)	(0.003)
	TMLE-2eps-cw-Hájek	0.999	0.308	0.861	1.179	1.000	0.236	0.881	0.553	1.000	0.216	0.695	0.266
			(0.025)	(0.008)	(0.020)		(0.011)	(0.007)	(0.009)		(0.003)	(0.010)	(0.003)
	TMLE-2eps-pw-Hájek	1.000	0.204	0.869	1.225	1.000	0.183	0.896	0.539	1.000	0.175	0.785	0.235
			(0.027)	(0.008)	(0.025)		(0.011)	(0.007)	(0.009)		(0.003)	(0.009)	(0.003)
C2	EEq-HT	0.997	0.006	0.765	1.041	1.000	0.014	0.799	0.472	1.000	−0.003	0.819	0.148
			(0.023)	(0.009)	(0.018)		(0.011)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	EEq-Hájek	0.997	0.007	0.855	1.166	1.000	0.013	0.892	0.539	1.000	−0.003	0.907	0.168
			(0.026)	(0.008)	(0.020)		(0.012)	(0.007)	(0.008)		(0.004)	(0.007)	(0.003)
	TMLE-cw-HT	0.981	0.012	0.629	1.316	1.000	0.014	0.640	0.629	1.000	−0.004	0.672	0.198
			(0.030)	(0.011)	(0.023)		(0.014)	(0.011)	(0.010)		(0.004)	(0.011)	(0.003)
	TMLE-cw-Hájek	0.981	0.021	0.669	1.627	1.000	0.013	0.757	0.718	1.000	−0.004	0.793	0.221
			(0.037)	(0.011)	(0.033)		(0.016)	(0.010)	(0.012)		(0.005)	(0.009)	(0.003)
	TMLE-pw-HT	0.997	0.012	0.627	1.375	1.000	0.017	0.646	0.630	1.000	−0.003	0.673	0.200
			(0.031)	(0.011)	(0.026)		(0.014)	(0.011)	(0.010)		(0.004)	(0.010)	(0.003)
	TMLE-pw-Hájek	0.997	0.022	0.693	1.660	1.000	0.015	0.765	0.714	1.000	−0.004	0.793	0.221
			(0.037)	(0.010)	(0.036)		(0.016)	(0.009)	(0.012)		(0.005)	(0.009)	(0.003)
	TMLE-2eps-cw-Hájek	0.981	0.025	0.635	1.754	1.000	0.014	0.728	0.756	1.000	−0.005	0.770	0.231
			(0.040)	(0.011)	(0.037)		(0.017)	(0.010)	(0.013)		(0.005)	(0.009)	(0.004)
	TMLE-2eps-pw-Hájek	0.997	0.024	0.661	1.788	1.000	0.015	0.741	0.754	1.000	−0.005	0.767	0.233
			(0.040)	(0.011)	(0.041)		(0.017)	(0.010)	(0.012)		(0.005)	(0.009)	(0.004)

Table J.3: Static design, continuous outcome: ES(1) = 1; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.982	−0.013	0.943	0.746	1.000	0.003	0.948	0.263	1.000	−0.005	0.942	0.082
			(0.017)	(0.005)	(0.015)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	EEq-Hájek	0.982	−0.012	0.936	0.672	1.000	0.003	0.949	0.262	1.000	−0.005	0.945	0.082
			(0.015)	(0.006)	(0.012)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-cw-HT	0.981	−0.009	0.919	0.622	1.000	0.003	0.943	0.260	1.000	−0.005	0.940	0.083
			(0.014)	(0.006)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-cw-Hájek	0.981	−0.007	0.912	0.628	1.000	0.003	0.942	0.260	1.000	−0.005	0.941	0.083
			(0.014)	(0.006)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-pw-HT	0.981	−0.008	0.919	0.642	1.000	0.003	0.943	0.261	1.000	−0.005	0.940	0.083
			(0.015)	(0.006)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-pw-Hájek	0.981	−0.007	0.916	0.646	1.000	0.003	0.942	0.261	1.000	−0.005	0.942	0.083
			(0.015)	(0.006)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-2eps-cw-Hájek	0.981	−0.005	0.906	0.630	1.000	0.003	0.938	0.260	1.000	−0.005	0.941	0.083
			(0.014)	(0.007)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.981	−0.004	0.907	0.648	1.000	0.003	0.940	0.262	1.000	−0.005	0.940	0.083
			(0.015)	(0.007)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	DR-nt	0.809	−0.039	0.895	0.592	1.000	−0.003	0.941	0.257	1.000	−0.004	0.951	0.087
			(0.015)	(0.008)	(0.012)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	DR-ny	0.809	−0.024	0.939	0.510	1.000	0.001	0.954	0.220	1.000	−0.003	0.940	0.075
			(0.013)	(0.006)	(0.010)		(0.005)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
B	EEq-HT	0.937	0.011	0.935	0.828	1.000	−0.007	0.940	0.320	1.000	0.001	0.940	0.099
			(0.019)	(0.006)	(0.014)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	EEq-Hájek	0.937	0.011	0.932	0.779	1.000	−0.007	0.944	0.326	1.000	0.001	0.946	0.102
			(0.018)	(0.006)	(0.013)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	TMLE-cw-HT	0.936	0.011	0.881	0.776	1.000	−0.007	0.909	0.338	1.000	0.002	0.920	0.106
			(0.018)	(0.007)	(0.013)		(0.008)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.936	0.013	0.873	0.806	1.000	−0.007	0.916	0.343	1.000	0.002	0.932	0.107
			(0.019)	(0.008)	(0.014)		(0.008)	(0.006)	(0.006)		(0.002)	(0.006)	(0.002)
	TMLE-pw-HT	0.936	0.012	0.887	0.791	1.000	−0.007	0.912	0.337	1.000	0.001	0.922	0.105
			(0.018)	(0.007)	(0.013)		(0.008)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-Hájek	0.936	0.011	0.883	0.820	1.000	−0.007	0.918	0.344	1.000	0.001	0.935	0.107
			(0.019)	(0.007)	(0.014)		(0.008)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-cw-Hájek	0.936	0.014	0.847	0.852	1.000	−0.007	0.901	0.351	1.000	0.002	0.928	0.109
			(0.020)	(0.008)	(0.016)		(0.008)	(0.007)	(0.006)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-pw-Hájek	0.936	0.011	0.859	0.855	1.000	−0.007	0.906	0.352	1.000	0.001	0.928	0.109
			(0.020)	(0.008)	(0.016)		(0.008)	(0.007)	(0.006)		(0.002)	(0.006)	(0.002)
	DR-nt	0.577	0.006	0.884	0.656	1.000	−0.001	0.940	0.331	0.999	0.002	0.943	0.119
			(0.019)	(0.009)	(0.015)		(0.007)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)
	DR-ny	0.577	0.006	0.914	0.604	1.000	0.001	0.933	0.302	1.000	0.002	0.949	0.099
			(0.018)	(0.008)	(0.014)		(0.007)	(0.006)	(0.006)		(0.002)	(0.005)	(0.002)
B+	EEq-HT	0.703	0.001	0.875	0.914	1.000	0.009	0.881	0.370	1.000	0.003	0.890	0.117
			(0.024)	(0.009)	(0.019)		(0.008)	(0.007)	(0.006)		(0.003)	(0.007)	(0.002)
	EEq-Hájek	0.703	−0.005	0.900	0.973	1.000	0.007	0.925	0.419	1.000	0.002	0.928	0.133
			(0.026)	(0.008)	(0.020)		(0.009)	(0.006)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-cw-HT	0.697	−0.005	0.735	1.031	1.000	0.004	0.771	0.460	1.000	0.002	0.792	0.146
			(0.028)	(0.012)	(0.021)		(0.010)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	TMLE-cw-Hájek	0.697	−0.015	0.731	1.215	1.000	0.002	0.844	0.498	1.000	0.002	0.868	0.156
			(0.033)	(0.012)	(0.029)		(0.011)	(0.008)	(0.008)		(0.003)	(0.008)	(0.003)
	TMLE-pw-HT	0.697	0.000	0.746	1.059	1.000	0.007	0.775	0.460	1.000	0.003	0.794	0.145
			(0.028)	(0.012)	(0.021)		(0.010)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	TMLE-pw-Hájek	0.697	−0.003	0.754	1.222	1.000	0.005	0.853	0.499	1.000	0.003	0.870	0.156
			(0.033)	(0.012)	(0.027)		(0.011)	(0.008)	(0.008)		(0.003)	(0.008)	(0.003)
	TMLE-2eps-cw-Hájek	0.697	−0.023	0.705	1.321	1.000	0.001	0.825	0.515	1.000	0.002	0.860	0.160
			(0.035)	(0.012)	(0.035)		(0.012)	(0.008)	(0.008)		(0.004)	(0.008)	(0.003)
	TMLE-2eps-pw-Hájek	0.697	−0.003	0.740	1.294	1.000	0.005	0.834	0.516	1.000	0.003	0.858	0.161
			(0.035)	(0.012)	(0.030)		(0.012)	(0.008)	(0.008)		(0.004)	(0.008)	(0.003)
	DR-nt	0.188	−0.014	0.806	0.936	0.999	0.010	0.888	0.482	0.998	0.000	0.939	0.180
			(0.048)	(0.020)	(0.037)		(0.011)	(0.007)	(0.009)		(0.004)	(0.005)	(0.003)
	DR-ny	0.188	−0.007	0.875	0.730	1.000	0.010	0.920	0.438	1.000	0.002	0.945	0.162
			(0.038)	(0.017)	(0.034)		(0.010)	(0.006)	(0.009)		(0.004)	(0.005)	(0.003)

Table J.3, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.984	−0.250	0.955	1.060	1.000	0.014	0.956	0.353	1.000	0.065	0.919	0.125
			(0.023)	(0.005)	(0.026)		(0.008)	(0.005)	(0.006)		(0.002)	(0.006)	(0.002)
	EEq-Hájek	0.984	−0.087	0.952	0.854	1.000	0.026	0.961	0.335	1.000	0.059	0.941	0.119
			(0.019)	(0.005)	(0.017)		(0.007)	(0.004)	(0.006)		(0.002)	(0.005)	(0.002)
	TMLE-cw-HT	0.984	0.046	0.928	0.698	1.000	0.040	0.941	0.301	1.000	0.044	0.933	0.103
			(0.016)	(0.006)	(0.012)		(0.007)	(0.005)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.984	0.035	0.921	0.697	1.000	0.039	0.941	0.297	1.000	0.041	0.939	0.101
			(0.016)	(0.006)	(0.012)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	TMLE-pw-HT	0.984	0.019	0.931	0.713	1.000	0.034	0.940	0.301	1.000	0.042	0.934	0.103
			(0.016)	(0.006)	(0.012)		(0.007)	(0.005)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-Hájek	0.984	0.004	0.927	0.714	1.000	0.033	0.941	0.298	1.000	0.040	0.940	0.101
			(0.016)	(0.006)	(0.013)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	TMLE-2eps-cw-Hájek	0.984	0.064	0.905	0.672	1.000	0.041	0.925	0.283	1.000	0.032	0.941	0.093
			(0.015)	(0.007)	(0.012)		(0.006)	(0.006)	(0.005)		(0.002)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.984	0.027	0.906	0.688	1.000	0.031	0.926	0.284	1.000	0.026	0.941	0.091
			(0.015)	(0.007)	(0.012)		(0.006)	(0.006)	(0.005)		(0.002)	(0.005)	(0.001)
	DR-nt	0.802	0.848	0.648	1.270	0.991	0.642	0.520	0.837	0.989	0.583	0.197	0.613
			(0.024)	(0.012)	(0.019)		(0.012)	(0.011)	(0.010)		(0.004)	(0.009)	(0.004)
	DR-ny	0.802	0.345	0.845	0.794	1.000	0.236	0.806	0.422	1.000	0.209	0.503	0.243
			(0.018)	(0.009)	(0.014)		(0.008)	(0.009)	(0.006)		(0.003)	(0.011)	(0.002)
C2	EEq-HT	0.891	−0.002	0.866	0.758	1.000	0.008	0.883	0.304	1.000	−0.001	0.866	0.097
			(0.018)	(0.008)	(0.013)		(0.007)	(0.007)	(0.005)		(0.002)	(0.008)	(0.002)
	EEq-Hájek	0.891	0.005	0.906	0.798	1.000	0.008	0.925	0.332	1.000	−0.001	0.911	0.106
			(0.019)	(0.007)	(0.014)		(0.007)	(0.006)	(0.006)		(0.002)	(0.006)	(0.002)
	TMLE-cw-HT	0.890	0.007	0.766	0.873	1.000	0.010	0.785	0.376	1.000	−0.002	0.779	0.121
			(0.021)	(0.010)	(0.016)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	TMLE-cw-Hájek	0.890	0.012	0.785	0.968	1.000	0.010	0.841	0.406	1.000	−0.002	0.843	0.129
			(0.023)	(0.010)	(0.019)		(0.009)	(0.008)	(0.007)		(0.003)	(0.008)	(0.002)
	TMLE-pw-HT	0.890	0.004	0.775	0.889	1.000	0.010	0.790	0.377	1.000	−0.001	0.783	0.121
			(0.021)	(0.010)	(0.017)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	TMLE-pw-Hájek	0.890	0.009	0.802	0.987	1.000	0.010	0.844	0.406	1.000	−0.002	0.839	0.130
			(0.023)	(0.009)	(0.021)		(0.009)	(0.008)	(0.007)		(0.003)	(0.008)	(0.002)
	TMLE-2eps-cw-Hájek	0.890	0.015	0.758	1.032	1.000	0.010	0.819	0.427	1.000	−0.002	0.821	0.136
			(0.024)	(0.010)	(0.021)		(0.010)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	TMLE-2eps-pw-Hájek	0.890	0.010	0.775	1.052	1.000	0.011	0.821	0.429	1.000	−0.002	0.818	0.136
			(0.025)	(0.010)	(0.024)		(0.010)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	DR-nt	0.451	0.001	0.887	0.786	1.000	0.004	0.937	0.356	1.000	−0.002	0.957	0.107
			(0.026)	(0.011)	(0.019)		(0.008)	(0.005)	(0.006)		(0.002)	(0.005)	(0.002)
	DR-ny	0.451	−0.003	0.906	0.639	1.000	0.006	0.944	0.292	1.000	−0.003	0.951	0.093
			(0.021)	(0.010)	(0.017)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)

Table J.4: Static design, continuous outcome: ES(1) = 1; no cross-fitting (proposed procedures).

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.997	−0.020	0.924	0.595	1.000	0.002	0.944	0.253	1.000	−0.005	0.945	0.082
			(0.013)	(0.006)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	EEq-Hájek	0.997	−0.019	0.927	0.587	1.000	0.002	0.946	0.253	1.000	−0.005	0.946	0.082
			(0.013)	(0.006)	(0.010)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-cw-HT	0.997	−0.017	0.907	0.587	1.000	0.002	0.940	0.254	1.000	−0.005	0.940	0.083
			(0.013)	(0.007)	(0.010)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-cw-Hájek	0.997	−0.016	0.906	0.595	1.000	0.002	0.940	0.255	1.000	−0.005	0.943	0.083
			(0.013)	(0.007)	(0.010)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-pw-HT	0.997	−0.019	0.909	0.597	1.000	0.001	0.939	0.255	1.000	−0.005	0.940	0.083
			(0.013)	(0.006)	(0.010)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-pw-Hájek	0.997	−0.018	0.907	0.605	1.000	0.001	0.940	0.255	1.000	−0.005	0.943	0.083
			(0.014)	(0.006)	(0.010)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-2eps-cw-Hájek	0.997	−0.014	0.895	0.615	1.000	0.002	0.940	0.257	1.000	−0.005	0.940	0.083
			(0.014)	(0.007)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.997	−0.017	0.898	0.629	1.000	0.001	0.938	0.258	1.000	−0.005	0.939	0.083
			(0.014)	(0.007)	(0.011)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)

Table J.4, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.991	0.006	0.920	0.679	1.000	−0.005	0.935	0.309	1.000	0.001	0.938	0.098
			(0.015)	(0.006)	(0.011)		(0.007)	(0.006)	(0.005)		(0.002)	(0.005)	(0.002)
	EEq-Hájek	0.991	0.007	0.921	0.687	1.000	−0.005	0.944	0.318	1.000	0.001	0.946	0.101
			(0.015)	(0.006)	(0.011)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	TMLE-cw-HT	0.991	0.008	0.860	0.722	1.000	−0.006	0.901	0.333	1.000	0.001	0.919	0.106
			(0.016)	(0.008)	(0.012)		(0.007)	(0.007)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.991	0.007	0.861	0.764	1.000	−0.006	0.907	0.339	1.000	0.001	0.932	0.107
			(0.017)	(0.008)	(0.013)		(0.008)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-HT	0.991	0.007	0.866	0.735	1.000	−0.005	0.904	0.332	1.000	0.001	0.919	0.105
			(0.017)	(0.008)	(0.012)		(0.007)	(0.007)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-Hájek	0.991	0.006	0.867	0.773	1.000	−0.005	0.911	0.339	1.000	0.001	0.932	0.107
			(0.017)	(0.008)	(0.014)		(0.008)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-cw-Hájek	0.991	0.009	0.831	0.834	1.000	−0.006	0.895	0.349	1.000	0.002	0.927	0.109
			(0.019)	(0.008)	(0.016)		(0.008)	(0.007)	(0.006)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-pw-Hájek	0.991	0.005	0.838	0.832	1.000	−0.006	0.897	0.351	1.000	0.001	0.925	0.109
			(0.019)	(0.008)	(0.015)		(0.008)	(0.007)	(0.006)		(0.002)	(0.006)	(0.002)
B+	EEq-HT	0.936	−0.007	0.841	0.806	1.000	0.010	0.871	0.363	1.000	0.003	0.892	0.117
			(0.019)	(0.008)	(0.014)		(0.008)	(0.007)	(0.005)		(0.003)	(0.007)	(0.002)
	EEq-Hájek	0.936	−0.010	0.884	0.894	1.000	0.008	0.923	0.412	1.000	0.002	0.927	0.133
			(0.021)	(0.007)	(0.015)		(0.009)	(0.006)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-cw-HT	0.936	−0.010	0.704	0.981	1.000	0.005	0.762	0.455	1.000	0.002	0.786	0.146
			(0.023)	(0.011)	(0.016)		(0.010)	(0.010)	(0.007)		(0.003)	(0.009)	(0.002)
	TMLE-cw-Hájek	0.936	−0.022	0.709	1.215	1.000	0.003	0.835	0.495	1.000	0.002	0.867	0.156
			(0.028)	(0.010)	(0.028)		(0.011)	(0.008)	(0.008)		(0.003)	(0.008)	(0.003)
	TMLE-pw-HT	0.936	−0.010	0.698	1.018	1.000	0.008	0.770	0.456	1.000	0.003	0.790	0.145
			(0.024)	(0.011)	(0.017)		(0.010)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	TMLE-pw-Hájek	0.936	−0.013	0.714	1.206	1.000	0.006	0.841	0.496	1.000	0.003	0.867	0.156
			(0.028)	(0.010)	(0.023)		(0.011)	(0.008)	(0.008)		(0.003)	(0.008)	(0.003)
	TMLE-2eps-cw-Hájek	0.936	−0.029	0.680	1.356	1.000	0.001	0.823	0.513	1.000	0.002	0.859	0.160
			(0.031)	(0.011)	(0.036)		(0.011)	(0.009)	(0.008)		(0.004)	(0.008)	(0.003)
	TMLE-2eps-pw-Hájek	0.936	−0.013	0.697	1.308	1.000	0.006	0.826	0.514	1.000	0.003	0.858	0.161
			(0.030)	(0.011)	(0.026)		(0.011)	(0.008)	(0.008)		(0.004)	(0.008)	(0.003)
C1	EEq-HT	0.999	0.173	0.918	0.756	1.000	0.094	0.935	0.342	1.000	0.073	0.911	0.129
			(0.016)	(0.006)	(0.013)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	EEq-Hájek	0.999	0.184	0.927	0.721	1.000	0.090	0.946	0.330	1.000	0.066	0.932	0.122
			(0.016)	(0.006)	(0.012)		(0.007)	(0.005)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-HT	0.999	0.155	0.898	0.666	1.000	0.074	0.927	0.301	1.000	0.048	0.926	0.105
			(0.015)	(0.007)	(0.011)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.999	0.130	0.907	0.661	1.000	0.068	0.935	0.297	1.000	0.045	0.936	0.102
			(0.015)	(0.006)	(0.011)		(0.006)	(0.006)	(0.005)		(0.002)	(0.005)	(0.002)
	TMLE-pw-HT	0.999	0.138	0.907	0.668	1.000	0.069	0.930	0.300	1.000	0.046	0.929	0.104
			(0.015)	(0.006)	(0.011)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-Hájek	0.999	0.111	0.910	0.664	1.000	0.063	0.937	0.296	1.000	0.043	0.937	0.102
			(0.015)	(0.006)	(0.011)		(0.006)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	TMLE-2eps-cw-Hájek	0.999	0.065	0.895	0.652	1.000	0.045	0.924	0.281	1.000	0.032	0.939	0.093
			(0.015)	(0.007)	(0.011)		(0.006)	(0.006)	(0.005)		(0.002)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.999	0.034	0.892	0.660	1.000	0.035	0.921	0.281	1.000	0.026	0.940	0.091
			(0.015)	(0.007)	(0.012)		(0.006)	(0.006)	(0.005)		(0.002)	(0.005)	(0.001)
C2	EEq-HT	0.981	0.000	0.844	0.689	1.000	0.009	0.880	0.299	1.000	−0.001	0.865	0.097
			(0.016)	(0.008)	(0.011)		(0.007)	(0.007)	(0.005)		(0.002)	(0.008)	(0.002)
	EEq-Hájek	0.981	0.004	0.885	0.738	1.000	0.009	0.919	0.325	1.000	−0.001	0.912	0.106
			(0.017)	(0.007)	(0.012)		(0.007)	(0.006)	(0.006)		(0.002)	(0.006)	(0.002)
	TMLE-cw-HT	0.981	0.007	0.750	0.816	1.000	0.011	0.782	0.369	1.000	−0.002	0.778	0.120
			(0.018)	(0.010)	(0.014)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	TMLE-cw-Hájek	0.981	0.013	0.773	0.918	1.000	0.011	0.835	0.400	1.000	−0.002	0.839	0.129
			(0.021)	(0.009)	(0.017)		(0.009)	(0.008)	(0.007)		(0.003)	(0.008)	(0.002)
	TMLE-pw-HT	0.981	0.004	0.746	0.834	1.000	0.012	0.783	0.371	1.000	−0.001	0.781	0.120
			(0.019)	(0.010)	(0.014)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	TMLE-pw-Hájek	0.981	0.010	0.777	0.934	1.000	0.012	0.840	0.401	1.000	−0.002	0.839	0.129
			(0.021)	(0.009)	(0.018)		(0.009)	(0.008)	(0.007)		(0.003)	(0.008)	(0.002)
	TMLE-2eps-cw-Hájek	0.981	0.018	0.737	0.990	1.000	0.011	0.809	0.421	1.000	−0.002	0.815	0.135
			(0.022)	(0.010)	(0.019)		(0.009)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)
	TMLE-2eps-pw-Hájek	0.981	0.011	0.753	1.005	1.000	0.012	0.810	0.423	1.000	−0.002	0.815	0.136
			(0.023)	(0.010)	(0.020)		(0.009)	(0.009)	(0.007)		(0.003)	(0.009)	(0.002)

Table J.5: Dynamic design, continuous outcome: ATT$(2,4) = 1$; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.986	−0.021	0.948	0.849	1.000	−0.001	0.944	0.296	1.000	−0.001	0.938	0.091
			(0.019)	(0.005)	(0.017)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.001)
	EEq-Hájek	0.986	−0.023	0.942	0.726	1.000	−0.002	0.945	0.293	1.000	−0.001	0.940	0.091
			(0.016)	(0.005)	(0.013)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.001)
	TMLE-cw-HT	0.943	−0.020	0.912	0.694	1.000	−0.002	0.923	0.295	1.000	−0.001	0.928	0.093
			(0.016)	(0.007)	(0.013)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.943	−0.019	0.890	0.723	1.000	−0.002	0.919	0.297	1.000	−0.001	0.932	0.094
			(0.017)	(0.007)	(0.016)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-HT	0.986	−0.029	0.921	0.703	1.000	−0.002	0.930	0.295	1.000	−0.001	0.934	0.093
			(0.016)	(0.006)	(0.012)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.001)
	TMLE-pw-Hájek	0.986	−0.026	0.905	0.725	1.000	−0.002	0.926	0.296	1.000	−0.001	0.935	0.093
			(0.016)	(0.007)	(0.015)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-cw-Hájek	0.943	−0.015	0.870	0.809	1.000	−0.002	0.916	0.305	1.000	−0.001	0.926	0.096
			(0.019)	(0.008)	(0.029)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-pw-Hájek	0.986	−0.028	0.891	0.769	1.000	−0.001	0.920	0.303	1.000	−0.001	0.929	0.095
			(0.017)	(0.007)	(0.024)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	DR-nt	0.549	0.152	0.910	0.769	1.000	0.175	0.904	0.377	1.000	0.182	0.574	0.212
			(0.023)	(0.009)	(0.020)		(0.007)	(0.007)	(0.006)		(0.002)	(0.011)	(0.002)
	DR-ny	0.549	0.051	0.933	0.653	1.000	0.088	0.928	0.316	1.000	0.090	0.828	0.131
			(0.020)	(0.008)	(0.015)		(0.007)	(0.006)	(0.006)		(0.002)	(0.008)	(0.002)
B	EEq-HT	0.925	0.020	0.947	1.047	1.000	0.000	0.939	0.344	1.000	0.004	0.945	0.105
			(0.024)	(0.005)	(0.022)		(0.008)	(0.005)	(0.006)		(0.002)	(0.005)	(0.002)
	EEq-Hájek	0.925	0.014	0.937	0.880	1.000	−0.001	0.943	0.348	1.000	0.004	0.947	0.109
			(0.020)	(0.006)	(0.016)		(0.008)	(0.005)	(0.006)		(0.002)	(0.005)	(0.002)
	TMLE-cw-HT	0.801	0.020	0.860	0.903	1.000	−0.002	0.892	0.370	1.000	0.005	0.900	0.119
			(0.023)	(0.009)	(0.019)		(0.008)	(0.007)	(0.006)		(0.003)	(0.007)	(0.002)
	TMLE-cw-Hájek	0.801	0.011	0.825	1.032	1.000	−0.002	0.889	0.383	1.000	0.005	0.908	0.122
			(0.026)	(0.010)	(0.031)		(0.009)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-pw-HT	0.925	0.011	0.880	0.906	1.000	−0.002	0.901	0.367	1.000	0.004	0.907	0.116
			(0.021)	(0.008)	(0.017)		(0.008)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-pw-Hájek	0.925	0.012	0.857	1.007	1.000	−0.002	0.900	0.378	1.000	0.005	0.914	0.120
			(0.023)	(0.008)	(0.026)		(0.008)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-2eps-cw-Hájek	0.801	−0.003	0.782	1.280	1.000	−0.003	0.856	0.414	1.000	0.005	0.893	0.128
			(0.032)	(0.010)	(0.050)		(0.009)	(0.008)	(0.007)		(0.003)	(0.007)	(0.002)
	TMLE-2eps-pw-Hájek	0.925	0.006	0.826	1.151	1.000	−0.004	0.874	0.401	1.000	0.005	0.900	0.126
			(0.027)	(0.009)	(0.038)		(0.009)	(0.007)	(0.007)		(0.003)	(0.007)	(0.002)
	DR-nt	0.233	0.109	0.890	0.887	1.000	0.193	0.898	0.439	1.000	0.195	0.654	0.233
			(0.041)	(0.014)	(0.036)		(0.009)	(0.007)	(0.007)		(0.003)	(0.011)	(0.003)
	DR-ny	0.233	0.042	0.901	0.791	1.000	0.101	0.922	0.379	1.000	0.107	0.848	0.158
			(0.037)	(0.014)	(0.033)		(0.008)	(0.006)	(0.007)		(0.003)	(0.008)	(0.002)
B+	EEq-HT	0.645	−0.015	0.931	1.082	1.000	0.001	0.920	0.378	1.000	−0.002	0.922	0.115
			(0.030)	(0.007)	(0.025)		(0.008)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)
	EEq-Hájek	0.645	−0.005	0.937	1.015	1.000	−0.002	0.939	0.434	1.000	−0.001	0.939	0.135
			(0.028)	(0.007)	(0.021)		(0.010)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-cw-HT	0.399	0.005	0.744	1.143	0.999	−0.006	0.760	0.517	1.000	−0.001	0.787	0.164
			(0.040)	(0.015)	(0.029)		(0.012)	(0.010)	(0.009)		(0.004)	(0.009)	(0.003)
	TMLE-cw-Hájek	0.399	0.042	0.648	1.701	0.999	−0.005	0.760	0.600	1.000	0.000	0.829	0.184
			(0.060)	(0.017)	(0.088)		(0.013)	(0.010)	(0.010)		(0.004)	(0.008)	(0.003)
	TMLE-pw-HT	0.645	−0.001	0.756	1.167	1.000	−0.004	0.787	0.505	1.000	−0.002	0.800	0.158
			(0.033)	(0.012)	(0.024)		(0.011)	(0.009)	(0.008)		(0.004)	(0.009)	(0.002)
	TMLE-pw-Hájek	0.645	0.015	0.714	1.545	1.000	−0.008	0.805	0.583	1.000	−0.001	0.838	0.178
			(0.043)	(0.013)	(0.054)		(0.013)	(0.009)	(0.010)		(0.004)	(0.008)	(0.003)
	TMLE-2eps-cw-Hájek	0.399	0.054	0.600	2.269	0.999	−0.005	0.721	0.663	1.000	0.000	0.800	0.195
			(0.080)	(0.017)	(0.151)		(0.015)	(0.010)	(0.012)		(0.004)	(0.009)	(0.003)
	TMLE-2eps-pw-Hájek	0.645	0.021	0.683	1.795	1.000	−0.010	0.769	0.631	1.000	−0.001	0.813	0.190
			(0.050)	(0.013)	(0.074)		(0.014)	(0.009)	(0.011)		(0.004)	(0.009)	(0.003)
	DR-nt	0.039	−0.014	0.821	1.061	1.000	0.223	0.880	0.549	1.000	0.223	0.722	0.283
			(0.121)	(0.043)	(0.087)		(0.011)	(0.007)	(0.011)		(0.004)	(0.010)	(0.004)
	DR-ny	0.039	0.097	0.821	0.945	1.000	0.136	0.907	0.487	1.000	0.129	0.880	0.213
			(0.107)	(0.043)	(0.074)		(0.010)	(0.006)	(0.011)		(0.004)	(0.007)	(0.004)

Table J.5, continued

Scen.	Procedure	$n = 100$ Compl.	Bias	Cov.	RMSE	$n = 500$ Compl.	Bias	Cov.	RMSE	$n = 5{,}000$ Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.989	−0.130	0.955	1.720	1.000	0.416	0.885	0.709	1.000	0.506	0.308	0.534
			(0.039)	(0.005)	(0.049)		(0.013)	(0.007)	(0.010)		(0.004)	(0.010)	(0.004)
	EEq-Hájek	0.989	0.209	0.959	1.316	1.000	0.389	0.922	0.656	1.000	0.432	0.478	0.462
			(0.029)	(0.004)	(0.025)		(0.012)	(0.006)	(0.010)		(0.004)	(0.011)	(0.004)
	TMLE-cw-HT	0.945	0.329	0.931	1.032	1.000	0.261	0.898	0.488	1.000	0.247	0.546	0.278
			(0.023)	(0.006)	(0.019)		(0.009)	(0.007)	(0.008)		(0.003)	(0.011)	(0.003)
	TMLE-cw-Hájek	0.945	0.242	0.934	1.009	1.000	0.209	0.922	0.452	1.000	0.198	0.696	0.234
			(0.023)	(0.006)	(0.019)		(0.009)	(0.006)	(0.008)		(0.003)	(0.010)	(0.003)
	TMLE-pw-HT	0.989	0.256	0.940	1.034	1.000	0.222	0.910	0.467	1.000	0.221	0.618	0.255
			(0.023)	(0.005)	(0.019)		(0.009)	(0.006)	(0.008)		(0.003)	(0.011)	(0.003)
	TMLE-pw-Hájek	0.989	0.124	0.940	1.030	1.000	0.170	0.935	0.439	1.000	0.173	0.762	0.213
			(0.023)	(0.005)	(0.019)		(0.009)	(0.006)	(0.007)		(0.003)	(0.010)	(0.003)
	TMLE-2eps-cw-Hájek	0.945	0.081	0.869	1.105	1.000	0.046	0.895	0.409	1.000	0.056	0.887	0.134
			(0.025)	(0.008)	(0.036)		(0.009)	(0.007)	(0.007)		(0.003)	(0.007)	(0.002)
	TMLE-2eps-pw-Hájek	0.989	−0.041	0.871	1.166	1.000	−0.028	0.890	0.439	1.000	−0.026	0.917	0.135
			(0.026)	(0.008)	(0.029)		(0.010)	(0.007)	(0.008)		(0.003)	(0.006)	(0.002)
	DR-nt	0.585	2.359	0.530	2.791	1.000	2.191	0.102	2.297	1.000	2.136	0.002	2.149
			(0.044)	(0.015)	(0.041)		(0.015)	(0.007)	(0.014)		(0.005)	(0.001)	(0.005)
	DR-ny	0.585	1.520	0.704	2.077	1.000	1.495	0.280	1.637	1.000	1.457	0.012	1.474
			(0.041)	(0.013)	(0.036)		(0.015)	(0.010)	(0.013)		(0.005)	(0.002)	(0.005)
C2	EEq-HT	0.986	0.008	0.919	0.674	1.000	0.002	0.909	0.248	1.000	−0.002	0.905	0.079
			(0.015)	(0.006)	(0.013)		(0.006)	(0.006)	(0.004)		(0.002)	(0.007)	(0.001)
	EEq-Hájek	0.986	0.010	0.928	0.664	1.000	0.003	0.929	0.257	1.000	−0.002	0.927	0.082
			(0.015)	(0.006)	(0.012)		(0.006)	(0.006)	(0.004)		(0.002)	(0.006)	(0.001)
	TMLE-cw-HT	0.949	0.016	0.849	0.723	1.000	0.005	0.854	0.292	1.000	−0.002	0.849	0.092
			(0.017)	(0.008)	(0.014)		(0.007)	(0.008)	(0.005)		(0.002)	(0.008)	(0.002)
	TMLE-cw-Hájek	0.949	0.018	0.839	0.809	1.000	0.006	0.860	0.316	1.000	−0.002	0.861	0.099
			(0.019)	(0.008)	(0.022)		(0.007)	(0.008)	(0.005)		(0.002)	(0.008)	(0.002)
	TMLE-pw-HT	0.986	0.007	0.858	0.731	1.000	0.004	0.854	0.286	1.000	−0.002	0.861	0.090
			(0.016)	(0.008)	(0.013)		(0.006)	(0.008)	(0.005)		(0.002)	(0.008)	(0.001)
	TMLE-pw-Hájek	0.986	0.004	0.850	0.808	1.000	0.005	0.870	0.308	1.000	−0.002	0.869	0.096
			(0.018)	(0.008)	(0.018)		(0.007)	(0.008)	(0.005)		(0.002)	(0.008)	(0.002)
	TMLE-2eps-cw-Hájek	0.949	0.024	0.792	0.972	1.000	0.007	0.788	0.373	1.000	−0.002	0.788	0.116
			(0.022)	(0.009)	(0.030)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	TMLE-2eps-pw-Hájek	0.986	0.001	0.801	0.937	1.000	0.006	0.798	0.363	1.000	−0.003	0.799	0.113
			(0.021)	(0.009)	(0.022)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	DR-nt	0.525	0.134	0.926	0.745	1.000	0.165	0.909	0.347	1.000	0.168	0.567	0.193
			(0.023)	(0.008)	(0.019)		(0.007)	(0.006)	(0.006)		(0.002)	(0.011)	(0.002)
	DR-ny	0.525	0.059	0.938	0.647	1.000	0.090	0.937	0.287	1.000	0.089	0.835	0.123
			(0.020)	(0.007)	(0.017)		(0.006)	(0.005)	(0.005)		(0.002)	(0.008)	(0.002)

Table J.6: Dynamic design, continuous outcome: ATT$(2, 4) = 1$; no cross-fitting (proposed procedures).

Scen.	Procedure	$n = 100$ Compl.	Bias	Cov.	RMSE	$n = 500$ Compl.	Bias	Cov.	RMSE	$n = 5{,}000$ Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.999	−0.029	0.931	0.633	1.000	−0.002	0.944	0.284	1.000	−0.001	0.938	0.090
			(0.014)	(0.006)	(0.011)		(0.006)	(0.005)	(0.005)		(0.002)	(0.005)	(0.001)
	EEq-Hájek	0.999	−0.029	0.935	0.622	1.000	−0.002	0.949	0.285	1.000	−0.001	0.940	0.091
			(0.014)	(0.005)	(0.011)		(0.006)	(0.005)	(0.005)		(0.002)	(0.005)	(0.001)
	TMLE-cw-HT	0.994	−0.030	0.893	0.652	1.000	−0.002	0.919	0.291	1.000	−0.001	0.929	0.093
			(0.015)	(0.007)	(0.011)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-cw-Hájek	0.994	−0.027	0.884	0.698	1.000	−0.002	0.920	0.295	1.000	−0.001	0.932	0.094
			(0.016)	(0.007)	(0.016)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-pw-HT	0.999	−0.032	0.892	0.657	1.000	−0.002	0.924	0.291	1.000	−0.001	0.932	0.092
			(0.015)	(0.007)	(0.011)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.001)
	TMLE-pw-Hájek	0.999	−0.030	0.891	0.691	1.000	−0.002	0.927	0.294	1.000	−0.001	0.935	0.093
			(0.015)	(0.007)	(0.015)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-cw-Hájek	0.994	−0.021	0.851	0.837	1.000	−0.003	0.908	0.308	1.000	−0.001	0.924	0.096
			(0.019)	(0.008)	(0.031)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)
	TMLE-2eps-pw-Hájek	0.999	−0.031	0.869	0.795	1.000	−0.001	0.914	0.306	1.000	−0.001	0.928	0.095
			(0.018)	(0.008)	(0.025)		(0.007)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)

Table J.6, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.990	0.017	0.921	0.757	1.000	−0.003	0.939	0.327	1.000	0.004	0.945	0.104
			(0.017)	(0.006)	(0.015)		(0.007)	(0.005)	(0.005)		(0.002)	(0.005)	(0.002)
	EEq-Hájek	0.990	0.016	0.929	0.751	1.000	−0.004	0.945	0.335	1.000	0.004	0.949	0.108
			(0.017)	(0.006)	(0.013)		(0.008)	(0.005)	(0.006)		(0.002)	(0.005)	(0.002)
	TMLE-cw-HT	0.968	0.015	0.823	0.858	1.000	−0.004	0.878	0.365	1.000	0.005	0.897	0.118
			(0.019)	(0.009)	(0.016)		(0.008)	(0.007)	(0.006)		(0.003)	(0.007)	(0.002)
	TMLE-cw-Hájek	0.968	0.013	0.798	1.090	1.000	−0.005	0.877	0.382	1.000	0.005	0.910	0.122
			(0.025)	(0.009)	(0.033)		(0.009)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-pw-HT	0.990	0.016	0.839	0.869	1.000	−0.004	0.888	0.361	1.000	0.004	0.905	0.116
			(0.020)	(0.008)	(0.016)		(0.008)	(0.007)	(0.006)		(0.003)	(0.007)	(0.002)
	TMLE-pw-Hájek	0.990	0.020	0.821	1.042	1.000	−0.005	0.886	0.376	1.000	0.004	0.919	0.119
			(0.023)	(0.009)	(0.028)		(0.008)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)
	TMLE-2eps-cw-Hájek	0.968	0.010	0.715	1.541	1.000	−0.005	0.837	0.422	1.000	0.005	0.890	0.128
			(0.035)	(0.010)	(0.065)		(0.009)	(0.008)	(0.007)		(0.003)	(0.007)	(0.002)
	TMLE-2eps-pw-Hájek	0.990	0.017	0.757	1.353	1.000	−0.006	0.856	0.409	1.000	0.004	0.900	0.126
			(0.030)	(0.010)	(0.045)		(0.009)	(0.008)	(0.007)		(0.003)	(0.007)	(0.002)
B+	EEq-HT	0.931	0.000	0.899	0.824	1.000	−0.001	0.913	0.360	1.000	−0.002	0.923	0.115
			(0.019)	(0.007)	(0.016)		(0.008)	(0.006)	(0.006)		(0.003)	(0.006)	(0.002)
	EEq-Hájek	0.931	0.008	0.922	0.892	1.000	−0.004	0.935	0.419	1.000	−0.001	0.937	0.135
			(0.021)	(0.006)	(0.016)		(0.009)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)
	TMLE-cw-HT	0.832	0.008	0.666	1.096	1.000	−0.007	0.731	0.513	1.000	−0.001	0.783	0.165
			(0.027)	(0.012)	(0.020)		(0.011)	(0.010)	(0.009)		(0.004)	(0.009)	(0.003)
	TMLE-cw-Hájek	0.832	0.047	0.569	2.319	1.000	−0.005	0.741	0.615	1.000	0.000	0.826	0.184
			(0.057)	(0.012)	(0.186)		(0.014)	(0.010)	(0.011)		(0.004)	(0.008)	(0.003)
	TMLE-pw-HT	0.931	−0.002	0.678	1.155	1.000	−0.007	0.766	0.503	1.000	−0.002	0.797	0.158
			(0.027)	(0.011)	(0.022)		(0.011)	(0.009)	(0.008)		(0.004)	(0.009)	(0.002)
	TMLE-pw-Hájek	0.931	0.016	0.606	1.830	1.000	−0.010	0.778	0.595	1.000	−0.001	0.839	0.179
			(0.042)	(0.011)	(0.055)		(0.013)	(0.009)	(0.010)		(0.004)	(0.008)	(0.003)
	TMLE-2eps-cw-Hájek	0.832	0.059	0.484	3.310	1.000	−0.004	0.696	0.692	1.000	0.000	0.795	0.196
			(0.081)	(0.012)	(0.310)		(0.015)	(0.010)	(0.013)		(0.004)	(0.009)	(0.003)
	TMLE-2eps-pw-Hájek	0.931	0.011	0.548	2.297	1.000	−0.013	0.736	0.656	1.000	−0.001	0.808	0.191
			(0.053)	(0.012)	(0.074)		(0.015)	(0.010)	(0.011)		(0.004)	(0.009)	(0.003)
C1	EEq-HT	0.999	0.682	0.905	1.405	1.000	0.559	0.842	0.777	1.000	0.520	0.276	0.547
			(0.027)	(0.007)	(0.025)		(0.012)	(0.008)	(0.010)		(0.004)	(0.010)	(0.004)
	EEq-Hájek	0.999	0.663	0.930	1.293	1.000	0.495	0.889	0.707	1.000	0.443	0.446	0.472
			(0.025)	(0.006)	(0.022)		(0.011)	(0.007)	(0.010)		(0.004)	(0.011)	(0.004)
	TMLE-cw-HT	0.994	0.421	0.909	1.023	1.000	0.294	0.883	0.500	1.000	0.251	0.536	0.282
			(0.021)	(0.006)	(0.019)		(0.009)	(0.007)	(0.008)		(0.003)	(0.011)	(0.003)
	TMLE-cw-Hájek	0.994	0.229	0.928	0.963	1.000	0.217	0.922	0.447	1.000	0.199	0.690	0.234
			(0.021)	(0.006)	(0.020)		(0.009)	(0.006)	(0.007)		(0.003)	(0.010)	(0.003)
	TMLE-pw-HT	0.999	0.347	0.921	0.991	1.000	0.257	0.897	0.477	1.000	0.225	0.599	0.259
			(0.021)	(0.006)	(0.018)		(0.009)	(0.007)	(0.008)		(0.003)	(0.011)	(0.003)
	TMLE-pw-Hájek	0.999	0.133	0.930	0.960	1.000	0.180	0.933	0.433	1.000	0.175	0.751	0.214
			(0.021)	(0.006)	(0.018)		(0.009)	(0.006)	(0.007)		(0.003)	(0.010)	(0.003)
	TMLE-2eps-cw-Hájek	0.994	−0.287	0.837	1.209	1.000	−0.033	0.883	0.405	1.000	0.048	0.892	0.131
			(0.026)	(0.008)	(0.057)		(0.009)	(0.007)	(0.007)		(0.003)	(0.007)	(0.002)
	TMLE-2eps-pw-Hájek	0.999	−0.428	0.815	1.305	1.000	−0.109	0.874	0.451	1.000	−0.034	0.909	0.137
			(0.028)	(0.009)	(0.038)		(0.010)	(0.007)	(0.008)		(0.003)	(0.006)	(0.002)
C2	EEq-HT	0.998	0.009	0.888	0.579	1.000	0.001	0.903	0.245	1.000	−0.002	0.905	0.079
			(0.013)	(0.007)	(0.010)		(0.005)	(0.007)	(0.004)		(0.002)	(0.007)	(0.001)
	EEq-Hájek	0.998	0.009	0.916	0.594	1.000	0.001	0.922	0.253	1.000	−0.002	0.928	0.081
			(0.013)	(0.006)	(0.010)		(0.006)	(0.006)	(0.004)		(0.002)	(0.006)	(0.001)
	TMLE-cw-HT	0.996	0.008	0.839	0.658	1.000	0.002	0.845	0.286	1.000	−0.002	0.851	0.091
			(0.015)	(0.008)	(0.012)		(0.006)	(0.008)	(0.005)		(0.002)	(0.008)	(0.002)
	TMLE-cw-Hájek	0.996	0.004	0.839	0.758	1.000	0.003	0.856	0.312	1.000	−0.002	0.864	0.098
			(0.017)	(0.008)	(0.019)		(0.007)	(0.008)	(0.005)		(0.002)	(0.008)	(0.002)
	TMLE-pw-HT	0.998	0.007	0.845	0.658	1.000	0.001	0.851	0.280	1.000	−0.002	0.862	0.089
			(0.015)	(0.008)	(0.012)		(0.006)	(0.008)	(0.004)		(0.002)	(0.008)	(0.001)
	TMLE-pw-Hájek	0.998	0.002	0.847	0.751	1.000	0.002	0.867	0.303	1.000	−0.002	0.873	0.096
			(0.017)	(0.008)	(0.018)		(0.007)	(0.008)	(0.005)		(0.002)	(0.007)	(0.002)
	TMLE-2eps-cw-Hájek	0.996	0.003	0.768	0.942	1.000	0.004	0.780	0.372	1.000	−0.002	0.785	0.116
			(0.021)	(0.009)	(0.028)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)
	TMLE-2eps-pw-Hájek	0.998	−0.001	0.782	0.912	1.000	0.002	0.794	0.361	1.000	−0.002	0.798	0.113
			(0.020)	(0.009)	(0.023)		(0.008)	(0.009)	(0.006)		(0.003)	(0.009)	(0.002)

Table J.7: Dynamic design, continuous outcome: ES(1) = 1; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.934	−0.018	0.954	0.525	1.000	0.001	0.949	0.160	1.000	0.000	0.940	0.049
			(0.012)	(0.005)	(0.012)		(0.004)	(0.005)	(0.003)		(0.001)	(0.005)	(0.001)
	EEq-Hájek	0.934	−0.019	0.942	0.425	1.000	0.001	0.947	0.157	1.000	0.000	0.940	0.049
			(0.010)	(0.005)	(0.008)		(0.004)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-cw-HT	0.907	−0.016	0.933	0.383	1.000	0.000	0.942	0.154	1.000	0.000	0.940	0.049
			(0.009)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-cw-Hájek	0.907	−0.016	0.913	0.387	1.000	0.000	0.941	0.154	1.000	0.000	0.941	0.049
			(0.009)	(0.007)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-pw-HT	0.929	−0.020	0.936	0.388	1.000	0.000	0.942	0.155	1.000	0.000	0.940	0.049
			(0.009)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-pw-Hájek	0.929	−0.019	0.914	0.392	1.000	0.000	0.942	0.154	1.000	0.000	0.940	0.049
			(0.009)	(0.006)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-cw-Hájek	0.907	−0.015	0.901	0.399	1.000	0.000	0.938	0.155	1.000	0.000	0.939	0.049
			(0.009)	(0.007)	(0.008)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.929	−0.018	0.909	0.395	1.000	0.000	0.939	0.154	1.000	0.000	0.939	0.049
			(0.009)	(0.007)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	DR-nt	0.549	0.070	0.891	0.443	1.000	0.098	0.900	0.211	1.000	0.099	0.590	0.115
			(0.013)	(0.009)	(0.011)		(0.004)	(0.007)	(0.003)		(0.001)	(0.011)	(0.001)
	DR-ny	0.549	0.001	0.908	0.369	1.000	0.028	0.936	0.154	1.000	0.028	0.907	0.056
			(0.011)	(0.009)	(0.008)		(0.003)	(0.005)	(0.003)		(0.001)	(0.006)	(0.001)
B	EEq-HT	0.786	0.015	0.954	0.646	1.000	0.001	0.943	0.199	1.000	0.002	0.946	0.058
			(0.016)	(0.005)	(0.015)		(0.004)	(0.005)	(0.003)		(0.001)	(0.005)	(0.001)
	EEq-Hájek	0.786	0.011	0.938	0.487	1.000	0.000	0.942	0.194	1.000	0.002	0.946	0.059
			(0.012)	(0.006)	(0.010)		(0.004)	(0.005)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-cw-HT	0.731	0.007	0.915	0.454	1.000	0.000	0.923	0.193	1.000	0.002	0.935	0.060
			(0.012)	(0.007)	(0.009)		(0.004)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-cw-Hájek	0.731	0.006	0.886	0.472	1.000	0.000	0.916	0.195	1.000	0.002	0.937	0.061
			(0.012)	(0.008)	(0.010)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-pw-HT	0.772	0.008	0.924	0.459	1.000	0.000	0.931	0.193	1.000	0.002	0.939	0.060
			(0.012)	(0.007)	(0.009)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-pw-Hájek	0.772	0.008	0.894	0.480	1.000	0.000	0.926	0.195	1.000	0.002	0.939	0.060
			(0.012)	(0.008)	(0.011)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-cw-Hájek	0.731	0.005	0.871	0.526	1.000	−0.001	0.899	0.202	1.000	0.002	0.929	0.062
			(0.014)	(0.009)	(0.014)		(0.005)	(0.007)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-2eps-pw-Hájek	0.772	0.007	0.881	0.517	1.000	−0.001	0.910	0.200	1.000	0.002	0.931	0.062
			(0.013)	(0.008)	(0.014)		(0.004)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
	DR-nt	0.232	0.075	0.892	0.466	1.000	0.106	0.900	0.242	0.998	0.106	0.673	0.127
			(0.021)	(0.014)	(0.017)		(0.005)	(0.007)	(0.004)		(0.002)	(0.011)	(0.001)
	DR-ny	0.232	0.015	0.929	0.380	1.000	0.031	0.935	0.187	1.000	0.035	0.907	0.067
			(0.018)	(0.012)	(0.013)		(0.004)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
B+	EEq-HT	0.382	−0.013	0.949	0.785	0.999	−0.002	0.931	0.243	1.000	−0.001	0.939	0.071
			(0.028)	(0.008)	(0.025)		(0.005)	(0.006)	(0.004)		(0.002)	(0.005)	(0.001)
	EEq-Hájek	0.382	−0.016	0.928	0.599	0.999	−0.003	0.936	0.252	1.000	−0.001	0.941	0.076
			(0.022)	(0.009)	(0.016)		(0.006)	(0.005)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-cw-HT	0.296	−0.013	0.844	0.606	0.999	−0.004	0.842	0.274	1.000	−0.001	0.870	0.085
			(0.025)	(0.015)	(0.019)		(0.006)	(0.008)	(0.005)		(0.002)	(0.008)	(0.001)
	TMLE-cw-Hájek	0.296	−0.021	0.775	0.712	0.999	−0.004	0.838	0.291	1.000	−0.001	0.892	0.089
			(0.029)	(0.017)	(0.029)		(0.007)	(0.008)	(0.005)		(0.002)	(0.007)	(0.001)
	TMLE-pw-HT	0.352	−0.018	0.865	0.592	0.999	−0.003	0.859	0.270	1.000	−0.001	0.882	0.083
			(0.022)	(0.013)	(0.016)		(0.006)	(0.008)	(0.005)		(0.002)	(0.007)	(0.001)
	TMLE-pw-Hájek	0.352	−0.017	0.808	0.666	0.999	−0.004	0.858	0.288	1.000	−0.001	0.898	0.088
			(0.025)	(0.015)	(0.021)		(0.006)	(0.008)	(0.005)		(0.002)	(0.007)	(0.001)
	TMLE-2eps-cw-Hájek	0.296	−0.035	0.746	0.860	0.999	−0.005	0.803	0.316	1.000	−0.001	0.870	0.093
			(0.035)	(0.018)	(0.046)		(0.007)	(0.009)	(0.006)		(0.002)	(0.008)	(0.002)
	TMLE-2eps-pw-Hájek	0.352	−0.023	0.781	0.731	0.999	−0.006	0.823	0.309	1.000	−0.001	0.875	0.092
			(0.028)	(0.016)	(0.026)		(0.007)	(0.009)	(0.005)		(0.002)	(0.007)	(0.002)
	DR-nt	0.039	0.037	0.821	0.593	0.999	0.127	0.879	0.318	0.997	0.131	0.723	0.164
			(0.067)	(0.043)	(0.046)		(0.007)	(0.007)	(0.006)		(0.002)	(0.010)	(0.002)
	DR-ny	0.039	−0.031	0.936	0.454	1.000	0.039	0.919	0.257	1.000	0.040	0.919	0.096
			(0.052)	(0.028)	(0.036)		(0.006)	(0.006)	(0.005)		(0.002)	(0.006)	(0.002)

Table J.7, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.932	−0.368	0.963	0.995	1.000	0.027	0.943	0.284	1.000	0.094	0.832	0.124
			(0.021)	(0.004)	(0.030)		(0.006)	(0.005)	(0.005)		(0.002)	(0.008)	(0.002)
	EEq-Hájek	0.932	−0.088	0.957	0.639	1.000	0.044	0.948	0.261	1.000	0.083	0.878	0.113
			(0.015)	(0.005)	(0.013)		(0.006)	(0.005)	(0.004)		(0.002)	(0.007)	(0.001)
	TMLE-cw-HT	0.910	0.057	0.937	0.478	1.000	0.045	0.930	0.212	1.000	0.048	0.895	0.080
			(0.011)	(0.006)	(0.008)		(0.005)	(0.006)	(0.003)		(0.001)	(0.007)	(0.001)
	TMLE-cw-Hájek	0.910	0.032	0.930	0.481	1.000	0.039	0.940	0.206	1.000	0.041	0.920	0.074
			(0.011)	(0.006)	(0.008)		(0.005)	(0.005)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-pw-HT	0.928	0.041	0.942	0.478	1.000	0.038	0.938	0.210	1.000	0.046	0.898	0.078
			(0.011)	(0.005)	(0.008)		(0.005)	(0.005)	(0.003)		(0.001)	(0.007)	(0.001)
	TMLE-pw-Hájek	0.928	0.003	0.936	0.485	1.000	0.032	0.944	0.205	1.000	0.039	0.925	0.073
			(0.011)	(0.006)	(0.008)		(0.005)	(0.005)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-2eps-cw-Hájek	0.910	0.018	0.895	0.519	1.000	−0.001	0.918	0.197	1.000	0.002	0.942	0.059
			(0.012)	(0.007)	(0.015)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.928	−0.005	0.902	0.516	1.000	−0.012	0.929	0.201	1.000	−0.011	0.937	0.061
			(0.012)	(0.007)	(0.014)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	DR-nt	0.585	1.029	0.513	1.258	0.994	0.946	0.169	1.005	0.993	0.920	0.006	0.928
			(0.021)	(0.015)	(0.019)		(0.008)	(0.008)	(0.007)		(0.003)	(0.002)	(0.003)
	DR-ny	0.585	0.479	0.779	0.732	1.000	0.423	0.503	0.502	1.000	0.409	0.034	0.419
			(0.016)	(0.012)	(0.014)		(0.006)	(0.011)	(0.005)		(0.002)	(0.004)	(0.002)
C2	EEq-HT	0.945	−0.005	0.940	0.393	1.000	0.001	0.928	0.139	1.000	−0.001	0.931	0.043
			(0.009)	(0.005)	(0.008)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	EEq-Hájek	0.945	−0.004	0.940	0.361	1.000	0.001	0.938	0.141	1.000	−0.001	0.942	0.043
			(0.008)	(0.005)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-cw-HT	0.919	−0.005	0.913	0.362	1.000	0.002	0.908	0.148	1.000	−0.001	0.915	0.045
			(0.008)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-cw-Hájek	0.919	−0.006	0.896	0.383	1.000	0.002	0.914	0.152	1.000	−0.001	0.923	0.046
			(0.009)	(0.007)	(0.009)		(0.003)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-pw-HT	0.937	−0.005	0.914	0.365	1.000	0.002	0.912	0.147	1.000	−0.001	0.919	0.045
			(0.008)	(0.006)	(0.007)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-pw-Hájek	0.937	−0.005	0.901	0.382	1.000	0.002	0.922	0.151	1.000	−0.002	0.926	0.046
			(0.009)	(0.007)	(0.007)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-2eps-cw-Hájek	0.919	−0.008	0.864	0.426	1.000	0.003	0.894	0.166	1.000	−0.002	0.890	0.050
			(0.010)	(0.008)	(0.012)		(0.004)	(0.007)	(0.003)		(0.001)	(0.007)	(0.001)
	TMLE-2eps-pw-Hájek	0.937	−0.006	0.869	0.416	1.000	0.002	0.896	0.164	1.000	−0.002	0.900	0.050
			(0.010)	(0.008)	(0.009)		(0.004)	(0.007)	(0.003)		(0.001)	(0.007)	(0.001)
	DR-nt	0.524	0.069	0.916	0.408	1.000	0.093	0.903	0.196	1.000	0.094	0.567	0.108
			(0.012)	(0.009)	(0.010)		(0.004)	(0.007)	(0.003)		(0.001)	(0.011)	(0.001)
	DR-ny	0.524	0.007	0.941	0.327	1.000	0.028	0.938	0.144	1.000	0.027	0.917	0.051
			(0.010)	(0.007)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.006)	(0.001)

Table J.8: Dynamic design, continuous outcome: ES(1) = 1; no cross-fitting (proposed procedures).

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.994	−0.021	0.931	0.367	1.000	0.000	0.947	0.152	1.000	0.000	0.940	0.049
			(0.008)	(0.006)	(0.006)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	EEq-Hájek	0.994	−0.021	0.931	0.360	1.000	0.000	0.950	0.152	1.000	0.000	0.940	0.049
			(0.008)	(0.006)	(0.006)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-cw-HT	0.989	−0.021	0.897	0.368	1.000	0.000	0.938	0.151	1.000	0.000	0.937	0.049
			(0.008)	(0.007)	(0.006)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-cw-Hájek	0.989	−0.021	0.897	0.378	1.000	0.000	0.938	0.152	1.000	0.000	0.937	0.049
			(0.008)	(0.007)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-pw-HT	0.994	−0.021	0.899	0.370	1.000	0.000	0.939	0.152	1.000	0.000	0.936	0.049
			(0.008)	(0.007)	(0.006)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-pw-Hájek	0.994	−0.021	0.901	0.379	1.000	0.000	0.941	0.152	1.000	0.000	0.938	0.049
			(0.008)	(0.007)	(0.007)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-cw-Hájek	0.989	−0.018	0.865	0.425	1.000	0.000	0.932	0.155	1.000	0.000	0.937	0.049
			(0.010)	(0.008)	(0.008)		(0.003)	(0.006)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.994	−0.019	0.875	0.419	1.000	0.000	0.932	0.155	1.000	0.000	0.936	0.049
			(0.009)	(0.007)	(0.008)		(0.003)	(0.006)	(0.002)		(0.001)	(0.005)	(0.001)

Table J.8, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.967	0.014	0.932	0.421	1.000	−0.001	0.939	0.185	1.000	0.002	0.944	0.058
			(0.010)	(0.006)	(0.008)		(0.004)	(0.005)	(0.003)		(0.001)	(0.005)	(0.001)
	EEq-Hájek	0.967	0.014	0.932	0.407	1.000	−0.001	0.940	0.185	1.000	0.002	0.946	0.058
			(0.009)	(0.006)	(0.007)		(0.004)	(0.005)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-cw-HT	0.956	0.011	0.879	0.439	1.000	−0.002	0.909	0.190	1.000	0.002	0.933	0.060
			(0.010)	(0.007)	(0.008)		(0.004)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-cw-Hájek	0.956	0.011	0.867	0.485	1.000	−0.002	0.909	0.193	1.000	0.002	0.936	0.061
			(0.011)	(0.008)	(0.010)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-pw-HT	0.967	0.011	0.877	0.446	1.000	−0.002	0.915	0.189	1.000	0.002	0.934	0.060
			(0.010)	(0.007)	(0.008)		(0.004)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-pw-Hájek	0.967	0.010	0.863	0.492	1.000	−0.002	0.916	0.192	1.000	0.002	0.936	0.060
			(0.011)	(0.008)	(0.010)		(0.004)	(0.006)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-cw-Hájek	0.956	0.014	0.809	0.633	1.000	−0.001	0.886	0.206	1.000	0.002	0.927	0.062
			(0.014)	(0.009)	(0.016)		(0.005)	(0.007)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-2eps-pw-Hájek	0.967	0.006	0.813	0.615	1.000	−0.002	0.892	0.204	1.000	0.002	0.927	0.062
			(0.014)	(0.009)	(0.015)		(0.005)	(0.007)	(0.003)		(0.001)	(0.006)	(0.001)
B+	EEq-HT	0.809	−0.006	0.902	0.501	1.000	−0.003	0.927	0.227	1.000	−0.001	0.937	0.070
			(0.012)	(0.007)	(0.010)		(0.005)	(0.006)	(0.004)		(0.002)	(0.005)	(0.001)
	EEq-Hájek	0.809	−0.006	0.915	0.510	1.000	−0.004	0.932	0.241	1.000	−0.001	0.944	0.076
			(0.013)	(0.007)	(0.010)		(0.005)	(0.006)	(0.004)		(0.002)	(0.005)	(0.001)
	TMLE-cw-HT	0.777	−0.006	0.748	0.599	1.000	−0.005	0.816	0.272	1.000	−0.001	0.868	0.085
			(0.015)	(0.011)	(0.011)		(0.006)	(0.009)	(0.005)		(0.002)	(0.008)	(0.001)
	TMLE-cw-Hájek	0.777	−0.003	0.693	0.848	1.000	−0.005	0.816	0.295	1.000	−0.001	0.890	0.089
			(0.022)	(0.012)	(0.024)		(0.007)	(0.009)	(0.005)		(0.002)	(0.007)	(0.001)
	TMLE-pw-HT	0.809	−0.013	0.758	0.609	1.000	−0.005	0.836	0.270	1.000	−0.001	0.880	0.083
			(0.015)	(0.011)	(0.012)		(0.006)	(0.008)	(0.005)		(0.002)	(0.007)	(0.001)
	TMLE-pw-Hájek	0.809	−0.016	0.711	0.801	1.000	−0.006	0.838	0.292	1.000	−0.001	0.898	0.088
			(0.020)	(0.011)	(0.020)		(0.007)	(0.008)	(0.005)		(0.002)	(0.007)	(0.001)
	TMLE-2eps-cw-Hájek	0.777	−0.006	0.630	1.178	1.000	−0.006	0.773	0.330	1.000	−0.001	0.866	0.093
			(0.030)	(0.012)	(0.041)		(0.007)	(0.009)	(0.006)		(0.002)	(0.008)	(0.002)
	TMLE-2eps-pw-Hájek	0.809	−0.025	0.660	1.022	1.000	−0.007	0.785	0.323	1.000	−0.001	0.871	0.093
			(0.025)	(0.012)	(0.028)		(0.007)	(0.009)	(0.005)		(0.002)	(0.007)	(0.001)
C1	EEq-HT	0.993	0.204	0.896	0.581	1.000	0.122	0.900	0.281	1.000	0.103	0.803	0.131
			(0.012)	(0.007)	(0.010)		(0.006)	(0.007)	(0.004)		(0.002)	(0.009)	(0.002)
	EEq-Hájek	0.993	0.212	0.913	0.536	1.000	0.114	0.922	0.263	1.000	0.091	0.855	0.119
			(0.011)	(0.006)	(0.009)		(0.005)	(0.006)	(0.004)		(0.002)	(0.008)	(0.001)
	TMLE-cw-HT	0.990	0.129	0.901	0.461	1.000	0.069	0.918	0.213	1.000	0.051	0.887	0.081
			(0.010)	(0.007)	(0.008)		(0.004)	(0.006)	(0.003)		(0.001)	(0.007)	(0.001)
	TMLE-cw-Hájek	0.990	0.068	0.922	0.451	1.000	0.052	0.935	0.203	1.000	0.042	0.917	0.075
			(0.010)	(0.006)	(0.008)		(0.004)	(0.006)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-pw-HT	0.993	0.112	0.908	0.456	1.000	0.063	0.922	0.210	1.000	0.049	0.891	0.080
			(0.010)	(0.006)	(0.008)		(0.004)	(0.006)	(0.003)		(0.001)	(0.007)	(0.001)
	TMLE-pw-Hájek	0.993	0.047	0.926	0.449	1.000	0.046	0.938	0.202	1.000	0.041	0.921	0.074
			(0.010)	(0.006)	(0.008)		(0.004)	(0.005)	(0.003)		(0.001)	(0.006)	(0.001)
	TMLE-2eps-cw-Hájek	0.990	−0.152	0.864	0.551	1.000	−0.042	0.905	0.200	1.000	−0.003	0.941	0.059
			(0.012)	(0.008)	(0.016)		(0.004)	(0.007)	(0.003)		(0.001)	(0.005)	(0.001)
	TMLE-2eps-pw-Hájek	0.993	−0.185	0.861	0.563	1.000	−0.055	0.903	0.207	1.000	−0.016	0.929	0.062
			(0.012)	(0.008)	(0.016)		(0.004)	(0.007)	(0.003)		(0.001)	(0.006)	(0.001)
C2	EEq-HT	0.995	−0.005	0.922	0.317	1.000	0.001	0.922	0.138	1.000	−0.001	0.929	0.042
			(0.007)	(0.006)	(0.005)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	EEq-Hájek	0.995	−0.005	0.935	0.321	1.000	0.001	0.938	0.140	1.000	−0.001	0.941	0.043
			(0.007)	(0.006)	(0.005)		(0.003)	(0.005)	(0.002)		(0.001)	(0.005)	(0.001)
	TMLE-cw-HT	0.993	−0.007	0.895	0.339	1.000	0.001	0.906	0.146	1.000	−0.001	0.915	0.045
			(0.008)	(0.007)	(0.006)		(0.003)	(0.007)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-cw-Hájek	0.993	−0.010	0.897	0.366	1.000	0.002	0.915	0.151	1.000	−0.001	0.920	0.046
			(0.008)	(0.007)	(0.008)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-pw-HT	0.995	−0.007	0.899	0.339	1.000	0.001	0.908	0.145	1.000	−0.001	0.918	0.044
			(0.008)	(0.007)	(0.006)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-pw-Hájek	0.995	−0.009	0.904	0.359	1.000	0.001	0.917	0.149	1.000	−0.001	0.926	0.045
			(0.008)	(0.007)	(0.007)		(0.003)	(0.006)	(0.002)		(0.001)	(0.006)	(0.001)
	TMLE-2eps-cw-Hájek	0.993	−0.013	0.858	0.428	1.000	0.002	0.884	0.165	1.000	−0.002	0.889	0.050
			(0.010)	(0.008)	(0.013)		(0.004)	(0.007)	(0.003)		(0.001)	(0.007)	(0.001)
	TMLE-2eps-pw-Hájek	0.995	−0.011	0.865	0.412	1.000	0.001	0.892	0.164	1.000	−0.002	0.899	0.049
			(0.009)	(0.008)	(0.009)		(0.004)	(0.007)	(0.003)		(0.001)	(0.007)	(0.001)

Table J.9: Static design, binary outcome: ATT$(2,4)$ by scenario: A, 0.058; B, 0.048; B+, 0.044; C1, 0.043; C2, 0.051; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.988	0.0013	0.949	0.1340	1.000	0.0024	0.940	0.0536	1.000	−0.0005	0.955	0.0163
			(0.0030)	(0.005)	(0.0026)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.988	0.0012	0.954	0.1325	1.000	0.0024	0.946	0.0542	1.000	−0.0005	0.959	0.0165
			(0.0030)	(0.005)	(0.0025)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.004)	(0.0003)
	TMLE-cw-HT	0.936	0.0012	0.923	0.1388	1.000	0.0023	0.933	0.0556	1.000	−0.0005	0.949	0.0169
			(0.0032)	(0.006)	(0.0028)		(0.0012)	(0.006)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.936	0.0014	0.924	0.1461	1.000	0.0022	0.937	0.0562	1.000	−0.0005	0.955	0.0170
			(0.0034)	(0.006)	(0.0034)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-HT	0.988	0.0006	0.942	0.1343	1.000	0.0024	0.936	0.0548	1.000	−0.0006	0.952	0.0167
			(0.0030)	(0.005)	(0.0025)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.988	0.0005	0.943	0.1391	1.000	0.0023	0.942	0.0555	1.000	−0.0006	0.956	0.0169
			(0.0031)	(0.005)	(0.0028)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.936	0.0020	0.915	0.1570	1.000	0.0021	0.934	0.0571	1.000	−0.0006	0.953	0.0172
			(0.0036)	(0.006)	(0.0042)		(0.0013)	(0.006)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-pw-Hájek	0.988	0.0003	0.936	0.1440	1.000	0.0023	0.940	0.0562	1.000	−0.0006	0.954	0.0170
			(0.0032)	(0.006)	(0.0030)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	DR-nt	0.809	−0.0446	0.933	0.1983	1.000	−0.0342	0.950	0.0885	1.000	−0.0330	0.761	0.0419
			(0.0048)	(0.006)	(0.0045)		(0.0018)	(0.005)	(0.0015)		(0.0006)	(0.010)	(0.0005)
	DR-ny	0.809	−0.0166	0.952	0.1471	1.000	−0.0171	0.947	0.0663	1.000	−0.0181	0.860	0.0271
			(0.0036)	(0.005)	(0.0028)		(0.0014)	(0.005)	(0.0011)		(0.0005)	(0.008)	(0.0004)
B	EEq-HT	0.983	0.0067	0.949	0.1525	1.000	−0.0031	0.945	0.0606	1.000	0.0001	0.953	0.0184
			(0.0034)	(0.005)	(0.0033)		(0.0014)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.983	0.0061	0.966	0.1635	1.000	−0.0036	0.964	0.0660	1.000	0.0000	0.963	0.0203
			(0.0037)	(0.004)	(0.0038)		(0.0015)	(0.004)	(0.0011)		(0.0005)	(0.004)	(0.0003)
	TMLE-cw-HT	0.925	0.0050	0.870	0.1856	1.000	−0.0041	0.883	0.0736	1.000	−0.0002	0.887	0.0226
			(0.0043)	(0.008)	(0.0046)		(0.0016)	(0.007)	(0.0012)		(0.0005)	(0.007)	(0.0003)
	TMLE-cw-Hájek	0.925	0.0029	0.860	0.2227	1.000	−0.0043	0.917	0.0784	1.000	−0.0003	0.925	0.0237
			(0.0052)	(0.008)	(0.0071)		(0.0018)	(0.006)	(0.0014)		(0.0005)	(0.006)	(0.0004)
	TMLE-pw-HT	0.983	0.0065	0.908	0.1741	1.000	−0.0041	0.904	0.0702	1.000	−0.0004	0.908	0.0215
			(0.0039)	(0.007)	(0.0040)		(0.0016)	(0.007)	(0.0012)		(0.0005)	(0.006)	(0.0003)
	TMLE-pw-Hájek	0.983	0.0043	0.916	0.2020	1.000	−0.0043	0.934	0.0753	1.000	−0.0004	0.934	0.0230
			(0.0046)	(0.006)	(0.0063)		(0.0017)	(0.006)	(0.0013)		(0.0005)	(0.006)	(0.0004)
	TMLE-2eps-cw-Hájek	0.925	0.0014	0.831	0.2528	1.000	−0.0045	0.905	0.0820	1.000	−0.0003	0.916	0.0245
			(0.0059)	(0.009)	(0.0088)		(0.0018)	(0.007)	(0.0015)		(0.0005)	(0.006)	(0.0004)
	TMLE-2eps-pw-Hájek	0.983	0.0036	0.902	0.2174	1.000	−0.0045	0.924	0.0782	1.000	−0.0005	0.925	0.0238
			(0.0049)	(0.007)	(0.0078)		(0.0017)	(0.006)	(0.0013)		(0.0005)	(0.006)	(0.0004)
	DR-nt	0.577	−0.0744	0.909	0.2596	1.000	−0.0767	0.896	0.1368	1.000	−0.0681	0.508	0.0767
			(0.0073)	(0.008)	(0.0065)		(0.0025)	(0.007)	(0.0024)		(0.0008)	(0.011)	(0.0008)
	DR-ny	0.577	−0.0396	0.942	0.1894	1.000	−0.0463	0.932	0.0999	1.000	−0.0406	0.696	0.0493
			(0.0055)	(0.007)	(0.0053)		(0.0020)	(0.006)	(0.0019)		(0.0006)	(0.010)	(0.0006)
B+	EEq-HT	0.895	−0.0077	0.909	0.2030	1.000	−0.0010	0.912	0.0775	1.000	0.0001	0.913	0.0246
			(0.0048)	(0.007)	(0.0040)		(0.0017)	(0.006)	(0.0012)		(0.0006)	(0.006)	(0.0004)
	EEq-Hájek	0.895	−0.0102	0.978	0.2836	1.000	−0.0037	0.972	0.1098	1.000	−0.0016	0.959	0.0347
			(0.0067)	(0.003)	(0.0061)		(0.0025)	(0.004)	(0.0019)		(0.0008)	(0.004)	(0.0005)
	TMLE-cw-HT	0.691	−0.0119	0.637	0.3276	1.000	−0.0051	0.703	0.1301	1.000	−0.0025	0.693	0.0412
			(0.0088)	(0.013)	(0.0079)		(0.0029)	(0.010)	(0.0023)		(0.0009)	(0.010)	(0.0007)
	TMLE-cw-Hájek	0.691	−0.0179	0.691	0.5272	1.000	−0.0066	0.833	0.1639	1.000	−0.0035	0.845	0.0498
			(0.0142)	(0.012)	(0.0235)		(0.0037)	(0.008)	(0.0030)		(0.0011)	(0.008)	(0.0008)
	TMLE-pw-HT	0.895	−0.0149	0.690	0.3197	1.000	−0.0044	0.731	0.1216	1.000	−0.0026	0.725	0.0389
			(0.0076)	(0.011)	(0.0069)		(0.0027)	(0.010)	(0.0020)		(0.0009)	(0.010)	(0.0006)
	TMLE-pw-Hájek	0.895	−0.0188	0.770	0.4557	1.000	−0.0067	0.852	0.1566	1.000	−0.0037	0.851	0.0484
			(0.0108)	(0.010)	(0.0116)		(0.0035)	(0.008)	(0.0028)		(0.0011)	(0.008)	(0.0008)
	TMLE-2eps-cw-Hájek	0.691	−0.0209	0.663	0.5889	1.000	−0.0069	0.820	0.1708	1.000	−0.0038	0.830	0.0515
			(0.0158)	(0.013)	(0.0290)		(0.0038)	(0.009)	(0.0032)		(0.0011)	(0.008)	(0.0008)
	TMLE-2eps-pw-Hájek	0.895	−0.0206	0.754	0.4858	1.000	−0.0071	0.836	0.1631	1.000	−0.0040	0.842	0.0502
			(0.0115)	(0.010)	(0.0131)		(0.0036)	(0.008)	(0.0029)		(0.0011)	(0.008)	(0.0008)
	DR-nt	0.188	−0.1899	0.811	0.4451	1.000	−0.1622	0.814	0.2451	1.000	−0.1522	0.285	0.1655
			(0.0208)	(0.020)	(0.0172)		(0.0041)	(0.009)	(0.0039)		(0.0015)	(0.010)	(0.0014)
	DR-ny	0.189	−0.1067	0.878	0.3090	1.000	−0.1239	0.859	0.2002	1.000	−0.1126	0.387	0.1247
			(0.0150)	(0.017)	(0.0128)		(0.0035)	(0.008)	(0.0033)		(0.0012)	(0.011)	(0.0012)

Table J.9, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.983	0.0009	0.957	0.0910	1.000	0.0014	0.951	0.0382	1.000	−0.0005	0.952	0.0117
			(0.0021)	(0.005)	(0.0020)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	EEq-Hájek	0.983	0.0006	0.959	0.0913	1.000	0.0015	0.954	0.0383	1.000	−0.0003	0.959	0.0117
			(0.0021)	(0.004)	(0.0021)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.004)	(0.0002)
	TMLE-cw-HT	0.935	0.0006	0.935	0.0961	1.000	0.0016	0.950	0.0388	1.000	−0.0002	0.953	0.0118
			(0.0022)	(0.006)	(0.0023)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	TMLE-cw-Hájek	0.935	0.0001	0.935	0.1028	1.000	0.0017	0.953	0.0390	1.000	−0.0001	0.956	0.0118
			(0.0024)	(0.006)	(0.0031)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-HT	0.983	0.0006	0.948	0.0915	1.000	0.0016	0.947	0.0385	1.000	−0.0003	0.953	0.0117
			(0.0021)	(0.005)	(0.0019)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.983	0.0004	0.952	0.0938	1.000	0.0016	0.954	0.0387	1.000	−0.0002	0.957	0.0117
			(0.0021)	(0.005)	(0.0021)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.935	−0.0004	0.919	0.1133	1.000	0.0018	0.951	0.0394	1.000	0.0000	0.957	0.0119
			(0.0026)	(0.006)	(0.0046)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.983	0.0004	0.947	0.0961	1.000	0.0017	0.953	0.0389	1.000	−0.0001	0.957	0.0118
			(0.0022)	(0.005)	(0.0022)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	DR-nt	0.802	−0.0313	0.956	0.1624	1.000	−0.0184	0.954	0.0611	1.000	−0.0189	0.827	0.0257
			(0.0040)	(0.005)	(0.0040)		(0.0013)	(0.005)	(0.0011)		(0.0004)	(0.008)	(0.0003)
	DR-ny	0.802	−0.0107	0.959	0.1129	1.000	−0.0063	0.943	0.0472	1.000	−0.0081	0.922	0.0160
			(0.0028)	(0.005)	(0.0024)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.006)	(0.0002)
C2	EEq-HT	0.955	−0.0041	0.906	0.1885	1.000	−0.0022	0.923	0.0684	1.000	0.0000	0.923	0.0211
			(0.0043)	(0.007)	(0.0036)		(0.0015)	(0.006)	(0.0011)		(0.0005)	(0.006)	(0.0003)
	EEq-Hájek	0.955	−0.0066	0.971	0.2344	1.000	−0.0027	0.976	0.0855	1.000	−0.0014	0.968	0.0261
			(0.0054)	(0.004)	(0.0049)		(0.0019)	(0.003)	(0.0014)		(0.0006)	(0.004)	(0.0004)
	TMLE-cw-HT	0.810	−0.0086	0.706	0.2858	1.000	−0.0035	0.729	0.1069	1.000	−0.0030	0.757	0.0328
			(0.0071)	(0.011)	(0.0064)		(0.0024)	(0.010)	(0.0018)		(0.0007)	(0.010)	(0.0005)
	TMLE-cw-Hájek	0.810	−0.0106	0.756	0.3776	1.000	−0.0039	0.843	0.1285	1.000	−0.0042	0.870	0.0385
			(0.0094)	(0.011)	(0.0099)		(0.0029)	(0.008)	(0.0023)		(0.0009)	(0.008)	(0.0006)
	TMLE-pw-HT	0.955	−0.0074	0.743	0.2781	1.000	−0.0035	0.749	0.1028	1.000	−0.0029	0.774	0.0315
			(0.0064)	(0.010)	(0.0056)		(0.0023)	(0.010)	(0.0017)		(0.0007)	(0.009)	(0.0005)
	TMLE-pw-Hájek	0.955	−0.0106	0.816	0.3581	1.000	−0.0038	0.856	0.1246	1.000	−0.0041	0.878	0.0377
			(0.0082)	(0.009)	(0.0091)		(0.0028)	(0.008)	(0.0021)		(0.0008)	(0.007)	(0.0006)
	TMLE-2eps-cw-Hájek	0.810	−0.0119	0.719	0.4164	1.000	−0.0042	0.819	0.1367	1.000	−0.0047	0.853	0.0408
			(0.0103)	(0.011)	(0.0113)		(0.0031)	(0.009)	(0.0024)		(0.0009)	(0.008)	(0.0007)
	TMLE-2eps-pw-Hájek	0.955	−0.0117	0.787	0.3871	1.000	−0.0040	0.837	0.1328	1.000	−0.0046	0.858	0.0401
			(0.0089)	(0.009)	(0.0101)		(0.0030)	(0.008)	(0.0023)		(0.0009)	(0.008)	(0.0006)
	DR-nt	0.451	−0.1792	0.865	0.4104	1.000	−0.1738	0.771	0.2297	1.000	−0.1657	0.062	0.1723
			(0.0123)	(0.011)	(0.0097)		(0.0034)	(0.009)	(0.0031)		(0.0011)	(0.005)	(0.0010)
	DR-ny	0.451	−0.1368	0.880	0.3111	1.000	−0.1368	0.783	0.1844	1.000	−0.1318	0.070	0.1373
			(0.0093)	(0.011)	(0.0085)		(0.0028)	(0.009)	(0.0026)		(0.0009)	(0.006)	(0.0009)

Table J.10: Static design, binary outcome: ATT(2, 4) by scenario: A, 0.058; B, 0.048; B+, 0.044; C1, 0.043; C2, 0.051; no cross-fitting (proposed procedures).

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.999	0.0003	0.943	0.1227	1.000	0.0024	0.939	0.0531	1.000	−0.0005	0.956	0.0163
			(0.0027)	(0.005)	(0.0023)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.999	0.0000	0.948	0.1245	1.000	0.0024	0.946	0.0538	1.000	−0.0005	0.959	0.0165
			(0.0028)	(0.005)	(0.0023)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.004)	(0.0003)
	TMLE-cw-HT	0.991	−0.0003	0.911	0.1331	1.000	0.0023	0.927	0.0553	1.000	−0.0005	0.948	0.0169
			(0.0030)	(0.006)	(0.0025)		(0.0012)	(0.006)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.991	−0.0002	0.917	0.1415	1.000	0.0022	0.937	0.0559	1.000	−0.0005	0.955	0.0170
			(0.0032)	(0.006)	(0.0030)		(0.0012)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-HT	0.999	−0.0003	0.928	0.1288	1.000	0.0024	0.932	0.0546	1.000	−0.0006	0.951	0.0167
			(0.0029)	(0.006)	(0.0024)		(0.0012)	(0.006)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.999	−0.0007	0.933	0.1345	1.000	0.0023	0.941	0.0552	1.000	−0.0006	0.956	0.0169
			(0.0030)	(0.006)	(0.0026)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.991	0.0001	0.900	0.1542	1.000	0.0021	0.931	0.0570	1.000	−0.0006	0.951	0.0172
			(0.0035)	(0.007)	(0.0038)		(0.0013)	(0.006)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-pw-Hájek	0.999	−0.0009	0.921	0.1416	1.000	0.0023	0.939	0.0561	1.000	−0.0006	0.955	0.0170
			(0.0032)	(0.006)	(0.0030)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)

Table J.10, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.998	0.0055	0.935	0.1382	1.000	−0.0031	0.943	0.0595	1.000	0.0001	0.951	0.0184
			(0.0031)	(0.006)	(0.0027)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.998	0.0049	0.962	0.1511	1.000	−0.0036	0.962	0.0649	1.000	0.0000	0.962	0.0202
			(0.0034)	(0.004)	(0.0033)		(0.0014)	(0.004)	(0.0011)		(0.0005)	(0.004)	(0.0003)
	TMLE-cw-HT	0.990	0.0031	0.843	0.1787	1.000	−0.0042	0.875	0.0728	1.000	−0.0002	0.886	0.0226
			(0.0040)	(0.008)	(0.0042)		(0.0016)	(0.007)	(0.0012)		(0.0005)	(0.007)	(0.0003)
	TMLE-cw-Hájek	0.990	−0.0007	0.851	0.2243	1.000	−0.0044	0.915	0.0779	1.000	−0.0003	0.927	0.0237
			(0.0050)	(0.008)	(0.0069)		(0.0017)	(0.006)	(0.0014)		(0.0005)	(0.006)	(0.0004)
	TMLE-pw-HT	0.998	0.0055	0.877	0.1685	1.000	−0.0042	0.896	0.0695	1.000	−0.0004	0.911	0.0214
			(0.0038)	(0.007)	(0.0037)		(0.0016)	(0.007)	(0.0012)		(0.0005)	(0.006)	(0.0003)
	TMLE-pw-Hájek	0.998	0.0026	0.894	0.2013	1.000	−0.0045	0.926	0.0748	1.000	−0.0004	0.936	0.0230
			(0.0045)	(0.007)	(0.0058)		(0.0017)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-2eps-cw-Hájek	0.990	−0.0031	0.814	0.2596	1.000	−0.0046	0.899	0.0818	1.000	−0.0003	0.914	0.0245
			(0.0058)	(0.009)	(0.0089)		(0.0018)	(0.007)	(0.0015)		(0.0005)	(0.006)	(0.0004)
	TMLE-2eps-pw-Hájek	0.998	0.0015	0.871	0.2224	1.000	−0.0047	0.913	0.0780	1.000	−0.0005	0.924	0.0238
			(0.0050)	(0.008)	(0.0070)		(0.0017)	(0.006)	(0.0013)		(0.0005)	(0.006)	(0.0004)
B+	EEq-HT	0.983	−0.0051	0.872	0.1822	1.000	−0.0004	0.913	0.0760	1.000	0.0001	0.912	0.0246
			(0.0041)	(0.008)	(0.0032)		(0.0017)	(0.006)	(0.0012)		(0.0005)	(0.006)	(0.0004)
	EEq-Hájek	0.983	−0.0071	0.964	0.2519	1.000	−0.0028	0.974	0.1078	1.000	−0.0015	0.959	0.0346
			(0.0057)	(0.004)	(0.0050)		(0.0024)	(0.004)	(0.0018)		(0.0008)	(0.004)	(0.0005)
	TMLE-cw-HT	0.934	−0.0089	0.607	0.3003	1.000	−0.0041	0.687	0.1287	1.000	−0.0025	0.694	0.0411
			(0.0069)	(0.011)	(0.0058)		(0.0029)	(0.010)	(0.0022)		(0.0009)	(0.010)	(0.0007)
	TMLE-cw-Hájek	0.934	−0.0146	0.633	0.5397	1.000	−0.0056	0.832	0.1643	1.000	−0.0036	0.842	0.0497
			(0.0125)	(0.011)	(0.0156)		(0.0037)	(0.008)	(0.0031)		(0.0011)	(0.008)	(0.0008)
	TMLE-pw-HT	0.983	−0.0123	0.643	0.2974	1.000	−0.0034	0.716	0.1203	1.000	−0.0026	0.724	0.0388
			(0.0067)	(0.011)	(0.0061)		(0.0027)	(0.010)	(0.0020)		(0.0009)	(0.010)	(0.0006)
	TMLE-pw-Hájek	0.983	−0.0177	0.717	0.4569	1.000	−0.0056	0.843	0.1565	1.000	−0.0037	0.853	0.0483
			(0.0103)	(0.010)	(0.0112)		(0.0035)	(0.008)	(0.0028)		(0.0011)	(0.008)	(0.0008)
	TMLE-2eps-cw-Hájek	0.934	−0.0172	0.604	0.6123	1.000	−0.0059	0.816	0.1716	1.000	−0.0038	0.826	0.0514
			(0.0142)	(0.011)	(0.0193)		(0.0038)	(0.009)	(0.0032)		(0.0011)	(0.008)	(0.0008)
	TMLE-2eps-pw-Hájek	0.983	−0.0203	0.689	0.4945	1.000	−0.0060	0.830	0.1634	1.000	−0.0040	0.837	0.0502
			(0.0111)	(0.010)	(0.0124)		(0.0037)	(0.008)	(0.0029)		(0.0011)	(0.008)	(0.0008)
C1	EEq-HT	1.000	0.0006	0.944	0.0874	1.000	0.0013	0.948	0.0381	1.000	−0.0005	0.953	0.0117
			(0.0020)	(0.005)	(0.0016)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	EEq-Hájek	1.000	0.0005	0.956	0.0883	1.000	0.0014	0.954	0.0382	1.000	−0.0004	0.959	0.0117
			(0.0020)	(0.005)	(0.0018)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.004)	(0.0002)
	TMLE-cw-HT	0.994	0.0002	0.927	0.0930	1.000	0.0016	0.948	0.0387	1.000	−0.0002	0.953	0.0118
			(0.0021)	(0.006)	(0.0021)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-cw-Hájek	0.994	−0.0003	0.935	0.0990	1.000	0.0017	0.953	0.0389	1.000	−0.0001	0.958	0.0118
			(0.0022)	(0.006)	(0.0026)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.004)	(0.0002)
	TMLE-pw-HT	1.000	0.0006	0.941	0.0895	1.000	0.0015	0.946	0.0384	1.000	−0.0003	0.954	0.0117
			(0.0020)	(0.005)	(0.0018)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-Hájek	1.000	0.0003	0.953	0.0920	1.000	0.0016	0.952	0.0385	1.000	−0.0002	0.958	0.0117
			(0.0021)	(0.005)	(0.0020)		(0.0009)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.994	−0.0008	0.922	0.1090	1.000	0.0018	0.951	0.0393	1.000	0.0000	0.958	0.0118
			(0.0024)	(0.006)	(0.0036)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	1.000	0.0003	0.944	0.0955	1.000	0.0017	0.952	0.0388	1.000	−0.0001	0.959	0.0118
			(0.0021)	(0.005)	(0.0022)		(0.0009)	(0.005)	(0.0007)		(0.0003)	(0.004)	(0.0002)
C2	EEq-HT	0.996	−0.0023	0.877	0.1688	1.000	−0.0019	0.917	0.0671	1.000	0.0000	0.923	0.0210
			(0.0038)	(0.007)	(0.0031)		(0.0015)	(0.006)	(0.0011)		(0.0005)	(0.006)	(0.0003)
	EEq-Hájek	0.996	−0.0042	0.960	0.2041	1.000	−0.0023	0.975	0.0833	1.000	−0.0014	0.968	0.0260
			(0.0046)	(0.004)	(0.0039)		(0.0019)	(0.004)	(0.0014)		(0.0006)	(0.004)	(0.0004)
	TMLE-cw-HT	0.974	−0.0081	0.695	0.2496	1.000	−0.0030	0.717	0.1044	1.000	−0.0030	0.758	0.0327
			(0.0057)	(0.010)	(0.0049)		(0.0023)	(0.010)	(0.0017)		(0.0007)	(0.010)	(0.0005)
	TMLE-cw-Hájek	0.974	−0.0124	0.745	0.3483	1.000	−0.0033	0.835	0.1265	1.000	−0.0041	0.871	0.0384
			(0.0079)	(0.010)	(0.0082)		(0.0028)	(0.008)	(0.0022)		(0.0009)	(0.007)	(0.0006)
	TMLE-pw-HT	0.996	−0.0047	0.721	0.2476	1.000	−0.0031	0.735	0.1006	1.000	−0.0028	0.775	0.0314
			(0.0055)	(0.010)	(0.0049)		(0.0022)	(0.010)	(0.0016)		(0.0007)	(0.009)	(0.0005)
	TMLE-pw-Hájek	0.996	−0.0073	0.797	0.3307	1.000	−0.0033	0.855	0.1225	1.000	−0.0041	0.878	0.0376
			(0.0074)	(0.009)	(0.0087)		(0.0027)	(0.008)	(0.0021)		(0.0008)	(0.007)	(0.0006)
	TMLE-2eps-cw-Hájek	0.974	−0.0144	0.697	0.3882	1.000	−0.0036	0.810	0.1349	1.000	−0.0046	0.851	0.0407
			(0.0088)	(0.010)	(0.0095)		(0.0030)	(0.009)	(0.0024)		(0.0009)	(0.008)	(0.0007)
	TMLE-2eps-pw-Hájek	0.996	−0.0082	0.762	0.3625	1.000	−0.0035	0.829	0.1310	1.000	−0.0046	0.857	0.0400
			(0.0081)	(0.010)	(0.0099)		(0.0029)	(0.008)	(0.0023)		(0.0009)	(0.008)	(0.0006)

Table J.11: Static design, binary outcome: ES(1) by scenario: A, 0.073; B, 0.066; B+, 0.066; C1, 0.056; C2, 0.075; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.936	0.0028	0.951	0.0966	1.000	0.0015	0.954	0.0377	1.000	−0.0001	0.950	0.0120
			(0.0022)	(0.005)	(0.0023)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	EEq-Hájek	0.936	0.0031	0.949	0.0918	1.000	0.0014	0.957	0.0377	1.000	−0.0001	0.951	0.0120
			(0.0021)	(0.005)	(0.0017)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-cw-HT	0.936	0.0032	0.947	0.0919	1.000	0.0014	0.954	0.0377	1.000	−0.0001	0.948	0.0120
			(0.0021)	(0.005)	(0.0017)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-cw-Hájek	0.936	0.0032	0.938	0.0932	1.000	0.0014	0.956	0.0378	1.000	−0.0001	0.949	0.0120
			(0.0022)	(0.006)	(0.0018)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-HT	0.936	0.0029	0.947	0.0913	1.000	0.0014	0.955	0.0377	1.000	−0.0002	0.948	0.0120
			(0.0021)	(0.005)	(0.0017)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.936	0.0029	0.942	0.0920	1.000	0.0014	0.956	0.0377	1.000	−0.0002	0.949	0.0120
			(0.0021)	(0.005)	(0.0017)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.936	0.0034	0.931	0.0972	1.000	0.0013	0.954	0.0379	1.000	−0.0002	0.949	0.0120
			(0.0022)	(0.006)	(0.0020)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.936	0.0028	0.935	0.0937	1.000	0.0013	0.956	0.0378	1.000	−0.0002	0.948	0.0120
			(0.0022)	(0.006)	(0.0018)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	DR-nt	0.809	−0.0205	0.933	0.1154	1.000	−0.0166	0.943	0.0509	1.000	−0.0165	0.808	0.0226
			(0.0028)	(0.006)	(0.0024)		(0.0011)	(0.005)	(0.0009)		(0.0003)	(0.009)	(0.0003)
	DR-ny	0.809	−0.0028	0.950	0.0861	1.000	−0.0033	0.947	0.0366	1.000	−0.0045	0.922	0.0122
			(0.0021)	(0.005)	(0.0015)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.006)	(0.0002)
B	EEq-HT	0.925	0.0077	0.962	0.1113	1.000	−0.0028	0.957	0.0439	1.000	0.0002	0.949	0.0137
			(0.0026)	(0.004)	(0.0026)		(0.0010)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	EEq-Hájek	0.925	0.0078	0.965	0.1079	1.000	−0.0028	0.962	0.0447	1.000	0.0001	0.959	0.0140
			(0.0025)	(0.004)	(0.0023)		(0.0010)	(0.004)	(0.0007)		(0.0003)	(0.004)	(0.0002)
	TMLE-cw-HT	0.923	0.0071	0.918	0.1156	1.000	−0.0031	0.939	0.0464	1.000	0.0000	0.932	0.0145
			(0.0027)	(0.006)	(0.0024)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.006)	(0.0002)
	TMLE-cw-Hájek	0.923	0.0065	0.918	0.1228	1.000	−0.0031	0.947	0.0471	1.000	0.0000	0.946	0.0146
			(0.0029)	(0.006)	(0.0027)		(0.0011)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-HT	0.923	0.0077	0.940	0.1117	1.000	−0.0031	0.943	0.0458	1.000	0.0000	0.937	0.0143
			(0.0026)	(0.006)	(0.0024)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.923	0.0070	0.943	0.1179	1.000	−0.0031	0.950	0.0466	1.000	0.0000	0.946	0.0145
			(0.0027)	(0.005)	(0.0026)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.923	0.0058	0.897	0.1362	1.000	−0.0033	0.939	0.0484	1.000	0.0000	0.940	0.0148
			(0.0032)	(0.007)	(0.0031)		(0.0011)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.923	0.0066	0.923	0.1257	1.000	−0.0033	0.946	0.0476	1.000	−0.0001	0.943	0.0147
			(0.0029)	(0.006)	(0.0030)		(0.0011)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	DR-nt	0.577	−0.0398	0.923	0.1548	1.000	−0.0412	0.913	0.0799	1.000	−0.0353	0.611	0.0414
			(0.0044)	(0.008)	(0.0041)		(0.0015)	(0.006)	(0.0015)		(0.0005)	(0.011)	(0.0005)
	DR-ny	0.577	−0.0071	0.956	0.1020	1.000	−0.0145	0.950	0.0460	1.000	−0.0100	0.891	0.0173
			(0.0030)	(0.006)	(0.0028)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.007)	(0.0003)
B+	EEq-HT	0.691	−0.0035	0.929	0.1661	1.000	0.0010	0.932	0.0625	1.000	0.0000	0.924	0.0200
			(0.0045)	(0.007)	(0.0036)		(0.0014)	(0.006)	(0.0010)		(0.0004)	(0.006)	(0.0003)
	EEq-Hájek	0.691	−0.0015	0.967	0.1839	1.000	0.0002	0.966	0.0735	1.000	−0.0006	0.956	0.0235
			(0.0050)	(0.005)	(0.0041)		(0.0016)	(0.004)	(0.0012)		(0.0005)	(0.005)	(0.0004)
	TMLE-cw-HT	0.687	−0.0029	0.790	0.2098	1.000	−0.0006	0.815	0.0831	1.000	−0.0011	0.815	0.0265
			(0.0057)	(0.011)	(0.0048)		(0.0019)	(0.009)	(0.0013)		(0.0006)	(0.009)	(0.0004)
	TMLE-cw-Hájek	0.687	−0.0041	0.809	0.2668	1.000	−0.0011	0.900	0.0930	1.000	−0.0014	0.896	0.0289
			(0.0072)	(0.011)	(0.0079)		(0.0021)	(0.007)	(0.0015)		(0.0006)	(0.007)	(0.0005)
	TMLE-pw-HT	0.687	−0.0058	0.825	0.2045	1.000	0.0000	0.840	0.0792	1.000	−0.0011	0.832	0.0255
			(0.0055)	(0.010)	(0.0048)		(0.0018)	(0.008)	(0.0012)		(0.0006)	(0.008)	(0.0004)
	TMLE-pw-Hájek	0.687	−0.0058	0.852	0.2527	1.000	−0.0006	0.910	0.0904	1.000	−0.0015	0.903	0.0284
			(0.0068)	(0.010)	(0.0070)		(0.0020)	(0.006)	(0.0014)		(0.0006)	(0.007)	(0.0004)
	TMLE-2eps-cw-Hájek	0.687	−0.0060	0.772	0.2994	1.000	−0.0014	0.882	0.0971	1.000	−0.0016	0.888	0.0297
			(0.0081)	(0.011)	(0.0093)		(0.0022)	(0.007)	(0.0016)		(0.0007)	(0.007)	(0.0005)
	TMLE-2eps-pw-Hájek	0.687	−0.0076	0.828	0.2724	1.000	−0.0008	0.897	0.0940	1.000	−0.0016	0.894	0.0294
			(0.0073)	(0.010)	(0.0079)		(0.0021)	(0.007)	(0.0015)		(0.0007)	(0.007)	(0.0005)
	DR-nt	0.188	−0.1245	0.824	0.3032	1.000	−0.0871	0.858	0.1564	1.000	−0.0851	0.467	0.0968
			(0.0143)	(0.020)	(0.0116)		(0.0029)	(0.008)	(0.0026)		(0.0010)	(0.011)	(0.0010)
	DR-ny	0.188	−0.0348	0.912	0.1797	1.000	−0.0421	0.924	0.0971	1.000	−0.0362	0.756	0.0465
			(0.0091)	(0.015)	(0.0072)		(0.0020)	(0.006)	(0.0018)		(0.0006)	(0.010)	(0.0006)

Table J.11, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.935	0.0022	0.956	0.0695	1.000	0.0015	0.952	0.0294	1.000	0.0001	0.956	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	EEq-Hájek	0.935	0.0020	0.951	0.0686	1.000	0.0015	0.953	0.0294	1.000	0.0001	0.957	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-cw-HT	0.935	0.0018	0.949	0.0697	1.000	0.0015	0.953	0.0294	1.000	0.0001	0.956	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-cw-Hájek	0.935	0.0018	0.942	0.0709	1.000	0.0015	0.952	0.0294	1.000	0.0001	0.957	0.0089
			(0.0016)	(0.005)	(0.0013)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-pw-HT	0.935	0.0018	0.951	0.0687	1.000	0.0015	0.954	0.0294	1.000	0.0001	0.956	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-pw-Hájek	0.935	0.0017	0.945	0.0693	1.000	0.0015	0.953	0.0294	1.000	0.0001	0.957	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-2eps-cw-Hájek	0.935	0.0018	0.938	0.0740	1.000	0.0015	0.953	0.0295	1.000	0.0001	0.956	0.0089
			(0.0017)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-2eps-pw-Hájek	0.935	0.0017	0.945	0.0705	1.000	0.0015	0.953	0.0294	1.000	0.0001	0.957	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	DR-nt	0.802	−0.0162	0.959	0.0959	1.000	−0.0093	0.956	0.0369	1.000	−0.0101	0.866	0.0148
			(0.0024)	(0.005)	(0.0024)		(0.0008)	(0.005)	(0.0006)		(0.0002)	(0.008)	(0.0002)
	DR-ny	0.802	−0.0003	0.956	0.0670	1.000	−0.0004	0.951	0.0286	1.000	−0.0016	0.947	0.0088
			(0.0017)	(0.005)	(0.0012)		(0.0006)	(0.005)	(0.0004)		(0.0002)	(0.005)	(0.0001)
C2	EEq-HT	0.810	−0.0018	0.935	0.1405	1.000	−0.0006	0.935	0.0515	1.000	0.0000	0.941	0.0159
			(0.0035)	(0.006)	(0.0029)		(0.0012)	(0.006)	(0.0008)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.810	−0.0035	0.964	0.1510	1.000	−0.0004	0.971	0.0573	1.000	−0.0006	0.975	0.0176
			(0.0038)	(0.005)	(0.0032)		(0.0013)	(0.004)	(0.0009)		(0.0004)	(0.003)	(0.0003)
	TMLE-cw-HT	0.809	−0.0054	0.830	0.1733	1.000	−0.0004	0.852	0.0668	1.000	−0.0013	0.848	0.0204
			(0.0043)	(0.009)	(0.0037)		(0.0015)	(0.008)	(0.0011)		(0.0005)	(0.008)	(0.0003)
	TMLE-cw-Hájek	0.809	−0.0093	0.856	0.2047	1.000	−0.0002	0.906	0.0739	1.000	−0.0017	0.914	0.0222
			(0.0051)	(0.009)	(0.0053)		(0.0017)	(0.007)	(0.0013)		(0.0005)	(0.006)	(0.0003)
	TMLE-pw-HT	0.809	−0.0029	0.858	0.1717	1.000	−0.0007	0.859	0.0651	1.000	−0.0012	0.857	0.0200
			(0.0043)	(0.009)	(0.0038)		(0.0015)	(0.008)	(0.0011)		(0.0004)	(0.008)	(0.0003)
	TMLE-pw-Hájek	0.809	−0.0062	0.886	0.2019	1.000	−0.0004	0.910	0.0726	1.000	−0.0017	0.915	0.0220
			(0.0050)	(0.008)	(0.0065)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.006)	(0.0003)
	TMLE-2eps-cw-Hájek	0.809	−0.0111	0.819	0.2260	1.000	−0.0003	0.888	0.0784	1.000	−0.0021	0.898	0.0234
			(0.0056)	(0.010)	(0.0061)		(0.0018)	(0.007)	(0.0014)		(0.0005)	(0.007)	(0.0004)
	TMLE-2eps-pw-Hájek	0.809	−0.0071	0.858	0.2196	1.000	−0.0004	0.887	0.0772	1.000	−0.0020	0.901	0.0233
			(0.0055)	(0.009)	(0.0073)		(0.0017)	(0.007)	(0.0013)		(0.0005)	(0.007)	(0.0004)
	DR-nt	0.451	−0.1000	0.871	0.2604	1.000	−0.0938	0.834	0.1396	1.000	−0.0911	0.188	0.0967
			(0.0080)	(0.011)	(0.0063)		(0.0023)	(0.008)	(0.0021)		(0.0007)	(0.009)	(0.0007)
	DR-ny	0.451	−0.0584	0.923	0.1638	1.000	−0.0555	0.872	0.0885	1.000	−0.0539	0.278	0.0579
			(0.0051)	(0.009)	(0.0044)		(0.0015)	(0.007)	(0.0014)		(0.0005)	(0.010)	(0.0005)

Table J.12: Static design, binary outcome: ES(1) by scenario: A, 0.073; B, 0.066; B+, 0.066; C1, 0.056; C2, 0.075; no cross-fitting (proposed procedures).

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.991	0.0023	0.942	0.0879	1.000	0.0014	0.954	0.0376	1.000	−0.0001	0.950	0.0120
			(0.0020)	(0.005)	(0.0015)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	EEq-Hájek	0.991	0.0024	0.945	0.0881	1.000	0.0014	0.956	0.0376	1.000	−0.0001	0.950	0.0120
			(0.0020)	(0.005)	(0.0015)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-cw-HT	0.991	0.0023	0.933	0.0902	1.000	0.0013	0.952	0.0377	1.000	−0.0001	0.948	0.0120
			(0.0020)	(0.006)	(0.0016)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-cw-Hájek	0.991	0.0024	0.933	0.0918	1.000	0.0013	0.954	0.0378	1.000	−0.0001	0.949	0.0120
			(0.0021)	(0.006)	(0.0017)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-HT	0.991	0.0022	0.934	0.0895	1.000	0.0013	0.953	0.0377	1.000	−0.0001	0.949	0.0120
			(0.0020)	(0.006)	(0.0016)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.991	0.0023	0.937	0.0908	1.000	0.0013	0.955	0.0378	1.000	−0.0001	0.950	0.0120
			(0.0020)	(0.005)	(0.0016)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.991	0.0024	0.929	0.0964	1.000	0.0012	0.953	0.0380	1.000	−0.0001	0.949	0.0120
			(0.0022)	(0.006)	(0.0019)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.991	0.0022	0.933	0.0938	1.000	0.0012	0.954	0.0379	1.000	−0.0002	0.950	0.0120
			(0.0021)	(0.006)	(0.0018)		(0.0008)	(0.005)	(0.0006)		(0.0003)	(0.005)	(0.0002)

Table J.12, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.990	0.0061	0.951	0.0995	1.000	−0.0028	0.950	0.0432	1.000	0.0002	0.950	0.0137
			(0.0022)	(0.005)	(0.0019)		(0.0010)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	EEq-Hájek	0.990	0.0063	0.960	0.1027	1.000	−0.0029	0.959	0.0442	1.000	0.0001	0.959	0.0140
			(0.0023)	(0.004)	(0.0020)		(0.0010)	(0.004)	(0.0007)		(0.0003)	(0.004)	(0.0002)
	TMLE-cw-HT	0.990	0.0056	0.901	0.1150	1.000	−0.0032	0.934	0.0462	1.000	0.0000	0.930	0.0145
			(0.0026)	(0.007)	(0.0025)		(0.0010)	(0.006)	(0.0008)		(0.0003)	(0.006)	(0.0002)
	TMLE-cw-Hájek	0.990	0.0045	0.909	0.1290	1.000	−0.0033	0.944	0.0469	1.000	0.0000	0.943	0.0146
			(0.0029)	(0.006)	(0.0046)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-HT	0.990	0.0060	0.915	0.1115	1.000	−0.0033	0.938	0.0456	1.000	0.0000	0.937	0.0143
			(0.0025)	(0.006)	(0.0025)		(0.0010)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.990	0.0050	0.922	0.1225	1.000	−0.0033	0.947	0.0464	1.000	0.0000	0.945	0.0145
			(0.0028)	(0.006)	(0.0037)		(0.0010)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.990	0.0033	0.870	0.1489	1.000	−0.0035	0.937	0.0484	1.000	0.0000	0.939	0.0149
			(0.0033)	(0.008)	(0.0063)		(0.0011)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.990	0.0039	0.906	0.1359	1.000	−0.0035	0.943	0.0477	1.000	−0.0001	0.940	0.0147
			(0.0031)	(0.007)	(0.0047)		(0.0011)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
B+	EEq-HT	0.934	−0.0030	0.907	0.1439	1.000	0.0013	0.928	0.0608	1.000	0.0000	0.921	0.0199
			(0.0033)	(0.007)	(0.0026)		(0.0014)	(0.006)	(0.0009)		(0.0004)	(0.006)	(0.0003)
	EEq-Hájek	0.934	−0.0033	0.962	0.1698	1.000	0.0006	0.966	0.0718	1.000	−0.0006	0.956	0.0235
			(0.0039)	(0.004)	(0.0034)		(0.0016)	(0.004)	(0.0011)		(0.0005)	(0.005)	(0.0004)
	TMLE-cw-HT	0.932	−0.0054	0.746	0.2011	1.000	−0.0001	0.812	0.0819	1.000	−0.0011	0.814	0.0264
			(0.0047)	(0.010)	(0.0039)		(0.0018)	(0.009)	(0.0013)		(0.0006)	(0.009)	(0.0004)
	TMLE-cw-Hájek	0.932	−0.0093	0.769	0.2817	1.000	−0.0005	0.896	0.0921	1.000	−0.0015	0.896	0.0288
			(0.0065)	(0.010)	(0.0093)		(0.0021)	(0.007)	(0.0015)		(0.0006)	(0.007)	(0.0005)
	TMLE-pw-HT	0.932	−0.0071	0.774	0.1970	1.000	0.0004	0.831	0.0782	1.000	−0.0011	0.831	0.0255
			(0.0046)	(0.010)	(0.0040)		(0.0017)	(0.008)	(0.0012)		(0.0006)	(0.008)	(0.0004)
	TMLE-pw-Hájek	0.932	−0.0087	0.808	0.2603	1.000	−0.0001	0.906	0.0895	1.000	−0.0015	0.902	0.0284
			(0.0060)	(0.009)	(0.0070)		(0.0020)	(0.007)	(0.0014)		(0.0006)	(0.007)	(0.0004)
	TMLE-2eps-cw-Hájek	0.932	−0.0119	0.729	0.3368	1.000	−0.0009	0.877	0.0967	1.000	−0.0016	0.887	0.0297
			(0.0078)	(0.010)	(0.0144)		(0.0022)	(0.007)	(0.0016)		(0.0007)	(0.007)	(0.0005)
	TMLE-2eps-pw-Hájek	0.932	−0.0106	0.775	0.2938	1.000	−0.0003	0.891	0.0936	1.000	−0.0017	0.892	0.0293
			(0.0068)	(0.010)	(0.0087)		(0.0021)	(0.007)	(0.0015)		(0.0007)	(0.007)	(0.0005)
C1	EEq-HT	0.994	0.0014	0.947	0.0673	1.000	0.0013	0.952	0.0293	1.000	0.0001	0.956	0.0089
			(0.0015)	(0.005)	(0.0011)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	EEq-Hájek	0.994	0.0014	0.950	0.0675	1.000	0.0013	0.954	0.0293	1.000	0.0001	0.958	0.0089
			(0.0015)	(0.005)	(0.0011)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-cw-HT	0.994	0.0015	0.942	0.0687	1.000	0.0014	0.953	0.0293	1.000	0.0001	0.956	0.0089
			(0.0015)	(0.005)	(0.0011)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-cw-Hájek	0.994	0.0014	0.943	0.0698	1.000	0.0014	0.954	0.0294	1.000	0.0001	0.956	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-pw-HT	0.994	0.0014	0.945	0.0680	1.000	0.0014	0.952	0.0293	1.000	0.0001	0.956	0.0089
			(0.0015)	(0.005)	(0.0011)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-pw-Hájek	0.994	0.0013	0.946	0.0686	1.000	0.0014	0.954	0.0293	1.000	0.0001	0.957	0.0089
			(0.0015)	(0.005)	(0.0011)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-2eps-cw-Hájek	0.994	0.0015	0.939	0.0729	1.000	0.0015	0.953	0.0294	1.000	0.0001	0.956	0.0089
			(0.0016)	(0.005)	(0.0014)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
	TMLE-2eps-pw-Hájek	0.994	0.0013	0.943	0.0702	1.000	0.0014	0.953	0.0294	1.000	0.0001	0.957	0.0089
			(0.0016)	(0.005)	(0.0012)		(0.0007)	(0.005)	(0.0005)		(0.0002)	(0.005)	(0.0001)
C2	EEq-HT	0.974	−0.0005	0.912	0.1235	1.000	−0.0005	0.931	0.0508	1.000	0.0000	0.941	0.0159
			(0.0028)	(0.006)	(0.0023)		(0.0011)	(0.006)	(0.0008)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.974	−0.0019	0.960	0.1357	1.000	−0.0004	0.969	0.0563	1.000	−0.0006	0.973	0.0176
			(0.0031)	(0.004)	(0.0026)		(0.0013)	(0.004)	(0.0009)		(0.0004)	(0.004)	(0.0003)
	TMLE-cw-HT	0.973	−0.0039	0.817	0.1579	1.000	−0.0005	0.845	0.0658	1.000	−0.0013	0.846	0.0204
			(0.0036)	(0.009)	(0.0030)		(0.0015)	(0.008)	(0.0011)		(0.0005)	(0.008)	(0.0003)
	TMLE-cw-Hájek	0.973	−0.0082	0.849	0.1915	1.000	−0.0004	0.901	0.0731	1.000	−0.0017	0.911	0.0222
			(0.0043)	(0.008)	(0.0050)		(0.0016)	(0.007)	(0.0012)		(0.0005)	(0.006)	(0.0003)
	TMLE-pw-HT	0.973	−0.0019	0.830	0.1577	1.000	−0.0007	0.858	0.0643	1.000	−0.0012	0.858	0.0200
			(0.0036)	(0.009)	(0.0031)		(0.0014)	(0.008)	(0.0011)		(0.0004)	(0.008)	(0.0003)
	TMLE-pw-Hájek	0.973	−0.0049	0.874	0.1875	1.000	−0.0005	0.906	0.0719	1.000	−0.0017	0.916	0.0220
			(0.0043)	(0.008)	(0.0054)		(0.0016)	(0.007)	(0.0012)		(0.0005)	(0.006)	(0.0003)
	TMLE-2eps-cw-Hájek	0.973	−0.0104	0.809	0.2141	1.000	−0.0005	0.880	0.0777	1.000	−0.0021	0.896	0.0234
			(0.0049)	(0.009)	(0.0058)		(0.0017)	(0.007)	(0.0013)		(0.0005)	(0.007)	(0.0004)
	TMLE-2eps-pw-Hájek	0.973	−0.0059	0.838	0.2068	1.000	−0.0005	0.886	0.0767	1.000	−0.0020	0.900	0.0233
			(0.0047)	(0.008)	(0.0062)		(0.0017)	(0.007)	(0.0013)		(0.0005)	(0.007)	(0.0004)

Table J.13: Dynamic design, binary outcome: ATT$(2,4)$ by scenario: A, 0.175; B, 0.154; B+, 0.133; C1, 0.149; C2, 0.185; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.993	0.0087	0.954	0.3057	1.000	0.0008	0.957	0.0946	1.000	0.0000	0.947	0.0300
			(0.0069)	(0.005)	(0.0070)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	EEq-Hájek	0.993	0.0055	0.944	0.2582	1.000	0.0009	0.956	0.0934	1.000	0.0000	0.947	0.0300
			(0.0058)	(0.005)	(0.0051)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-HT	0.965	0.0033	0.917	0.2483	1.000	0.0009	0.949	0.0935	1.000	0.0000	0.939	0.0303
			(0.0057)	(0.006)	(0.0048)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-Hájek	0.965	0.0030	0.901	0.2556	1.000	0.0009	0.947	0.0938	1.000	0.0000	0.939	0.0304
			(0.0058)	(0.007)	(0.0055)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-pw-HT	0.993	0.0036	0.931	0.2491	1.000	0.0011	0.952	0.0933	1.000	0.0000	0.941	0.0302
			(0.0056)	(0.006)	(0.0048)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-pw-Hájek	0.993	0.0034	0.919	0.2520	1.000	0.0011	0.951	0.0934	1.000	0.0000	0.942	0.0303
			(0.0057)	(0.006)	(0.0051)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-cw-Hájek	0.965	0.0018	0.883	0.2863	1.000	0.0008	0.932	0.0964	1.000	0.0000	0.940	0.0307
			(0.0065)	(0.007)	(0.0085)		(0.0022)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-pw-Hájek	0.993	0.0019	0.906	0.2636	1.000	0.0013	0.942	0.0947	1.000	0.0000	0.941	0.0305
			(0.0059)	(0.007)	(0.0067)		(0.0021)	(0.005)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	DR-nt	0.479	−0.0097	0.900	0.2447	1.000	−0.0034	0.947	0.0942	1.000	−0.0032	0.948	0.0303
			(0.0079)	(0.010)	(0.0061)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	DR-ny	0.479	−0.0050	0.920	0.2228	1.000	−0.0024	0.948	0.0863	1.000	−0.0024	0.942	0.0277
			(0.0072)	(0.009)	(0.0056)		(0.0019)	(0.005)	(0.0014)		(0.0006)	(0.005)	(0.0004)
B	EEq-HT	0.941	0.0062	0.960	0.3596	1.000	0.0043	0.953	0.1162	1.000	−0.0012	0.956	0.0343
			(0.0083)	(0.005)	(0.0092)		(0.0026)	(0.005)	(0.0020)		(0.0008)	(0.005)	(0.0005)
	EEq-Hájek	0.941	0.0052	0.953	0.2955	1.000	0.0042	0.950	0.1166	1.000	−0.0012	0.958	0.0352
			(0.0068)	(0.005)	(0.0055)		(0.0026)	(0.005)	(0.0020)		(0.0008)	(0.004)	(0.0006)
	TMLE-cw-HT	0.843	0.0027	0.890	0.3055	1.000	0.0044	0.905	0.1240	1.000	−0.0013	0.932	0.0375
			(0.0074)	(0.008)	(0.0062)		(0.0028)	(0.007)	(0.0022)		(0.0008)	(0.006)	(0.0006)
	TMLE-cw-Hájek	0.843	0.0032	0.855	0.3504	1.000	0.0045	0.901	0.1281	1.000	−0.0013	0.937	0.0381
			(0.0085)	(0.009)	(0.0118)		(0.0029)	(0.007)	(0.0023)		(0.0009)	(0.005)	(0.0006)
	TMLE-pw-HT	0.941	0.0037	0.910	0.3080	1.000	0.0039	0.919	0.1209	1.000	−0.0012	0.942	0.0367
			(0.0071)	(0.007)	(0.0064)		(0.0027)	(0.006)	(0.0021)		(0.0008)	(0.005)	(0.0006)
	TMLE-pw-Hájek	0.941	0.0051	0.882	0.3288	1.000	0.0042	0.918	0.1240	1.000	−0.0013	0.946	0.0374
			(0.0076)	(0.007)	(0.0076)		(0.0028)	(0.006)	(0.0022)		(0.0008)	(0.005)	(0.0006)
	TMLE-2eps-cw-Hájek	0.843	0.0052	0.823	0.4652	1.000	0.0048	0.871	0.1411	1.000	−0.0013	0.926	0.0398
			(0.0113)	(0.009)	(0.0257)		(0.0032)	(0.008)	(0.0027)		(0.0009)	(0.006)	(0.0006)
	TMLE-2eps-pw-Hájek	0.941	0.0050	0.869	0.3693	1.000	0.0042	0.899	0.1317	1.000	−0.0013	0.936	0.0388
			(0.0085)	(0.008)	(0.0101)		(0.0029)	(0.007)	(0.0024)		(0.0009)	(0.005)	(0.0006)
	DR-nt	0.150	0.0179	0.913	0.2783	1.000	−0.0053	0.940	0.1172	1.000	−0.0091	0.937	0.0375
			(0.0161)	(0.016)	(0.0134)		(0.0026)	(0.005)	(0.0020)		(0.0008)	(0.005)	(0.0006)
	DR-ny	0.150	−0.0038	0.926	0.2555	1.000	−0.0035	0.935	0.1071	1.000	−0.0067	0.945	0.0330
			(0.0148)	(0.015)	(0.0119)		(0.0024)	(0.006)	(0.0018)		(0.0007)	(0.005)	(0.0005)
B+	EEq-HT	0.692	−0.0018	0.961	0.4343	1.000	−0.0039	0.950	0.1356	1.000	−0.0009	0.942	0.0419
			(0.0117)	(0.005)	(0.0105)		(0.0030)	(0.005)	(0.0022)		(0.0009)	(0.005)	(0.0007)
	EEq-Hájek	0.692	−0.0014	0.966	0.3844	1.000	−0.0051	0.960	0.1542	1.000	−0.0010	0.952	0.0493
			(0.0103)	(0.005)	(0.0087)		(0.0034)	(0.004)	(0.0025)		(0.0011)	(0.005)	(0.0008)
	TMLE-cw-HT	0.457	0.0031	0.778	0.4303	0.999	−0.0067	0.797	0.1872	1.000	−0.0010	0.797	0.0599
			(0.0142)	(0.014)	(0.0123)		(0.0042)	(0.009)	(0.0032)		(0.0013)	(0.009)	(0.0009)
	TMLE-cw-Hájek	0.457	0.0030	0.714	0.6119	0.999	−0.0082	0.812	0.2153	1.000	−0.0010	0.851	0.0654
			(0.0203)	(0.015)	(0.0304)		(0.0048)	(0.009)	(0.0042)		(0.0015)	(0.008)	(0.0010)
	TMLE-pw-HT	0.692	0.0016	0.825	0.4430	1.000	−0.0045	0.828	0.1794	1.000	−0.0013	0.825	0.0578
			(0.0119)	(0.010)	(0.0113)		(0.0040)	(0.008)	(0.0029)		(0.0013)	(0.008)	(0.0009)
	TMLE-pw-Hájek	0.692	0.0048	0.788	0.5711	1.000	−0.0066	0.848	0.2021	1.000	−0.0013	0.866	0.0637
			(0.0154)	(0.011)	(0.0278)		(0.0045)	(0.008)	(0.0035)		(0.0014)	(0.008)	(0.0010)
	TMLE-2eps-cw-Hájek	0.457	0.0060	0.663	0.8126	0.999	−0.0093	0.775	0.2415	1.000	−0.0009	0.823	0.0696
			(0.0269)	(0.016)	(0.0434)		(0.0054)	(0.009)	(0.0053)		(0.0016)	(0.009)	(0.0011)
	TMLE-2eps-pw-Hájek	0.692	0.0070	0.750	0.6690	1.000	−0.0071	0.818	0.2189	1.000	−0.0014	0.839	0.0679
			(0.0180)	(0.012)	(0.0363)		(0.0049)	(0.009)	(0.0038)		(0.0015)	(0.008)	(0.0011)
	DR-nt	0.030	−0.0060	0.950	0.2735	1.000	−0.0192	0.932	0.1648	1.000	−0.0154	0.946	0.0553
			(0.0356)	(0.028)	(0.0277)		(0.0037)	(0.006)	(0.0034)		(0.0012)	(0.005)	(0.0010)
	DR-ny	0.030	−0.0002	0.983	0.2536	1.000	−0.0208	0.937	0.1473	1.000	−0.0128	0.947	0.0502
			(0.0330)	(0.017)	(0.0240)		(0.0033)	(0.005)	(0.0028)		(0.0011)	(0.005)	(0.0009)

Table J.13, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.987	−0.0101	0.956	0.2532	1.000	0.0010	0.945	0.0810	1.000	0.0003	0.948	0.0244
			(0.0057)	(0.005)	(0.0094)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	EEq-Hájek	0.987	−0.0072	0.949	0.2184	1.000	0.0012	0.943	0.0808	1.000	0.0003	0.949	0.0244
			(0.0049)	(0.005)	(0.0056)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-cw-HT	0.960	−0.0067	0.927	0.2206	1.000	0.0016	0.934	0.0822	1.000	0.0003	0.946	0.0245
			(0.0050)	(0.006)	(0.0065)		(0.0018)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-cw-Hájek	0.960	−0.0068	0.908	0.2266	1.000	0.0016	0.936	0.0826	1.000	0.0003	0.947	0.0245
			(0.0052)	(0.007)	(0.0069)		(0.0018)	(0.005)	(0.0014)		(0.0005)	(0.005)	(0.0004)
	TMLE-pw-HT	0.987	−0.0060	0.940	0.2126	1.000	0.0012	0.940	0.0814	1.000	0.0003	0.947	0.0245
			(0.0048)	(0.005)	(0.0042)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-pw-Hájek	0.987	−0.0061	0.925	0.2173	1.000	0.0013	0.940	0.0817	1.000	0.0003	0.948	0.0245
			(0.0049)	(0.006)	(0.0053)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-2eps-cw-Hájek	0.960	−0.0059	0.896	0.2562	1.000	0.0019	0.927	0.0852	1.000	0.0002	0.947	0.0247
			(0.0058)	(0.007)	(0.0093)		(0.0019)	(0.006)	(0.0014)		(0.0006)	(0.005)	(0.0004)
	TMLE-2eps-pw-Hájek	0.987	−0.0053	0.915	0.2306	1.000	0.0014	0.936	0.0831	1.000	0.0003	0.947	0.0246
			(0.0052)	(0.006)	(0.0062)		(0.0019)	(0.005)	(0.0014)		(0.0005)	(0.005)	(0.0004)
	DR-nt	0.465	−0.0129	0.945	0.1973	1.000	−0.0051	0.945	0.0850	1.000	−0.0038	0.954	0.0258
			(0.0065)	(0.007)	(0.0053)		(0.0019)	(0.005)	(0.0014)		(0.0006)	(0.005)	(0.0004)
	DR-ny	0.465	−0.0101	0.954	0.1731	1.000	−0.0037	0.940	0.0791	1.000	−0.0027	0.952	0.0239
			(0.0057)	(0.007)	(0.0046)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
C2	EEq-HT	0.995	0.0089	0.957	0.2671	1.000	0.0000	0.939	0.0937	1.000	0.0013	0.943	0.0280
			(0.0060)	(0.005)	(0.0061)		(0.0021)	(0.005)	(0.0015)		(0.0006)	(0.005)	(0.0004)
	EEq-Hájek	0.995	0.0079	0.950	0.2469	1.000	−0.0001	0.945	0.0944	1.000	0.0013	0.950	0.0284
			(0.0055)	(0.005)	(0.0047)		(0.0021)	(0.005)	(0.0015)		(0.0006)	(0.005)	(0.0004)
	TMLE-cw-HT	0.977	0.0084	0.907	0.2516	1.000	−0.0002	0.920	0.0973	1.000	0.0013	0.938	0.0293
			(0.0057)	(0.007)	(0.0048)		(0.0022)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-Hájek	0.977	0.0082	0.900	0.2621	1.000	−0.0002	0.926	0.0983	1.000	0.0013	0.944	0.0295
			(0.0059)	(0.007)	(0.0054)		(0.0022)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-pw-HT	0.995	0.0085	0.930	0.2511	1.000	−0.0001	0.921	0.0972	1.000	0.0014	0.937	0.0291
			(0.0056)	(0.006)	(0.0047)		(0.0022)	(0.006)	(0.0015)		(0.0006)	(0.005)	(0.0005)
	TMLE-pw-Hájek	0.995	0.0087	0.925	0.2567	1.000	−0.0001	0.928	0.0981	1.000	0.0014	0.945	0.0293
			(0.0058)	(0.006)	(0.0050)		(0.0022)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-cw-Hájek	0.977	0.0082	0.872	0.2860	1.000	−0.0002	0.917	0.1016	1.000	0.0014	0.940	0.0302
			(0.0065)	(0.008)	(0.0066)		(0.0023)	(0.006)	(0.0017)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-pw-Hájek	0.995	0.0090	0.901	0.2712	1.000	−0.0001	0.916	0.1015	1.000	0.0014	0.938	0.0301
			(0.0061)	(0.007)	(0.0059)		(0.0023)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	DR-nt	0.683	−0.0006	0.908	0.2428	1.000	−0.0036	0.937	0.1005	1.000	−0.0033	0.948	0.0308
			(0.0066)	(0.008)	(0.0053)		(0.0022)	(0.005)	(0.0017)		(0.0007)	(0.005)	(0.0005)
	DR-ny	0.683	−0.0004	0.919	0.2122	1.000	−0.0030	0.942	0.0877	1.000	−0.0025	0.949	0.0271
			(0.0057)	(0.007)	(0.0045)		(0.0020)	(0.005)	(0.0014)		(0.0006)	(0.005)	(0.0004)

Table J.14: Dynamic design, binary outcome: ATT$(2, 4)$ by scenario: A, 0.175; B, 0.154; B+, 0.133; C1, 0.149; C2, 0.185; no cross-fitting (proposed procedures).

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	1.000	0.0047	0.929	0.2275	1.000	0.0005	0.955	0.0909	1.000	0.0000	0.946	0.0299
			(0.0051)	(0.006)	(0.0048)		(0.0020)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	EEq-Hájek	1.000	0.0052	0.931	0.2230	1.000	0.0006	0.954	0.0907	1.000	0.0000	0.946	0.0299
			(0.0050)	(0.006)	(0.0043)		(0.0020)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-HT	0.998	0.0055	0.898	0.2313	1.000	0.0009	0.944	0.0921	1.000	0.0000	0.939	0.0302
			(0.0052)	(0.007)	(0.0043)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-Hájek	0.998	0.0060	0.891	0.2427	1.000	0.0010	0.943	0.0927	1.000	0.0000	0.940	0.0303
			(0.0054)	(0.007)	(0.0049)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-pw-HT	1.000	0.0057	0.905	0.2322	1.000	0.0009	0.946	0.0919	1.000	0.0000	0.942	0.0301
			(0.0052)	(0.007)	(0.0046)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-pw-Hájek	1.000	0.0058	0.903	0.2399	1.000	0.0010	0.945	0.0923	1.000	0.0000	0.942	0.0302
			(0.0054)	(0.007)	(0.0051)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-cw-Hájek	0.998	0.0080	0.862	0.2869	1.000	0.0011	0.927	0.0967	1.000	0.0000	0.937	0.0306
			(0.0064)	(0.008)	(0.0070)		(0.0022)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-pw-Hájek	1.000	0.0069	0.881	0.2686	1.000	0.0013	0.933	0.0951	1.000	0.0000	0.941	0.0305
			(0.0060)	(0.007)	(0.0065)		(0.0021)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)

Table J.14, continued

Scen.	Procedure	$n = 100$				$n = 500$				$n = 5{,}000$			
		Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.993	−0.0003	0.933	0.2584	1.000	0.0038	0.944	0.1106	1.000	−0.0013	0.953	0.0341
			(0.0058)	(0.006)	(0.0051)		(0.0025)	(0.005)	(0.0018)		(0.0008)	(0.005)	(0.0005)
	EEq-Hájek	0.993	−0.0001	0.943	0.2537	1.000	0.0039	0.950	0.1130	1.000	−0.0013	0.955	0.0350
			(0.0057)	(0.005)	(0.0044)		(0.0025)	(0.005)	(0.0019)		(0.0008)	(0.005)	(0.0006)
	TMLE-cw-HT	0.975	−0.0007	0.840	0.2955	1.000	0.0042	0.889	0.1233	1.000	−0.0013	0.932	0.0374
			(0.0067)	(0.008)	(0.0059)		(0.0028)	(0.007)	(0.0021)		(0.0008)	(0.006)	(0.0006)
	TMLE-cw-Hájek	0.975	−0.0010	0.822	0.3792	1.000	0.0045	0.893	0.1287	1.000	−0.0013	0.940	0.0380
			(0.0086)	(0.009)	(0.0130)		(0.0029)	(0.007)	(0.0023)		(0.0009)	(0.005)	(0.0006)
	TMLE-pw-HT	0.993	0.0011	0.854	0.2976	1.000	0.0039	0.902	0.1203	1.000	−0.0013	0.937	0.0366
			(0.0067)	(0.008)	(0.0062)		(0.0027)	(0.007)	(0.0020)		(0.0008)	(0.005)	(0.0006)
	TMLE-pw-Hájek	0.993	0.0011	0.853	0.3409	1.000	0.0042	0.904	0.1246	1.000	−0.0013	0.942	0.0373
			(0.0077)	(0.008)	(0.0087)		(0.0028)	(0.007)	(0.0021)		(0.0008)	(0.005)	(0.0006)
	TMLE-2eps-cw-Hájek	0.975	0.0007	0.756	0.5523	1.000	0.0051	0.854	0.1450	1.000	−0.0014	0.926	0.0398
			(0.0125)	(0.010)	(0.0244)		(0.0032)	(0.008)	(0.0027)		(0.0009)	(0.006)	(0.0006)
	TMLE-2eps-pw-Hájek	0.993	0.0032	0.809	0.4319	1.000	0.0044	0.876	0.1358	1.000	−0.0013	0.935	0.0389
			(0.0097)	(0.009)	(0.0126)		(0.0030)	(0.007)	(0.0024)		(0.0009)	(0.006)	(0.0006)
B+	EEq-HT	0.938	−0.0036	0.921	0.2945	1.000	−0.0041	0.943	0.1280	1.000	−0.0010	0.941	0.0417
			(0.0068)	(0.006)	(0.0053)		(0.0029)	(0.005)	(0.0021)		(0.0009)	(0.005)	(0.0007)
	EEq-Hájek	0.938	−0.0040	0.956	0.3226	1.000	−0.0054	0.959	0.1486	1.000	−0.0011	0.951	0.0491
			(0.0075)	(0.005)	(0.0060)		(0.0033)	(0.004)	(0.0025)		(0.0011)	(0.005)	(0.0008)
	TMLE-cw-HT	0.852	−0.0037	0.678	0.4195	1.000	−0.0066	0.763	0.1860	1.000	−0.0011	0.795	0.0599
			(0.0102)	(0.011)	(0.0084)		(0.0042)	(0.010)	(0.0032)		(0.0013)	(0.009)	(0.0009)
	TMLE-cw-Hájek	0.852	−0.0101	0.605	0.8373	1.000	−0.0078	0.786	0.2211	1.000	−0.0011	0.850	0.0656
			(0.0203)	(0.012)	(0.0642)		(0.0049)	(0.009)	(0.0044)		(0.0015)	(0.008)	(0.0010)
	TMLE-pw-HT	0.938	−0.0037	0.705	0.4403	1.000	−0.0050	0.802	0.1789	1.000	−0.0014	0.820	0.0578
			(0.0102)	(0.011)	(0.0088)		(0.0040)	(0.009)	(0.0030)		(0.0013)	(0.009)	(0.0009)
	TMLE-pw-Hájek	0.938	−0.0006	0.682	0.7387	1.000	−0.0071	0.827	0.2065	1.000	−0.0014	0.862	0.0638
			(0.0171)	(0.011)	(0.0365)		(0.0046)	(0.008)	(0.0037)		(0.0014)	(0.008)	(0.0010)
	TMLE-2eps-cw-Hájek	0.852	−0.0151	0.510	1.2043	1.000	−0.0084	0.736	0.2528	1.000	−0.0011	0.820	0.0699
			(0.0292)	(0.012)	(0.0788)		(0.0057)	(0.010)	(0.0056)		(0.0016)	(0.009)	(0.0011)
	TMLE-2eps-pw-Hájek	0.938	0.0032	0.606	0.9448	1.000	−0.0077	0.789	0.2279	1.000	−0.0015	0.836	0.0682
			(0.0218)	(0.011)	(0.0444)		(0.0051)	(0.009)	(0.0042)		(0.0015)	(0.008)	(0.0011)
C1	EEq-HT	1.000	−0.0062	0.930	0.1916	1.000	0.0009	0.941	0.0793	1.000	0.0002	0.950	0.0243
			(0.0043)	(0.006)	(0.0035)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	EEq-Hájek	1.000	−0.0059	0.935	0.1913	1.000	0.0010	0.942	0.0795	1.000	0.0002	0.952	0.0243
			(0.0043)	(0.005)	(0.0036)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-cw-HT	0.998	−0.0065	0.900	0.2057	1.000	0.0015	0.931	0.0812	1.000	0.0002	0.947	0.0244
			(0.0046)	(0.007)	(0.0051)		(0.0018)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-cw-Hájek	0.998	−0.0058	0.892	0.2203	1.000	0.0015	0.934	0.0818	1.000	0.0002	0.951	0.0245
			(0.0049)	(0.007)	(0.0067)		(0.0018)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-pw-HT	1.000	−0.0081	0.904	0.2000	1.000	0.0010	0.934	0.0804	1.000	0.0002	0.947	0.0244
			(0.0045)	(0.007)	(0.0036)		(0.0018)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-pw-Hájek	1.000	−0.0075	0.901	0.2098	1.000	0.0011	0.936	0.0809	1.000	0.0002	0.948	0.0244
			(0.0047)	(0.007)	(0.0048)		(0.0018)	(0.005)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-2eps-cw-Hájek	0.998	−0.0061	0.874	0.2704	1.000	0.0019	0.926	0.0849	1.000	0.0002	0.949	0.0246
			(0.0061)	(0.007)	(0.0129)		(0.0019)	(0.006)	(0.0014)		(0.0006)	(0.005)	(0.0004)
	TMLE-2eps-pw-Hájek	1.000	−0.0095	0.885	0.2408	1.000	0.0012	0.934	0.0828	1.000	0.0002	0.948	0.0245
			(0.0054)	(0.007)	(0.0073)		(0.0019)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
C2	EEq-HT	1.000	0.0046	0.931	0.2123	1.000	−0.0004	0.936	0.0911	1.000	0.0013	0.943	0.0279
			(0.0047)	(0.006)	(0.0037)		(0.0020)	(0.005)	(0.0014)		(0.0006)	(0.005)	(0.0004)
	EEq-Hájek	1.000	0.0048	0.939	0.2140	1.000	−0.0005	0.945	0.0922	1.000	0.0013	0.948	0.0283
			(0.0048)	(0.005)	(0.0037)		(0.0021)	(0.005)	(0.0015)		(0.0006)	(0.005)	(0.0004)
	TMLE-cw-HT	0.999	0.0057	0.875	0.2311	1.000	−0.0008	0.916	0.0958	1.000	0.0013	0.935	0.0292
			(0.0052)	(0.007)	(0.0042)		(0.0021)	(0.006)	(0.0015)		(0.0007)	(0.006)	(0.0005)
	TMLE-cw-Hájek	0.999	0.0055	0.877	0.2467	1.000	−0.0009	0.925	0.0970	1.000	0.0013	0.944	0.0295
			(0.0055)	(0.007)	(0.0048)		(0.0022)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-pw-HT	1.000	0.0048	0.892	0.2298	1.000	−0.0006	0.918	0.0955	1.000	0.0013	0.934	0.0290
			(0.0051)	(0.007)	(0.0042)		(0.0021)	(0.006)	(0.0015)		(0.0006)	(0.006)	(0.0005)
	TMLE-pw-Hájek	1.000	0.0050	0.895	0.2403	1.000	−0.0007	0.925	0.0966	1.000	0.0013	0.943	0.0293
			(0.0054)	(0.007)	(0.0045)		(0.0022)	(0.006)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-cw-Hájek	0.999	0.0055	0.845	0.2807	1.000	−0.0011	0.910	0.1010	1.000	0.0013	0.941	0.0302
			(0.0063)	(0.008)	(0.0062)		(0.0023)	(0.006)	(0.0017)		(0.0007)	(0.005)	(0.0005)
	TMLE-2eps-pw-Hájek	1.000	0.0052	0.868	0.2667	1.000	−0.0008	0.913	0.1008	1.000	0.0014	0.939	0.0301
			(0.0060)	(0.008)	(0.0055)		(0.0023)	(0.006)	(0.0016)		(0.0007)	(0.005)	(0.0005)

Table J.15: Dynamic design, binary outcome: ES(1) by scenario: A, 0.184; B, 0.168; B+, 0.152; C1, 0.162; C2, 0.193; five-fold cross-fitting.

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.953	0.0088	0.961	0.1945	1.000	0.0009	0.952	0.0613	1.000	−0.0004	0.946	0.0188
			(0.0045)	(0.004)	(0.0040)		(0.0014)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.953	0.0059	0.953	0.1595	1.000	0.0009	0.950	0.0606	1.000	−0.0004	0.946	0.0188
			(0.0037)	(0.005)	(0.0029)		(0.0014)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-HT	0.926	0.0048	0.949	0.1481	1.000	0.0010	0.951	0.0599	1.000	−0.0004	0.946	0.0188
			(0.0034)	(0.005)	(0.0025)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.926	0.0049	0.933	0.1497	1.000	0.0010	0.949	0.0598	1.000	−0.0004	0.947	0.0188
			(0.0035)	(0.006)	(0.0027)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-HT	0.945	0.0040	0.955	0.1494	1.000	0.0010	0.952	0.0601	1.000	−0.0004	0.947	0.0188
			(0.0034)	(0.005)	(0.0027)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.945	0.0041	0.940	0.1506	1.000	0.0010	0.949	0.0600	1.000	−0.0004	0.946	0.0188
			(0.0035)	(0.005)	(0.0028)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.926	0.0040	0.924	0.1567	1.000	0.0010	0.948	0.0599	1.000	−0.0004	0.948	0.0188
			(0.0036)	(0.006)	(0.0032)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-pw-Hájek	0.945	0.0029	0.933	0.1535	1.000	0.0010	0.949	0.0600	1.000	−0.0004	0.946	0.0188
			(0.0035)	(0.006)	(0.0033)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	DR-nt	0.479	−0.0026	0.944	0.1345	1.000	0.0005	0.952	0.0550	1.000	−0.0020	0.948	0.0178
			(0.0043)	(0.007)	(0.0031)		(0.0012)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	DR-ny	0.479	0.0004	0.946	0.1242	1.000	0.0002	0.950	0.0528	1.000	−0.0010	0.946	0.0167
			(0.0040)	(0.007)	(0.0028)		(0.0012)	(0.005)	(0.0008)		(0.0004)	(0.005)	(0.0003)
B	EEq-HT	0.791	0.0050	0.963	0.2398	1.000	0.0009	0.945	0.0731	1.000	−0.0005	0.948	0.0209
			(0.0060)	(0.005)	(0.0055)		(0.0016)	(0.005)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	EEq-Hájek	0.791	0.0035	0.953	0.1841	1.000	0.0008	0.947	0.0719	1.000	−0.0005	0.948	0.0210
			(0.0046)	(0.005)	(0.0036)		(0.0016)	(0.005)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-cw-HT	0.722	0.0054	0.929	0.1740	1.000	0.0006	0.934	0.0721	1.000	−0.0005	0.946	0.0212
			(0.0046)	(0.007)	(0.0035)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.722	0.0059	0.906	0.1788	1.000	0.0005	0.929	0.0726	1.000	−0.0005	0.946	0.0213
			(0.0047)	(0.008)	(0.0037)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-pw-HT	0.770	0.0048	0.941	0.1767	1.000	0.0006	0.937	0.0719	1.000	−0.0005	0.947	0.0212
			(0.0045)	(0.006)	(0.0037)		(0.0016)	(0.005)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.770	0.0055	0.919	0.1808	1.000	0.0006	0.934	0.0722	1.000	−0.0005	0.947	0.0212
			(0.0046)	(0.007)	(0.0039)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.722	0.0060	0.891	0.2022	1.000	0.0004	0.921	0.0758	1.000	−0.0005	0.944	0.0216
			(0.0053)	(0.008)	(0.0053)		(0.0017)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-2eps-pw-Hájek	0.770	0.0073	0.906	0.1925	1.000	0.0005	0.925	0.0742	1.000	−0.0005	0.945	0.0215
			(0.0049)	(0.007)	(0.0047)		(0.0017)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0004)
	DR-nt	0.150	−0.0026	0.920	0.1567	1.000	−0.0038	0.941	0.0653	1.000	−0.0043	0.946	0.0208
			(0.0091)	(0.016)	(0.0062)		(0.0015)	(0.005)	(0.0011)		(0.0005)	(0.005)	(0.0003)
	DR-ny	0.150	−0.0030	0.923	0.1382	1.000	−0.0014	0.945	0.0607	1.000	−0.0020	0.944	0.0185
			(0.0080)	(0.015)	(0.0055)		(0.0014)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
B+	EEq-HT	0.373	0.0013	0.965	0.3059	1.000	−0.0006	0.950	0.0940	1.000	−0.0001	0.942	0.0285
			(0.0112)	(0.007)	(0.0101)		(0.0021)	(0.005)	(0.0015)		(0.0006)	(0.005)	(0.0004)
	EEq-Hájek	0.373	−0.0022	0.954	0.2319	1.000	−0.0003	0.956	0.0966	1.000	−0.0001	0.946	0.0302
			(0.0085)	(0.008)	(0.0076)		(0.0022)	(0.005)	(0.0016)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-HT	0.282	0.0001	0.888	0.2361	0.998	0.0004	0.874	0.1063	1.000	−0.0001	0.892	0.0333
			(0.0099)	(0.013)	(0.0092)		(0.0024)	(0.007)	(0.0017)		(0.0007)	(0.007)	(0.0005)
	TMLE-cw-Hájek	0.282	−0.0010	0.856	0.2713	0.998	0.0007	0.879	0.1124	1.000	−0.0001	0.912	0.0345
			(0.0114)	(0.015)	(0.0140)		(0.0025)	(0.007)	(0.0019)		(0.0008)	(0.006)	(0.0006)
	TMLE-pw-HT	0.343	−0.0015	0.913	0.2364	1.000	0.0007	0.891	0.1047	1.000	−0.0002	0.897	0.0328
			(0.0090)	(0.011)	(0.0083)		(0.0023)	(0.007)	(0.0017)		(0.0007)	(0.007)	(0.0005)
	TMLE-pw-Hájek	0.343	−0.0019	0.872	0.2631	1.000	0.0009	0.896	0.1103	1.000	−0.0001	0.916	0.0342
			(0.0101)	(0.013)	(0.0095)		(0.0025)	(0.007)	(0.0019)		(0.0008)	(0.006)	(0.0005)
	TMLE-2eps-cw-Hájek	0.282	−0.0043	0.812	0.3367	0.998	0.0013	0.849	0.1235	1.000	−0.0001	0.897	0.0360
			(0.0142)	(0.016)	(0.0207)		(0.0028)	(0.008)	(0.0022)		(0.0008)	(0.007)	(0.0006)
	TMLE-2eps-pw-Hájek	0.343	−0.0026	0.854	0.3027	1.000	0.0013	0.871	0.1185	1.000	−0.0002	0.901	0.0357
			(0.0116)	(0.013)	(0.0123)		(0.0026)	(0.007)	(0.0021)		(0.0008)	(0.007)	(0.0006)
	DR-nt	0.029	−0.0004	0.931	0.1694	1.000	−0.0091	0.921	0.1018	1.000	−0.0084	0.941	0.0325
			(0.0224)	(0.033)	(0.0120)		(0.0023)	(0.006)	(0.0020)		(0.0007)	(0.005)	(0.0006)
	DR-ny	0.029	−0.0125	0.845	0.2013	1.000	−0.0070	0.932	0.0871	1.000	−0.0044	0.948	0.0286
			(0.0266)	(0.048)	(0.0214)		(0.0019)	(0.006)	(0.0015)		(0.0006)	(0.005)	(0.0005)

Table J.15, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
C1	EEq-HT	0.952	0.0048	0.963	0.1618	1.000	−0.0009	0.966	0.0507	1.000	0.0000	0.950	0.0160
			(0.0037)	(0.004)	(0.0034)		(0.0011)	(0.004)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	EEq-Hájek	0.952	0.0028	0.955	0.1352	1.000	−0.0008	0.963	0.0504	1.000	0.0000	0.950	0.0160
			(0.0031)	(0.005)	(0.0025)		(0.0011)	(0.004)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	TMLE-cw-HT	0.921	0.0004	0.948	0.1299	1.000	−0.0007	0.963	0.0505	1.000	0.0000	0.950	0.0161
			(0.0030)	(0.005)	(0.0023)		(0.0011)	(0.004)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	TMLE-cw-Hájek	0.921	0.0004	0.934	0.1309	1.000	−0.0007	0.961	0.0505	1.000	0.0000	0.950	0.0161
			(0.0031)	(0.006)	(0.0023)		(0.0011)	(0.004)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	TMLE-pw-HT	0.942	0.0010	0.953	0.1301	1.000	−0.0009	0.964	0.0504	1.000	0.0000	0.950	0.0160
			(0.0030)	(0.005)	(0.0023)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.942	0.0011	0.941	0.1306	1.000	−0.0008	0.963	0.0504	1.000	0.0000	0.949	0.0160
			(0.0030)	(0.005)	(0.0023)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.921	−0.0004	0.929	0.1388	1.000	−0.0007	0.960	0.0511	1.000	0.0000	0.952	0.0161
			(0.0032)	(0.006)	(0.0026)		(0.0011)	(0.004)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.942	0.0003	0.939	0.1347	1.000	−0.0009	0.963	0.0507	1.000	0.0000	0.951	0.0161
			(0.0031)	(0.005)	(0.0025)		(0.0011)	(0.004)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	DR-nt	0.465	0.0002	0.940	0.1221	1.000	−0.0040	0.951	0.0496	1.000	−0.0019	0.952	0.0155
			(0.0040)	(0.008)	(0.0030)		(0.0011)	(0.005)	(0.0008)		(0.0003)	(0.005)	(0.0002)
	DR-ny	0.465	0.0026	0.940	0.1127	1.000	−0.0020	0.952	0.0467	1.000	−0.0007	0.951	0.0147
			(0.0037)	(0.008)	(0.0026)		(0.0010)	(0.005)	(0.0007)		(0.0003)	(0.005)	(0.0002)
C2	EEq-HT	0.953	−0.0017	0.964	0.1717	1.000	0.0013	0.946	0.0592	1.000	0.0009	0.951	0.0175
			(0.0039)	(0.004)	(0.0031)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.953	−0.0017	0.952	0.1539	1.000	0.0013	0.948	0.0591	1.000	0.0009	0.953	0.0176
			(0.0035)	(0.005)	(0.0026)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-HT	0.933	−0.0014	0.939	0.1503	1.000	0.0013	0.943	0.0595	1.000	0.0009	0.949	0.0178
			(0.0035)	(0.006)	(0.0027)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.933	−0.0009	0.921	0.1527	1.000	0.0012	0.944	0.0597	1.000	0.0009	0.951	0.0178
			(0.0035)	(0.006)	(0.0027)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-HT	0.951	−0.0013	0.951	0.1512	1.000	0.0013	0.943	0.0595	1.000	0.0009	0.949	0.0177
			(0.0035)	(0.005)	(0.0027)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.951	−0.0012	0.932	0.1533	1.000	0.0013	0.943	0.0596	1.000	0.0009	0.951	0.0178
			(0.0035)	(0.006)	(0.0028)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.933	−0.0008	0.914	0.1620	1.000	0.0012	0.939	0.0607	1.000	0.0009	0.948	0.0180
			(0.0038)	(0.006)	(0.0032)		(0.0014)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-pw-Hájek	0.951	−0.0012	0.920	0.1600	1.000	0.0012	0.938	0.0605	1.000	0.0009	0.949	0.0180
			(0.0037)	(0.006)	(0.0031)		(0.0014)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	DR-nt	0.683	−0.0040	0.926	0.1388	1.000	−0.0004	0.946	0.0576	1.000	−0.0014	0.954	0.0175
			(0.0038)	(0.007)	(0.0030)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	DR-ny	0.683	−0.0047	0.933	0.1227	1.000	0.0004	0.950	0.0510	1.000	−0.0003	0.952	0.0157
			(0.0033)	(0.007)	(0.0023)		(0.0011)	(0.005)	(0.0008)		(0.0004)	(0.005)	(0.0002)

Table J.16: Dynamic design, binary outcome: ES(1) by scenario: A, 0.184; B, 0.168; B+, 0.152; C1, 0.162; C2, 0.193; no cross-fitting (proposed procedures).

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
A	EEq-HT	0.995	0.0051	0.946	0.1394	1.000	0.0008	0.950	0.0595	1.000	−0.0004	0.946	0.0188
			(0.0031)	(0.005)	(0.0025)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.995	0.0050	0.948	0.1369	1.000	0.0008	0.949	0.0594	1.000	−0.0004	0.946	0.0188
			(0.0031)	(0.005)	(0.0023)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-HT	0.991	0.0055	0.930	0.1404	1.000	0.0010	0.946	0.0594	1.000	−0.0004	0.946	0.0188
			(0.0032)	(0.006)	(0.0024)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.991	0.0056	0.926	0.1442	1.000	0.0010	0.945	0.0595	1.000	−0.0004	0.946	0.0188
			(0.0032)	(0.006)	(0.0026)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-HT	0.994	0.0054	0.936	0.1405	1.000	0.0009	0.946	0.0595	1.000	−0.0004	0.947	0.0188
			(0.0031)	(0.006)	(0.0024)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.994	0.0060	0.931	0.1458	1.000	0.0009	0.945	0.0595	1.000	−0.0004	0.946	0.0188
			(0.0033)	(0.006)	(0.0033)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.991	0.0065	0.906	0.1605	1.000	0.0010	0.941	0.0603	1.000	−0.0004	0.947	0.0188
			(0.0036)	(0.007)	(0.0031)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-pw-Hájek	0.994	0.0069	0.912	0.1610	1.000	0.0010	0.944	0.0602	1.000	−0.0004	0.947	0.0188
			(0.0036)	(0.006)	(0.0048)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)

Table J.16, continued

		$n = 100$				$n = 500$				$n = 5{,}000$			
Scen.	Procedure	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE	Compl.	Bias	Cov.	RMSE
B	EEq-HT	0.972	−0.0016	0.935	0.1610	1.000	0.0007	0.942	0.0696	1.000	−0.0005	0.949	0.0208
			(0.0037)	(0.006)	(0.0027)		(0.0016)	(0.005)	(0.0011)		(0.0005)	(0.005)	(0.0003)
	EEq-Hájek	0.972	−0.0017	0.940	0.1595	1.000	0.0006	0.943	0.0698	1.000	−0.0005	0.950	0.0209
			(0.0036)	(0.005)	(0.0026)		(0.0016)	(0.005)	(0.0011)		(0.0005)	(0.005)	(0.0003)
	TMLE-cw-HT	0.944	−0.0001	0.886	0.1736	1.000	0.0005	0.920	0.0716	1.000	−0.0005	0.944	0.0212
			(0.0040)	(0.007)	(0.0030)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.944	−0.0002	0.875	0.1929	1.000	0.0005	0.920	0.0725	1.000	−0.0005	0.944	0.0212
			(0.0044)	(0.008)	(0.0038)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-pw-HT	0.958	−0.0013	0.891	0.1751	1.000	0.0005	0.926	0.0712	1.000	−0.0005	0.945	0.0211
			(0.0040)	(0.007)	(0.0032)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.958	−0.0016	0.887	0.1938	1.000	0.0005	0.927	0.0720	1.000	−0.0005	0.945	0.0212
			(0.0044)	(0.007)	(0.0047)		(0.0016)	(0.006)	(0.0012)		(0.0005)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.944	−0.0011	0.837	0.2532	1.000	0.0005	0.905	0.0774	1.000	−0.0005	0.943	0.0216
			(0.0058)	(0.008)	(0.0066)		(0.0017)	(0.007)	(0.0013)		(0.0005)	(0.005)	(0.0004)
	TMLE-2eps-pw-Hájek	0.958	−0.0016	0.851	0.2475	1.000	0.0005	0.915	0.0758	1.000	−0.0005	0.943	0.0215
			(0.0057)	(0.008)	(0.0095)		(0.0017)	(0.006)	(0.0013)		(0.0005)	(0.005)	(0.0004)
B+	EEq-HT	0.818	−0.0026	0.938	0.1976	1.000	−0.0009	0.946	0.0879	1.000	−0.0001	0.940	0.0283
			(0.0049)	(0.006)	(0.0040)		(0.0020)	(0.005)	(0.0014)		(0.0006)	(0.005)	(0.0004)
	EEq-Hájek	0.818	−0.0024	0.951	0.2036	1.000	−0.0007	0.956	0.0929	1.000	−0.0001	0.945	0.0301
			(0.0050)	(0.005)	(0.0041)		(0.0021)	(0.005)	(0.0015)		(0.0007)	(0.005)	(0.0005)
	TMLE-cw-HT	0.737	0.0020	0.801	0.2481	1.000	0.0000	0.856	0.1056	1.000	−0.0001	0.891	0.0333
			(0.0065)	(0.010)	(0.0054)		(0.0024)	(0.008)	(0.0017)		(0.0007)	(0.007)	(0.0005)
	TMLE-cw-Hájek	0.737	0.0074	0.741	0.3958	1.000	0.0003	0.864	0.1140	1.000	−0.0001	0.910	0.0345
			(0.0103)	(0.011)	(0.0277)		(0.0025)	(0.008)	(0.0020)		(0.0008)	(0.006)	(0.0006)
	TMLE-pw-HT	0.777	0.0003	0.813	0.2503	1.000	0.0002	0.874	0.1042	1.000	−0.0002	0.895	0.0328
			(0.0064)	(0.010)	(0.0054)		(0.0023)	(0.007)	(0.0018)		(0.0007)	(0.007)	(0.0005)
	TMLE-pw-Hájek	0.777	0.0032	0.774	0.3562	1.000	0.0004	0.883	0.1117	1.000	−0.0002	0.915	0.0342
			(0.0090)	(0.011)	(0.0150)		(0.0025)	(0.007)	(0.0019)		(0.0008)	(0.006)	(0.0005)
	TMLE-2eps-cw-Hájek	0.737	0.0097	0.691	0.5609	1.000	0.0007	0.824	0.1287	1.000	−0.0001	0.893	0.0362
			(0.0146)	(0.012)	(0.0345)		(0.0029)	(0.009)	(0.0023)		(0.0008)	(0.007)	(0.0006)
	TMLE-2eps-pw-Hájek	0.777	0.0045	0.717	0.4713	1.000	0.0008	0.850	0.1234	1.000	−0.0002	0.897	0.0359
			(0.0120)	(0.011)	(0.0208)		(0.0028)	(0.008)	(0.0022)		(0.0008)	(0.007)	(0.0006)
C1	EEq-HT	0.996	0.0016	0.941	0.1243	1.000	−0.0009	0.962	0.0499	1.000	0.0000	0.951	0.0160
			(0.0028)	(0.005)	(0.0022)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	EEq-Hájek	0.996	0.0013	0.943	0.1230	1.000	−0.0009	0.961	0.0499	1.000	0.0000	0.951	0.0160
			(0.0028)	(0.005)	(0.0021)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-cw-HT	0.991	0.0008	0.931	0.1259	1.000	−0.0008	0.959	0.0502	1.000	0.0000	0.951	0.0160
			(0.0028)	(0.006)	(0.0021)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-cw-Hájek	0.991	0.0007	0.927	0.1287	1.000	−0.0008	0.959	0.0502	1.000	0.0000	0.951	0.0160
			(0.0029)	(0.006)	(0.0022)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-pw-HT	0.992	0.0008	0.932	0.1259	1.000	−0.0010	0.960	0.0501	1.000	0.0000	0.951	0.0160
			(0.0028)	(0.006)	(0.0022)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-pw-Hájek	0.992	0.0007	0.932	0.1283	1.000	−0.0009	0.960	0.0501	1.000	0.0000	0.951	0.0160
			(0.0029)	(0.006)	(0.0022)		(0.0011)	(0.004)	(0.0007)		(0.0004)	(0.005)	(0.0002)
	TMLE-2eps-cw-Hájek	0.991	−0.0001	0.916	0.1444	1.000	−0.0008	0.957	0.0510	1.000	0.0000	0.951	0.0161
			(0.0032)	(0.006)	(0.0028)		(0.0011)	(0.005)	(0.0008)		(0.0004)	(0.005)	(0.0002)
	TMLE-2eps-pw-Hájek	0.992	0.0002	0.920	0.1407	1.000	−0.0010	0.958	0.0507	1.000	0.0000	0.951	0.0161
			(0.0032)	(0.006)	(0.0027)		(0.0011)	(0.005)	(0.0008)		(0.0004)	(0.005)	(0.0002)
C2	EEq-HT	0.995	−0.0024	0.938	0.1370	1.000	0.0013	0.944	0.0577	1.000	0.0009	0.952	0.0175
			(0.0031)	(0.005)	(0.0023)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	EEq-Hájek	0.995	−0.0022	0.941	0.1370	1.000	0.0013	0.947	0.0578	1.000	0.0009	0.954	0.0176
			(0.0031)	(0.005)	(0.0022)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-HT	0.995	−0.0018	0.911	0.1421	1.000	0.0013	0.937	0.0585	1.000	0.0009	0.948	0.0177
			(0.0032)	(0.006)	(0.0023)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-cw-Hájek	0.995	−0.0018	0.906	0.1477	1.000	0.0013	0.943	0.0588	1.000	0.0009	0.950	0.0178
			(0.0033)	(0.007)	(0.0025)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-HT	0.995	−0.0020	0.914	0.1427	1.000	0.0013	0.938	0.0585	1.000	0.0009	0.949	0.0177
			(0.0032)	(0.006)	(0.0024)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-pw-Hájek	0.995	−0.0021	0.913	0.1471	1.000	0.0013	0.941	0.0587	1.000	0.0009	0.950	0.0178
			(0.0033)	(0.006)	(0.0025)		(0.0013)	(0.005)	(0.0009)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-cw-Hájek	0.995	−0.0017	0.890	0.1646	1.000	0.0012	0.936	0.0602	1.000	0.0009	0.949	0.0181
			(0.0037)	(0.007)	(0.0034)		(0.0013)	(0.005)	(0.0010)		(0.0004)	(0.005)	(0.0003)
	TMLE-2eps-pw-Hájek	0.995	−0.0021	0.896	0.1608	1.000	0.0012	0.935	0.0600	1.000	0.0009	0.949	0.0180
			(0.0036)	(0.007)	(0.0030)		(0.0013)	(0.006)	(0.0010)		(0.0004)	(0.005)	(0.0003)